



Nova Lake-S Processor

External Design Specification, Volume 1 of 2

Rev. 1.0

November 2025

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Revision History

Document Number	Revision Number	Description	Revision Date
844276	0.4	Initial Release	March 2025
844276	0.5	Updated Introduction on page 15 Processor and Device IDs on page 26 • Updated Table 8 on page 29 Pin Straps on page 37 • Updated Pin Straps on page 37 Security Technologies on page 40 • Updated Intel® Total Storage Encryption (Intel® TSE) on page 46 and Intel® Partner Security Engine on page 47 Platform Environmental Control Interface (PECI) on page 56 • Updated Nova Lake Peci over eSPI Implementation Guide document number Updated System Clocks on page 153 Power Management on page 62 • Removed xDCI from Table 22 on page 72 Graphics on page 180 • Updated Table 81 on page 181 Display on page 184 • Updated Table 83 on page 184, General Capabilities on page 187 USB-C Sub System on page 166 • Updated Table 74 on page 167 and notes in IO Signal Planes and States on page 170	April 2025
844276	0.7	Introduction on page 15 • Updated MH to HUB, Form factors to Processor Series • Updated Related Documents on page 25 Processor and Device IDs on page 26 • Updated CPUID and Stepping ID for NVL-S 16+32 and NVL-S 8+16 in Table 4 on page 26 Pin Straps on page 37 • Added GPP_SD13 and GPP_SD14 • Updated GPP_SA20 sampling description Intel® Neural Processing Unit (Intel® NPU) on page 57 • Updated — The NPU components in Deep Learning (Neural Compute Engine) on page 57 and Figure 4 on page 58, — Number of Neural Compute Tiles in Deep Learning Accelerators (NCE) on page 58 and Feature Set on page 59 — Number of tile configuration in NCE Tile on page 59 — The Supported Data types in Feature Set on page 60 Power Management on page 62 • Updated EPD_ON signals and their description in Signal Description on page 82 and IA Core Level 0, Level 1, Level 2 and Level 3 Caches on page 73 Thermal Management on page 114	August 2025

continued...

Document Number	Revision Number	Description	Revision Date
		• Updated maximum temperature range in Table 52 on page 131 and Table 57 on page 135 • Graphics on page 180 • Updated BT2020 into the encoding support in Hardware Accelerated Video Encode on page 181 • Removed RawCodec support model from Hardware Accelerated Video Decode on page 180 • General Purpose Input and Output on page 193 • Updated Signal Description on page 195 • Removed Native Function Signals Supporting Dynamic Termination Override table • Direct eSPI on page 199 • Added PECI over eSPI on page 201 • Testability and Monitoring on page 203 • Added a note • Added Electrical Specifications on page 86	
844276	0.9	Introduction on page 15 • Updated Table 1 • Updated Figure 1 • Updated Supported Technologies to include support for AVX10.2 • Updated Table 2 in Terminology and Special Marks • Updated Related Documents • Memory Mapping • Updated Table 12 in Memory Map • Intel® Neural Processing Unit (Intel® NPU) • Updated Deep Learning Acceleration (NCE) • Updated Feature Set List in Data Processing Unit (DPU) • Power Management • Updated Table 16 • Updated Intel® Smart 2D Display Technology (Intel® S2DDT) in Display Power Savings Technologies • Updated Table 22 in TCSS Power State • Updated Intel® Advanced Vector Extensions • Updated Table in Signal Description • Power Delivery • Updated Current Excursion Protection • Electrical Specifications on page 86 • Updated Processor Power Rails DC Specifications • Updated Table 26 in VCC_CORE DC Specifications • Updated Table 27 in VCCGT DC Specifications • Updated Processor AC Timing Waveforms • Updated Direct eSPI Timing Specifications • Thermal Management on page 114 • Updated Table 47 in Assured Power (CTDP Modes) • Updated Voltage Regulator Protection using FORCEPR# • Memory • Updated Table 58 in Processor SKU Support Matrix • USB-C Sub System on page 166 • Added support for TBT5 • Removed Note in USB Type-C Signals • Graphics • Updated Processor Graphics • Display on page 184	October 2025

continued...

Document Number	Revision Number	Description	Revision Date
		• Removed points in General Capabilities • Removed point in Multi-Stream Transport • Removed Display Resolutions and Link Bandwidth for Multi-Stream Transport Calculations table in Multi-Stream Transport • Updated Table 83 in Multi-Stream Transport • Updated High-Definition Multimedia Interface (HDMI*) Direct eSPI on page 199 • Updated Processor-PCH eSPI Return Clock Support Miscellaneous Signals • Removed CPU_ID in Ground and Reserved Signals	
844276	1.0	Updated USB-C Sub System • Updated USB-C Sub-System General Capabilities • Updated USB Type-C* Port Configuration table • Updated USB4v2 Router • Removed USB Type-C* Non-Supported Lane Configuration table Thermal Management • Added Processor Frequency vs. Temperature section • Updated Operating Temperature Specifications (S Processor Line) table	November 2025

1.0 Introduction

This document is intended for Original Equipment Manufacturers (OEMs), Original Design Manufacturers (ODMs) and BIOS vendors creating products based on the Nova Lake Desktop Processor.

This document assumes a working knowledge of the vocabulary and principles of interfaces and architectures such as PCI Express* (PCIe*), Universal Serial Bus (USB), eXtensible Host Controller Interface (xHCI), and so on.

This document abbreviates buses as Bn, devices as Dn and functions as Fn. For example, Device 31 Function 0 is abbreviated as D31:F0, Bus 1 Device 8 Function 0 is abbreviated as B1:D8:F0. Generally, the bus number will not be used, and can be considered to be Bus 0.

The Nova Lake (NVL) desktop processor is a 64-bit, multi-core processor.

- The Nova Lake processor series is offered in a single package platform that includes the Compute, HUB, PCD, and GPU tiles on the same package.

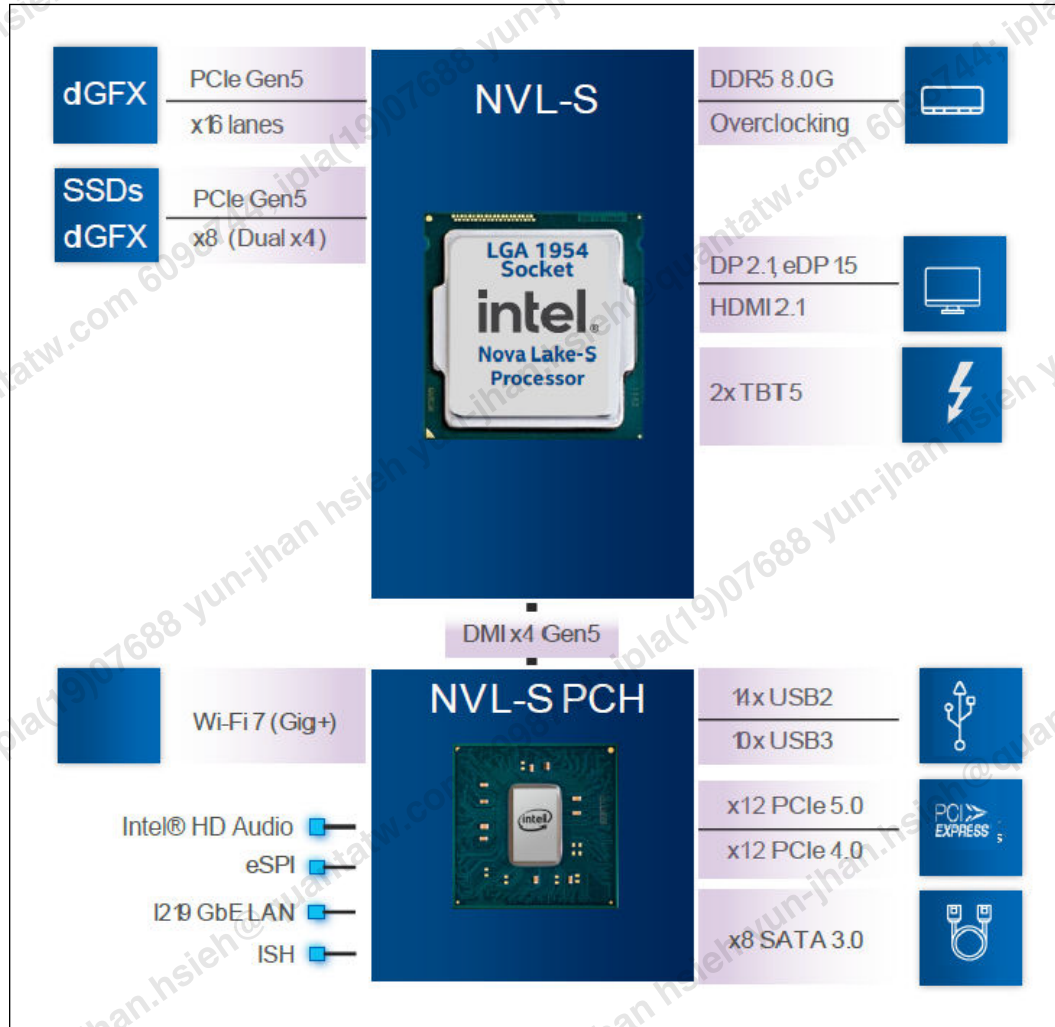
The following table describes the offered Nova Lake desktop processor series:

Table 1. NVL Processor Series¹

Processor Series ¹	Package	Processor Base Power ^{3, 4}	Processor Max P-cores ²	Processor Max E-cores ²	LP E-Cores ²	Graphics Configuration Max Xe3 - cores	Platform Type
NVL-S 52C DS	LGA1954	175W	16	32	4	2	2-Chip
NVL-S 28C DS	LGA1954	125W	8	16	4	2	2-Chip
NVL-S 28C	LGA1954	65W/125W	8	16	4	2	2-Chip
NVL-S 16C	LGA1954	35W/65W	4	8	4	2	2-Chip
NVL-S 8C	LGA1954	35W/65W	4	0	4	2	2-Chip

Notes: 1. Processor Series offering may change.
 2. Core count refers to native die with maximum physical cores. Some SKU may have reduced functional cores.
 3. For additional Processor Base Power Configurations, refer to [Processor Base Power Thermal and Power Specifications](#) on page 127, for adjustment to the Processor Base Power required to preserve base frequency associated with the sustained long-term thermal capability.
 4. Processor Base Power workload does not reflect I/O connectivity cases such as Thunderbolt. For power adders estimation for various I/O connectivity scenarios, refer to Nova Lake-S Platform Design Guide (#830661).

Figure 1. NVL-S Processor Line Platform Diagram



NOTE

Not all processor or PCH interfaces and features are present in all processor or PCH SKUs. The presence of various interfaces and features will be indicated within the relevant sections and tables.

1.1 Processor Volatility Statement

The processor families do not retain any end-user data when powered down and/or when the processor is physically removed.

NOTE

Powered down refers to the state which all processor power rails are off.

1.2 Package Support

1.2.1 Package Support

The S-Processor line is available in the following package:

LGA1954

- 45 mm X 37.5 mm
- Package Z-Height = 4.454 ± 0.125 mm

1.3 Supported Technologies

- PCI Express* (PCIe*)
- Xe Graphics Core Based Processor Graphics
- Display
 - DisplayPort* 2.1 (DP* 2.1)
 - Embedded DisplayPort* 1.4 (eDP* 1.4)
 - Embedded DisplayPort* 1.5 (eDP* 1.5)
 - High-Definition Multimedia Interface* 2.1 (HDMI* 2.1)
- Intel® Neural Processing Unit (Intel® NPU)
- Memory
 - DDR5
 - Error Code Correction (ECC)
 - In Band Error Code Correction (IB ECC)
- Platform Environmental Control Interface (PECI)
- Intel® Volume Management Device (Intel® VMD)
- Integrated Clock Controller (ICC)/Integrated Reference Clock PLL
- Real Time Clock Controller (RTCC)
- General Purpose Input Output (GPIO)
- Virtualization
 - Intel® Virtualization Technology (Intel® VT-x)
 - Intel® Virtualization Technology for Directed I/O (Intel® VT-d)
 - Intel® APIC Virtualization Technology (Intel® APICv)
- Security Technologies
 - Intel® Trusted Execution Technology (Intel® TXT)
 - Intel® Converged Boot Guard and Intel® Trusted Execution Technology (Intel® CBnT)
 - Intel® Secure Key
 - Execute Disable Bit
 - Intel® Supervisor Mode Execution Protection (Intel® SMEP)

- Intel® Supervisor Mode Access Protection (Intel® SMAP)
- User Mode Instruction Prevention (UMIP)
- Read Processor ID (RDPID)
- Intel® Total Memory Encryption (Intel® TME)
- Intel® Multi-Key Total Memory Encryption (Intel® MK-TME)
- Intel® Control-flow Enforcement Technology (Intel® CET)
- Linear Address Space Separation (LASS)
- KeyLocker Technology
- Intel® System Resources Defense and Intel® System Security Report
- Intel® BIOS Guard
- Intel® Boot Guard
- Intel® Platform Trust Technology (Intel® PTT)
- Security Technologies - Crypto Acceleration Instructions
 - Intel® Advanced Encryption Standard New Instructions (Intel® AES-NI)
 - Perform Carry-Less Multiplication Quad Word Instruction (PCLMULQDQ)
 - Intel® Secure Hash Algorithm - 512 (Intel® SHA - 512)
 - Intel® Secure Hash Algorithm Extensions (Intel® SHA Extensions)
- Security Technologies - Security Firmware Engines
 - Intel® Converged Security and Management Engine (Intel® CSME)
 - Intel® Silicon Security Engine
 - Intel® Graphics System Controller (Intel® GSC)
 - Intel® Active Management Technology (Intel® AMT)
- Testability and Monitoring
 - JTAG Boundary Scan
 - Intel® Software Toolkit
 - Platform Crashlog
 - Platform Monitoring Technology (PMT)
 - Intel® Processor Trace
 - Intel® Trace Hub (Intel® TH)
 - Direct Connect Interface (DCI) for debug
 - Debug Island
 - Early Boot Debug
- Power Management Technologies
 - Advanced Configuration and Power Interface (ACPI) Power Management Logic Support
 - Intel® Smart Cache Technology
 - Power and Efficient Cores Level 1 and Level 2 Caches
 - Cache Line Write Back (CLWB)

- Intel® Thread Director
- Intel® Hybrid Technology
- Intel® Turbo Boost Technology 2.0
- Intel® Turbo Boost Max Technology 3.0
- Intel® Adaptive Boost Technology
- Intel® Thermal Velocity Boost (Intel® TVB)
- Intel SpeedStep® Technology
- Intel® System Agent Enhanced SpeedStep Technology (Intel® SAGV)
- Intel® Speed Shift Technology
- Intel® Advanced Vector Extensions 2 (Intel® AVX2)
- Intel® Advanced Vector Extensions 10 Version 2 (Intel® AVX10.2)
- Intel® AVX2 Vector Neural Network Instructions (Intel® AVX2 VNNI)
- Intel® Advanced Programmable Interrupt Controller (APIC)
- Intel® 64 Architecture x2APIC
- Intel® Dynamic Tuning technology (Intel® DTT)
- Power Delivery Technologies
 - Digital Linear Voltage Regulator (DLVR)
 - Fast V-Mode (FVM)
 - Current Excursion Protection (CEP)

NOTE

The availability of the features above may vary between different processor SKUs.

1.3.1 API Support Windows

- Direct3D* 2015, Direct3D 12, Direct3D 11.2, Direct3D 11.1, Direct3D 9, Direct3D 10, Direct2D
- OpenGL* 4.5
- Open CL* 3.0

DirectX* extensions:

- PixelSync, Instant Access, Conservative Rasterization, Render Target Reads, Floating-point De-norms, Shared a Virtual memory, Floating Point atomics, MSAA sample-indexing, Fast Sampling (Coarse LOD), Quilted Textures, GPU Enqueue Kernels, GPU Signals processing unit. Other enhancements include color compression.

Gen 12 architecture delivers hardware acceleration of Direct X* 12 Render pipeline comprising of the following stages: Vertex Fetch, Vertex Shader, Hull Shader, Tessellation, Domain Shader, Geometry Shader, Rasterizer, Pixel Shader, Pixel Output.

1.3.2 Firmware Resiliency

Intel's NVMe based recovery supports recovery of all firmware from NVMe storage boot partition in a secure manner.

Firmware Resiliency and Recovery in-field is critical to keep PCs up and running while preventing the requirement of additional space on SPI flash to keep a backup firmware. Therefore, it decreases the Platform BOM cost.

1.4 Power Management Support

1.4.1 Processor Core Power Management

Full support of ACPI C-states as implemented by the following processor C-states:

- C0, C2, C3, C6, C8, and C10

Refer to [Processor IA Core Power Management](#) on page 64 for more information.

1.4.2 System Power Management

Supports the following power management system states:

- Modern Standby (S0ix)
- S3
- S4
- S5

Refer to [Power Management](#) on page 62 for more information.

1.4.3 Memory Controller Power Management

- Disabling Unused System Memory Outputs
- DRAM Power Management and Initialization
- Clock Enable (CKE)
- Conditional Self-Refresh
- Dynamic Power Down
- DRAM I/O Power Management
- DDR Electrical Power Gating (EPG)
- Power Training

Refer to [Integrated Memory Controller Power Management](#) on page 162 for more information.

1.4.4 Processor Graphics Power Management

Memory Power Savings Technologies

- Intel® Rapid Memory Power Management (Intel® RMPM)

Display Power Savings Technologies

- Intel® (Seamless and Static) Display Refresh Rate Switching (DRRS) with eDP* port
- Intel® Display Power Saving Technology (Intel® DPST 8.0)
- Intel® OLED Power Saving Technology (Intel® OPST 1.1)
- Panel Self-Refresh 2 (PSR 2)
- Low-Power Single Pipe (LPSP)
- Low-Power Dual Pipe (LPDP)
- Intel® Smart 2D Display Technology (Intel® S2DDT)
- Intel® Low Refresh Rate (LRR)

Graphics Core Power Savings Technologies

- Graphics Dynamic Frequency
- Intel® Graphics Render Standby Technology (Intel® GRST)
- Intel Capped Frames Per Second (CFPS)

1.5 Thermal Management Support

- Intel® Adaptive Thermal Monitor
- Digital Thermal Sensor
- FORCEPR# support
- Critical Temperature Detection
- Software Controlled Clock Modulation (On-Demand Mode)
- Memory Thermal Throttling
- Render Thermal Throttling
- Fan Speed Control with DTS

Refer to Thermal Management chapter for more information.

1.6 Ballout Information

For more information on ballout, refer to Nova Lake S Series LGA Package Ballout Mechanical Specification ([#847619](#)).

1.7 Processor Testability

Refer to Nova Lake-S Platform Design Guide ([#830661](#)) for more information.

The processor includes boundary-scan for board and system level testability. Refer to the appropriate processor Testability Information - Boundary Scan Description Language (BSDL) file.

1.8 Operating Systems Support

Windows* 10	Windows* 11	Chrome* OS	Linux* OS
No	Yes	No	Yes

NOTE

Refer to OS Vendor site for more information regarding latest OS revision support.

1.9 Terminology and Special Marks

This document uses the term Processor to indicate the Compute Tile + HUB Tile + PCD Tile + GPU Tile. Individually, Compute Tile, PCD Tile, HUB Tile, and GPU Tile are used.

Table 2. Terminology

Term	Description
4K	Ultra High Definition (UHD)
AES	Advanced Encryption Standard
AGC	Adaptive Gain Control
AI	Artificial Intelligence
API	Application Programming Interface
AVC	Advanced Video Coding
BLT	Block Level Transfer
BPP	Bits per Pixel
CDR	Clock and Data Recovery
CTLE	Continuous Time Linear Equalizer
DDC	Digital Display Channel
DDI	Digital Display Interface for DisplayPort or HDMI/DVI
DDR5	Fifth-Generation Double Data Rate SDRAM Memory Technology
DFE	Decision Feedback Equalizer
DMA	Direct Memory Access
DPPM	Dynamic Power Performance Management
DP*	DisplayPort*
DSC	Display Stream Compression
DSI	Display Serial Interface
DTS	Digital Thermal Sensor
ECC	Error Correction Code - used to fix DDR transactions errors
eDP*	Embedded DisplayPort*
EU	Execution Unit in the Graphics Processor
GSA	Graphics in System Agent
continued...	

Term	Description
GNA	Gaussian & Neural-Network Accelerator
HDCP	High-Bandwidth Digital Content Protection
HDMI*	High Definition Multimedia Interface
IMC	Integrated Memory Controller
Intel® 64 Technology	64-bit memory extensions to the IA-32 architecture
Intel® DPST	Intel® Display Power Saving Technology
Intel® PTT	Intel® Platform Trust Technology
Intel® TXT	Intel® Trusted Execution Technology
Intel® VT	Intel® Virtualization Technology. Processor Virtualization, when used in conjunction with Virtual Machine Monitor software, enables multiple, robust independent software environments inside a single platform.
Intel® VT-d	Intel® Virtualization Technology for Directed I/O (Intel® VT-d) is a hardware assist, under system software (Virtual Machine Manager or OS) control, for enabling I/O device Virtualization. Intel® VT-d also brings robust security by providing protection from errant DMAs by using DMA remapping, a key feature of Intel® VT-d.
Intel® TH	Intel® Trace Hub
IOV	I/O Virtualization
IPU	Image Processing Unit
LFM	Low Frequency Mode. corresponding to the Enhanced Intel SpeedStep® Technology's lowest voltage/frequency pair. It can be read at MSR CEh [47:40]. For more information, refer to appropriate BIOS Specification.
LLC	Last Level Cache
LPDDR5/x	Low Power Double Data Rate SDRAM memory technology /x- additional power save.
LPSP	Low-Power Single Pipe
LSF	Lowest Supported Frequency. This frequency is the lowest frequency where manufacturing confirms logical functionality under the set of operating conditions.
LTR	The Latency Tolerance Reporting (LTR) mechanism enables Endpoints to report their service latency requirements for Memory Reads and Writes to the Root Complex, so that power management policies for central platform resources (such as main memory, RC internal interconnects, and snoop resources) can be implemented to consider Endpoint service requirements.
MFM	Minimum Frequency Mode. MFM is the minimum ratio supported by the processor and can be read from MSR CEh [55:48]. For more information, refer to the appropriate BIOS specification.
MLC	Mid-Level Cache
MPEG	Motion Picture Expert Group, international standard body JTC1/SC29/WG11 under ISO/IEC that has defined audio and video compression standards such as MPEG-1, MPEG-2, and MPEG-4, etc.
NCTF	Non-Critical to Function. NCTF locations are typically redundant ground or non-critical reserved balls/lands, so the loss of the solder joint continuity at end of life conditions will not affect the overall product functionality.
NPU	Neural Processing Unit
PECI	Platform Environment Control Interface
PEG	PCI Express* Graphics
continued...	

Term	Description
PCD	Platform Controller Die
PCH	Platform Controller Hub
PL1, PL2, PL3, PL4	Power Limit 1, Power Limit 2, Power Limit 3, Power Limit 4
PMIC	Power Management Integrated Circuit
Processor	The 64-bit multi-core component (package)
Processor Core	The term "processor core" refers to the Si tile itself, which can contain multiple execution cores.
Processor Graphics	Intel® Processor Graphics
PSR	Panel Self-Refresh
PSx	Power Save States (PS0, PS1, PS2, PS3, PS4)
Rank	A unit of DRAM corresponding to four to eight devices in parallel, ignoring ECC. These devices are usually, but not always, mounted on a single side of a DIMM.
SCI	System Control Interrupt. SCI is used in the ACPI protocol.
SDP	Scenario Design Power
SHA	Secure Hash Algorithm
SSC	Spread Spectrum Clock
Storage Conditions	Refer to Package Storage Specifications on page 30.
STR	Suspend to RAM
TAC	Thermal Averaging Constant
TBT	Thunderbolt™ Interface
TCC	Thermal Control Circuit
TMTV Processor Base Power	Thermal Test Vehicle Processor Base Power
VLD	Variable Length Decoding
VPID	Virtual Processor ID
V _{SS}	Processor Ground
D0ix-states	USB controller power states ranging from D0i0 to D0i3, where D0i0 is fully powered on and D0i3 is primarily powered off. Controlled by SW.
S0ix-states	Processor and PCH residency idle standby power states.

Table 3. Special Marks

Mark	Definition
[]	Brackets ([]) sometimes follow a ball, pin, registers or a bit name. These brackets enclose a range of numbers, for example, TCP[2:0]_TXRX_P[1:0] may refer to six USB-C* pins or EAX[7:0] may indicate a range that is 8 bits length.
_N / #	A suffix of _N or # indicates an active low signal. For example, CATERR# or CATERR_N does not refer to a differential pair of signals such as CLK_P, CLK_N
h	Hexadecimal numbers are identified with 'h' appended at the end of the number. For example, CF9h. All numbers are decimal (base 10) unless otherwise specified. Non-obvious binary numbers have 'b' appended at the end of the number. For example, 0101b.

1.10 Related Documents

Document	Document Number
Nova Lake S External Design Specification (EDS) Volume 2 of 2	855519
Nova Lake-S Platform Design Guide	830661
Nova Lake S Series LGA Package Ballout Mechanical Specification	847619
Nova Lake-S Processor Line (LGA) - GPIO Implementation Summary	853123
Nova Lake-S Platform Thermal and Mechanical Design Guide	853008
Nova Lake-S PCH External Design Specification (EDS) Volume 1 of 2	835009
Nova Lake-S Platform Controller Hub (PCH) BGA Package Ballout Mechanical Specification	844066
Nova Lake S Package Mechanical Drawings and Models	853013
Nova Lake PCH S External Design Specification (EDS) – Volume 2 of 2	845995

2.0 Processor and Device IDs

2.1 CPUID

Table 4. CPUID Format

SKU	CPUID	Reserved [31:28]	Extended Family [27:20]	Extended Model [19:16]	Reserved [15:14]	Processor Type [13:12]	Family Code [11:8]	Model Number [7:4]	Stepping ID [3:0]
NVL-S 52C	300F12h	Reserved	3h	0h	Reserved	0h	Fh	1h	2h
NVL-S 28C	300F10h	Reserved	3h	0h	Reserved	0h	Fh	1h	0h
NVL-S 16C	300F11h	Reserved	3h	0h	Reserved	0h	Fh	1h	1h
NVL-S 8C	300F13h	Reserved	3h	0h	Reserved	0h	Fh	1h	3h

- The Extended Family, Bits [27:20] are used in conjunction with the Family Code, specified in Bits[11:8], to indicate whether the processor belongs to Intel® Core™ processor family.
- The Extended Model, Bits [19:16] in conjunction with the Model Number, specified in Bits [7:4], are used to identify the model of the processor within the processor's family.
- The Family Code corresponds to Bits [11:8] of the EDX register after RESET, Bits [11:8] of the EAX register after the CPUID instruction is executed with a 1 in the EAX register, and the generation field of the Device ID register accessible through Boundary Scan.
- The Model Number corresponds to Bits [7:4] of the EDX register after RESET, Bits [7:4] of the EAX register after the CPUID instruction is executed with a 1 in the EAX register, and the model field of the Device ID register accessible through Boundary Scan.
- The Stepping ID in Bits [3:0] indicates the revision number of that model.
- Refer to the Nova Lake Processor BIOS Specification for additional information. When EAX is initialized to a value of '1', the CPUID instruction returns the Extended Family, Extended Model, Processor Type, Family Code, Model Number and Stepping ID value in the EAX register.

NOTE

The EDX processor signature value after reset is equivalent to the processor signature output value in the EAX register.

Cache and TLB descriptor parameters are provided in the EAX, EBX, ECX and EDX registers after the CPUID instruction is executed with a 2 in the EAX register.

2.2 PCI Configuration Header

Every PCI-compatible function has a standard PCI configuration header, as shown in the table below. This includes mandatory registers (Bold) to determine which driver to load for the device. Some of these registers define ID values for the PCI function, which are described in this chapter.

Table 5. PCI Configuration Header

Byte3	Byte2	Byte1	Byte0	Address
Device ID		Vendor ID (8086h)		00h
Status		Command		04h
Class Code			Revision ID	08h
BIST	Header Type	Latency Timer	Cache Line Size	0Ch
Base Address Register0 (BAR0)				10h
Base Address Register1 (BAR1)				14h
Base Address Register2 (BAR2)				18h
Base Address Register3 (BAR3)				1Ch
Base Address Register4 (BAR4)				20h
Base Address Register5 (BAR5)				24h
Card-bus CIS Pointer				28h
Subsystem ID		Subsystem Vendor ID		2Ch
Expansion ROM Base Address				30h
Reserved			Capabilities Pointer	34h
Reserved				38h
Maximum Latency	Minimum Grant	Interrupt Pin	Interrupt Line	3Ch

2.3 Device IDs

This section specifies the device IDs of the processor.

NOTE

For more information on other device IDs, refer to Nova Lake-S Processor Sightings Report - NDA User Guide ([#866163](#)).

Table 6. Host Device ID (DID0) and Processor Graphics Device ID (DID2)

Processor Line	Package	P-Cores	E-Cores	LP E-Cores	Graphics Configuration Xe-Cores	Host Device ID (DID0)	Processor Graphics Device ID (DID2)
NVL-S 52C DS	LGA1954	16	32	4	2	D706h	D740h
NVL-S 28C DS	LGA1954	8	16	4	2	D70Ah	D740h
continued...							

Processor Line	Package	P-Cores	E-Cores	LP E-Cores	Graphics Configuration X ^e -Cores	Host Device ID (DID0)	Processor Graphics Device ID (DID2)
NVL-S 28C	LGA1954	8	16	4	2	D707h	D740h
NVL-S 16C	LGA1954	4	8	4	2	D708h	D740h
NVL-S 8C	LGA1954	4	0	4	2	D709h	D740h

Table 7. Other Device ID

Device	Bus / Device / Function	NVL-S DID
Dynamic Tuning Technology (DTT)	0 / 4 / 0	D703h
PCI Express Root Port #1 (Gen5)	0 / 6 / 0	D461h
PCI Express Root Port #2 (Gen5)	0 / 6 / 1	D465h
PCI Express Root Port #3 (Gen5)	0 / 6 / 2	D467h
PCI Express Root Port #4 (Gen5)	0 / 6 / 3	D468h
PCI Express Root Port #5 (Gen5)	0 / 6 / 4	D469h
PCI Express Root Port #6 (Gen5)	0 / 6 / 5	D46Ah
USB Type-C Subsystem PCIe Root Port #21	0 / 7 / 0	D44Eh
USB Type-C Subsystem PCIe Root Port #22	0 / 7 / 1	D44Fh
Crash Log & Telemetry	0 / 10 / 0	D70Dh
NPU	0 / 11 / 0	D71Dh
IAA	0 / 12 / 0	D72Dh
Type-C Subsystem xHCI	0 / 13 / 0	D431h
Thunderbolt™ DMA0	0 / 13 / 2	D433h
Intel® Volume Management Device (VMD)	0 / 14 / 0	D70Bh
IEH #1	0 / 18 / 0	D445h
P2SB #1	0 / 18 / 1	D44Ch
Intel® PSE: HECI #1	0 / 19 / 0	D462h
Intel® PSE: HECI #2	0 / 19 / 1	D463h
Intel® PSE: HECI #3	0 / 19 / 2	D464h
Shared SRAM	0 / 20 / 0	D47Eh
IEH #0	0 / 20 / 5	D444h
Intel® ESE++: HECI #1	0 / 22 / 0	D470h
Intel® ESE++: HECI #2	0 / 22 / 1	D471h
Intel® ESE++: HECI #3	0 / 22 / 4	D472h
Processor Direct eSPI Controller	0 / 31 / 0	D400h - D41Fh
P2SB	0 / 31 / 1	D420h
PMC	0 / 31 / 2	D421h
SPI (flash) Controller	0 / 31 / 5	D423h
Intel® Trace Hub (Intel® TH) North Peak	0 / 31 / 7	D424h

Table 8. ACPI Device ID for GPIO Controller

Controller	ACPI ID
NVL-S GPIO Controller	INTC10DD

2.4 Revision IDs

The Revision ID (RID) register is an 8-bit register located at offset 08h in the PCI header of every PCI/PCIe* function. The RID register is used by software to identify a particular component stepping when a driver change or patch unique to that stepping is needed.

The RID register reports one of the two possible values:

- Stepping Revision Identification (SRID)
- Compatible Revision ID (CRID)

The default power-on value for the RID register is SRID. The assigned value is based on the product's stepping. CRID is intended for the corporate Intel® Stable Image Platform Program (Intel® SIPP). CRID is normally identical to the SRID value of a previous production stepping of the product with which the new stepping is deemed compatible. Intel® SIPP allows an OS image built on the earlier stepping to be used on any new compatible stepping(s). Three CRID values are possible and can be used to manage software images.

NOTE

SRID and CRID are not addressable PCI registers. The SRID and CRID value are reflected through the RID register when appropriately selected.

Following reset, the SRID value can be read from the RID registers of all Processor devices and functions.

To select either SRID or CRID to be reflected in the RID registers:

- BIOS needs to write appropriate value into the Configured Revision ID (CRID) register located in the PMC MMIO space.
- BIOS must write this register with the appropriate value after S4/S5 states and after PLTRST# events.

After CRID is selected and applied by BIOS, software will not be able to obtain the original SRID value of the Processor by reading the RID registers. Customers implementing CRID who also want to determine the SRID in runtime may develop their own tool. For example, BIOS can capture the SRID value before BIOS applies CRID and store that value in a runtime accessible place (that is, SMBIOS, ACPI Type 4 Memory, NVRAM, CMOS) so that it can be read by the customer tool later. Alternatively, the BIOS can store the SRID value and display this information in BIOS setup while reporting that CRID is enabled.

BIOS needs to check CRID_UIP bit (in PMC MMIO space) as a part of the update flow. PMC HW sets this bit to indicate that SetID broadcast flow has been requested by BIOS. This bit is cleared by PMC FW only when the completion/s of SetIDVal message is received by PMC. BIOS is required to read this bit as cleared before writing to the CRID register (to request a CRID update). BIOS is also required to poll on reads to this bit until it detects the bit as cleared after BIOS has written to the CRID register.

3.0 Package Mechanical Specifications

3.1 Package Mechanical Attributes

The S LGA Processor Lines use a Flip Chip technology available in a Land Grid Array (LGA) package. The following table provides an overview of the package mechanical attributes. For specific package dimensions, refer to the Nova Lake S Package Mechanical Drawings and Models (#853013).

Table 9. S LGA Package Mechanical Attributes

Package	Parameter	S Processor Line
Package Technology	Package Type	Flip Chip Land Grid Array
	Interconnect	Land Grid Array (LGA)
	Lead Free	Yes
	Halogenated Flame Retardant Free	Yes
Package Configuration	Solder Ball Composition	SAC405
	Ball/Pin Count	1954
	Grid Array Pattern	Grid Array
	Land Side Capacitors	Yes
	Tile Side Capacitors	Yes
	Tile Configuration	Foveros
Package Dimensions	Nominal Package Size	45 x 37.5 mm
	Maximum Package Z-Height	4.454 ± 0.125 mm
	Minimum Ball/Pin pitch	0.8 mm

3.2 Package Storage Specifications

Parameter	Description	Minimum	Maximum	Notes
T _{ABSOLUTE STORAGE}	The non-operating device storage temperature. Damage (latent or otherwise) may occur when subjected to this temperature for any length of time in Intel Original sealed moisture barrier bag and / or box.	-25°C	125°C	
T _{SUSTAINED STORAGE}	The ambient storage temperature limit (in shipping media) for the sustained period of time	-5°C	40°C	

continued...

Parameter	Description	Minimum	Maximum	Notes
RH _{SUSTAINED STORAGE}	The maximum device storage relative humidity for the sustained period of time as specified below in Intel Original sealed moisture barrier bag and / or box	60%@24°C		
TIME _{SUSTAINED STORAGE}	Maximum time; associated with customer shelf life in Intel Original sealed moisture barrier bag and / or box	NA	Moisture Sensitive Devices: 60 months from bag seal date; Non-moisture sensitive devices: 60 months from lot date	
Storage Conditions	Processors in a non-operational state may be installed in a platform, in a tray, boxed, or loose and may be sealed in airtight package or exposed to free air. Under these conditions, processor landings should not be connected to any supply voltages, have any I/Os biased, or receive any clocks. Upon exposure to "free air" (that is, unsealed packaging or a device removed from packaging material), the processor should be handled in accordance with moisture sensitivity labeling (MSL) as indicated on the packaging material. Boxed Land Grid Array packaged (LGA) processors are MSL 1 ('unlimited' or unaffected) as they are not heated in order to be inserted in the socket.			
Notes: 1. T _{ABSOLUTE STORAGE} applies to the un-assembled component only and does not apply to the shipping media, moisture barrier bags or desiccant. Refers to a component device that is not assembled in a board or socket that is not to be electrically connected to a voltage reference or I/O signals. 2. Specified temperatures are based on data collected. Exceptions for surface mount re-flow are specified by applicable JEDEC J-STD-020 and MAS documents. The JEDEC, J-STD-020 moisture level rating and associated handling practices apply to all moisture sensitive devices removed from the moisture barrier bag. 3. Post board attaches storage temperature limits are not specified for non-Intel branded boards. Consult your board manufacturer for storage specifications.				

4.0 Memory Mapping

This chapter describes (from the processor perspective) the memory ranges that the Processor decodes.

4.1 Functional Description

4.1.1 PCI Devices and Functions

The Processor incorporates a variety of PCI devices and functions, as shown in the following table. If for some reason, the particular system platform does not want to support any one of the Device Functions, with the exception of D30:F0, they can individually be disabled. When a function is disabled, it does not appear to the software. A disabled function will not respond to any register reads or writes, ensuring that these devices appear hidden to software.

4.1.2 Fixed IO Address Ranges

The following table shows the Fixed I/O decode ranges from the processor perspective.

NOTE

For each I/O range, there may be separate behavior for reads and writes.

I/O cycles that go to target ranges that are marked as Reserved will be handled as follow : writes are ignored and reads will return all 1's. The P2SB will claim many of the fixed I/O accesses and forward those transactions over IOSF-SB to their functional target.

Address ranges that are not listed or marked Reserved are NOT positively decoded (unless assigned to one of the variable ranges) and will be internally terminated.

Table 10. Fixed I/O Ranges Decoded by Processor

I/O Address	Read Target	Write Target	Internal Unit (Unless[E]: External) ²	Separate Enable/Disable
20h – 21h	Interrupt Controller	Interrupt Controller	Interrupt	None
24h – 25h	Interrupt Controller	Interrupt Controller	Interrupt	None
28h – 29h	Interrupt Controller	Interrupt Controller	Interrupt	None
2Ch – 2Dh	Interrupt Controller	Interrupt Controller	Interrupt	None
30h – 31h	Interrupt Controller	Interrupt Controller	Interrupt	None
34h – 35h	Interrupt Controller	Interrupt Controller	Interrupt	None
38h – 39h	Interrupt Controller	Interrupt Controller	Interrupt	None
continued...				

I/O Address	Read Target	Write Target	Internal Unit (Unless[E]: External) ²	Separate Enable/Disable
3Ch – 3Dh	Interrupt Controller	Interrupt Controller	Interrupt	None
40h	Timer/Counter	Timer/Counter	8254 Timer	None
42h-43h	Timer/Counter	Timer/Counter	8254 Timer	None
50h	Timer/Counter	Timer/Counter	8254 Timer	None
52h-53h	Timer/Counter	Timer/Counter	8254 Timer	None
61h	NMI Controller	NMI Controller	CPU I/F	None
63h	NMI Controller ¹	NMI Controller ¹	CPU I/F	Yes, alias to 61h. GIC.P61AE
65h	NMI Controller ¹	NMI Controller ¹	CPU I/F	Yes, alias to 61h. GIC.P61AE
67h	NMI Controller ¹	NMI Controller ¹	CPU I/F	Yes, alias to 61h. GIC.P61AE
70h	RTC Controller	NMI and RTC Controller	RTC	None
80h ³	eSPI or PCIe	eSPI or PCIe	Read: [E] eSPI or PCIe Write: [E] eSPI or [E] PCIe	None. PCIe if GCS.RPR='1', else eSPI
84h - 86h	eSPI or PCIe	eSPI or PCIe	Read: [E] eSPI or PCIe Write: [E] eSPI or [E] PCIe	None. PCIe if GCS.RPR='1', else eSPI
88h	eSPI or PCIe	eSPI or PCIe	Read: [E] eSPI or PCIe Write: [E] eSPI or [E] PCIe	None. PCIe if GCS.RPR='1', else eSPI
8Ch - 8Eh	eSPI or PCIe	eSPI or PCIe	Read: [E] eSPI or PCIe Write: [E] eSPI or [E] PCIe	None. PCIe if GCS.RPR='1', else eSPI
92h	Reset Generator	Reset Generator	CPU I/F	None
A0h - A1h	Interrupt Controller	Interrupt Controller	Interrupt	None
A4h - A5h	Interrupt Controller	Interrupt Controller	Interrupt	None
A8h - A9h	Interrupt Controller	Interrupt Controller	Interrupt	None
ACh - ADh	Interrupt Controller	Interrupt Controller	Interrupt	None
B0h - B1h	Interrupt Controller	Interrupt Controller	Interrupt	None
B2h - B3h	Power Management	Power Management	Power Management	None
B4h - B5h	Interrupt Controller	Interrupt Controller	Interrupt	None
continued...				

I/O Address	Read Target	Write Target	Internal Unit (Unless[E]: External) ²	Separate Enable/Disable
B8h - B9h	Interrupt Controller	Interrupt Controller	Interrupt	None
BCh - BDh	Interrupt Controller	Interrupt Controller	Interrupt	None
4D0h - 4D1h	Interrupt Controller	Interrupt Controller	Interrupt Controller	None
CF9h	Reset Generator	Reset Generator	Interrupt controller	None
Notes: 1. Only if the Port 61 Alias Enable bit (GIC.P61AE) bit is set. Otherwise, the cycle is internally terminated by the Processor. 2. Destination of eSPI when eSPI Disabled pin strap is 0. 3. This includes byte, word or double-word (DW) access at I/O address 80h.				

4.1.3 Variable IO Decode Ranges

The following Table shows the Variable I/O Decode Ranges. They are set using Base Address Registers (BARs) or other configuration bits in the various configuration spaces. The PnP software (PCI or ACPI) can use their configuration mechanisms to set and adjust these values.

WARNING

The Variable I/O Ranges should not be set to conflict with the Fixed I/O Ranges. There may some unpredictable results if the configuration software allows conflicts to occur. The Processor does not perform any checks for conflicts.

Table 11. Variable I/O Decode Ranges

Range Name ¹	Mappable	Size (Bytes)	Target
ACPI	Anywhere in 64K I/O Space	256	Power Management
IO Trapping Ranges	Anywhere in 64K I/O Space	1 to 256 Bytes	Trap
DMI General Purpose I/O Ranges (1 to 3)	Anywhere in 64K I/O Space	4 to 256 Bytes	General Purpose
Note: All ranges are decoded directly from DMI.			

4.2 Memory Map

The following table shows (from the processor perspective) the memory ranges that the processor will decode. Cycles that are not directed to any of the internal memory targets, will be host aborted.

PCIe cycles generated by external PCIe hosts will be positively decoded unless they fall in the PCI-PCI bridge memory forwarding ranges (those addresses are reserved for PCI peer-to-peer traffic). Software must not attempt locks to the processor's memory-mapped I/O ranges.

NOTE

Total ports are different for the different SKUs.

Table 12. Processor Memory Decode Ranges (Processor Perspective)

Memory Range	Target	Dependency/Comments
FECX X000 - FECX X040	I/O(x)APIC inside the processor, or behind the processor's PCIe root-port	XX controlled via APIC Range Select (ASEL) field and APIC Enable (AEN) bit. For PCIe root port, I/OxAPIC Enable (PAE) bit
FFFC 0000 - FFFF FFFF	Intel® CSME	Always enabled.
FED0 X000 - FED0 X3FF	HPET	BIOS determines "fixed" location which is one of four 1 KB ranges where X (in the first column) is 0h, 1h, 2h, or 3h.
FED4 0000 - FED4 7FFF	CSME or OSSE	TPM and Trusted Mobile KBC
FED4 8000 - FED4 BFFF	LPC	Always enabled - Intel® TXT address range NOTE: the LPC "bus" interface is removed, but the LPC HW is retained to support this register set.
FED4 C000 - FED4 FFFF	Internal (PSF Error Handler)	Always enabled
FED5 0000 - FED5 FFFF	ESE++	Always enabled
FED6 0000 - FED6 1FFF	xHCI	NOT positively decoded in PCH (DMI/PSF)
FED6 2000 - FED6 3FFF	xHCI (Processor)	Fixed range in the processor - never forwarded to PCH.
FED7 0000 - FED7 4FFF	Internal Device	Security feature related
DMI General Purpose Memory ranges (1 to 3)	General purpose	Enable via setting the Decode Enable bit of the respective Memory Range register.
MMIO resources for VMD managed devices	Storage devices	Enable through the VMD device in the processor.

4.2.1 Boot Block Update Scheme

The Processor supports a "Top-Block Swap" mode that has the Processor swap the top block in the SPI flash (the boot block) with another location. This allows for safe update of the Boot Block (even if a power failure occurs). When the "top-swap" enable bit is set, the Processor will invert A16 for cycles going to the upper two 64-KB blocks in the FWH or appropriate address lines as selected in Boot Block Size (BOOT_BLOCK_SIZE) soft strap for SPI.

For SPI when top swap is enabled, the behavior is as described below. When the Top Swap Enable bit is 0, the Processor will not invert any address bit.

Table 13. Boot Block Update Scheme

BOOT_BLOCK_SIZE Value	Accesses to	Being Directed to
000 (64KB)	FFFF_0000h - FFFF_FFFFh	FFFE_0000h - FFFE_FFFFh and vice versa
001 (128KB)	FFFE_0000h - FFFF_FFFFh	FFFC_0000h - FFFD_FFFFh and vice versa
010 (256KB)	FFFC_0000h - FFFF_FFFFh	FFF8_0000h - FFFB_FFFFh and vice versa
011 (512KB)	FFF8_0000h - FFFF_FFFFh	FFF0_0000h - FFF7_FFFFh and vice versa
100 (1MB)	FFF0_0000h - FFFF_FFFFh	FFE0_0000h - FFEF_FFFFh and vice versa
101 (2MB)	FFE0_0000h - FFFF_FFFFh	FFC0_0000h - FFDF_FFFFh and vice versa
110 (4MB)	FFC0_0000h - FFFF_FFFFh	FF80_0000h - FFBF_FFFFh and vice versa
111 (8MB)	FF80_0000h - FFFF_FFFFh	FF00_0000h - FF7F_FFFFh and vice versa
Note: This bit is automatically set to 0 by RTEST#, but not by PLTRST#.		

The scheme is based on the concept that the top block is reserved as the “boot” block, and the block immediately below the top block is reserved for doing boot-block updates.

The algorithm is:

1. Software copies the top block to the block immediately below the top
2. Software checks that the copied block is correct. This could be done by performing a checksum calculation.
3. Software sets the “Top-Block Swap” bit. This will invert the appropriate address bits for the cycles going to the SPI.
4. Software erases the top block
5. Software writes the new top block
6. Software checks the new top block
7. Software clears the top-block swap bit
8. Software sets the Top_Swap Lock-Down bit

If a power failure occurs at any point after step 3, the system will be able to boot from the copy of the boot block that is stored in the block below the top. This is because the top-swap bit is backed in the RTC well.

There is one remaining unusual case that could occur if the RTC battery is not sufficiently high to maintain the RTC well. To avoid the potentially fatal case (where the Top-Swap bit is NOT set, but the top block is not valid), a pin strap will allow forcing the top-swap bit to be set. This would be a last resort to allow the user to get the system to boot (and avoid having to de-solder the system flash).

When the top-swap strap is used, the top-swap bit will be forced to 1 (cannot be cleared by software).

5.0 Pin Straps

The following signals are used for static configuration. They are sampled at the rising edge of either RSMRST# or PLT_PWROK to select configuration and then revert later to their normal usage. To invoke the associated mode, the signal should meet both set up and hold time of 1us, with respect to the rising edge of the sampling signal.

The processor also implements soft straps, which are used to configure specific functions within the processor very early in the boot process before BIOS or software intervention. The processor will read soft strap data out of the SPI device before the de-assertion of reset to both the Intel® Management Engine and the Host system.

NOTE

Internal pull-down/pull-up for default pin strap mode is released ~2 RTC clocks after the strap sampling event.

Table 14. Pin Straps

Signal	Usage	When Sampled	Comment
GPP_SA14/ DDSP_HPD0#/ DISP_MISC0/ TIME_SYNC0	PCIe_B Configurati on Select Enable	Rising edge of PCH_PWROK	0 – Disable PEG0CFGSEL[1:0]. Softstrap (mFIT) to set PEG16 bifurcation. (Default) 1 – Enable PEG0CFGSEL[1:0]. PinStrap to set PEG16 bifurcation. <i>Note:</i> The internal pull-down is disabled ~2 RTC clocks after the strap sampling event.
GPP_SA15/ DDSP_HPD1#/ DISP_MISC1/ TIME_SYNC1	PCIe_B[15 :0] PEG0CFGSEL[0]	Rising edge of PCH_PWROK	This is bit 0 (LSB) of the 2-bit encoded pin straps for PCIe Port Bifurcation configuration. This strap is used in conjunction with PCIe Lanes 1-16 Configuration Select 1 on GPP_SA16 pin. The 2-bit strap encoding: 00 = x16 (Default) 10 = x8, x8 01 = x4, x4, x4, x4 11 = x8, x4, x4
GPP_SA16/ DDSP_HPD3#/ DISP_MISC3	PCIe_B[15 :0] PEG0CFGSEL[1]	Rising edge of PCH_PWROK	This is bit 1 (MSB) of the 2-bit encoded pin straps for PCIe PEG Bifurcation configuration. This strap is used in conjunction with PCIe Lanes 1-16 Configuration Select 0 on GPP_SA15 pin. The 2-bit strap encoding: 00 = x16 (Default) 10 = x8, x8 01 = x4, x4, x4, x4 11 = x8, x4, x4
GPP_SA20/ DDSP_HPD2#/ DISP_MISC2	Flash Descriptor Security Override	Rising edge of PCH_PWROK	0 = Enable security measures defined in the Flash Descriptor (Default) . 1 = Disable Flash Descriptor Security (<u>override</u>). This strap should only be asserted high using external Pull-up in manufacturing/debug environments ONLY.

continued...

Signal	Usage	When Sampled	Comment
GPP_SD00/TIME_SYNC0	Boot BIOS Strap (BBS)	Rising edge of RSMRST#	BIOS fetches are routed based on this strap. 0 = BIOS fetches are routed to SPI (MAF) or the eSPI Flash Channel (SAF) (Default) 1 = BIOS fetches are routed to the eSPI Peripheral Channel.
GPP_SD01/TIME_SYNC1	Reserved	Rising edge of RSMRST#	This strap has a 20 kohm $\pm$ 30% internal pull-down. This strap should sample LOW. There should NOT be any onboard device driving it to opposite direction during strap sampling. <i>Note:</i> The internal pull-down is disabled $\sim$ 2 RTC clocks after the strap sampling event.
GPP_SD02	eSPI Disable	Rising edge of RSMRST#	This strap has a 20 kohm $\pm$ 30% internal pull-down. This strap should sample LOW. There should NOT be any onboard device driving it to opposite direction during strap sampling. 0 = Enable eSPI (Default). 1 = Disable eSPI. <i>Note:</i> The internal pull-down is disabled $\sim$ 2 RTC clocks after the strap sampling event.
GPP_SD04	JTAG ODT Enable	Rising edge of RSMRST#	This strap has a 20 kohm $\pm$ 30% internal pull-up. This strap should sample HIGH. There should NOT be any onboard device driving it to opposite direction during strap sampling. 0=> JTAG ODT is disabled. 1=> JTAG ODT is enabled (Default). <i>Notes:</i> For ES1_A_step, JTAG ODT must be disabled - The internal pull-up is disabled $\sim$2 RTC clocks after the strap sampling event.
GPP_SD05	Reserved	Rising edge of RSMRST#	This strap has a 20 kohm $\pm$ 30% internal pull-up. This strap should sample HIGH. There should NOT be any onboard device driving it to opposite direction during strap sampling. <i>Note:</i> The internal pull-up is disabled $\sim$ 2 RTC clocks after the strap sampling event.
GPP_SD06/SRCCLKREQ1#	Reserved	Rising edge of RSMRST#	This strap has a 20 kohm $\pm$ 30% internal pull-up. This strap should sample HIGH. There should NOT be any onboard device driving it to opposite direction during strap sampling. <i>Note:</i> The internal pull-up is disabled $\sim$ 2 RTC clocks after the strap sampling event.
GPP_SD08	Reserved	Rising edge of RSMRST#	This strap has a 20 kohm $\pm$ 30% internal pull-up. This strap should sample HIGH. There should NOT be any onboard device driving it to opposite direction during strap sampling. <i>Note:</i> The internal pull-up is disabled $\sim$ 2 RTC clocks after the strap sampling event.
GPP_SD12/SRCCLKREQ3#	Enable/Disable SPI Flash Descriptor Recovery	Rising edge of RSMRST#	This strap has a 20 kohm $\pm$ 30% internal pull-down. This strap should sample LOW. There should NOT be any onboard device driving it to opposite direction during strap sampling. It is used to enable/disable Flash Descriptor Recovery to support SPI descriptor resiliency feature per NIST SP800-193 Guidance. 0 - Flash descriptor recovery disable (Default) 1 - Flash descriptor recovery enable <i>Note:</i> The internal pull-down is disabled $\sim$ 2 RTC clocks after the strap sampling event.
GPP_SD13/DDSP_HPD3#/DISP_MISC3	Reserved	Rising edge of RSMRST#	This strap has a 20 kohm $\pm$ 30% internal pull-down. This strap should sample LOW. There should NOT be any onboard device driving it to opposite direction during strap sampling. <i>Note:</i> The internal pull-down is disabled $\sim$ 2 RTC clocks after the strap sampling event.

continued...

Signal	Usage	When Sampled	Comment
GPP_SD14/ DDSP_HP2#/ DISP_MISC2	Reserved	Rising edge of RSMRST#	This strap has a 20 kohm $\pm$ 30% internal pull-down. This strap should sample LOW. There should NOT be any onboard device driving it to opposite direction during strap sampling. <i>Note:</i> The internal pull-down is disabled $\sim$ 2 RTC clocks after the strap sampling event.
GPP_SD15/ DDSP_HP0#/ DISP_MISC0	Reserved	Rising edge of RSMRST#	This strap has a 20 kohm $\pm$ 30% internal pull-down. This strap should sample LOW. There should NOT be any onboard device driving it to opposite direction during strap sampling. <i>Note:</i> The internal pull-down is disabled $\sim$ 2 RTC clocks after the strap sampling event.
GPP_SD16/ DDSP_HP1#/ DISP_MISC1	Reserved	Rising edge of RSMRST#	This strap has a 20 kohm $\pm$ 30% internal pull-down. This strap should sample LOW. There should NOT be any onboard device driving it to opposite direction during strap sampling. <i>Note:</i> The internal pull-down is disabled $\sim$ 2 RTC clocks after the strap sampling event.

6.0 Security Technologies

6.1 Intel® Converged Boot Guard and TXT

Intel® Converged Boot Guard and Intel® TXT (Intel® CBnT) is an unification of Intel® Trusted Execution Technology (Intel® TXT) and Intel® Platform Protection Technology with Intel® Boot Guard. Intel® CBnT merges elements of Intel® TXT and Intel® Boot Guard to enhance platform boot security, while also simplifying the implementation. Although Intel® CBnT implements some architectural changes, it is not fundamentally a new technology, but rather a fusion of existing Intel® Boot Guard and Intel® TXT technologies.

Intel® CBnT has been designed to allow greater commonality between implementations for client platforms and server platforms. Previously, the architectural implementation of Intel® TXT was somewhat different between client and server platforms, which necessitated some differences in BIOS implementation depending on the platform. With Intel® CBnT, Intel has largely combined features across client and server providing greater alignment in design of the BIOS and ACMS.

Intel® Converged Boot Guard and Intel® TXT provides both a static root of trust for verifying the BIOS initial boot block and measuring the boot path, as well as a dynamic root of trust for measuring the OS or VMM.

The purpose of Intel® Boot Guard is to verify that the initial BIOS startup code is good, i.e., BIOS has not been maliciously nor inadvertently modified. Several different Boot Profiles are supported, which primarily differ in:

- **Enforcement Policy:** what actions are taken if BIOS cannot be verified.
- **Measurement Policy:** whether BIOS startup code is measured into the TPM for attestation.

The primary objective of Intel® TXT is to provide a dynamic root of trust for measuring the OS or VMM enabling platform boot into a secure measured launch environment (MLE). Intel® TXT relies on the static root of trust provided by Intel® Boot Guard to ensure validity of the MLE Trusted Compute Base (TCB), which is the BIOS code that is trusted to configure the platform. Intel® TXT provides the ability to allow only a known good OS/VMM to launch into a trusted environment via a Launch Control Policy (LCP). And once an OS/VMM is in a trusted environment, Intel® TXT protects memory secrets against surprise reset attacks.

With the modifications made to the Intel® TXT architecture in Intel® CBnT, it is now required that some of the verifications performed by Intel® Boot Guard be implemented for Intel® TXT support. Verifications of pre-boot objects such as FIT, key and policy manifests, and of Startup BIOS.

Still formally all four combinations of constituent technologies are supported at OEM choice:

- Intel® Boot Guard only enabled.
- Intel® TXT only enabled.

- Both Intel® Boot Guard and Intel® TXT enabled.

Refer to the Intel® Trusted Execution Technology (Intel® TXT): Software Development Guide (#315168) for more details.

6.2 Crypto Acceleration Instructions

6.2.1 Intel® Advanced Encryption Standard New Instructions (Intel® AES-NI)

The processor supports Intel® Advanced Encryption Standard New Instructions (Intel® AES-NI) that are a set of Single Instruction Multiple Data (SIMD) instructions that enable fast and secure data encryption and decryption based on the Advanced Encryption Standard (AES). Intel® AES-NI is valuable for a wide range of cryptographic applications, such as applications that perform bulk encryption/decryption, authentication, random number generation, and authenticated encryption. AES is broadly accepted as the standard for both government and industrial applications and is widely deployed in various protocols.

Intel® AES-NI consists of six Intel® Streaming SIMD Extensions (Intel® SSE) instructions. Four instructions, AESENC, AESENCLAST, AESDEC, and AESDELAST facilitate high-performance AES encryption and decryption. The other two, AESIMC and AESKEYGENASSIST, support the AES key expansion procedure. Together, these instructions provide full hardware for supporting AES; offering security, high performance, and a great deal of flexibility.

The Intel® AES-NI specifications and functional descriptions are included in the *Intel® 64 Architectures Software Developer's Manual, Volume 2*. Available at:

<http://www.intel.com/products/processor/manuals>

NOTE

Intel® AES-NI Technology may not be available on all SKUs.

6.2.2 Perform Carry-Less Multiplication Quad Word Instruction (PCLMULQDQ)

The processor supports the carry-less multiplication instruction, PCLMULQDQ. PCLMULQDQ is a Single Instruction Multiple Data (SIMD) instruction that computes the 128-bit carry-less multiplication of two 64-bit operands without generating and propagating carries. Carry-less multiplication is an essential processing component of several cryptographic systems and standards. Hence, accelerating carry-less multiplication can significantly contribute to achieving high-speed secure computing and communication.

PCLMULQDQ specifications and functional descriptions are included in the *Intel® 64 Architectures Software Developer's Manual, Volume 2*. Available at:

<http://www.intel.com/products/processor/manuals>

6.2.3 Intel® Secure Hash Algorithm Extensions (Intel® SHA Extensions)

The Secure Hash Algorithm (SHA) is one of the most commonly employed cryptographic algorithms. Primary usages of SHA include data integrity, message authentication, digital signatures, and data de-duplication. As the pervasive use of security solutions continues to grow, SHA can be seen in more applications now than ever. The Intel® SHA Extensions are designed to improve the performance of these compute-intensive algorithms on Intel® architecture-based processors.

The Intel® SHA Extensions are a family of seven instructions based on the Intel® Streaming SIMD Extensions (Intel® SSE) that are used together to accelerate the performance of processing SHA-1 and SHA-256 on Intel architecture-based processors. Given the growing importance of SHA in our everyday computing devices, the instructions are designed to provide a needed boost of performance to hashing a single buffer of data. The performance benefits will not only help improve responsiveness and lower power consumption for a given application, but they may also enable developers to adopt SHA in new applications to protect data while delivering to their user experience goals. The instructions are defined in a way that simplifies their mapping into the algorithm processing flow of most software libraries, thus enabling easier development.

Information on Intel® SHA can be found at: <http://software.intel.com/en-us/artTGLes/intel-sha-extensions>

6.2.4 New Cryptographic Acceleration Instructions

The processor supports new extensions for acceleration of some common or emerging cryptographic algorithms:

1. AVX2 version of VPMSQB for acceleration of RSA signature verification
2. SHA2-512 (or 384)
3. Chinese crypto standards SM3 and SM4

6.3 Intel® Secure Key

The processor supports Intel® Secure Key (formerly known as Digital Random Number Generator or DRNG), a software visible random number generation mechanism supported by a high-quality entropy source. This capability is available to programmers through the RDRAND and RDSEED instructions. The resultant random number generation capability is designed to comply with existing industry standards in this regard (ANSI X9.82 and NIST SP 800-90).

Some possible usages of the RDRAND and RDSEED instructions include cryptographic key generation as used in a variety of applications, including communication, digital signatures, secure storage, etc.

RDRAND and RDSEED instructions specifications and functional descriptions are included in the *Intel® 64 Architectures Software Developer's Manual, Volume 2*. Available at:

<http://www.intel.com/products/processor/manuals>

6.4 Execute Disable Bit

The Execute Disable Bit allows memory to be marked as non-executable when combined with a supporting operating system. If code attempts to run in non-executable memory, the processor raises an error to the operating system. This feature can prevent some classes of viruses or worms that exploit buffer overrun vulnerabilities and can, thus, help improve the overall security of the system.

6.5 Intel® Supervisor Mode Execution Protection (Intel® SMEP)

Intel® Supervisor Mode Execution Protection (Intel® SMEP) is a mechanism that provides the next level of system protection by blocking malicious software attacks from user mode code when the system is running in the highest privilege level. This technology helps to protect from virus attacks and unwanted code from harming the system. For more information, refer to *Intel® 64 Architectures Software Developer's Manual, Volume 3* at:

<http://www.intel.com/products/processor/manuals>

6.6 Intel® Supervisor Mode Access Protection (Intel® SMAP)

Intel® Supervisor Mode Access Protection (Intel® SMAP) is a mechanism that provides next level of system protection by blocking a malicious user from tricking the operating system into branching off user data. This technology shuts down very popular attack vectors against operating systems.

For more information, refer to *Intel® 64 Architectures Software Developer's Manual, Volume 3*:

<http://www.intel.com/products/processor/manuals>

6.7 User Mode Instruction Prevention (UMIP)

User Mode Instruction Prevention (UMIP) provides additional hardening capability to the OS kernel by allowing certain instructions to execute only in supervisor mode (Ring 0).

If the OS opt-in to use UMIP, the following instruction are enforced to run in supervisor mode:

- **SGDT** - Store the GDTR register value
- **SIDT** - Store the IDTR register value
- **SLDT** - Store the LDTR register value
- **SMSW** - Store Machine Status Word
- **STR** - Store the TR register value

An attempt at such execution in user mode causes a general protection exception (#GP).

UMIP specifications and functional descriptions are included in the *Intel® 64 Architectures Software Developer's Manual, Volume 3*. Available at:

<http://www.intel.com/products/processor/manuals>

6.8 Read Processor ID (RDPID)

A companion instruction that returns the current logical processor's ID and provides a faster alternative to using the RDTSCP instruction.

RDPID specifications and functional descriptions are included in the *Intel® 64 Architectures Software Developer's Manual, Volume 2*. Available at:

<http://www.intel.com/products/processor/manuals>

6.9 Intel System Resources Defense and Intel System Security Report

Intel® System Resources Defense is the collection of techniques and code within the BIOS used to create and enforce HW access policy for the SMI handler. It consists of a collection of policy mechanisms that are configured by POST before the SMI handler is locked down. Once the SMI handler is locked, all accesses into the system must be compliant with the policy established during POST.

Intel® Runtime BIOS Resilience is a subset of Intel® System Resources Defense covering SMM memory policy only. Intel® Runtime BIOS Resilience Protection hardens the SMI handler via hardware enforced BIOS policy regarding SMI handler access to memory using an enhanced paging policy. This paging policy covers SMI handler access to both BIOS and MLE resources. Intel® Runtime BIOS Resilience Protection is extended using a technology codenamed Intel® System Security Report.

The Platform Properties Assessment Module (PPAM) is the primary component of Intel® System Security Report. It collects and reports information about platform SMM implementation and configuration, in order to provide trustworthy attestation of the resulting SMI memory policy regarding SMM secure configuration and access to MLE owned memory. Intel® System Security Report is used to create a trustworthy report describing the SMM policy. PPAM is a major/core component of Intel® System Security Report 1.0/1.1 technology

6.10 Intel® Total Memory Encryption - Multi-Key

This technology encrypts the platform's entire memory with multiple encryption keys. Intel® Total Memory Encryption (Intel® TME), when enabled via BIOS configuration, ensures that all memory accessed from the Intel processor is encrypted.

Intel TME encrypts memory accesses using the AES XTS algorithm with 256-bit keys. The global encryption key used for memory encryption is generated using a hardened random number generator in the processor and is not exposed to software.

Software (OS/VMM) manages the use of keys and can use each of the available keys for encrypting any page of the memory. Thus, Intel® Multi-key Total Memory Encryption (Intel® MK-TME) allows page granular encryption of memory. By default Intel MK-TME uses the Intel TME encryption key unless explicitly specified by software.

Data in-memory and on the external memory buses is encrypted and exists in plain text only inside the processor. This allows existing software to operate without any modification while protecting memory using Intel TME. Intel TME does not protect memory from modifications.

Intel TME allows the BIOS to specify a physical address range to remain unencrypted. Software running on Intel TME enabled system has full visibility into all portions of memory that are configured to be unencrypted by reading a configuration register in the processor.

NOTES

- More information on Intel MK-TME can be found at [854686](#)
 - A cold boot is required when enable/ disable Intel TME feature on this platform.
-

6.11 Control-flow Enforcement Technology (Intel® CET)

Return-oriented Programming (ROP), and similarly CALL/JMP-oriented programming (COP/JOP), have been the prevalent attack methodology for stealth exploit writers targeting vulnerabilities in programs.

Intel® CET provides the following components to defend against ROP/JOP style control-flow subversion attacks:

6.11.1 Shadow Stack

A shadow stack is a second stack for the program that is used exclusively for control transfer operations. This stack is separate from the data stack and can be enabled for operation individually in user mode or supervisor mode.

The shadow stack is protected from tamper through the page table protections such that regular store instructions cannot modify the contents of the shadow stack. To provide this protection the page table protections are extended to support an additional attribute for pages to mark them as "Shadow Stack" pages. When shadow stacks are enabled, control transfer instructions/flows such as near call, far call, call to interrupt/exception handlers, etc. store their return addresses to the shadow stack. The RET instruction pops the return address from both stacks and compares them. If the return addresses from the two stacks do not match, the processor signals a control protection exception (#CP). Stores from instructions such as MOV, XSAVE, etc. are not allowed to the shadow stack.

6.11.2 Indirect Branch Tracking

The ENDBR32 and ENDBR64 (collectively ENDBRANCH) are two instructions that are used to mark valid indirect CALL/JMP target locations in the program. This instruction is a NOP on legacy processors for backward compatibility.

The processor implements a state machine that tracks indirect JMP and CALL instructions. When one of these instructions is seen, the state machine moves from IDLE to WAIT_FOR_ENDBRANCH state. In WAIT_FOR_ENDBRANCH state the next instruction in the program stream must be an ENDBRANCH. If an ENDBRANCH is not seen the processor causes a control protection exception (#CP), otherwise the state machine moves back to IDLE state.

More information on Intel® CET can be found at Intel® 64 and IA-32 Architectures Software Developer's Manual, Volume 1, Chapter 18:

<https://www.intel.com/content/www/us/en/developer/articles/technical/intel-sdm.html>

6.12 BIOS Guard

The platform must implement hardware controls to provide the platform manufacturer a robust mechanism to prevent unauthorized flash updates, while still allowing platform manufacturer approved updates. Intel® Platform Protection Technology with Intel® BIOS Guard accomplishes this by providing a very robust environment from which signed update images can be cryptographically verified and host flash writes can be done. Furthermore, a Intel® BIOS Guard enabled system does not allow host flash writes from any other environment.

6.13 Intel® Platform Trust Technology

Intel® Platform Trust Technology (Intel® PTT) offers the capabilities of discrete TPM 2.0. Intel PTT is a platform functionality for credential storage and key management used by Windows* 8, Windows* 10 and Windows* 11. Intel PTT supports BitLocker* for hard drive encryption and supports all Microsoft* requirements for Trusted Platform Module (TPM) 2.0.

6.14 Linear Address Space Separation (LASS)

Linear Address Space Separation (LASS) can harden an OS kernel against specific classes of side channel exploit techniques.

6.15 Intel® Total Storage Encryption (Intel® TSE)

Intel® Total Storage Encryption provides a security measure for a PCIe-NVMe device by encrypting the data in the device using the Intel inline encryption.

The Intel® Total Storage Encryption, which is enabled over BIOS configuration, ensures that the data in the storage device and through external PCIe buses are encrypted and exist in plaintext only inside the Intel processor.

The Intel® Total Storage Encryption encrypts the data using the AES XTS algorithm with a 256-bit data encryption key and 256-bit tweak key when the data is written to the storage from system memory and decrypted when read from the storage to the system memory.

The software can wrap the 256-bit keys using Platform Bind Key BLOB (PBNDKB) instruction and get the key handle wrapped by a platform-specific wrapping key. Once the software obtains the handle, the software can delete the original keys from memory. The software can also program the 256-bit keys using either the key handles or plaintext keys through the Platform Config (PCONFIG) instruction.

The Intel® Total Storage Encryption driver or the UEFI Inline Cryptographic Interface Protocol programs the keys over the PBNDKB/PCONFIG instructions and creates the Intel® Total Storage Encryption table in the system memory so that the Intel® Total Storage Encryption HW can look up the table to identify the Key ID with tweak value based on the physical address accessed through the storage driver of the PCIe-NVMe device. BitLocker could use the Intel® Total Storage Encryption as security enhancement through BitLocker Drive Encryption configuration besides OS managed software encryption using AES-NI instruction sets.

NOTES

- Intel® TSE is currently a vPRO only feature and should be enabled on any NVMe attached to the platform PCIe port, including PCH ports.
 - Intel® TSE works with PCIe NVMe storage devices off CPU and PCH Root Ports.
 - Intel® TSE works with up to a max of 8 PCIe NVMe storage devices (regardless of Processor/PCH).
-

6.16 Security Firmware Engines

6.16.1 Intel® Converged Security and Management Engine (Intel® CSME)

CSxS is security engine which provides security firmware authentication and loading, secure boot, platform debug control and manageability via Intel® Active Management Technology (Intel® AMT).

CSxS has a standalone small x86 processor, memory, crypto engine and I/O's.

CSxS is isolated in a secured hardware and firmware environment from the host processors.

6.16.2 Intel® Silicon Security Engine

A Security engine which is HW IP is based on CSxS HW IP and FW IP design to be silicon Root of Trust providing secure FW loading, measurements and on-tile certification authority.

The firmware is based on a new design which focus on security, simplicity of architecture and isolated environment.

6.16.3 Intel® Graphics System Controller (Intel® GSC)

Graphics System Controller (GSC) is a HW IP block embedded within the media IP block of the graphics component to support content and display protection services such as DRM and HDCP.

NOTE

All graphics security functionalities are handled by GSC which was previously implemented by CSxS.

6.16.4 Intel® Partner Security Engine

Intel® Partner Security Engine is a new security engine whose HW IP block is based on Intel® Silicon Security Engine. Its IP is located on the PCD tile and its purpose is to run or offload security-sensitive flows of the operating system.

Intel® Partner Security Engine HW components are provided by Intel while the IP FW is provided by third party. Initial target for NVL is to implement Microsoft Pluton FW on Intel® Partner Security Engine IP.

7.0 Intel Virtualization Technology

Intel® Virtualization Technology (Intel® VT) makes a single system appear as multiple independent systems to software. This allows multiple, independent operating systems to run simultaneously on a single system. Intel® VT comprises technology components to support Virtualization of platforms based on Intel® architecture microprocessors and chipsets.

Intel® Virtualization Technology (Intel® VT) Intel® 64 and Intel® Architecture (Intel® VT-x) added hardware support in the processor to improve the Virtualization performance and robustness. Intel® Virtualization Technology for Directed I/O (Intel® VT-d) extends Intel® VT-x by adding hardware assisted support to improve I/O device Virtualization performance.

Intel® VT-x specifications and functional descriptions are included in the *Intel® 64 Architectures Software Developer's Manual, Volume 3*. Available at:

<http://www.intel.com/products/processor/manuals>

The Intel® VT-d specification and other VT documents can be referenced at:

<http://www.intel.com/content/www/us/en/virtualization/virtualization-technology/>.

7.1 Intel® Virtualization Technology (Intel® VT) for Intel® 64 and Intel® Architecture (Intel® VT-x)

Intel® VT-x Objectives

Intel® VT-x provides hardware acceleration for virtualization of IA platforms. Virtual Machine Monitor (VMM) can use Intel® VT-x features to provide an improved reliable virtualization platform. By using Intel® VT-x, a VMM is:

- **Robust:** VMMs no longer need to use para-virtualization or binary translation. This means that VMMs will be able to run off-the-shelf operating systems and applications without any special steps.
- **Enhanced:** Intel® VT enables VMMs to run 64-bit guest operating systems on IA x86 processors.
- **More Reliable:** Due to the hardware support, VMMs can now be smaller, less complex, and more efficient. This improves reliability and availability and reduces the potential for software conflicts.
- **More Secure:** The use of hardware transitions in the VMM strengthens the isolation of VMs and further prevents corruption of one VM from affecting others on the same system.

Intel® VT-x Key Features

The processor supports the following Intel® VT-x features:

- **Mode-based Execute Control for EPT (MBEC)**

A mode of EPT operation which enables different controls for executability of Guest Physical Address (GPA) based on Guest specified mode (User/ Supervisor) of linear address translating to the GPA.

- **Extended Page Table (EPT) Accessed and Dirty Bits**

EPT A/D bits enabled VMMs to efficiently implement memory management and page classification algorithms to optimize VM memory operations, such as defragmentation, paging, live migration, and check-pointing. Without hardware support for EPT A/D bits, VMMs may need to emulate A/D bits by marking EPT paging-structures as not-present or read-only, and incur the overhead of EPT page-fault VM exits and associated software processing.

- **EPTP (EPT pointer) switching**

EPTP switching is a specific VM function. EPTP switching allows guest software (in VMX non-root operation, supported by EPT) to request a different EPT paging-structure hierarchy. This is a feature by which software in VMX nonroot operation can request a change of EPTP without a VM exit. The software will be able to choose among a set of potential EPTP values determined in advance by software in VMX root operation.

- **Pause loop exiting**

Support VMM schedulers seeking to determine when a virtual processor of a multiprocessor virtual machine is not performing useful work. This situation may occur when not all virtual processors of the virtual machine are currently scheduled and when the virtual processor in question is in a loop involving the PAUSE instruction. The feature allows detection of such loops and is thus called PAUSE-loop exiting.

- **Extended Page Tables (EPT)**

- EPT is hardware assisted page table virtualization.
- It eliminates VM exits from guest OS to the VMM for shadow page-table maintenance.

- **Virtual Processor IDs (VPID)**

- Ability to assign a VM ID to tag processor P/LP E core hardware structures (such as TLBs).
- This avoids flushes on VM transitions to give a lower-cost VM transition time and an overall reduction in virtualization overhead.

- **Guest Preemption Timer**

- The mechanism for a VMM to preempt the execution of a guest OS after an amount of time specified by the VMM. The VMM sets a timer value before entering a guest.
- The feature aids VMM developers in flexibility and Quality of Service (QoS) guarantees.

- **Descriptor-Table Exiting**

- Descriptor-table exiting allows a VMM to protect a guest OS from internal (malicious software based) attack by preventing the relocation of key system data structures like IDT (interrupt descriptor table), GDT (global descriptor table), LDT (local descriptor table), and TSS (task segment selector).
- A VMM using this feature can intercept (by a VM exit) attempts to relocate these data structures and prevent them from being tampered by malicious software.

- **Hypervisor-Managed Linear Address Translation (HLAT)**

- HLAT is active when the “enable HLAT” VM-execution control is 1. The processor looks up the HLAT if, during a guest linear address translation, the guest linear address matches the Protected Linear Range. The lookup from guest linear addresses to the guest physical address and attributes is determined by a set of HLAT paging structures.
- The guest paging structure managed by the guest OS specifies the ordinary translation of a guest linear address to the guest physical address and attributes that the guest ring-0 software has programmed, whereas HLAT specifies the alternate translation of the guest linear address to guest physical address and attributes that the Secure Kernel and VMM seek to enforce. A logical processor uses HLAT to translate guest linear addresses only when those guest linear addresses are used to access memory (both for code fetch and data load/store) and the guest linear addresses match the PLR programmed by the VMM/Secure Kernel.
- HLAT specifications and functional descriptions are included in the Intel® Architecture Instruction Set Extensions Programming Reference. Available at: <https://software.intel.com/en-us/download/intel-architecture-instruction-set-extensions-programming-reference>

- **Virtualization Exceptions**

A virtualization exception is a new processor exception. It uses vector 20 and is abbreviated #VE. A virtualization exception can occur only in VMX non-root operation. Virtualization exceptions occur only with certain settings of certain VM-execution controls. Generally, these settings imply that certain conditions that would normally cause VM exits instead cause virtualization exceptions

- **Translation of Guest-Physical Addresses Used by Intel Processor Trace**

With the “Intel PT uses guest physical addresses” feature, the addresses used by Intel PT can be treated as guest-physical addresses and translated using EPT. These addresses include the addresses of the output regions as well as the addresses of the ToPA entries that contain the output-region addresses.

7.2 Intel® Virtualization Technology (Intel® VT) for Directed IO (Intel® VT-d)

Intel® VT-d Objectives

The key Intel® VT-d objectives are domain-based isolation and hardware-based virtualization. A domain can be abstractly defined as an isolated environment in a platform to which a subset of host physical memory is allocated. Intel® VT-d provides accelerated I/O performance for a Virtualization platform and provides software with the following capabilities:

- **I/O Device Assignment and Security:** for flexibly assigning I/O devices to VMs and extending the protection and isolation properties of VMs for I/O operations.
- **DMA Remapping:** for supporting independent address translations for Direct Memory Accesses (DMA) from devices.
- **Interrupt Remapping:** for supporting isolation and routing of interrupts from devices and external interrupt controllers to appropriate VMs.
- **Reliability:** for recording and reporting to system software DMA and interrupt errors that may otherwise corrupt memory or impact VM isolation.

Intel® VT-d accomplishes address translation by associating transaction from a given I/O device to a translation table associated with the Guest to which the device is assigned. It does this by means of the data structure in the following illustration. This table creates an association between the device's PCI Express* Bus/Device/Function (B/D/F) number and the base address of a translation table. This data structure is populated by a VMM to map devices to translation tables in accordance with the device assignment restrictions above and to include a multi-level translation table (VT-d Table) that contains Guest specific address translations.

Figure 2. Device to Domain Mapping Structure in Legacy Mode

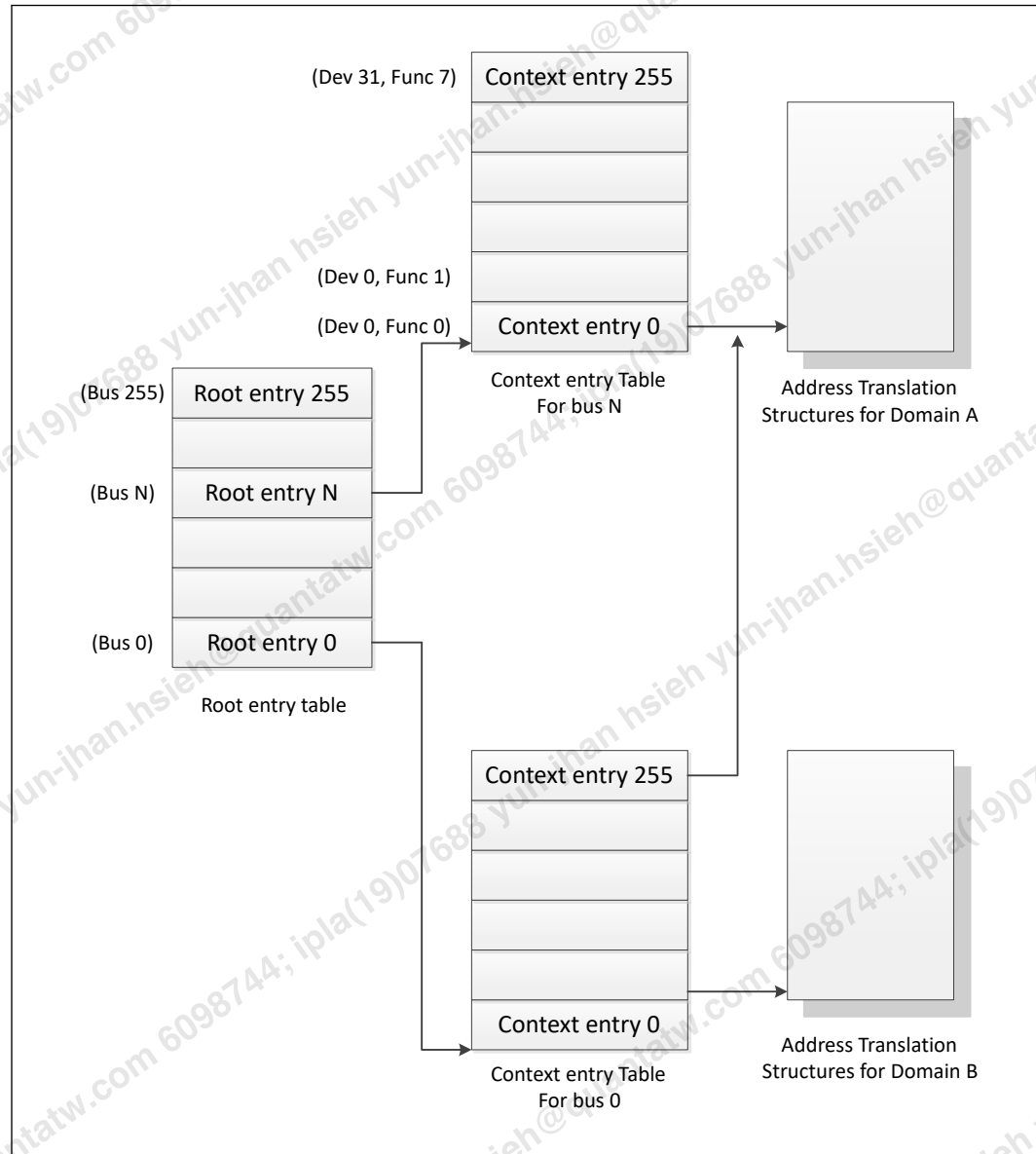
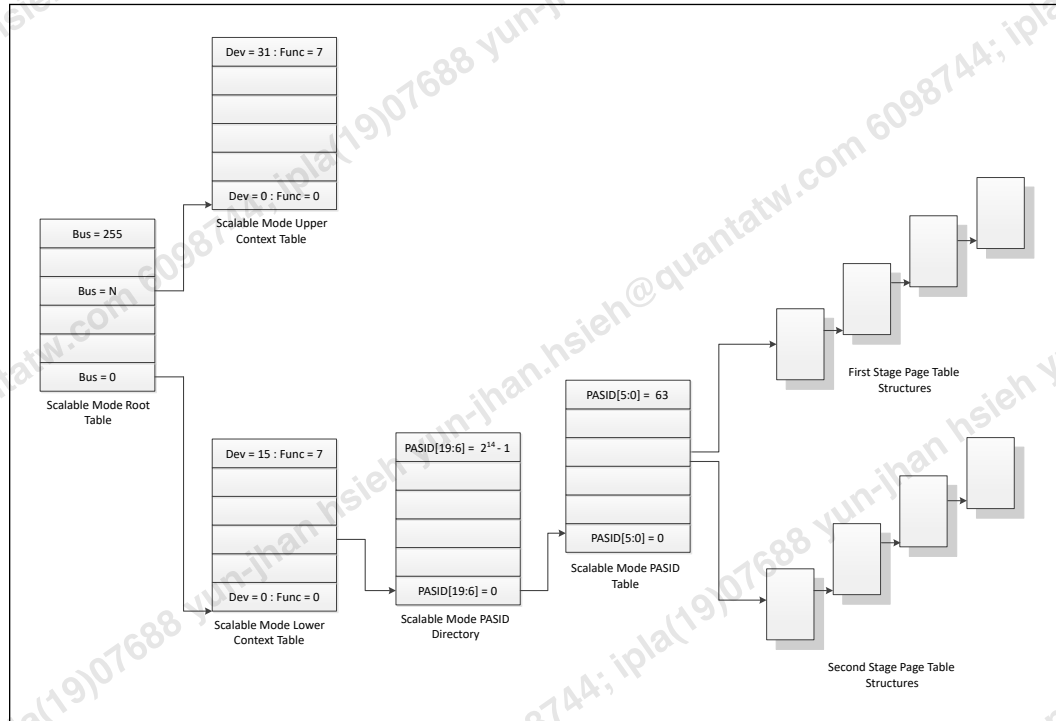


Figure 3. Device to Domain Mapping Structure in Scalable Mode


Intel® VT-d functionality often referred to as an Intel® VT-d Engine, has typically been implemented at or near a PCI Express* host bridge component of a computer system. This might be in a chipset component or in the PCI Express functionality of a processor with integrated I/O. When one such VT-d engine receives a PCI Express transaction from a PCI Express bus, it uses the B/D/F number associated with the transaction to search for an Intel® VT-d translation table. In doing so, it uses the B/D/F number to traverse the data structure shown in the above figure.

- If it finds a valid Intel® VT-d table in this data structure, it uses that table to translate the address provided on the PCI Express bus.
- If it does not find a valid translation table for a given translation, this results in an Intel® VT-d fault.

If Intel® VT-d translation is required, the Intel® VT-d engine performs an N-level table walk.

For more information, refer to Intel® Virtualization Technology for Directed I/O Architecture Specification:

<http://www.intel.com/content/dam/www/public/us/en/documents/product-specifications/vt-directed-io-spec.pdf>

Intel® VT-d Key Features

The processor supports the following Intel® VT-d features:

- Memory controller and processor graphics comply with the Intel® VT-d 4.0 Specification.
- 3 Intel® VT-d DMA Remapping Hardware Units:

- **GFX** DMA Remapping Hardware Unit: servicing iGD (Processor Graphics Dev2)
- **Non-GFX** DMA Remapping Hardware Unit: servicing NPU (Dev11), PMCS (dev4)
- Default DMA remap engine rest of PCI compatible devices and IOxAPIC + HPET
- 42-bit guest physical address and host physical address widths
- 4-level Intel® VT-d Page walk - all VTd engines support 4-level tables only (adjusted guest address width of 48 bits)
- Support Device-TLB for both engines for integrated accelerators (that is, GFX and Non-GFX)
- Support for register-based fault recording only (for single entry only) and support for MSI interrupts for faults
- Support for both leaf and non-leaf caching
- Support for non-caching of invalid page table entries
- Support for hardware-based flushing of translated but pending writes and pending reads, upon invalidation
- Support for all Queue based Invalidation descriptor types
- Support for Interrupt Remapping and Posted Interrupt
- Support Abort DMA Mode
- Support Performance Monitoring
- Intel® VT-d - All VTd engines support 4K, 2M and 1G page sizes
- Scalable Mode - All VTd engines support Scalable mode operation (using RID_PASID only)
- Nested - All Intel® VT-d engines support Nested translation

7.3 Intel® APIC Virtualization Technology (Intel® APICv)

APIC virtualization is a collection of features that can be used to support the virtualization of interrupts and the Advanced Programmable Interrupt Controller (APIC).

When APIC virtualization is enabled, the processor emulates many accesses to the APIC, tracks the state of the virtual APIC, and delivers virtual interrupts — all in VMX non-root operation without a VM exit.

The following are the VM-execution controls relevant to APIC virtualization and virtual interrupts:

- **Virtual-interrupt Delivery:** This control enables the evaluation and delivery of pending virtual interrupts. It also enables the emulation of writes (memory-mapped or MSR-based, as enabled) to the APIC registers that control interrupt prioritization.
- **Use TPR Shadow:** This control enables emulation of accesses to the APIC's task-priority register (TPR) via CR8 and, if enabled, via the memory-mapped or MSR-based interfaces.

- **Virtualize APIC Accesses:** This control enables virtualization of memory-mapped accesses to the APIC by causing VM exits on accesses to a VMM-specified APIC-access page. Some of the other controls, if set, may cause some of these accesses to be emulated rather than causing VM exits.
- **Virtualize x2APIC Mode:** This control enables virtualization of MSR-based accesses to the APIC.
- **APIC-register Virtualization:** This control allows memory-mapped and MSR-based reads of most APIC registers (as enabled) by satisfying them from the virtual-APIC page. It directs memory-mapped writes to the APIC-access page to the virtual-APIC page, following them by VM exits for VMM emulation.
- **Process Posted Interrupts:** This control allows software to post virtual interrupts in a data structure and send a notification to another logical processor; upon receipt of the notification, the target processor will process the posted interrupts by copying them into the virtual-APIC page.

NOTE

Intel® APIC Virtualization Technology may not be available on all SKUs.

Intel® APIC Virtualization specifications and functional descriptions are included in the *Intel® 64 Architectures Software Developer's Manual, Volume 3*. Available at:

<http://www.intel.com/products/processor/manuals>

8.0 Instructions Set Enhancements

8.1 CMPccXADD

CMPccXADD is an instruction set that can be used to optimize certain highly contended synchronization scenarios that today use CMPXCHG (semaphores, shared queues, etc).

8.2 LAM

LAM repurposes the upper (untranslated) linear address bits to make them available for software use (for example, as metadata). This is accomplished by removing exiting canonical address checks when LAM is enabled.

In 64-bit mode, linear address have 64 bits and are translated either with 4-level paging, which translates the low 48 bits of each linear address, or with 5-level paging, which translates 57 bits. The upper linear-address bits are reserved through the concept of canonicity. A linear address is 48-bit canonical if bits 63:47 of the address are identical; it is 57-bit canonical if bits 63:56 are identical. (Clearly, any linear address that is 48-bit canonical is also 57-bit canonical.)

When 4-level paging is active, the processor requires all linear addresses used to access memory to be 48-bit canonical; similarly, 5-level paging ensures that all linear addresses are 57-bit canonical.

Software usages that associate metadata with a pointer might benefit from being able to place metadata in the upper (untranslated) bits of the pointer itself. However, the canonicity enforcement mentioned earlier implies that software would have to mask the metadata bits in a pointer (making it canonical) before using it as a linear address to access memory or alternatively create Paging translation tables with redundancies.

LAM allows software to use pointers with metadata without having to mask the metadata bits. A LAM enabled processor internally ignores the metadata bits in a pointer before using it as a linear address to access memory.

NOTE

LAM is supported only in 64-bit mode and applies only to addresses used for data accesses. LAM does not apply to addresses used for instruction fetches or to those that specify the targets of jump and call instructions.

9.0 Platform Environmental Control Interface (PECI)

PECI is an Intel proprietary interface that provides a communication channel between Intel processors and external components such as Super IO (SIO) and Embedded Controllers (EC) to provide processor temperature, Turbo, Assured Power (cTDP), and Memory Throttling Control mechanisms and many other services. PEFI is used for platform thermal management and real-time control and configuration of processor features and performance.

NOTE

- PEFI is supported **only** over eSPI interface
-

10.0 Intel® Neural Processing Unit (Intel® NPU)

The NPU IP targets general Deep Learning inferencing applications in AI PC, connected devices and edge servers. It delivers the high processing throughput necessary to satisfy the demands of such applications.

The NPU technology is applicable to personal computing devices such as tablets, laptops and PCs as a way to introduce AI-based applications and services in power and performance sensitive platforms.

10.1 Functional Description

The NPU IP comprises several individual components grouped into two major subsystems:

1. Host Control
2. Deep Learning Accelerator

Details of these blocks are provided in the next sections.

Host Control

The functionality of the NPU is exposed to a SoC via a base set of registers (enumerated as a PCIe device or directly memory mapped into the address space of the Host).

These registers provide access to control and data path interfaces and reside in the Host Subsystem. All host communications are consumed by the NPU scheduler, a 64-bit RISC-V micro-controller. As well as responding to control messages it manages all the job submission/completion FIFOs that make up the data path of the NPU.

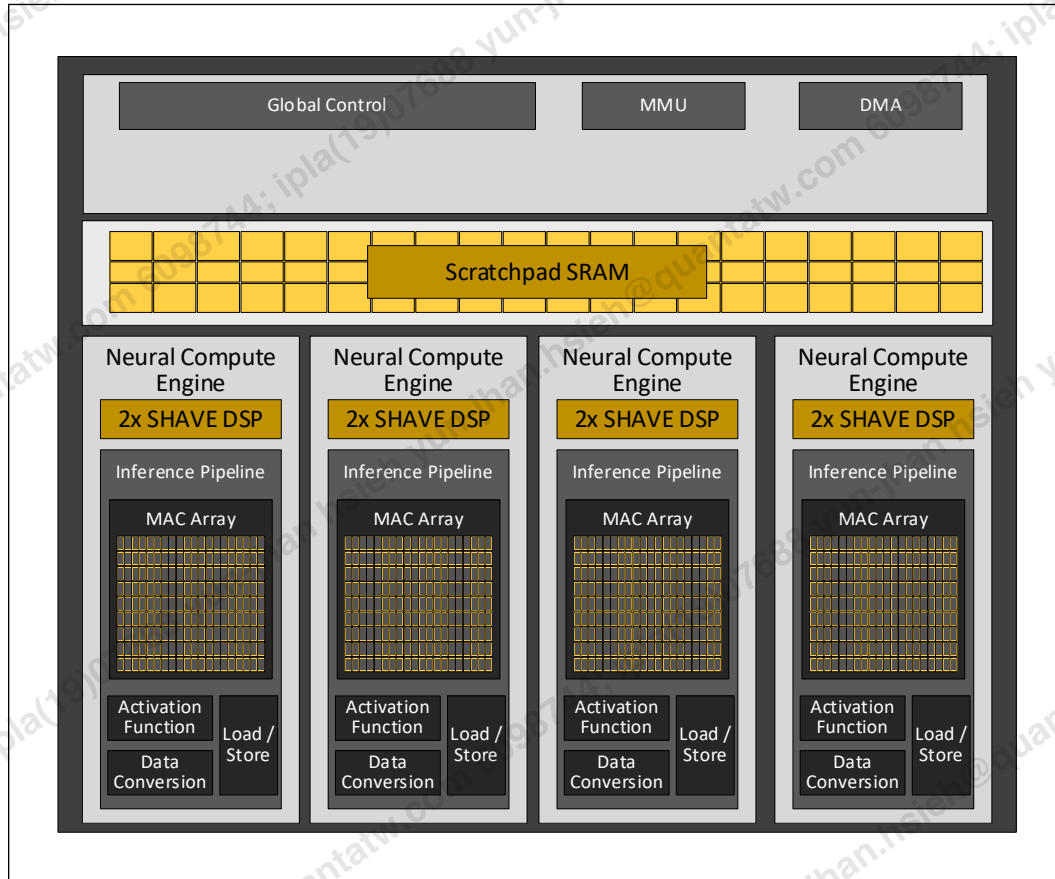
Deep Learning (Neural Compute Engine)

The NPU IP Deep Learning capability is provided by a configurable number of Neural Compute Engine (NCE) Tiles. The NCE Tiles are managed by the NPU Scheduler.

Each Tile includes a configurable amount of near-compute SRAM, one Data Processing Unit (DPU) with a configurable number of multiply-accumulates, and two DSPs (SHAVE-512) for optimal processing of custom deep learning operations. Global barriers and task FIFOs are also available for job synchronization and dispatch. The NPU comprises of 4 NCE Tiles, totaling 16k DPU INT8 MACs, 8 DSPs and 8 MB of associated near-compute memory.

Below is the block diagram of NPU IP:

Figure 4. NPU IP Block Diagram



10.1.1 HOST Control

The IP comprises several individual components such as CPU Subsystem, Host Subsystem and Host Interface.

The Host Subsystem is the main downstream interface between the NPU and the processor.

The Host Interface is the upstream interface between the NPU and the processor/buttruss.

The primary functions of the CPU subsystem are job scheduling and NPU resource management.

10.1.2 Deep Learning Accelerators (NCE)

The Neural Compute Engine (NCE) is a hardware accelerator for Deep Neural Network (DNN) workloads. It features a highly configurable pipeline for maximum support of DNN operations, such as Long Short-Term memory (LSTM) and Local Response Norm (LRN). It also leverages low precision for optimal performance.

The Neural Compute Engine is built from 4 **Neural Compute Tiles**, where each tile is a primary unit of computing. Each NCE Tile incorporates its own memory, and DPU and ACT-SHAVE compute resources, and can independently work on a single workload at a given time or be aggregated as a cluster of tiles running the same workload.

The NCE Subsystem features efficient Inter-Tile-Interconnect (ITI) with multicasting and broadcasting capability to allow NCE Tile Compute Units to efficiently share data between tiles (in case NCE Tiles are aggregated to split workload).

The NCE Subsystem incorporates DMA engine with compression unit and broadcasting/multicasting capability for populating multiple NCE Tiles Memory concurrently.

For hardware assisted task synchronization, the NCE Subsystem provides barriers and workload FIFOs. Barriers remove as much software overhead as possible through ISR loops and programming sequences to keep the compute and data-movement pipelines full.

10.1.2.1 Feature Set

- 4 Neural Compute Engine Tiles, where each tile consist of:
 - 2 MB CMX Memory
 - Two 512-bit ACT-SHAVES
 - DPU (4K MACs)
- DMA Engine:
 - 2x 64B AXI Interface to DDR
 - Bit Compactor Compression unit for weights decompression and activation compression and decompression
 - Broadcasting/Multicasting Capability to allow DMA to feed multiple NCE Tiles simultaneously.
 - Address Translation Prefetching capability in the DMA controller. A dedicated prefetch machine provides single-read accesses to pages at a configurable offset from the current transfer. All prefetch commands are carried on the read channel of the converged 64B AXI DMA data bus.
 - Address Patching capability for DRAM Accesses
- Inter-Tile-Interconnect (ITI) for output activation sharing (write-only). ITI provides broadcasting, multicasting, and unicasting capability,
- 256 KB of SHAVE L2 Cache for Data and Instruction shared between ACT-SHAVES.
- Barriers for hardware and assisted task synchronization and pipelining
- Programmable HW FIFO Block for Work Descriptors and IPC
- Tile Yield Recovery due to process defect
- Virtual Addressing for all resources used during Inference (Memory, Barriers, FIFOs and DMA/M2I Interrupt IDs)

10.1.2.2 NCE Tile

The NCE Tile is the building block of the NCE Subsystem. The NCE subsystem supports four tile configuration. Each NCE tile contains the following support:

- Maximum of 4 MB and minimum of 512 kB of NCE Tile Internal Storage.
- Single Data Processing Units (DPUs) where each DPU supports 4096 MACs.
- Two ACT-SHAVE DSPs with shared data and instruction L2 Cache (256 kB) used for flexible tensor compute operation.

10.1.2.3 ACTIVATION-SHAVE (ACT-SHAVE)

ACT-SHAVE is an Intel Movidius DSP Processor which supports 512 bit vector operations. Two ACT-SHAVE DSPs are placed in each NCE Tile and are used for custom layer and standard layers that do not map well to the DPU.

10.1.2.4 Data Processing Unit (DPU)

The Data Processing Unit (DPU) supports 4k MACs built from 256 MAC Processing Engines (MPE) with 16 Fused-MACs in each MPE.

Feature Set

- SprLUT feature - SprLUT was added for complex non-linear activation functions.
- Optimized HW support for standard and depth-wise convolutions,
- Convolution Kernel size of $M \times N$ where N, M are up to 11,
- Convolution Stride of up to 8,
- Support configurable padding of activations,
- Supports reuse of activations and weights to reduce CMX Memory read bandwidth,
- Support for both Dense and Sparse operations,
- Sparse Element-wise operations,
- Supported Data Types
 - FP8
 - F16
 - I8
 - U8
 - INT16
- Precision
 - FP32 and INT32 accumulators
 - Floating point and integer scaling supported
 - Integer floating point inline conversion
 - Floating point subnormal support
- Supports statistics gathering (Hardware Profiling)
- Features 64 Post Processing Elements (PPE) where each support:
 - Quantization
 - Activation function
 - Element-Wise functions
- Features Four 256-bit wide Read-Only Ports and four 256-bit wide Read/Write Ports

- Support for compression to increase effective TOPs by up to 2x.
- DPU State Machine is responsible for loading tensor workloads and micro-scheduling.
- New Feature in NPU that allows splitting Input channels (ICs) across MPE to improve the overall utilization of the PE array.

11.0 Power Management

For more information about Power Management Controller (PMC), refer to Nova Lake-S Platform Controller Hub (PCH) External Design Specification, Volume 1 of 2 (#835009).

NOTE

In this chapter, Sx refers to S3/S4/S5 states.

Table 15. Acronyms

Acronyms	Description
VR	Voltage Regulator

Table 16. References

Specification	Location
Advanced Configuration and Power Interface (ACPI)	Advanced Configuration and Power Interface (ACPI) Specification

11.1 Advanced Configuration and Power Interface States

This section describes System Power States and ACPI states supported by the processor.

For more information, refer to Nova Lake-S Platform Controller Hub (PCH) External Design Specification, Volume 1 of 2 (#835009).

Table 17. General System Power States

State	Description
G0/S0/C0	Full On: CPU operating. Individual devices may be shut to save power. The different CPU operating levels are defined by Cx states.
G0/S0/Cx	Cx state: CPU manages C-states by itself and can be in low power state
G0/S0ix/Cx	S0ix: The south supports an S0ix state which also requires the CPU be in a Cx state. Additional south power actions such as voltage reduction, chip-wide voltage rail removal may occur in this state.
G1/S3	Suspend-To-RAM (STR): The system context is maintained in system DRAM, but power is shut to non-critical circuits. Memory is retained, and refreshes continue. All external clocks are shut off; RTC clock and internal ring oscillator clocks are still toggling. In S3, the SLP_S3# signal stays asserted, the SLP_S4# and SLP_S5# signals are inactive until a wake occurs.
G1/S4	Suspend-To-Disk (STD): The context of the system is maintained on the disk. All power is then shut to the system except to the logic required to resume. Externally appears same as S5 but may have different wake events.
continued...	

State	Description
	In S4, SLP_S3# and SLP_S4# both stay asserted and SLP_S5# is inactive until a wake occurs.
G2/S5	Soft Off: System context not maintained. All power is shut except for the logic required to restart. A full boot is required when waking. Here, SLP_S3#, SLP_S4#, and SLP_S5# are all active until a wake occurs.
G3	Mechanical OFF: System context not maintained. All power shut except for the RTC. No "Wake" events are possible because the system does not have any power. This state occurs if the user removes the batteries, turns off a mechanical switch, or if the system power supply is at a level that is insufficient to power the "waking" logic. When system power returns the transition will depend on the state just prior to the entry to G3.

The table below shows the transitions rules among the various states.

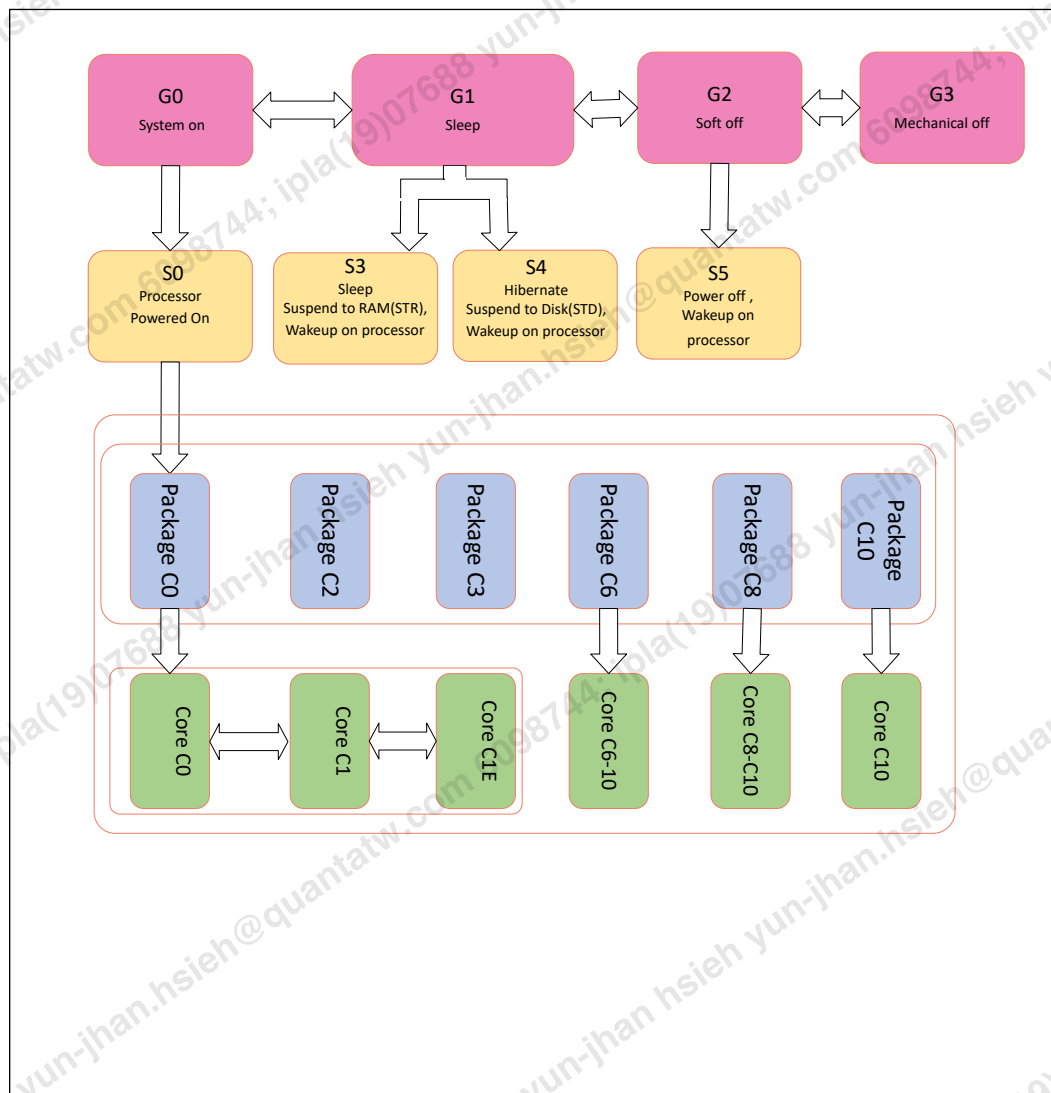
NOTE

Transitions among the various states may appear to temporarily transition through intermediate states. For example, in going from S0 to S5, it may appear to pass through the G1/S3/S4 state. These intermediate transitions and states are not listed in the table below.

Table 18. State Transition Rules for the Processor

Present State	Transition Trigger	Next State
G0/S0/C0	- SLP_EN bit set - Power Button Override³ - Mechanical Off/Power Failure	- G0/S0/Cx - G1/S3, G1/S4, or G2/S5 state - G2/S5 - G3
G0/S0/Cx	- Power Button Override³ - Mechanical Off/Power Failure	- G0/S0/C0 - S5 - G3
G0/S0ix/Cx	- Any south action which is blocked from occurring while in S0ix - CPU or south IP request for CPU C-state exit	- G0/S0ix/Cx - G0/S0/Cx - G0/S0/C2(or C0)
G1/S3, G1/S4	- Any Enabled Wake Event - Power Button Override³ - Mechanical Off/Power Failure	- G0/S0/C0² - G2/S5 - G3
G2/S5	- Any Enabled Wake Event - Mechanical Off/Power Failure	- G0/S0/C0² - G3
G3	Power Returns	S0/C0 (reboot) or G2/S5⁴ (stay off until power button pressed or other wake event)^{1,2}
Notes: 1. Some wake events can be preserved through power failure. 2. Transitions from the S3–S4–S5 states to the S0 state are deferred until BATLOW# is inactive. 3. Includes all other applicable types of events that force the host into and stay in G2/S5. 4. If the system was in G1/S4 before G3 entry, then the system will go to S0/C0 or G1/S4. 5. On G3 exit, prior to the first transition to S0, S5 power may be higher than S5 power after the first S0 to S5 transition. Some processor settings required to achieve minimum S5 power are loaded during first boot to S0 after a G3 exit. Consequently, entry into S5 from S0 will result in a more power-optimized S5 state than entry into S5 from G3 without an S5–S0–S5 transition. The difference is expected to be in the few mW range		

Figure 5. Power State Block Diagram



11.2 Processor IA Core Power Management

While executing code, Enhanced Intel SpeedStep® Technology and Intel® Speed Shift technology optimizes the processor's IA core frequency and voltage based on workload. Each frequency and voltage operating point is defined by ACPI as a P-state. When the processor is not executing code, it is idle. A low-power idle state is defined by ACPI as a C-state. In general, deeper power C-states have longer entry and exit latencies.

11.2.1 OS/HW Controlled P-states

11.2.1.1 Enhanced Intel SpeedStep Technology

Enhanced Intel SpeedStep® Technology enables OS to control and select P-state. For more information, refer to [Enhanced Intel SpeedStep® Technology](#) on page 78.

11.2.1.2 Intel® Speed Shift Technology

Intel® Speed Shift Technology is an energy efficient method of frequency control by the hardware rather than relying on OS control. For more details, refer to [Intel® Speed Shift Technology](#) on page 78.

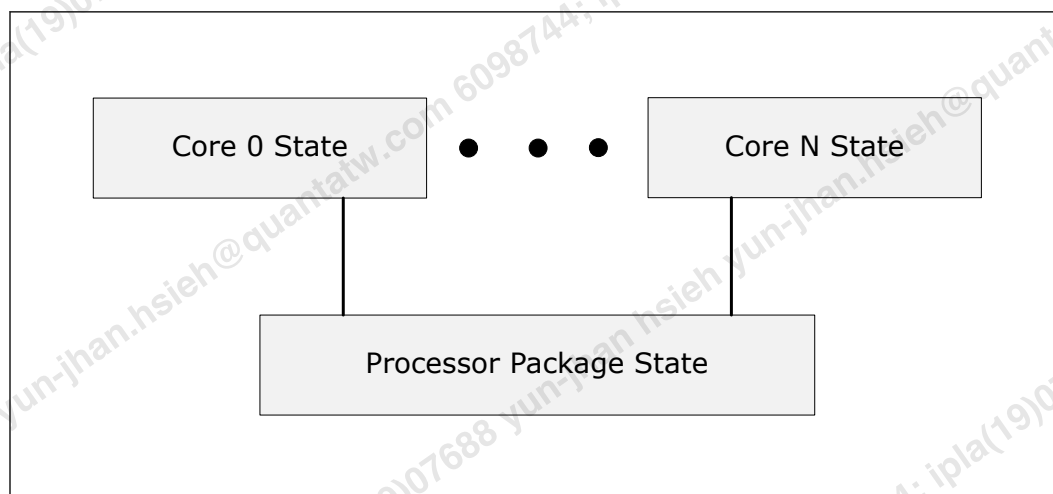
11.2.2 Low-Power Idle States

When the processor is idle, low-power idle states (C-states) are used to save power. More power savings actions are taken for numerically higher C-states. However, deeper C-states have longer exit and entry latencies. Resolution of C-states occurs at the processor IA core and processor package level.

CAUTION

Long-term reliability cannot be assured unless all the Low-Power Idle States are enabled.

Figure 6. Idle Power Management Breakdown of the Processor IA Cores



Processor IA core C-states are automatically resolved by the processor. For processor IA core C-states, a transition to and from C0 state is required before entering any other C-state.

11.2.3 Requesting the Low-Power Idle States

The primary software interfaces for requesting low-power idle states are through the MWAIT instruction with sub-state hints and the HLT instruction (for C1 and C1E). However, the software may make C-state requests using the legacy method of I/O reads from the ACPI-defined processor clock control registers, referred to as P_LVLx. This method of requesting C-states provides legacy support for operating systems that initiate C-state transitions using I/O reads.

For legacy operating systems, P_LVLx I/O reads are converted within the processor to the equivalent MWAIT C-state request. Therefore, P_LVLx reads do not directly result in I/O reads to the system. The feature, known as I/O MWAIT redirection, should be enabled in the BIOS. To enable it, refer to the appropriate processor family BIOS Specification.

The BIOS can write to the C-state range field of the PMG_IO_CAPTURE MSR to restrict the range of I/O addresses that are trapped and emulate MWAIT like functionality. Any P_LVLx reads outside of this range do not cause an I/O redirection to MWAIT(Cx) like the request. They fall through like a normal I/O instruction.

When P_LVLx I/O instructions are used, MWAIT sub-states cannot be defined. The MWAIT sub-state is always zero if I/O MWAIT redirection is used. By default, P_LVLx I/O redirections enable the MWAIT 'break on EFLAGS.IF' feature that triggers a wake up on an interrupt, even if interrupts are masked by EFLAGS.IF.

11.2.4 Processor IA Core C-State Rules

The following are general rules for all processor IA core C-states unless specified otherwise:

- A processor IA core C-State is determined by the lowest numerical thread state (such as Thread 0 requests C1E while Thread 1 requests C6 state, resulting in a processor IA core C1E state). Refer to the *G, S, and C Interface State Combinations* table.
- A processor IA core transitions to C0 state when:
 - An interrupt occurs
 - There is an access to the monitored address if the state was entered using an MWAIT/Timed MWAIT instruction
 - The deadline corresponding to the Timed MWAIT instruction expires
- An interrupt directed toward a single thread wakes up only that thread.
- If any thread in a processor IA core is active (in C0 state), the core's C-state will resolve to C0.
- Any interrupt coming into the processor package may wake any processor IA core.
- A system reset re-initializes all processor IA cores.

Table 19. Core C-States

Core C-State	C-State Request Instruction	Description
C0	N/A	The normal operating state of a processor IA core where a code is being executed
C1	MWAIT(C1)	AutoHalt - core execution stopped, autonomous clock gating (package in C0 state)
C1E	MWAIT(C1E)	Core C1 + lowest frequency and voltage operating point (package in C0 state)
C6-C10	MWAIT(C6/C8/10) or IO read=P_LVL3//6/8	Processor IA, flush their L1 instruction cache, the L1 data cache, and L2 cache to the LLC shared cache cores save their architectural state to an SRAM before reducing IA cores voltage, if possible may also be reduced to 0V. Core clocks are off.

This feature is disabled by default. BIOS should enable it in the PMG_CST_CONFIG_CONTROL register. The auto-demotion policy is also configured by this register. Refer to the appropriate processor family BIOS Specification for more details.

11.2.5 Package C-States

The processor supports C0, C2, C3, C6, C8, and C10 package states. The following is a summary of the general rules for package C-state entry. These apply to all package C-states, unless specified otherwise:

- A package C-state request is determined by the lowest numerical processor IA core C-state amongst all processor IA cores.
- A package C-state is automatically resolved by the processor depending on the processor IA core idle power states and the status of the platform components.
 - Each processor IA core can be at a lower idle power state than the package if the platform does not grant the processor permission to enter a requested package C-state.
 - The platform may allow additional power savings to be realized in the processor.
 - For package C-states, the processor is not required to enter C0 before entering any other C-state.
 - Entry into a package C-state may be subject to auto-demotion – that is, the processor may keep the package in a deeper package C-state than requested by the operating system if the processor determines, using heuristics, that the deeper C-state results in better power/performance.

The processor exits a package C-state when a break event is detected. Depending on the type of break event, the processor does the following:

- If a processor IA core break event is received, the target processor IA core is activated and the break event message is forwarded to the target processor IA core.
 - If the break event is not masked, the target processor IA core enters the processor IA core C0 state and the processor enters package C0.
 - If the break event is masked, the processor attempts to re-enter its previous package state.
- If the break event was due to a memory access or snoop request,
 - But the platform did not request to keep the processor in a higher package C-state, the package returns to its previous C-state.
 - And the platform requests a higher power C-state, the memory access or snoop request is serviced and the package remains in the higher power C-state.

PKG C2 and C3 can not be requested explicitly by the software.

Table 20. Package C-States

Package C- State	Description	Dependencies
PKG C0	Processor active state.	-
continued...		

Package C- State	Description	Dependencies
	At least one IA core in C0. Processor Graphic in RC0 (Graphics active state) or RC6 (Graphics Core power down state).	
PKG C2	Cannot be requested explicitly by the Software. All processor IA cores in C6 or deeper + Processor Graphic cores in RC6, memory path may be open. The processor will enter Package C2 when: - Transitioning from Package C0 to deep Package C - state or from deep Package C - state to Package C0. - All IA cores requested C6 or deeper + Processor Graphic cores in RC6 but there are constraints (LTR, programmed timer events in the near future and so forth) prevent entry to any state deeper than C2 state. - All IA cores requested C6 or deeper + Processor Graphic cores in RC6 but a device memory access request is received. Upon completion of all outstanding memory requests, the processor transitions back into a deeper package C-state.	All processor IA cores in C6 or deeper. Processor Graphic cores in RC6.
PKG C3	Cannot be requested explicitly by the Software. All cores in C6 or deeper + Processor Graphics in RC6, LLC may be flushed and turned off, memory in self refresh, memory clock stopped. The processor will enter Package C3 when: - All IA cores in C6 or deeper + Processor Graphic cores in RC6. - The platform components/devices allows proper LTR for entering Package C3.	All processor IA cores in C6 or deeper. Processor Graphics in RC6. memory in self refresh, memory clock stopped. LLC may be flushed and turned off.
PKG C6	Package C3 + BCLK is off + IMVP VRs voltage reduction/PSx state is possible. The processor will enter Package C6 when: - All IA cores in C6 or deeper + Processor Graphic cores in RC6. - The platform components/devices allow proper LTR for entering Package C6.	Package C3. BCLK is off. IMVP VRs voltage reduction/PSx state is possible.
PKG C8	Of all IA cores requested C8 + LLC should be flushed at once, voltage will be removed from the LLC. The processor will enter Package C8 when: - All IA cores in C8 or deeper + Processor Graphic cores in RC6. - The platform components/devices allow proper LTR for entering Package C8.	Package C6 If all IA cores requested C8, LLC is flushed in a single step, voltage will be removed from the LLC.
PKG C10	Display in PSR or powered off, all VRs at PS4 or LPM + crystal clock off. The processor will enter Package C10 when: - All IA cores in C10 + Processor Graphic cores in RC6. - The platform components/devices allow proper LTR for entering Package C10.	Package C8. All IA cores in C8 or deeper. Display in PSR or powered off¹. All VRs at PS4 or LPM. Crystal clock off.
<i>Note:</i> Display In PSR is only on single embedded panel configuration and panel support PSR feature.		

Package C-State Auto-Demotion

The Processor may demote the Package C-State to a shallower Package C-State to enable better performance, for example instead of going into Package C10, processor will demote to Package C6 (and shallower as required).

The processor's decision to demote the Package C-State is based on Power management parameters such as required C states latencies, entry/exit energy/power, Core wake rates, and device LTR (Latency Tolerance Report). This means that the processor is optimized to minimize platform energy for scenarios with low idle time.

Processor deeper Package C-State entry frequency is controlled to minimize platform energy. When Package C-State Auto-Demotion enabled, a reduced residency in a deeper Package C-State is expected during system runs with high wake rates. For example, some USB/Bluetooth* audio devices may request high wake rates to keep audio quality of service, this audio behavior may result in Package C-State Demotion and impact power consumption.

No change at IDLE scenario power consumption due to this feature. Package C-State Auto-Demotion is enabled by default and controlled through BIOS menu.

Modern Standby

Modern Standby is a platform state. On display time out the OS requests the processor to enter package C10 and platform devices at RTD3 (or disabled) in order to attain low power in idle. Modern Standby requires proper BIOS (refer to BIOS specification) and OS configuration.

Dynamic LLC Sizing

When all processor IA cores request C8 or deeper C-state, internal heuristics dynamically flushes the LLC. Once the processor IA cores enter a deep C-state, depending on their MWAIT sub-state request, the LLC is either gradually flushed N-ways at a time or flushed all at once. Upon the processor IA cores exiting to C0 state, the LLC is gradually expanded based on internal heuristics.

11.2.6 Package C-States and Display Resolutions

The integrated graphics engine has the frame buffer located in system memory. When the display is updated, the graphics engine fetches display data from system memory. Different screen resolutions and refresh rates have different memory latency requirements. These requirements may limit the deepest Package C-state the processor can enter. Other elements that may affect the deepest Package C-state available are the following:

- Display is on or off
- Single or multiple displays
- Native or non-native resolution
- Panel Self Refresh (PSR) technology

NOTE

Display resolution is not the only factor influencing the deepest Package C-state the processor can get into. Device latencies, interrupt response latencies, and core C-states are among other factors that influence the final package C-state the processor can enter.

The following table lists display resolutions and deepest available package C-State. The display resolutions are examples using common values for blanking and pixel rate. Actual results will vary. The table shows the deepest possible Package C-state. System workload, system idle, and AC or DC power also affect the deepest possible Package C-state.

Table 21. Deepest Package C-State Available

		S Processor line	
Resolution	Number of Displays	PSR Enabled	PSR Disabled
Up to 5120x3200 60 Hz ³	Single	PC10	PC8
Notes: 1. All Deep states are with Display ON. 2. The deepest C-state has variance, dependent various parameters such SW and Platform devices. 3. Partial data based on Pre-Silicon estimation, expected to be update in a future EDS release.			

11.3 Processor Graphics Power Management

11.3.1 Memory Power Savings Technologies

Intel® Rapid Memory Power Management (Intel® RMPM)

Intel® Rapid Memory Power Management (Intel® RMPM) conditionally places memory into self-refresh when the processor is in package C3 or deeper power state to allow the system to remain in the deeper power states longer for memory not reserved for graphics memory. Intel® RMPM functionality depends on graphics/display state (relevant only when processor graphics is being used), as well as memory traffic patterns generated by other connected I/O devices.

11.3.2 Display Power Savings Technologies

Intel® Seamless Display Refresh Rate Switching Technology (Intel® SDRRS Technology) with eDP* Port

Intel® DRRS provides a mechanism where the monitor is placed in a slower refresh rate (the rate at which the display is updated). The system is smart enough to know that the user is not displaying either 3D or media like a movie where specific refresh rates are required. The technology is very useful in an environment such as a plane where the user is in battery mode doing E-mail, or other standard office applications. It is also useful where the user may be viewing web pages or social media sites while in battery mode. Intel® DRRS supported on eDP ports only.

Intel® Display Power Saving Technology (Intel® DPST) 8.0

The Intel® DPST technique achieves back-light power savings while maintaining a good visual experience. This is accomplished by adaptively enhancing the displayed image while decreasing the back-light brightness simultaneously. The goal of this technique is to provide equivalent end-user-perceived image quality at a decreased back-light power level. Intel® DPST supported on eDP ports only.

1. The original (input) image produced by the operating system or application is analyzed by the Intel® DPST subsystem. An interrupt to Intel® DPST software is generated whenever a meaningful change in the image attributes is detected. (A meaningful change is when the Intel® DPST determines if the brightness of the displaying images and the image enhancement and back-light control needs to be altered.)
2. Intel® DPST subsystem applies an image-specific enhancement to increase image brightness.

3. A corresponding decrease to the back-light brightness is applied simultaneously to produce an image with similar user-perceived quality (such as brightness) as the original image.

Intel® OLED Power Saving Technology (Intel® OPST) 2.0

Intel® OPST solution uses same HW infrastructure as Intel® DPST. Frames are processed using frame change threshold based interrupt mechanism similar to Intel® DPST. Intel® OPST SW algorithm determines which pixels in the frame should be dimmed to save power keeping visual quality (such as contrast, color) impact to acceptable level. Since there is no backlight for OLED panels, the power savings come solely from pixel dimming.

Intel® Low Refresh Rate (Intel® LRR)

Intel® LRR is a combination of PSR2 and Dynamic Refresh Rate Switching. LRR uses the following mechanism for switching the refresh rate.

- VTOTAL Change (VRR/Adaptive Sync - VESA Standard)

LRR is classified into different versions based on the RR switching technique, Intel platform support/capabilities, and eDP* panel support/capabilities.

Panel Self-Refresh (PSR)

Panel Self-Refresh feature allows the Processor Graphics core to enter low-power state when the frame buffer content is not changing constantly. This feature is available on panels capable of supporting Panel Self-Refresh.

Low-Power Single Pipe (LPSP)

Low-power single pipe is a power conservation feature that helps save power by keeping the inactive pipes powered OFF. LPSP is achieved by keeping a pipe enabled during eDP* only with minimal display pipeline support.

Low-Power Dual Pipe (LPDP)

This feature is similar to LPSP and is applicable for designs with dual eDP* panels.

Intel® Smart 2D Display Technology (Intel® S2DDT)

Intel® S2DDT reduces display refresh memory traffic by reducing memory reads required for display refresh. Power consumption is reduced by less accesses to the IMC.

Intel® S2DDT is most effective with:

- Display images well suited to compression, such as text windows, slide shows, and so on. Poor examples are 3D games.
- Static screens such as screens with significant portions of the background showing 2D applications, processor benchmarks, and so on, or conditions when the processor is idle. Poor examples are full-screen 3D games and benchmarks that flip the display image at or near display refresh rates.

For more information, refer to Nova Lake Platform Graphics Software Product Requirement Documents (PRD) (#TBD), Nova Lake Intel® Graphics Power Management and Power Conservation Software Product Specification (SPS) (#TBD), Nova Lake Processor Intel® Graphics Driver Display Software Product Specification (SPS) (#TBD).

11.3.3 Processor Graphics Core Power Savings Technologies

Intel® Graphics Dynamic Frequency

Intel® Turbo Boost Technology 2.0 is the ability of the processor IA cores and graphics (Graphics Dynamic Frequency) cores to opportunistically increase frequency and/or voltage above the guaranteed processor and graphics frequency for the given part. Intel® Graphics Dynamic Frequency is a performance feature that makes use of unused package power and thermals to increase application performance. The increase in frequency is determined by how much power and thermal budget is available in the package, and the application demand for additional processor or graphics performance. The processor IA core control is maintained by an embedded controller. The graphics driver dynamically adjusts between P-States to maintain optimal performance, power, and thermals. The graphics driver will always place the graphics engine in its lowest possible P-State. Intel® Graphics Dynamic Frequency requires BIOS support. Additional power and thermal budget should be available.

Intel® Graphics Render Standby Technology (Intel® GRST)

Intel® Graphics Render Standby Technology is a technique designed to optimize the average power of the graphics part. The Graphics Render engine will be put in a sleep state, or Render Standby (RS), during times of inactivity or basic video modes. While in Render Standby state, the graphics part will place the VR (Voltage Regulator) into a low voltage state. Hardware will save the render context to the allocated context buffer when entering RS state and restore the render context upon exiting RS state.

Intel Capped Frames Per Second (CFPS)

Intel Capped Frames Per Second is a feature developed to save power during High FPS Gaming workloads while also achieving a tear and stutter free visual experience.

This feature ensures that the frame rate of the game does not exceed the panel refresh rate by matching screen updates to the Vertical Sync. That results fewer wakeups of graphics core and saves power.

When enabled, this feature works on any display panel, AC or DC mode and on any gaming workload.

11.4 USB Type-C Subsystem (TCSS) Power States

Table 22. USB Type-C Subsystem (TCSS) Power States

TCSS Power State	SOC PM State	Device Attached	Description
TC0	S0	Yes	xHCI, USB4 controllers may be active. USB4 DMA / PCIe may be active.
TC7	S*i2.1	Yes	xHCI and xDCI are in D3. USB4 controller is in D3 or D0 idle.
continued...			

TCSS Power State	SOC PM State	Device Attached	Description
			USB4 PCIe is inactive.
TC10	S*i2.2	No	Deepest Power state xHCI / xDCI / USB4 controllers are in D3.. USB4 DMA / USB4 PCIe are in D3 IOM is in low power state.
Note: "S*i2.1/S*i2.1" - See Naming Convention in Power Saving Features chapter. IOM - TCSS Input Output Manager: - The IOM interacts with the processor to perform power management, boot, reset, connect and disconnect devices to TYPE-C sub-system TCSS Devices (xHCI / xDCI / TBT Controllers) - power states: - D0 - Device at Active state. - D3 - Device at lowest-powered state.			

11.5 Power and Performance Technologies

11.5.1 Intel® Thread Director

Intel® Thread Director helps monitor and analyze performance data in real time to seamlessly place the right application thread on the right core and optimize performance per watt.

Built directly into the hardware, Intel® Thread Director uses machine learning to schedule tasks on the right core at the right time (as opposed to relying on static rules). This helps to ensure that Performance-cores and Efficient-cores work in concert, background tasks do not slow you down, and you can have more applications open simultaneously.

- Monitors the runtime instruction mix of each thread and the state of each core with nanosecond precision.
- Provides runtime feedback to the OS to make the optimal decision for any workload.
- Dynamically adapts its guidance according to the Assured Based Power (ABP) of the system, operating conditions, and power settings.

11.5.2 Intel® Smart Cache Technology

The Intel® Smart Cache Technology is a shared Last Level Cache (LLC).

- The LLC is shared between all Compute tile cores (of any type). The maximal size of LLC is 3MB (12 ways, set associative) per P-core or E-core module (bundle of 4 E-Cores).
- The LLC is non-inclusive.
- The LLC may also be referred to as a 3rd level cache.

11.5.3 IA Core Level 0, Level 1, Level 2 and Level 3 Caches

The 1st level cache is not shared between physical cores and each physical core has a separate set of caches.

The P-Core 1st level cache hierarchy is divided into:

- A Data Cache (DL0, DL1)
- An Instruction Cache (IL1)

On the data side, it is built as two-level cache, with L0 of 48KB and L1 of 192KB, both of which are 12-way set-associative.

On the instruction side, there is a single L1 cache of 64KB, which is 16-way set associative.

The E-Core 1st level cache hierarchy is divided into:

- A Data Cache (DL1)
- An Instruction Cache (IL1)

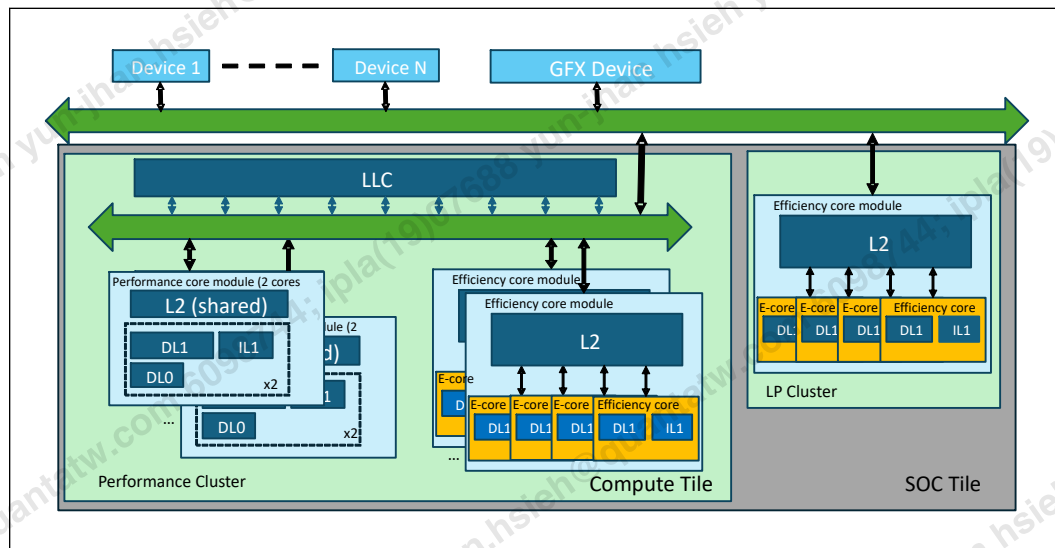
On the data side, it is built as one-level cache, with L1 of 48KB, 12-way set-associative.

On the instruction side, there is a single L1 cache of 64KB, which is 16-way set associative.

The 2nd level cache holds both data and instructions. It is also referred to as mid-level cache or MLC.

- The P-core 2nd level cache is shared between P-Cores within a module of 2 P-Cores in the Compute tile. Its size is 4MB and it is a 16-way associative non-inclusive cache.
- The E-core 2nd level cache is shared between E-Cores within a module of 4 E-Cores in the Compute tile. Its size is 4MB and it is a 16-way associative non-inclusive cache.
- The LP E-core 2nd level cache is shared between LP E-Cores within a module of 4 LP E-Cores in the HUB tile. Its size is 4MB and it is a 16-way associative non-inclusive cache.

Figure 7. P-core, E-core and LP E-core Cache Hierarchy



NOTE

The above figure does not represent the exact number of cores.

Cache	P-core	E-core	LP E-core
L0 DLO	48KB 12-way set-associative per core	None	None
L1 DL1	192KB 12-way set-associative per core	48KB 12-way set-associative per core	48KB 12-way set-associative per core
L1 IL1	64KB 16-way set-associative per core	64KB 16-way set-associative per core	64KB 16-way set-associative per core
L2	4MB 16-way set-associative shared between 2 physical cores	4MB 16-way set-associative within a module of 4 Compute tile E-cores	4MB 16-way set-associative within a module of 4 HUB tile E-cores
L3	Maximum of 3MB 12-way set-associative per P-core / bundle of 4 E-cores shared across Compute Die		

Note: Refer to Nova Lake Processor Family BIOS Specification (# 861572) for cache hierarchy reporting.

11.5.4 Ring Interconnect

The Ring is a high speed, wide interconnect that links the processor cores, processor graphics and the System Agent.

The Ring shares frequency and voltage with the Last Level Cache (LLC).

The Ring's frequency dynamically changes. Its frequency is relative to both processor cores and processor graphics frequencies.

11.5.5 Intel Hybrid Technology

The processor contains two types of cores, denoted as Performance and Efficiency Cores.

The big and small cores share the same instruction set and model specific registers (MSRs).

The available instruction sets, when hybrid computing is enabled, is limited compared to the instruction sets available to the big core.

The **FP16 support** instruction set is available only when the big cores are enabled.

NOTE

Hybrid Computing may not be available on all SKUs.

11.5.6 Intel Turbo Boost Technology 2.0

The Intel® Turbo Boost Technology 2.0 allows the processor IA core/processor graphics core to opportunistically and automatically run faster than the processor IA core base frequency/processor graphics base frequency if it is operating below power, temperature, and current limits. The Intel® Turbo Boost Technology 2.0 feature is designed to increase the performance of both multi-threaded and single-threaded workloads.

Compared with previous generation products, Intel® Turbo Boost Technology 2.0 will increase the ratio of application power towards Processor Base Power and also allows to increase power above Processor Base Power as high as PL2 for short periods of time. Thus, thermal solutions and platform cooling that are designed to less than thermal design guidance might experience thermal and performance issues since more applications will tend to run at the maximum power limit for significant periods of time.

NOTE

Intel® Turbo Boost Technology 2.0 may not be available on all SKUs.

11.5.6.1 Intel® Turbo Boost Technology 2.0 Power Monitoring

When operating in turbo mode, the processor monitors its own power and adjusts the processor and graphics frequencies to maintain the average power within limits over a thermally significant time period. The processor estimates the package power for all components on the package. In the event that a workload causes the temperature to exceed program temperature limits, the processor will protect itself using the Adaptive Thermal Monitor.

11.5.6.2 Intel® Turbo Boost Technology 2.0 Power Control

Illustration of Intel® Turbo Boost Technology 2.0 power control is shown in the following sections and figures. Multiple controls operate simultaneously allowing customization for multiple systems thermal and power limitations. These controls allow for turbo optimizations within system constraints and are accessible using MSR, MMIO, and PECI interfaces.

11.5.6.3 Intel® Turbo Boost Technology 2.0 Frequency

To determine the highest performance frequency amongst active processor IA cores, the processor takes the following into consideration:

- The number of processor IA cores operating in the C0 state.
- The estimated processor IA core current consumption and ICCMax settings.
- The estimated package prior and present power consumption and turbo power limits.
- The package temperature.

Any of these factors can affect the maximum frequency for a given workload. If the power, current, or thermal limit is reached, the processor will automatically reduce the frequency to stay within its Processor Base Power limit. Turbo processor frequencies are only active if the operating system is requesting the P0 state. For more information on P-states and C-states, refer to Power Management.

11.5.7 Intel Turbo Boost Max Technology 3.0

The Intel® Turbo Boost Max Technology 3.0 (ITBMT 3.0) grants a different maximum Turbo frequency for individual processor cores.

To enable ITBMT 3.0 the processor exposes individual core capabilities; including diverse maximum turbo frequencies.

An operating system that allows for varied per core frequency capability can then maximize power savings and performance usage by assigning tasks to the faster cores, especially on low core count workloads.

Processors enabled with these capabilities can also allow software (most commonly a driver) to override the maximum per-core Turbo frequency limit and notify the operating system via an interrupt mechanism.

NOTE

Intel® Turbo Boost Max 3.0 Technology is supported when processor voltage and temperature allows.

11.5.8 Intel® Adaptive Boost Technology

Intel® Adaptive Boost Technology (Intel® ABT) opportunistically increases the multicore turbo frequency while operating within IccMAX and temperature spec limitations.

Intel® ABT opportunistically delivers in-spec performance gains that are incremental to existing Turbo technologies. In systems equipped with performance spec power delivery, Intel® ABT allows additional multi-core turbo frequency while still operating within specified current and temperature limits.

11.5.9 Intel System Agent Enhanced SpeedStep® Technology

Intel® System Agent Enhanced SpeedStep® Technology

Intel® System Agent Enhanced SpeedStep® Technology also known as SAGV (System Agent Geyserville) is a dynamic voltage frequency scaling of the System Agent clock based on memory utilization. Unlike processor core and package Enhanced Intel SpeedStep® Technology, System Agent Enhanced SpeedStep® Technology has four valid operating points. When running light workload and Intel System Agent Enhanced SpeedStep® Technology is enabled, the DDR data rate may change as follows:

BIOS/MRC DDR training at high performance, maximum, mid and minimum frequencies sets I/O and timing parameters.

NOTE

In order to achieve the optimal levels of performance and power, the memory initialization and training process performed during first system boot or after CMOS clear or after a BIOS update will take a longer time than a typical boot. During this initialization and training process, end users may see a blank screen. More information on the memory initialization process can be found in the industry standard JEDEC Specifications found on www.JEDEC.org.

Before changing the DDR data rate, the processor sets DDR to self-refresh and changes the needed parameters. The DDR voltage remains stable and unchanged.

11.5.10 Enhanced Intel SpeedStep® Technology

Enhanced Intel SpeedStep® Technology enables OS to control and select P-state. The following are the key features of Enhanced Intel SpeedStep® Technology:

- Multiple frequencies and voltage points for optimal performance and power efficiency. These operating points are known as P-states.
- Frequency selection is software controlled by writing to processor MSRs. The voltage is optimized based on the selected frequency and the number of active processors IA cores.
 - Once the voltage is established, the PLL locks on to the target frequency.
 - All active processor IA cores share the same frequency and voltage. In a multi-core processor, the highest frequency P-state requested among all active IA cores is selected.
 - Software-requested transitions are accepted at any time. If a previous transition is in progress, the new transition is deferred until the previous transition is completed.
- The processor controls voltage ramp rates internally to ensure glitch-free transitions.

NOTE

Because there is low transition latency between P-states, a significant number of transitions per-second are possible.

11.5.11 Intel® Thermal Velocity Boost

Intel® Thermal Velocity Boost allows the processor IA core to opportunistically and automatically increase the Intel® Turbo Boost Technology 2.0 frequency speed bins whenever processor temperature and voltage allows. The Intel® Thermal Velocity Boost feature is designed to increase performance of both multi-threaded and single threaded workloads.

NOTE

Intel® Thermal Velocity Boost (Intel® TVB) may not be available on all SKUs.

11.5.12 Intel® Speed Shift Technology

Intel® Speed Shift Technology is an energy efficient method of frequency control by the hardware rather than relying on OS control. OS is aware of available hardware P-states and requests the desired P-state or it can let the hardware determine the P-state. The OS request is based on its workload requirements and awareness of processor capabilities. Processor decision is based on the different system constraints for example Workload demand, thermal limits while taking into consideration the minimum and maximum levels and activity window of performance requested by the Operating System.

For more details refer to:

- Intel® 64 Architectures Software Developer's Manual (SDM), volume 3B.
- Appropriate BIOS Specification.
- Turbo Implementation Guide.

11.5.13 Intel® Advanced Vector Extensions

Nova Lake systems support Intel® AVX10 Version 2 (Intel® AVX10.2). Intel® AVX10.2 includes a suite of new instructions delivering new AI features and performance, accelerated media processing, expanded Web Assembly, and Cryptography support, along with enhancements to existing legacy instructions for completeness and efficiency.

For additional information refer to: <https://www.intel.com/content/www/us/en/content-details/856721/intel-advanced-vector-extensions-10-2-intel-avx10-2-architecture-specification.html>

Intel® Advanced Vector Extensions 2.0 (Intel® AVX2) with 256-bit integer instructions, floating-point fused multiply-add (FMA) instructions, and gather operations. The 256-bit integer vectors benefit math, codec, image, and digital signal processing software. FMA improves performance in face detection, professional imaging, and high-performance computing. Gather operations increase vectorization opportunities for many applications. In addition to the vector extensions, this generation of Intel processors adds bit manipulation instructions useful in compression, encryption, and general purpose software.

Intel® Advanced Vector Extensions (Intel® AVX) are designed to achieve higher throughput to certain integer and floating point operation. Due to varying processor power characteristics, utilizing AVX instructions may cause a) parts to operate below the base frequency b) some parts with Intel® Turbo Boost Technology 2.0 to not achieve any or maximum turbo frequencies. Performance varies depending on hardware, software and system configuration and you should consult your system manufacturer for more information.

Intel® Advanced Vector Extensions refers to Intel® AVX or Intel® AVX2.

11.5.13.1 Intel® AVX2 Vector Neural Network Instructions

Vector instructions for deep learning extension for AVX2.

11.5.14 Intel® 64 Architecture x2APIC

The x2APIC architecture extends the xAPIC architecture that provides key mechanisms for interrupt delivery. This extension is primarily intended to increase processor addressability.

Specifically, x2APIC:

- Retains all key elements of compatibility to the xAPIC architecture:
 - Delivery modes
 - Interrupt and processor priorities
 - Interrupt sources
 - Interrupt destination types

- Provides extensions to scale processor addressability for both the logical and physical destination modes
- Adds new features to enhance the performance of interrupt delivery
- Reduces the complexity of logical destination mode interrupt delivery on link based architectures

The key enhancements provided by the x2APIC architecture over xAPIC are the following:

- Support for two modes of operation to provide backward compatibility and extensibility for future platform innovations:
 - In xAPIC compatibility mode, APIC registers are accessed through memory mapped interface to a 4K-Byte page, identical to the xAPIC architecture.
 - In the x2APIC mode, APIC registers are accessed through the Model Specific Register (MSR) interfaces. In this mode, the x2APIC architecture provides significantly increased processor addressability and some enhancements on interrupt delivery.
- Increased range of processor addressability in x2APIC mode:
 - Physical xAPIC ID field increases from 8 bits to 32 bits, allowing for interrupt processor addressability up to 4G-1 processors in physical destination mode. A processor implementation of x2APIC architecture can support fewer than 32-bits in a software transparent fashion.
 - Logical xAPIC ID field increases from 8 bits to 32 bits. The 32-bit logical x2APIC ID is partitioned into two sub-fields – a 16-bit cluster ID and a 16-bit logical ID within the cluster. Consequently, $(2^{20} - 16)$ processors can be addressed in logical destination mode. Processor implementations can support fewer than 16 bits in the cluster ID sub-field and logical ID sub-field in a software agnostic fashion.
- More efficient MSR interface to access APIC registers:
 - To enhance inter-processor and self-directed interrupt delivery as well as the ability to virtualize the local APIC, the APIC register set can be accessed only through MSR-based interfaces in x2APIC mode. The Memory Mapped IO (MMIO) interface used by xAPIC is not supported in x2APIC mode.
- The semantics for accessing APIC registers have been revised to simplify the programming of frequently-used APIC registers by system software. Specifically, the software semantics for using the Interrupt Command Register (ICR) and End Of Interrupt (EOI) registers have been modified to allow for more efficient delivery and dispatching of interrupts.
- The x2APIC extensions are made available to system software by enabling the local x2APIC unit in the “x2APIC” mode. To benefit from x2APIC capabilities, operating system support and a new BIOS are both needed, with special support for the x2APIC mode.
- The x2APIC architecture provides backward compatibility to the xAPIC architecture and forwards extensible for future Intel platform innovations.

NOTE

Intel® x2APIC Technology may not be available on all SKUs.

For more information, refer to Intel® 64 Architecture x2APIC Specification at <http://www.intel.com/products/processor/manuals/>.

11.5.15 Intel® Dynamic Tuning Technology

Intel® Dynamic Tuning consists of a set of software drivers and applications that allow a system manufacturer to optimize system performance and usability by:

- Dynamically optimize turbo settings of IA processors, power and thermal states of the platform for optimal performance
- Dynamically adjust the processor's peak power based on the current power delivery capability for optimal system usability
- Dynamically mitigate radio frequency interference for better RF throughput

11.5.16 Cache Line Write Back

Writes back to memory the cache line (if dirty) that contains the linear address specified with the memory operand from any level of the cache hierarchy in the cache coherence domain. The line may be retained in the cache hierarchy in the non-modified state. Retaining the line in the cache hierarchy is a performance optimization (treated as a hint by hardware) to reduce the possibility of a cache miss on a subsequent access. Hardware may choose to retain the line at any of the levels in the cache hierarchy, and in some cases, may invalidate the line from the cache hierarchy. The source operand is a byte memory location.

The CLWB instruction is documented in the Intel® Architecture Instruction Set Extensions Programming Reference (future architectures):

<https://software.intel.com/sites/default/files/managed/b4/3a/319433-024.pdf>

11.5.17 User Mode Wait Instructions

The *UMONITOR* and *UMWAIT* are user mode (Ring 3) instructions similar to the supervisor mode (Ring 0) *MONITOR*/*MWAIT* instructions without the C-state management capability.

TPAUSE is an enhanced *PAUSE* instruction.

The mnemonics for the three new instructions are:

- **UMONITOR**: operates just like *MONITOR* but allowed in all rings.
- **UMWAIT**: allowed in all rings, and no specification of target C-state.
- **TPAUSE**: similar to *PAUSE* but with a software-specified delay. Commonly used in spin loops.

11.6 Deprecated Technology

The processor has deprecated the following technology and it is no longer supported:

- DDR Running Average Power Limit (DDR RAPL)

11.7 Power and Internal Signals

11.7.1 Signal Description

Signal Name	Type	Description
PLT_PWROK	I	Platform Power OK: When asserted, PLT_PWROK is an indication to the processor that all of its core power rails have been stable. The platform may drive asynchronously. When PLT_PWROK is de-asserted, the PCH asserts PLTRST#. <i>Notes:</i> - PLT_PWROK must not glitch, even if RSMRST# is low - An external pull-down resistor is required.
RSMRST#	I	Primary Well Reset: This signal is used for resetting the primary power plane logic. This signal must be asserted for at least 10ms before de-asserting. <i>Note:</i> An external pull down resistor is required.
SLP_S0#	O	S0 Sleep Control: When the processor is in C10 state, this pin will assert to indicate VR controller can go into a light load mode. This signal can also be connected to EC for other power management related optimizations. <i>Note:</i> An external pull-up resistor is required.
GPP_SD00/ TIME_SYNC0 GPP_SD01/ TIME_SYNC1 GPP_SA14/DDSP_HPD0#/ DISP_MISC0/ TIME_SYNC0 GPP_SA15/DDSP_HPD1#/ DISP_MISC1/ TIME_SYNC1	I	Time Synchronization: Used for synchronization both input (latch time when pin asserted) and output (toggle pin when programmed time is hit).
PS_ON#	O	Used to indicate to PSU when to turn off its main rails
RESET_SYNC#	I/O	Bidirectional signal used to synchronize reset events between the processor and PCH. Sync reset exits with the PCH in following two steps: - The processor floats the RESET_SYNC# pin to allow it to be pulled high by the platform - The processor waits until it sees the RESET_SYNC# pin go high before proceeding to reset exit. The PCH will stop driving the pin low when the eSPI controller in the PCH is out of reset and ready to receive traffic over the eSPI link. It is required that the PCH eSPI controller is ready before the processor eSPI controller is released from reset.
PROC_C10_GATE#	O	When asserted, PROC_C10_GATE# is the indication to the system that the processor is entering C10 and can handle the voltages on the VCCPRIM_IO and VCCPRIM_VNNAON rails being lowered to 0 V. When de-asserted must ramp back up to their operational voltage levels. The power good indicators for these rails must still be asserted high when these rails are lowered to 0 V during PROC_C10_GATE# assertion and while these rails ramp back up to their operational levels after PROC_C10_GATE# de-assertion.
VDD2PWRGOOD_IN	I	Power Good signal from processor. This signal should be shorted to VDD2PWRGOOD_OUT on the platform Refer to PDG for implementation guidance.
VDD2PWRGOOD_OUT	O	Power Good signal from processor. This signal should be shorted to VDD2PWRGOOD_IN on the platform. Refer to PDG for implementation guidance.
EPD_ON_IN	I	Input signal to the processor, this signal should be shorted with EPD_ON_OUT on the platform. Refer to PDG for implementation guidance.
EPD_ON_OUT	O	Output signal from the processor, this signal should be shorted with EPD_ON_IN signals on the platform. Refer to PDG for implementation guidance.

11.7.2 Power Sequencing Signals

Table 23. Power Sequencing Signals

Signal Name	Description	Dir.	Link Type	Availability
VIDSOUT	VIDSOUT, VIDSCK, VIDALERT#: These signals comprise a three-signal serial synchronous interface used to transfer power management information between the processor and the voltage regulator controllers.	I/O	SE	S Processor Lines
VIDSCK		O		
VIDALERT#		I		

11.7.3 IO Signal Planes and States

Signal Name	Power Plane	During Reset	Immediately after Reset	S4/S5
PROC_C10_GATE#	Primary	Driven High	Driven High	Driven High
PLT_PWROK	RTC	Undriven	Undriven	Undriven
RSMRST#	RTC	Undriven	Undriven	Undriven
PS_ON#	Primary	Indeterministic, no deglitch PD.	Depending on the PMC HW default value	Depending on PMC HW default value
SLP_S0# ¹	Primary	Driven High	Driven High	Driven High
RESET_SYNC#	Primary	Hi-Z	Driven High	Driven High
Notes: 1. Driven High during S0 and driven Low during S0i3 when all criteria for assertion are met. 2. SLP_S4# is driven low in S4/S5. 3. SLP_S5# is driven high in S4, driven low in S5. 4. .PRIMPWRDNACK is always '0' while in M0 or M3, but can be driven to '0' or '1' while in M0ff state. .PRIMPWRDNACK is the default mode of operation. 5. The pad should only be pulled low momentarily when the corresponding buffer power supply is not stable. 6. Based on wake event and Intel CSME state. 7. Internal weak pull-down resistor is enabled during power sequencing. 8. Pin state is a function of whether the platform is configured to have Intel CSME on or off in Sx. 9. Output High-Z, not glitch free. 10. Output High-Z, glitch free with ~1 kohm Pull-down during respective power sequencing. 11. Output High-Z, glitch free with ~20 kohm Pull-down during respective power sequencing. 12. Output High-Z, glitch free with ~20 kohm Pull-up during respective power sequencing.				

12.0 Power Delivery

12.1 Power and Ground Signals

This section describes the power rails.

Table 24. Power Rail Description

Signal Name	Description
VCCPRIM_1P8_PROC	Fixed 1.8 V
VCCPRIM_1P8_PROC_COMPUTE	Fixed 1.8 V for the CPU
VCCPRIM_1P8_PROC_FLTR	VCCPRIM_1P8_PROC with filter requirements. Refer to the PDG for filter requirements.
VCCPRIM_IO	Fixed 1.25 V for IO blocks.
VCCPRIM_IO_OUT_SVID	VCCPRIM_IO voltage source for IMVP SVID buses pull up voltage
VCCPRIM_IO_FLTR	VCCPRIM_IO with filter requirements. Refer to the PDG for filter requirements.
VCCPRIM_VNNAON	Fixed 0.77 V for digital core blocks.
VCCPRIM_VNNAON_FLTR	VCCPRIM_VNNAON with filter requirements. Refer to the PDG for filter requirements.
VCC_CORE	Dynamic SVID power rail for IA cores.
VCCGT	Dynamic SVID power rail for graphics.
VCCSA	Dynamic SVID power rail for system agent.
VDD2	Fixed 1.1 V power rail for memory host controller.
VSS	Ground

Table 25. Power Rail Sense Signals

Signal Name	Description
VCC_CORE_SENSE	VCCCORE sense pin.
VCCGT_SENSE	VCCGT sense pin.
VCCPRIM_1P8_PROC_SENSE	VCCPRIM_1P8_PROC sense pin.
VCCPRIM_IO_SENSE	VCCPRIM_IO sense pin.
VCCPRIM_VNNAON_SENSE	VCCPRIM_VNNAON sense pin.
VCCSA_SENSE	VCCSA sense pin.
VDD2_SENSE	VDD2 sense pin.
VCC_CORE_VSS_SENSE	VCCCORE VSS sense pin.
continued...	

Signal Name	Description
VCCGT_VSS_SENSE	VCCGT VSS sense pin.
VCCPRIM_IO_VSS_SENSE	VCCIO VSS sense pin.
VCCPRIM_VNNAON_VSS_SENSE	VCCPRIM_VNNAON_VSS sense pin.
VCCSA_VSS_SENSE	VCCSA VSS sense pin.

12.2 Digital Linear Voltage Regulator

Digital Linear Voltage Regulator (DLVR) is implemented on Processor internal power rails for power saving, by gating power for Cores and digital IPs. DLVR mitigate EMI/RFI using Spread Spectrum Clock (SSC).

12.3 Fast V-Mode

This power management feature insulates VR FETs / Inductors from observing full PL4 current as well as upstream input power devices from observing full PL4 power.

IccMAX.APP represents the real PL4 workload maximum expected current when FVM enabled, which is less than IccMAX (PL4 current when FVM disabled).

Fast V-mode allows for platform power subsystems to be designed to IccMAX.APP, instead of IccMAX, while providing a performance improvement over proactive PL4 and Icc MAX reduction.

For more information, refer to Nova Lake-S Platform Design Guide (#830661).

12.4 Current Excursion Protection

This power management is a Processor integrated detector which senses when the Processor load current exceeds a preset threshold by monitoring for a Processor power domain voltage droop at the Processor power domain IMVPVR sense point. The Processor compares the IMVPVR output voltage with a preset threshold voltage (VTRIP) and when the IMVPVR output voltage is equal to or less than VTRIP, the Processor internally throttles itself to reduce the Processor load current and the power.

IMVP9.3 VRs enhance the CEP detector by adding a cycle by cycle current limiting feature where the IMVPVR quickly enters cycle by cycle current limit (becomes a current source) with the VR output current limited to a preset value (ITRIP) as set in the ICC_limit register.

For more information, refer to Nova Lake-S Platform Design Guide (#830661).

13.0 Electrical Specifications

13.1 Processor Power Rails

Power Rail	Description	S Processor Line Controls
VCC_CORE	Dynamic SVID power rail for IA cores.	SVID
VCCGT	Dynamic SVID power rail for graphics.	SVID
VCCSA	Dynamic SVID power rail for system agent.	SVID
VCCPRIM_1P8_PROC	Fixed 1.8 V.	Fixed
VCCPRIM_1P8_PROC_COMPUTE	Fixed 1.8 V for the CPU.	Fixed
VCCPRIM_1P8_PROC_FLTR	VCCPRIM_1P8_PROC with filter requirements. Refer to the PDG for filter requirements.	Fixed
VCCPRIM_IO	Fixed 1.25 V for IO blocks.	Fixed
VCCPRIM_IO_OUT_SVID	VCCPRIM_IO voltage source for IMVP SVID buses pull up voltage.	Fixed
VCCPRIM_IO_FLTR	VCCPRIM_IO with filter requirements. Refer to the PDG for filter requirements.	Fixed
VCCPRIM_VNNAON	Fixed 0.77 V for digital core blocks.	Fixed
VCCPRIM_VNNAON_FLTR	VCCPRIM_VNNAON with filter requirements. Refer to the PDG for filter requirements.	Fixed
VDD2	Fixed 1.05/1.10 V power rail for memory host controller.	Fixed (Memory technology dependent)
Note: 1. For details regarding each rail's VR, refer to the appropriate PDG.		

13.1.1 Power and Ground Pins

All power pins should be connected to their respective processor power planes, while all VSS pins should be connected to the system ground plane. Use of multiple power and ground planes is recommended to reduce I*R drop.

13.1.2 VCC Voltage Identification

Intel processors/chipsets are individually calibrated in the factory to operate on a specific voltage/frequency and operating-condition curve specified for that individual processor. In normal operation, the processor autonomously issues voltage control requests according to this calibrated curve using the serial voltage-identifier (SVID) interface. Altering the voltage applied at the processor/chipset causing operation outside of this calibrated curve is considered out-of-specification operation.

The SVID bus consists of three open-drain signals: clock, data, and alert# to both set voltage-levels and gather telemetry data from the voltage regulators. Voltages are controlled per an 8-bit integer value, called a VID, that maps to an analog voltage

level. An offset field also exists that allows altering the VID table. Alert can be used to inform the processor that a voltage-change request has been completed or to interrupt the processor with a fault notification.

For VID coding and further information, refer to the IMVP9.3 *PWM Specification* and *Serial VID (SVID) Protocol Specification*.

13.2 DC Specifications

The processor DC specifications in this section are defined at the processor signal pins, unless noted otherwise. For pin listing, refer to Nova Lake S Processor Line LGA Package Ballout Mechanical Specification (#847619).

- The DC specifications for the DDR5 signals are listed in the *Voltage and Current Specifications* section.
- The *Voltage and Current Specifications* section lists the DC specifications for the processor and are valid only while meeting specifications for operating temperature, clock frequency, and input voltages. Read all notes associated with each parameter.
- IccMAX is the maximum current processor can draw, typically seen running a virus application (stress applications specifically designed to push the processor to maximum Power).
- With Fast V-Mode enabled, the output decoupling would see this IccMAX current and would need to be able to take transient load step up to IccMAX whereas the power stage (FET/Inductor) would only see ITRIP_MAX current. Refer to the PDG for more details on Fast V-Mode.
- IccMAX.App is less than IccMax and is the electrical current drawn by the processor (per power rail) while running a typical user realistic application(s) scenario at P0nmax and Maximum Operating Temperature.

The processor VR must be able to sustain this current for at least 10ms.

- AC tolerances for all rails include voltage transients and voltage regulator voltage ripple up to 1 MHz. Refer to additional guidance for each rail.

13.2.1 Processor Power Rails DC Specifications

NOTE

Customers who desire to utilize higher power delivery settings (that is, Extreme Power Delivery (PD) configuration) than recommended (baseline, performance configurations) can still do so at their own risk as these higher power delivery settings may void warranty or affect system health. Refer to Nova Lake-S Platform Design Guide, Power maps (#830661) for 125W Extreme PD configuration guidance.

Altering clock frequency, power delivery, or voltage may void any product warranties and reduce stability, security, performance, and life of the processor and other components.

13.2.1.1 VCC_CORE DC Specifications

Table 26. Processor VCC_CORE Active and Idle Mode DC Voltage and Current Specifications (S Processor Line)

Processor	Native GPU Configuration	CPU Configuration	Symbol	Parameter	Minimum	Typical	Maximum	Unit	Note 1
All	All	All	Operating Voltage	Active Voltage Range for VCC_CORE	0	-	1.77	V	1,2,3,7,12,15,17
All	All	16P+32E Core 175W	IccMAX (S Processor)	Max. Current for Processor Core Rail	-	-	TBD	A	4,5,6,7,11
All	All	16P+32E Core 175W	IccMAX.App (S Processor)	Max. Application Current for Processor Core Rail	-	-	TBD	A	4,5,6,7,11
All	All	8P+16E Core 125W	IccMAX (S Processor)	Max. Current for Processor Core Rail	-	-	400	A	4,5,6,7,11
All	All	8P+16E Core 125W	IccMAX.App (S Processor)	Max. Application Current for Processor Core Rail	-	-	280	A	4,5,6,7,11
All	All	8P+16E Core 65W	IccMAX (S Processor)	Max. Current for Processor Core Rail	-	-	276	A	4,5,6,7,11
All	All	8P+16E Core 65W	IccMAX.App (S Processor)	Max. Application Current for Processor Core Rail	-	-	193	A	4,5,6,7,11
All	All	4P+8E Core 65W	IccMAX (S Processor)	Max. Current for Processor Core Rail	-	-	158	A	4,5,6,7,11
All	All	4P+8E Core 65W	IccMAX.App (S Processor)	Max. Application Current for Processor Core Rail	-	-	111	A	4,5,6,7,11
All	All	4P+8E Core 35W	IccMAX (S Processor)	Max. Current for Processor Core Rail	-	-	159	A	4,5,6,7,11
All	All	4P+8E Core 35W	IccMAX.App (S Processor)	Max. Application Current for Processor Core Rail	-	-	111	A	4,5,6,7,11
All	All	4P+0E Core 65W	IccMAX (S Processor)	Max. Current for Processor Core Rail	-	-	128	A	4,5,6,7,11
All	All	4P+0E Core 65W	IccMAX.App (S Processor)	Max. Application Current for Processor Core Rail	-	-	90	A	4,5,6,7,11
All	All	4P+0E Core 35W	IccMAX (S Processor)	Max. Current for Processor Core Rail	-	-	128	A	4,5,6,7,11
All	All	4P+0E Core 35W	IccMAX.App (S Processor)	Max. Application Current for Processor Core Rail	-	-	90	A	4,5,6,7,11
All	All	All	Icc TDC	Thermal Design Current for VCC_CORE	-	-	Refer to VR_TDC in the Platform Power Map	A	9
All	All	S-Processor Line: PS0, PS1, PS2, PS3	TOB VCC	DC Voltage Tolerance	-	-	±20	mV	3, 6, 8
All	All	S-Processor Lines: PS0, PS1, PS2, PS3	TOB VCC +Ripple	DC + Ripple Voltage Tolerance	-	-	-35 / +50	mV	3, 6, 8,16
All	All	S-Processor Line (125W)	DC_LL (S Processor)	DC Loadline	0	-	1.2	mΩ	10,13,14,18
All	All	S-Processor Line (65W,35W)	DC_LL (S Processor)	DC Loadline	0	-	1.7	mΩ	10,13,14,18
All	All	S-Processor Line (125W)	AC_LL (S Processor)	AC Loadline	0	-	TBD	mΩ	10,13,14,18
All	All	S-Processor Line (65W,35W)	AC_LL (S Processor)	AC Loadline	0	-	TBD	mΩ	10,13,14,18
All	All	All	V_OVS_MAX	Max Overshoot Allowance from IccMAX	-	-	200	mV	
All	All	All	T_OVS_MAX	Max Overshoot Time from IccMAX	-	-	500	us	

continued...

Processor	Native GPU Configuration	CPU Configuration	Symbol	Parameter	Minimum	Typical	Maximum	Unit	Note 1
All	All	All	V_OVS_MAX_APP	Max Overshoot Allowance from IccMAX.App	-	-	100	mV	
All	All	All	T_OVS_MAX_APP	Max Overshoot Time from IccMAX.App	-	-	50	us	
Notes: 1. All specifications in this table are based on estimates and simulations or empirical data. These specifications will be updated with characterized data from silicon measurements at a later date. 2. Maximum operating voltage given for motherboard design purposes. Each processor is programmed with a maximum valid voltage identification value (VID) that is set at manufacturing and cannot be altered. Individual maximum VID values are calibrated during manufacturing such that two processors at the same frequency may have different settings within the VID range. Note that this differs from the VID employed by the processor during a power management event (Adaptive Thermal Monitor, Enhanced Intel Speed-step Technology, or low-power states). Failure of product operation, including potential irreversible damage to the part, can occur from operating the part at voltages above the individual VID that is programmed. 3. The voltage specification requirements are measured across Vcc_SENSE and Vss_SENSE as near as possible to the processor. The measurement needs to be performed with a 20MHz bandwidth limit on the oscilloscope, 1.5pF maximum probe capacitance, and 1Ω minimum impedance. The maximum length of the ground wire on the probe should be less than 5mm. Ensure external noise from the system is not coupled into the oscilloscope probe. 4. Processor VccCORE VR to be designed to electrically support this current. 5. Processor VccCORE VR to be designed to thermally support this current indefinitely. 6. Long term reliability cannot be assured if tolerance, ripple, and core noise parameters are violated. 7. Long term reliability cannot be assured in conditions above or below Maximum/Minimum functional limits. 8. PSx refers to the voltage regulator power state as set by the SVID protocol. Refer to the IMVP9.3 Specification for more information. 9. Refer to the Platform Power Map for Thermal Design Current (TDC) guidance. 10. LL measured at sense points. 11. Typ column represents IccMAX for commercial application. It is NOT a specification but rather a characterization of limited samples using a limited set of benchmarks that can be exceeded. 12. Operating voltage range in steady state. 13. LL spec values should not be exceeded. If exceeded, power, performance and a reliability penalty are expected. 14. Load Line (AC/DC) should be measured by the VRTT tool and programmed accordingly via the BIOS Load Line override setup options. AC/DC Load Line BIOS programming directly affects operating voltages (AC) and power measurements (DC). A superior board design with a shallower AC Load Line can improve on power, performance and thermals compared to boards designed for POR impedance. 15. An IMVP9.3 controller to support VccCORE needs to have offset voltage capability to potentially support voltages (VID+Offset) higher than 1.5V. Refer to IMVP9.3 Pulse Width Modulation VR Vendor Enabling Specification (#637348) for more information. 16. Ripple can be higher if DC TOB is below 20mV, as long as Total TOB is within TOBVCC+Ripple spec. 17. 160 mV leakage may be observed when rail is powered off. There are no known functional or power implications due to this leakage. 18. 1.2mOhm is the design limit for -S, achieving a lower design value will improve performance and power efficiency.									

13.2.1.2 VCCGT DC Specifications

Table 27. Processor Graphics (VccGT) Supply DC Voltage and Current Specifications (S Processor Line)

Processor	Native GPU Configuration	CPU Configuration	Symbol	Parameter	Minimum	Typical	Maximum	Unit	Note 1
All	All	All	VCCGT	Active Voltage Range for VCCGT	0	-	1.52	V	2, 3, 5, 7
All	All	16P+32E Core 175W	IccMAX (S Processor)	Max. Current for Processor Graphics Rail	-	-	TBD	A	5
All	All	16P+32E Core 175W	IccMAX.App (S Processor)	Max. Application Current for Processor Graphics Rail	-	-	TBD	A	5
All	All	8P+16E Core 125W	IccMAX (S Processor)	Max. Current for Processor Graphics Rail	-	-	40	A	5
All	All	8P+16E Core 125W	IccMAX.App (S Processor)	Max. Application Current for Processor Graphics Rail	-	-	35	A	5
All	All	8P+16E Core 65W	IccMAX (S Processor)	Max. Current for Processor Graphics Rail	-	-	40	A	5
All	All	8P+16E Core 65W	IccMAX.App (S Processor)	Max. Application Current for Processor Graphics Rail	-	-	35	A	5
All	All	6P+8E Core 65W	IccMAX (S Processor)	Max. Current for Processor Graphics Rail	-	-	40	A	5
All	All	6P+8E Core 65W	IccMAX.App (S Processor)	Max. Application Current for Processor Graphics Rail	-	-	35	A	5

continued...

Processor	Native GPU Configuration	CPU Configuration	Symbol	Parameter	Minimum	Typical	Maximum	Unit	Note 1
All	All	6P+8E Core 35W	IccMAX (S Processor)	Max. Current for Processor Graphics Rail	-	-	40	A	5
All	All	6P+8E Core 35W	IccMAX.App (S Processor)	Max. Application Current for Processor Graphics Rail	-	-	35	A	5
All	All	4P+8E Core 65W	IccMAX (S Processor)	Max. Current for Processor Graphics Rail	-	-	40	A	5
All	All	4P+8E Core 65W	IccMAX.App (S Processor)	Max. Application Current for Processor Graphics Rail	-	-	35	A	5
All	All	4P+8E Core 35W	IccMAX (S Processor)	Max. Current for Processor Graphics Rail	-	-	40	A	5
All	All	4P+8E Core 35W	IccMAX.App (S Processor)	Max. Application Current for Processor Graphics Rail	-	-	35	A	5
All	All	4P+0E Core 65W	IccMAX (S Processor)	Max. Current for Processor Graphics Rail	-	-	40	A	5
All	All	4P+0E Core 65W	IccMAX.App (S Processor)	Max. Application Current for Processor Graphics Rail	-	-	35	A	5
All	All	4P+0E Core 35W	IccMAX (S Processor)	Max. Current for Processor Graphics Rail	-	-	40	A	5
All	All	4P+0E Core 35W	IccMAX.App (S Processor)	Max. Application Current for Processor Graphics Rail	-	-	35	A	5
All	All	All	Icc TDC _GT	Thermal Design Current (TDC) for Processor Graphics Rail	-	-	Refer to VccGT IPL2 in the Platform Power Map	A	5
All	All	S-Processor Line: PS0, PS1, PS2, PS3	TOB VCC GT	DC Voltage Tolerance	-	-	±20	mV	3,4
All	All	S-Processor Lines: PS0, PS1, PS2, PS3	TOBVCC GT+Ripple	DC + Ripple Voltage Tolerance	-	-	-35 /+50	mV	3, 4
All	All	All	DC_LL (S Processor)	DC Loadline	-	-	0	mΩ	6,8
All	All	All	AC_LL (S Processor)	AC Loadline	-	-	TBD	mΩ	6,8,9
All	All	All	V_OVS_MAX	Max Overshoot Allowance from IccMAX	-	-	70	mV	
All	All	All	T_OVS_MAX	Max Overshoot Time from IccMAX	-	-	10	μs	

Notes: 1. All specifications in this table are based on estimates and simulations or empirical data. These specifications will be updated with characterized data from silicon measurements at a later date.

2. Maximum operating voltage given for motherboard design purposes. Each processor is programmed with a maximum valid voltage identification value (VID) that is set at manufacturing and cannot be altered. Individual maximum VID values are calibrated during manufacturing such that two processors at the same frequency may have different settings within the VID range. Note that this differs from the VID employed by the processor during a power management event (Adaptive Thermal Monitor, Enhanced Intel Speed-step Technology, or low-power states). Failure of product operation, including potential irreversible damage to the part, can occur from operating the part at voltages above the individual VID that is programmed.

3. The voltage specification requirements are measured across VccGT_SENSE and VssGT_SENSE as near as possible to the processor. The measurement needs to be performed with a 20MHz bandwidth limit on the oscilloscope, 1.5 pF maximum probe capacitance, and 1 MΩ minimum impedance. The maximum length of ground wire on the probe should be less than 5 mm. Ensure external noise from the system is not coupled into the oscilloscope probe.

4. PSx refers to the voltage regulator power state as set by the SVID protocol. Refer to the IMVP9.3 Specification for more information.

5. Refer to the Platform Power Map for Thermal Design Current (TDC) guidance.

6. Operating voltage range in steady state.

7. LL spec values should not be exceeded. If exceeded, power, performance and a reliability penalty are expected.

8. Load Line measured at the sense point.

9. Ripple can be higher if DC TOB is below 20mV, as long as Total TOB is within TOBVCCGT+Ripple spec.

13.2.1.3 VCCSA DC Specification

Table 28. VccSA Supply DC Voltage and Current Specifications (S Processor Line)

Processor	Native GPU Configuration	CPU Configuration	Symbol	Parameter	Minimum	Typical	Maximum	Unit	Notes
All	All	All	VCCSA	Active Voltage Range for VCCSA	0	-	1.52	V	1,2,3,7
S	2Xe	16P+32E Core 175W	IccMAX (S Processor)	Max. Current for Processor System Agent Graphics Rail	0	-	TBD	A	1,2
S	2Xe	16P+32E Core 175W	IccMAX.App (S Processor)	Max. Application Current for Processor System Agent Rail	0	-	TBD	A	1,2
S	2Xe	8P+16E Core 125W	IccMAX (S Processor)	Max. Current for Processor System Agent Graphics Rail	0	-	99	A	1,2
S	2Xe	8P+16E Core 125W	IccMAX.App (S Processor)	Max. Application Current for Processor System Agent Rail	0	-	70	A	1,2
S	2Xe	8P+16E Core 65W	IccMAX (S Processor)	Max. Current for Processor System Agent Graphics Rail	0	-	99	A	1,2
S	2Xe	8P+16E Core 65W	IccMAX.App (S Processor)	Max. Application Current for Processor System Agent Rail	0	-	70	A	1,2
S	2Xe	4P+8E Core 65W	IccMAX (S Processor)	Max. Current for Processor System Agent Graphics Rail	0	-	99	A	1,2
S	2Xe	4P+8E Core 65W	IccMAX.App (S Processor)	Max. Application Current for Processor System Agent Rail	0	-	70	A	1,2
S	2Xe	4P+8E Core 35W	IccMAX (S Processor)	Max. Current for Processor System Agent Graphics Rail	0	-	99	A	1,2
S	2Xe	4P+8E Core 35W	IccMAX.App (S Processor)	Max. Application Current for Processor System Agent Rail	0	-	70	A	1,2
S	2Xe	4P+0E Core 65W	IccMAX (S Processor)	Max. Current for Processor System Agent Graphics Rail	0	-	99	A	1,2
S	2Xe	4P+0E Core 65W	IccMAX.App (S Processor)	Max. Application Current for Processor System Agent Rail	0	-	70	A	1,2
S	2Xe	4P+0E Core 35W	IccMAX (S Processor)	Max. Current for Processor System Agent Graphics Rail	0	-	99	A	1,2
S	2Xe	4P+0E Core 35W	IccMAX.App (S Processor)	Max. Application Current for Processor System Agent Rail	0	-	70	A	1,2
All	All	All	Icc_TDC_SA	Thermal Design Current (TDC) for Processor System Agent Rail	-	-	Refer to VccSA IPL2 in the Platform Power Map	A	5

continued...

Processor	Native GPU Configuration	CPU Configuration	Symbol	Parameter	Minimum	Typical	Maximum	Unit	Notes
All	All	S-Processor Line PS0, PS1, PS2, PS3	TOB _{VCCSA}	DC Voltage Tolerance	-	-	±20	mV	1,3,6
All	All	S-Processor Lines: PS0, PS1, PS2, PS3	TOB _{VCCSA} + Ripple	DC + Ripple Voltage Tolerance	-	-	-35 / +50	mV	3, 6, 8,16
All	All	S-Processor Line	DC_LL	DC Loadline	-	-	3.8	mΩ	4,5
All	All	S-Processor Line	AC_LL	AC Loadline	-	-	TBD	mΩ	4,5
All	All	All	V_OVS_MAX	Max Overshoot Allowance from IccMAX	-	-	70	mV	
All	All	All	T_OVS_MAX	Max Overshoot Time from IccMAX	-	-	10	μs	

- Notes:**
- All specifications in this table are based on estimates and simulations or empirical data. These specifications will be updated with characterized data from silicon measurements at a later date.
 - Maximum operating voltage given for motherboard design purposes. Each processor is programmed with a maximum valid voltage identification value (VID) that is set at manufacturing and cannot be altered. Individual maximum VID values are calibrated during manufacturing such that two processors at the same frequency may have different settings within the VID range. Note that this differs from the VID employed by the processor during a power management event (Adaptive Thermal Monitor, Enhanced Intel Speed-step Technology, or low-power states). Failure of product operation, including potential irreversible damage to the part, can occur from operating the part at voltages above the individual VID that is programmed.
 - Long term reliability cannot be assured in conditions above or below Maximum/Minimum functional limits.
 - The voltage specification requirements are measured on package pins as near as possible to the processor with an oscilloscope set to 100 MHz bandwidth, 1.5 pF maximum probe capacitance, and 1 MΩ minimum impedance. The maximum length of ground wire on the probe should be less than 5 mm. Ensure external noise from the system is not coupled into the oscilloscope probe.
 - LL measured at sense points. LL specification values should not be exceeded. If exceeded, power, performance, and reliability penalty are expected.
 - The LL values are for reference. Must still need to meet the voltage tolerance specification.**
 - Voltage Tolerance budget values Include ripples
 - Vcc_{SA} is having few point of voltage define by CPU VID, refer to Platform Design Guide for more details

13.2.1.4 VDD2 DC Specifications

Table 29. Memory Controller (VDD2) Supply DC Voltage and Current Specifications

Process or	Native GPU Configuration	CPU Configuration	Symbol	Parameter	Minimum	Typical	Maximum	Unit	Note1
All	All	All	VDD2 (DDR5)	Processor I/O Supply Voltage for DDR5	-	1.1	-	V	3,4,5
All	All	All	TOB _{VDD2}	Voltage Tolerance	-	-	± 5	%	3,4
All	All	All	IccMAX_VDD2	Maximum Current for VDD2 Rail	-	-	4.00	A	2
Notes: 1. All specifications in this table are based on estimates and simulations or empirical data. These specifications will be updated with characterized data from silicon measurements at a later date. 2. The current supplied to the DIMM modules is not included in this specification. 3. Includes AC and DC error, where the AC noise is bandwidth limited to under 1 MHz, measured on package pins. 4. No requirement on the breakdown of AC versus DC noise. 5. The voltage specification requirements are measured on package pins as near as possible to the processor with an oscilloscope set to 100 MHz bandwidth, 1.5 pF maximum probe capacitance, and 1 MΩ minimum impedance. The maximum length of ground wire on the probe should be less than 5 mm. Ensure external noise from the system is not coupled into the oscilloscope probe.									

13.2.1.5 VCCPRIM_VNNAON DC Specifications

Table 30. VCCPRIM_VNNAON Supply DC Voltage and Current Specifications

Process or	Native GPU Configuration	CPU Configuration	Symbol	Parameter	Minimum	Typical	Maximum	Unit	Note
All	All	All	VCCPRIM_VNNAON	VCCPRIM_VNNAON Power Rail Voltage	-	0.77	-	V	2,3,4
All	All	All	TOB _{VCCPRIM_VNNAON}	Voltage Tolerance	-	-	± 50	mV	2,3
All	All	All	IccMAX_VNNAON	Maximum Current	-	-	27.7	A	3
Notes: 1. All specifications in this table are based on estimates and simulations or empirical data. These specifications will be updated with characterized data from silicon measurements at a later date. 2. Includes AC and DC error, where the AC noise is bandwidth limited to under 1 MHz, measured on package pins. Keep 1st harmonic ripple ≤ 1MHz. 3. No requirement on the breakdown of AC versus DC noise. 4. The voltage specification requirements are measured on package pins as near as possible to the processor with an oscilloscope set to 100 MHz bandwidth, 1.5 pF maximum probe capacitance, and 1 MΩ minimum impedance. The maximum length of ground wire on the probe should be less than 5 mm. Ensure external noise from the system is not coupled into the oscilloscope probe									

Table 31. VCCPRIM_VNNAON_FLTR Supply DC Voltage and Current Specifications

Process or	Native GPU Configuration	CPU Configuration	Symbol	Parameter	Minimum	Typical	Maximum	Unit	Note
All	All	All	VCCPRIM_VNNAON_FLTR	VCCPRIM_VNNAON Power Rail Voltage	-	0.77	-	V	2,3,4
All	All	All	TOB VCCPRIM_VNNAON_FLTR	Voltage Tolerance	-	-	± 50	mV	2,3
All	All	All	IccMAX_VNNAON_FLTR	Maximum Current	-	-	0.1022	A	3
Notes: 1. All specifications in this table are based on estimates and simulations or empirical data. These specifications will be updated with characterized data from silicon measurements at a later date. 2. Includes AC and DC error, where the AC noise is bandwidth limited to under 1 MHz, measured on package pins. Keep 1st harmonic ripple ≤ 1MHz. 3. No requirement on the breakdown of AC versus DC noise. 4. The voltage specification requirements are measured on package pins as near as possible to the processor with an oscilloscope set to 100 MHz bandwidth, 1.5 pF maximum probe capacitance, and 1 MΩ minimum impedance. The maximum length of ground wire on the probe should be less than 5 mm. Ensure external noise from the system is not coupled into the oscilloscope probe									

13.2.1.6 VCCPRIM_IO DC Specifications

Table 32. VCCPRIM_IO Supply DC Voltage and Current Specifications

Process or	Native GPU Configuration	CPU Configuration	Symbol	Parameter	Minimum	Typical	Maximum	Units	Notes 1,2,5
All	All	All	VCCPRIM_IO	Processor I/O Power Rail Voltage	-	1.25	-	V	2,3,4
All	All	All	TOB VCCPRIM_IO	Voltage Tolerance	-	-	± 5	%	2,3
All	All	All	IccMAX_VCCPRIM_IO	Maximum Current	-	-	9.7	A	3
Notes: 1. All specifications in this table are based on estimates and simulations or empirical data. These specifications will be updated with characterized data from silicon measurements at a later date. 2. Includes AC and DC error, where the AC noise is bandwidth limited to under 1 MHz, measured on package pins. Keep 1st harmonic ripple ≤ 1MHz. 3. No requirement on the breakdown of AC versus DC noise. 4. The voltage specification requirements are measured on package pins as near as possible to the processor with an oscilloscope set to 100 MHz bandwidth, 1.5 pF maximum probe capacitance, and 1 MΩ minimum impedance. The maximum length of ground wire on the probe should be less than 5 mm. Ensure external noise from the system is not coupled into the oscilloscope probe									

Table 33. VCCPRIM_IO_OUT_SVID Supply DC Voltage and Current Specifications

Process or	Native GPU Configuration	CPU Configuration	Symbol	Parameter	Minimum	Typical	Maximum	Units	Notes 1,2,5
All	All	All	VCCPRIM_IO_OUT_SVID	Processor I/O Power Rail Voltage	-	1.25	-	V	2,3,4
All	All	All	TOB_VCCPRIM_IO_OUT_SVID	Voltage Tolerance	-	-	± 5	%	2,3
All	All	All	IccMAX_VCCPRIM_IO_OUT_SVID	Maximum Current	-	-	N/A	A	3
Notes: 1. All specifications in this table are based on estimates and simulations or empirical data. These specifications will be updated with characterized data from silicon measurements at a later date. 2. Includes AC and DC error, where the AC noise is bandwidth limited to under 1 MHz, measured on package pins. Keep 1st harmonic ripple ≤ 1MHz. 3. No requirement on the breakdown of AC versus DC noise. 4. The voltage specification requirements are measured on package pins as near as possible to the processor with an oscilloscope set to 100 MHz bandwidth, 1.5 pF maximum probe capacitance, and 1 MΩ minimum impedance. The maximum length of ground wire on the probe should be less than 5 mm. Ensure external noise from the system is not coupled into the oscilloscope probe									

Table 34. VCCPRIM_IO_FLTR Supply DC Voltage and Current Specifications

Process or	Native GPU Configuration	CPU Configuration	Symbol	Parameter	Minimum	Typical	Maximum	Units	Notes 1,2,5
All	All	All	VCCPRIM_IO_FLTR	Processor I/O Power Rail Voltage	-	1.25	-	V	2,3,4
All	All	All	TOB_VCCPRIM_IO_FLTR	Voltage Tolerance	-	-	± 5	%	2,3
All	All	All	IccMAX_VCCPRIM_IO_FLTR	Maximum Current	-	-	0.04	A	3
Notes: 1. All specifications in this table are based on estimates and simulations or empirical data. These specifications will be updated with characterized data from silicon measurements at a later date. 2. Includes AC and DC error, where the AC noise is bandwidth limited to under 1 MHz, measured on package pins. Keep 1st harmonic ripple ≤ 1MHz. 3. No requirement on the breakdown of AC versus DC noise. 4. The voltage specification requirements are measured on package pins as near as possible to the processor with an oscilloscope set to 100 MHz bandwidth, 1.5 pF maximum probe capacitance, and 1 MΩ minimum impedance. The maximum length of ground wire on the probe should be less than 5 mm. Ensure external noise from the system is not coupled into the oscilloscope probe									

13.2.1.7 VCCPRIM_1P8 DC Specifications

Table 35. VCCPRIM_1P8_PROC Supply DC Voltage and Current Specifications

Processor	Native GPU Configuration	CPU Configuration	Symbol	Parameter	Minimum	Typical	Maximum	Unit	Note
All	All	All	VCCPRIM_1P8_PROC	VCCPRIM_1P8_PROC Power Rail Voltage	-	1.8	-	V	2,3,4
All	All	All	TOB_VCCPRIM_1P8_PROC	Voltage Tolerance	-	-	±5	%	2,3
All	All	All	IccMAX_VCCPRIM_1P8_PROC	Maximum Current	-	-	1.5	A	3

Notes: 1. All specifications in this table are based on estimates and simulations or empirical data. These specifications will be updated with characterized data from silicon measurements at a later date.
2. Includes AC and DC error, where the AC noise is bandwidth limited to under 1 MHz, measured on package pins.
3. No requirement on the breakdown of AC versus DC noise.
4. The voltage specification requirements are measured on package pins as near as possible to the processor with an oscilloscope set to 100 MHz bandwidth, 1.5 pF maximum probe capacitance, and 1 MΩ minimum impedance. The maximum length of ground wire on the probe should be less than 5 mm. Ensure external noise from the system is not coupled into the oscilloscope probe

Table 36. VCCPRIM_1P8_PROC_COMPUTE Supply DC Voltage and Current Specifications

Processor	Native GPU Configuration	CPU Configuration	Symbol	Parameter	Minimum	Typical	Maximum	Unit	Note
All	All	All	VCCPRIM_1P8_PROC_COMPUTE	VCCPRIM_1P8_PROC_COMPUTE Power Rail Voltage	-	1.8	-	V	2,3,4
All	All	All	TOB_VCCPRIM_1P8_PROC_COMPUTE	Voltage Tolerance	-	-	±5	%	2,3
All	All	All	IccMAX_VCCPRIM_1P8_PROC_COMPUTE	Maximum Current	-	-	0.42	A	3

Notes: 1. All specifications in this table are based on estimates and simulations or empirical data. These specifications will be updated with characterized data from silicon measurements at a later date.
2. Includes AC and DC error, where the AC noise is bandwidth limited to under 1 MHz, measured on package pins.
3. No requirement on the breakdown of AC versus DC noise.
4. The voltage specification requirements are measured on package pins as near as possible to the processor with an oscilloscope set to 100 MHz bandwidth, 1.5 pF maximum probe capacitance, and 1 MΩ minimum impedance. The maximum length of ground wire on the probe should be less than 5 mm. Ensure external noise from the system is not coupled into the oscilloscope probe

Table 37. VCCPRIM_1P8_PROC_FLTR Supply DC Voltage and Current Specifications

Processor	Native GPU Configuration	CPU Configuration	Symbol	Parameter	Minimum	Typical	Maximum	Unit	Note
All	All	All	VCCPRIM_1P8_PROC_FLTR	VCCPRIM_1P8_PROC_FLTR Power Rail Voltage	-	1.8	-	V	2,3,4
All	All	All	TOBVCCPRIM_1P8_PROC_FLTR	Voltage Tolerance	-	-	±5	%	2,3
All	All	All	IccMAX_VCCPRIM_1P8_PROC_FLTR	Maximum Current	-	-	0.02	A	3

Notes: 1. All specifications in this table are based on estimates and simulations or empirical data. These specifications will be updated with characterized data from silicon measurements at a later date.
 2. Includes AC and DC error, where the AC noise is bandwidth limited to under 1 MHz, measured on package pins.
 3. No requirement on the breakdown of AC versus DC noise.
 4. The voltage specification requirements are measured on package pins as near as possible to the processor with an oscilloscope set to 100 MHz bandwidth, 1.5 pF maximum probe capacitance, and 1 MΩ minimum impedance. The maximum length of ground wire on the probe should be less than 5 mm. Ensure external noise from the system is not coupled into the oscilloscope probe

13.2.2 Processor Interfaces DC Specifications

13.2.2.1 DDR5 DC Specifications

Symbol	Minimum	Units
V _{IL}	-	V
V _{IH}	0.85*V _{dd2}	V
R _{ON_UP(DQ)}	30	Ω
R _{ON_DN(DQ)}	30	
R _{ODT(DQ)}	30	Ω
V _{ODT(DC)}	0.4*V _{dd2}	V
R _{ON_UP(CK)}	30	Ω
R _{ON_DN(CK)}	30	Ω
R _{ON_UP(CMD)}	30	Ω
R _{ON_DN(CMD)}	30	Ω
R _{ON_UP(CTL)}	30	Ω
R _{ON_DN(CTL)}	30	Ω
R _{ON_UP}		Ω
(SM_PG_CNTL1)		
R _{ON_DN}		Ω
(SM_PG_CNTL1)		
I _{LI}		mA
SM_RCOMP[0]	99	Ω

continued...

Symbol	Minimum	Units
SM_RCOMP[1]	99	Ω
SM_RCOMP[2]	99	Ω
Notes: 1. All specifications in this table apply to all processor frequencies. Timing specifications only depend on the operating frequency of the memory channel and not the maximum rated frequency. 2. VIL is defined as the maximum voltage level at a receiving agent that will be interpreted as a logical low value. 3. VIH is defined as the minimum voltage level at a receiving agent that will be interpreted as a logical high value. 4. VIH and VOH may experience excursions above VDD2. However, input signal drivers should comply with the signal quality specifications. 5. Pull up/down resistance after compensation (assuming $\pm 5\%$ COMP inaccuracy). Note that BIOS power training may change these values significantly based on margin/power trade-off. Refer to processor I/O Buffer Models for I/V characteristics. 6. ODT values after COMP (assuming $\pm 5\%$ inaccuracy). BIOS MRC can reduce ODT strength towards 7. The minimum and maximum values for these signals are programmable by BIOS to one of the two sets. 8. SM_RCOMP[x] resistance should be provided on the system board with 1% resistors. SM_RCOMP[x] resistors are to VSS. Values are pre-silicon estimations and are subject to change. 9. SM_DRAMPWROK must have a maximum of 15 ns rise or fall time over VDD2 * 0.30 ± 100 mV and the edge must be monotonic. 10. RON tolerance is preliminary and might be subject to change. 11. Maximum-minimum range is correct but center point is subject to change during MRC boot training. 12. Processor may be damaged if VIH exceeds the maximum voltage for extended periods.		

13.2.2.2 PCIe DC and Timing Specifications

The PCIe Controller(s) and Transmit/Receive Physical Layer PHYs are compliant with the **PCI Express* Base Specification Revision 5.0 Version 1.0, 22 May 2019**.

For PCIe* electrical specifications, refer to the PCI Express* Base Specification Revision 5.0 Version 1.0, 22 May 2019; which is available at <https://pcisig.com/>.

13.2.2.3 Single Ended DC Spec

Signals	Operating Voltage (V)	Input				Output	
		Input High Voltage Threshold (V _{IH})	Input Low Voltage Threshold (V _{IL})	Input Leakage Current (I _{IL})	Input Pin Capacitance (C _{IN})	Output High Voltage Threshold (V _{OH})	Output Low Voltage Threshold (V _{OL})
SOC_JTAG_TDO SOC_JTAG_TDI SOC_JTAG_TMS PRDY# PREQ# BPM[0] BPM[1] EPD_ON_OUT	1.25	0.8VDD VHYS=60 mV	0.48VDD	2mA	7.2pF	0.8VDD	0.2VDD
continued...							

Signals	Operating Voltage (V)	Input				Output	
		Input High Voltage Threshold (V_{IH})	Input Low Voltage Threshold (V_{IL})	Input Leakage Current (I_{IL})	Input Pin Capacitance (C_{IN})	Output High Voltage Threshold (V_{OH})	Output Low Voltage Threshold (V_{OL})
VDD2PWRGOOD_OUT							
SOC_JTAG_TCK SOC_JTAG_TRST#	1.25	0.5VDD VHYS=80 mV	0.2VDD	2mA	7.2pF	0.8VDD	0.2VDD
DIR_ESPI_RCLK \ DMI_PERST# \ GPP_SA16\DDSP_HPD3# \ \DISP_MISC3 \ GPP_SA17 \ GPP_SA18 \ GPP_SA21\VDDEN \ \SRCCLKREQ1# \ GPP_SA22 \ BKLTEN\SRCCLKREQ2# \ GPP_SA23\BKLTCTL \ \SRCCLKREQ3# \ RTC_CLK_IN \ RESET_SYNC# \ PLT_PWROK \ GPP_SA24\SNDW_CLK \ AUDCLK \ GPP_SA25\SNDW_DATA0_TX \ \AUDIN \ GPP_SA26 \ SNDW_DATA0_RX \ AUDOUT \ GPP_SA27\SNDW_DATA1_TX \ GPP_SA28\SNDW_DATA1_RX \ GPP_SB02\DDP3_CTRLCLK \ GPP_SB03\DDP3_CTRLDATA \ GPP_SB04\DDP2_CTRLCLK \ GPP_SB05\DDP2_CTRLDATA \ GPP_SB12\DDP0_CTRLCLK \ TBT_LSX0_TXD \ GPP_SB13\DDP0_CTRLDATA \ TBT_LSX0_RXD \ GPP_SB15\DDP1_CTRLCLK \ TBT_LSX1_TXD \ GPP_SB16 \ DDP1_CTRLDATA \ TBT_LSX1_RXD \ GPP_SD00\TIME_SYNC0 \ GPP_SD01\TIME_SYNC1 \ GPP_SD02 \ GPP_SD03 \ GPP_SD04 \ GPP_SD05 \ GPP_SD06\SRCCLKREQ1# \ GPP_SD07\SRCCLKREQ2# \ GPP_SD08 \ GPP_SD09\VDDEN \ GPP_SD10\BKLTEN \ GPP_SD11 \ BKLTCTL \ GPP_SD12\SRCCLKREQ3# \ GPP_SD13\DDSP_HPD3# \ DISP_MISC3 \ GPP_SD14\DDSP_HPD2# \ \DISP_MISC2 \ GPP_SD15 \ DDSP_HPD0# \ DISP_MISC0 \ GPP_SD16\DDSP_HPD1# \ \DISP_MISC1 \	1.8	0.65VDD VHYS=0.1 VDD	0.35VDD	2mA	5pF	VDD-0.45V	0.45V

continued...

Signals	Operating Voltage (V)	Input				Output	
		Input High Voltage Threshold (V _{IH})	Input Low Voltage Threshold (V _{IL})	Input Leakage Current (I _{IL})	Input Pin Capacitance (C _{IN})	Output High Voltage Threshold (V _{OH})	Output Low Voltage Threshold (V _{OL})
GPP_SD17\PCIE_LINK_DOWN \SRCCLKREQ3# \ GPP_SD19 \ GPP_SD20							
DIR_ESPI_IO0 \ DIR_ESPI_IO1 \ DIR_ESPI_IO2 \ DIR_ESPI_IO3 \ DIR_ESPI_CLK \ GPP_SB18 \ GPP_SB19	1.8	0.65VDD VHYS=0.1 VDD	0.35VDD	2mA	7.8pF	VDD-0.45V	0.45V
DIR_ESPI_CS0# \ DIR_ESPI_RESET# \ SLP_S0# \ CATERR# \ THERMTRIP# \ PROC_C10_GATE# \ PS_ON# \ GPP_SA14\DDSP_HPD0# \DISP_MISC0\TIME_SYNC0 \ GPP_SA15\DDSP_HPD1# \DISP_MISC1\TIME_SYNC1 \ GPP_SA19 \ GPP_SA20\DDSP_HPD2# \DISP_MISC2 \ GPP_SB07\DMI_CLKREQ# \ GPP_SB08\SRCCLKREQ4# \ GPP_SB09\SRCCLKREQ5# \ GPP_SB14\SRCCLKREQ0# \ FORCEPR# \ NA \ GPP_SD18\ BOOTHALT#	1.8	0.65VDD VHYS=0.1 VDD	0.35VDD	1mA	1.8pF	VDD-0.45V	0.45V

13.3 AC Specifications

The processor timings specified in this section are defined at the processor pads. Therefore, proper simulation of the signals is the only means to verify proper timing and signal quality.

The following tables list the AC specifications associated with the processor. The timings specified in this section should be used in conjunction with the processor signal integrity models provided by Intel. Read all table notes associated with the timing parameters.

Table 38. Differential Clock Jitter

Symbol	Description	Maximum	Units
T _{38.4MHz_HF_TIE_PTP} ^{1,2,3,4}	>1.5 MHz to Nyquist Jitter	60	ps Peak-to-Peak
T _{38.4MHz_Cycle_to_Cycle_Jitter}	Cycle to Cycle Jitter	150	ps
T _{38.4MHz_Duty_Cycle}	Duty Cycle	50% +/- 8%	%

Notes: 1. Measured 100K ref clock cycles using High Pass 1.5M 1st order filter.
 2. Measured differentially at CLK_NSSC_P/N Signal VIAs directly under the Processor.
 3. Time Interval Error (TIE) Jitter data can be measured using jitter analysis software such as DPOJET* from Tektronix.
 4. Time Interval Error (TIE) Jitter is the discrete time domain representation of phase noise expressed in seconds or pico-seconds.

Table 39. Differential Clocks (SSC off)

Signal Name	1 CLK -Jitter c-c Abs Per Minimum	0.1 S -ppm Long Avg Minimum	Ideal DC Target	0.1 S +ppm Long Avg Maximum	CLK +Jitter c-c Abs Per Maximum	Units
BCLK	9.925	9.999000	10.00000	10.00100	10.075	ns
PCI_BCLK	9.925	9.999000	10.00000	10.00100	10.075	ns

Notes: 1. Ideal DC Target: This serves only as an ideal reference target (0 ppm) to use for calculating the rest of the period measurement values.
 2. 0.1 second - Measurement Window (frequency counter): Valuable measurement done using a frequency counter to determine near DC average frequency (filtering out all jitter including SSC and cycle to cycle). This is used to determine if the system has a frequency static offset caused usually by incorrect crystal, crystal loading or incorrect clock configuration.
 3. 1CLK - No Filter: Any 1 Period measured with a scope. Measured on a real time Oscilloscope using no filters, a simple period measurement (or a Jit3 period measurement - more accurate), provides absolute Minimum/Maximum timing information.

Table 40. System Reference Clocks DC and AC Specifications

Symbol	Parameter	Signal	Minimum	Maximum	Unit	Measurement	Figure	Notes
Slew_rise	Rising Slew Rate	Diff	1.3	4	V/ns	Avg		2,3
Slew_fall	Falling Slew Rate	Diff	1.3	4	V/ns	Avg		2,3
Slew_var	Slew Rate Matching	Single Ended	-	20	%	Avg	Refer to Figure 9 on page 106	1,9
V _{SWING}	Differential Output Swing	Diff	300	-	mV	RT		2
V _{CROSS}	Crossing Point Voltage	Single Ended	250	500	mV	RT	Refer to Figure 9 on page 106	1, 4, 5
V _{CROSS_DELTA}	Variation of V _{CROSS}	Single Ended	-	140	mV	RT	Refer to Figure	1, 4, 8

continued...

Symbol	Parameter	Signal	Minimum	Maximum	Unit	Measurement	Figure	Notes
							9 on page 106	
V _{MAX}	Maximum Output Voltage	Single Ended	-	1.15	V	RT	Refer to Figure 9 on page 106	1, 6
V _{MIN}	Minimum Output Voltage	Single Ended	-0.3	-	V	RT	Refer to Figure 9 on page 106	1, 7
Notes: 1. Measurement taken from single-ended waveform on a component test board. 2. Measurement taken from differential waveform on a component test board. 3. Slew rate measured through VSWING voltage range centered about differential zero. 4. V _{CROSS} is defined as the voltage where Clock = Clock#. 5. Only applies to the differential rising edge (that is, Clock rising and Clock# falling). 6. The maximum voltage including overshoot. 7. The minimum voltage including undershoot. 8. The total variation of all V _{CROSS} measurements in any particular system. Note that this is a subset of V _{CROSS_MIN/MAX} (V _{CROSS} absolute) allowed. The intent is to limit V _{CROSS} induced modulation by setting V _{CROSS_DELTA} to be smaller than V _{CROSS} absolute. 9. Matching applies to rising edge rate for Clock and falling edge rate for Clock#. It is measured using a ±75 mV window centered on the average cross point where Clock rising meets Clock# falling (Refer to Figure 9 on page 106"). The median cross point is used to calculate the voltage thresholds the oscilloscope is to use for the edge rate calculations.								

13.3.1 DDR5 AC Specifications

Table 41. DDR5 Electrical Characteristics and AC Timings

Speed	Symbol	T _{CK} (ns)	T _{CMD_CO} (ps)	T _{CTRL_CO} (ps)	T _{DVB + T_{DVA}} (ps)	T _{SU+HD} (ps)
		System Memory Clock Timings	System Memory Command Signal Timings	System Memory Control Signal Timings	System Memory Data and Strobe Signal Timings	
		CK Period	CA[9:0] Edge Placement Accuracy	CS_#[1:0], CKE[3:0], ODT Edge Placement Accuracy	DQ[63:0] Valid before DQS[7:0] Rising or Falling Edge	DQ Input Setup Plus Hold Time to DQS Rising or Falling Edge
2000 MT/s	Max	1	0.15*TCK	0.15*TCK	0.15*TCK	0.15*TCK
3600 MT/s	Max	0.555	0.15*TCK	0.15*TCK	0.15*TCK	0.15*TCK
4000 MT/s	Max	0.5	0.15*TCK	0.15*TCK	0.15*TCK	0.15*TCK
4400 MT/s	Max	0.454	0.15*TCK	0.15*TCK	0.15*TCK	0.15*TCK
4800 MT/s	Max	0.416	0.15*TCK	0.15*TCK	0.15*TCK	0.15*TCK
5600 MT/s	Max	0.357	0.15*TCK	0.15*TCK	0.15*TCK	0.15*TCK
6000 MT/s	Max	0.333	0.15*TCK	0.15*TCK	0.15*TCK	0.15*TCK
6400 MT/s	Max	0.312	0.15*TCK	0.15*TCK	0.15*TCK	0.15*TCK
continued...						

Speed	Symbol	T _{CK} (ns)	T _{CMD_CO} (ps)	T _{CTRL_CO} (ps)	T _{DVB +T DVA} (ps)	T _{SU+HD} (ps)
		System Memory Clock Timings	System Memory Command Signal Timings	System Memory Control Signal Timings	System Memory Data and Strobe Signal Timings	
Figure		-	Refer to Figure 10 on page 107	Refer to Figure 10 on page 107	-	Refer to Figure 10 on page 107
Note 1,7		-	3,5,8	3,5,8	6,8	1,2,4,6,8

Notes: 1. All specifications in this table apply to all processor frequencies. Timing specifications only depend on the operating frequency of the memory channel and not on the maximum frequency

2. When the single ended slew rate of the input Data or Strobe signals, within a byte group, are below 1.0 V/ns, the TSU and THD specifications should be increased by a de-rating factor. The input single ended slew rate is measured DC to AC levels; VIL_DC to VIH_AC for rising edges, and VIH_DC to VIL_AC for falling edges. Use the worst case minimum slew rate measured between Data and Strobe, within a byte group, to determine the required de-rating value. No de-rating is required for single ended slew rates equal to or greater than 1.0 V/ns.

3. Edge Placement Accuracy (EPA): The silicon contains digital logic that automatically adjusts the timing relationship between the DDR reference clocks and DDR signals. The BIOS initiates a training procedure that will place a given signal appropriately within the clock period. The difference in delay between the signal and clock is accurate to within ±EPA. This EPA includes jitter, skew, within die variation and several other effects.

4. Data to Strobe read setup and Data from Strobe read hold minimum requirements specified at the processor pad are determined with the minimum Read DQS/DQS# delay.

5. The system memory clock outputs are differential (CLK and CLK#), the CLK rising edge is referenced at the crossing point where CLK is rising and CLK# is falling.

6. The system memory strobe outputs are differential (DQS and DQS#), the DQS rising edge is referenced at the crossing point where DQS is rising and DQS# is falling, and the DQS falling edge is referenced at the crossing point where DQS is falling and DQS# is rising.

7. These are pre-silicon estimates and are subject to change.

8. Maximum range is correct but center point is subject to change during MRC boot training.

13.3.2 DisplayPort AC Specifications

Intel follows VESA AC specifications, refer to VESA's specifications.

13.3.3 HDMI AC Specifications

Intel follows HDMI.org AC specifications, refer to HDMI.org's specifications.

13.3.4 Testability AC Specifications

Table 42. Testability Signal Group AC Specifications

T# Parameter	Minimum	Maximum	Unit	Figure	Notes
T14: PROC_JTAG_TCK Period	10	-	ns	-	1, 2, 3, 4
T15: PROC_JTAG_TDI, PROC_JTAG_TMS Setup Time	3	-	ns	Figure 12 on page 108	1, 2, 3, 4
T16: PROC_JTAG_TDI, PROC_JTAG_TMS Hold Time	3	-	ns	Figure 12 on page 108	1, 2, 3, 4
continued...					

T# Parameter	Minimum	Maximum	Unit	Figure	Notes
T17: PROC_JTAG_TDO Clock to Output Delay	1	9	ns	Figure 12 on page 108	1, 2, 3, 4
T18: PROC_JTAG_TRST# Assert Time	2	-	T _{TCK}	Figure 13 on page 108	1, 2, 3, 4, 5
Notes: 1. All specifications in this table apply to all processor frequencies. 2. Not 100% tested. Specified by design characterization. 3. It is recommended that SOC_JTAG_TMS be asserted while SOC_JTAG_TRST# is being de-asserted(dose not apply for all usage cases). 4. TMS, TDI and TDO are sampled on the rising edge of TCK but driven on the falling edge of TCK. 5. Required a minimum of an asynchronous 20 ns (there is no synchronizing to TCK for TRST_N). 6. Tco min/max are measured at 0.5Vcc with 2pF test load. 7. Tsetup/Thold are measured with a. Data input signal rise/fall time of 1ns to 3ns, measured across 0.2Vcc to 0.8Vcc. b. Clock being an internal loopback					

13.3.5 SVID AC Specifications

Table 43. SVID Signal Group AC Specifications

T # Parameter	Minimum	Maximum	Unit	Figure	Notes ^{1, 2}
VIDSCLK period	52.08	-	ns	-	-
VIDSOUT output valid delay with respect to VIDSCLK	-1.5	2.5	ns	-	-
VIDSOUT input setup time	5	-	ns	-	3, 4
VIDSOUT input hold time	0	-	ns	-	3, 4
VIDSCLK Rise Time	-	5.5	ns	-	7
VIDSCLK Fall Time	-	5.5	ns	-	8
Duty Cycle	40	60	%	-	-
Notes: 1. Refer to the IMPV9.3 voltage regulator design guidelines for additional information. 2. Platform support for VID transitions is required for the processor to operate within specifications. 3. Referenced to rising edge of VIDSCLK. 4. Setup and Hold times are measured with respect 0.5 X VCCPRIM_IO_OUT_SVID 5. Rise time is measured from 0.45V to 0.65V. 6. Fall time is measured from 0.65V to 0.45V. 7. Period and duty cycle are measured with respect to 0.5 X VCCPRIM_IO_OUT_SVID. 8. VIDSOUT output valid delay with respect to VIDSCLK min/max are measured at 0.5 X VCCPRIM_IO_OUT_SVID with 2pF test load.					

13.3.6 Miscellaneous AC Specifications

Table 44. Miscellaneous AC Specifications

T# Parameter	Minimum	Maximum	Unit	Figure	Notes
T1: Asynchronous CMOS input pulse width	10	-	BCLKs	-	-
T4: FORCEPR# output pulse width	500	-	μs	Figure 13 on page 108	-
T4: FORCEPR# assertion until VCC removed	-	500	ms	-	-

13.4 Test Access Port Connection

Due to the voltage levels supported by other components in the Test Access Port (TAP) logic, Intel recommends the processor be first in the TAP chain, followed by any other components within the system. Refer to the appropriate processor Testability Information - Boundary Scan Description Language (BSDL) File for more details. A translation buffer should be used to connect to the rest of the chain unless one of the other components is capable of accepting an input of the appropriate voltage. Two copies of each signal may be required with each driving a different voltage level.

The processor supports Boundary Scan (JTAG) IEEE 1149.1-2001, IEEE 1149.6-2003 and IEEE 1149.1-2013 standards.

13.5 Processor AC Timing Waveforms

The following figures are used in conjunction with the AC timing tables in [AC Specifications](#) on page 100.

For the following figures these notes apply:

- All common clock AC timings signals are referenced to the Crossing Voltage (VCROSS) of the EXT_INJ_BCLK_P/EXT_INJ_BCLK_N at rising edge of EXT_INJ_BCLK_P.
- All source synchronous AC timings are referenced to their associated strobe (address or data). Source synchronous data signals are referenced to the falling edge of their associated data strobe. Source synchronous address signals are referenced to the rising and falling edge of their associated address strobe.
- All AC timings for the TAP signals are referenced to the TCK at $0.5 * VCCPRIM_IO$ at the processor lands. All TAP signal timings (TMS, TDI, and so on) are referenced at $0.5 * VCCPRIM_IO$ at the processor die (pads).
- All CMOS signal timings are referenced at $0.5 * VCCPRIM_IO$ at the processor pins.

Figure 8. Differential Clock – Differential Measurements

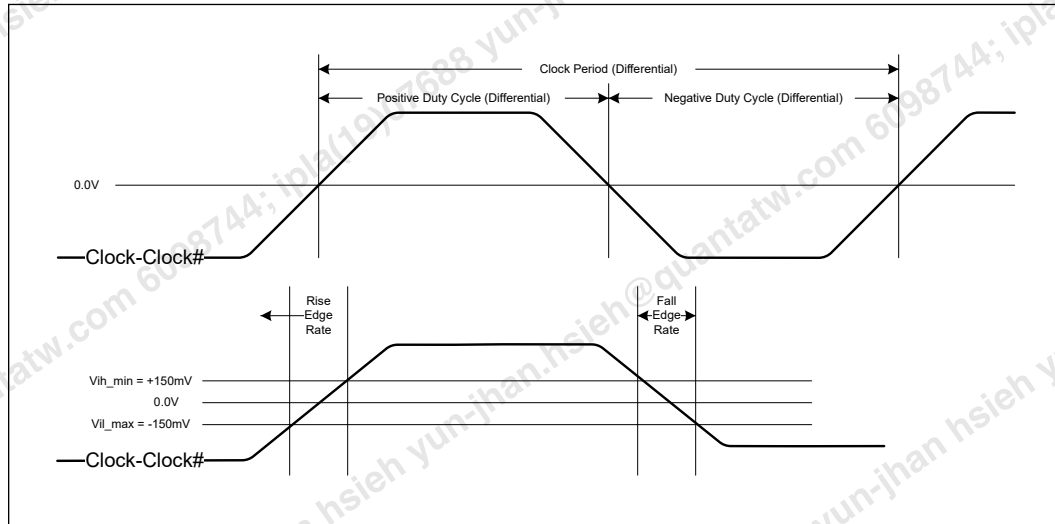


Figure 9. Differential Clock – Single-Ended Measurements

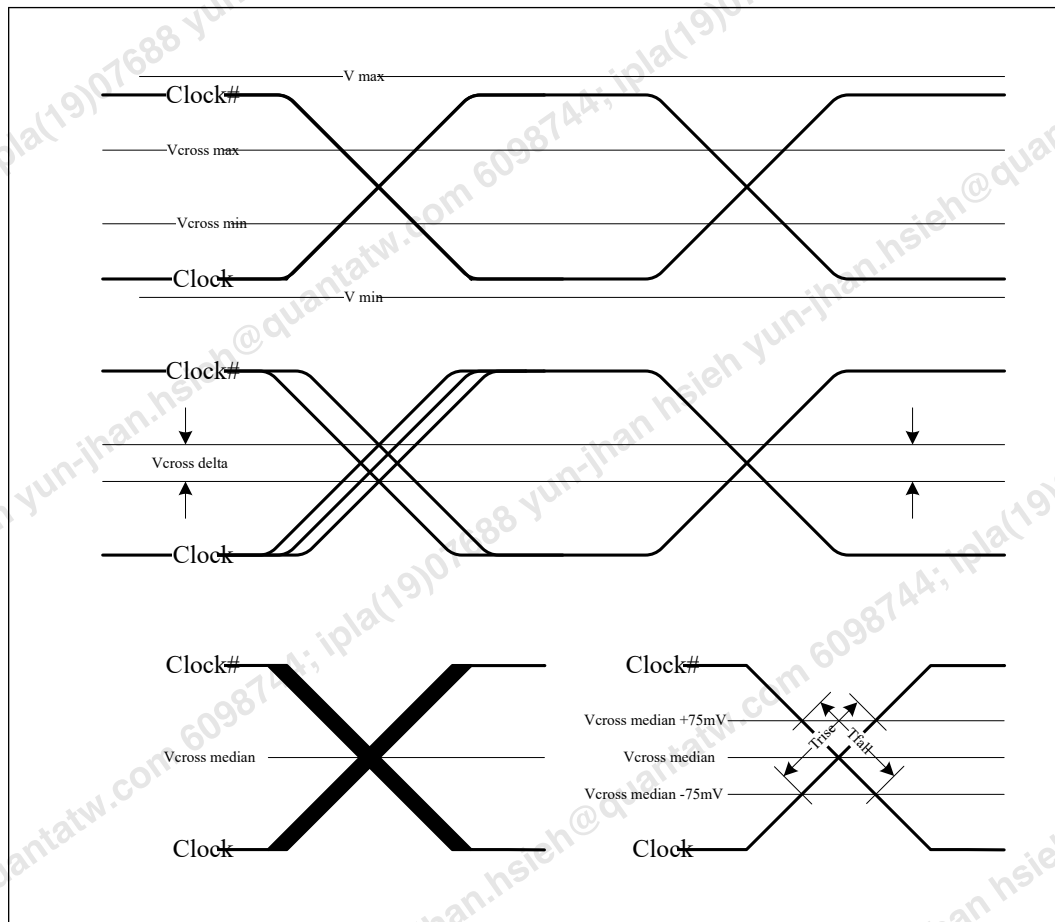


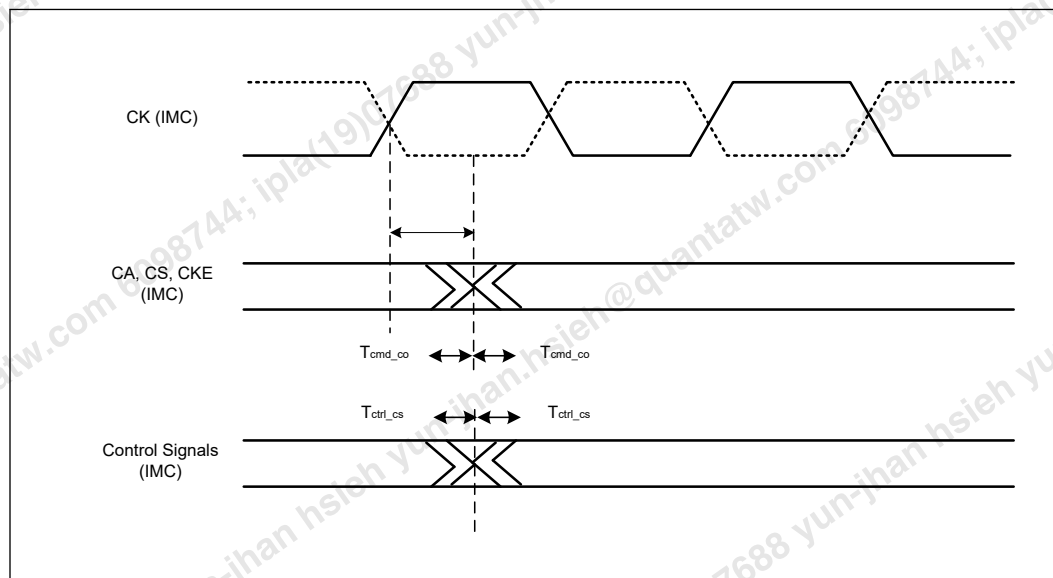
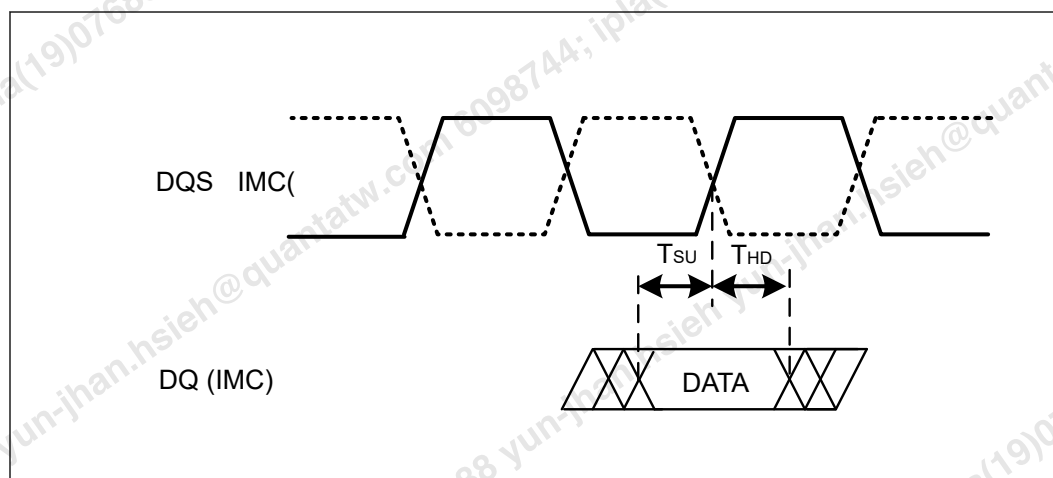
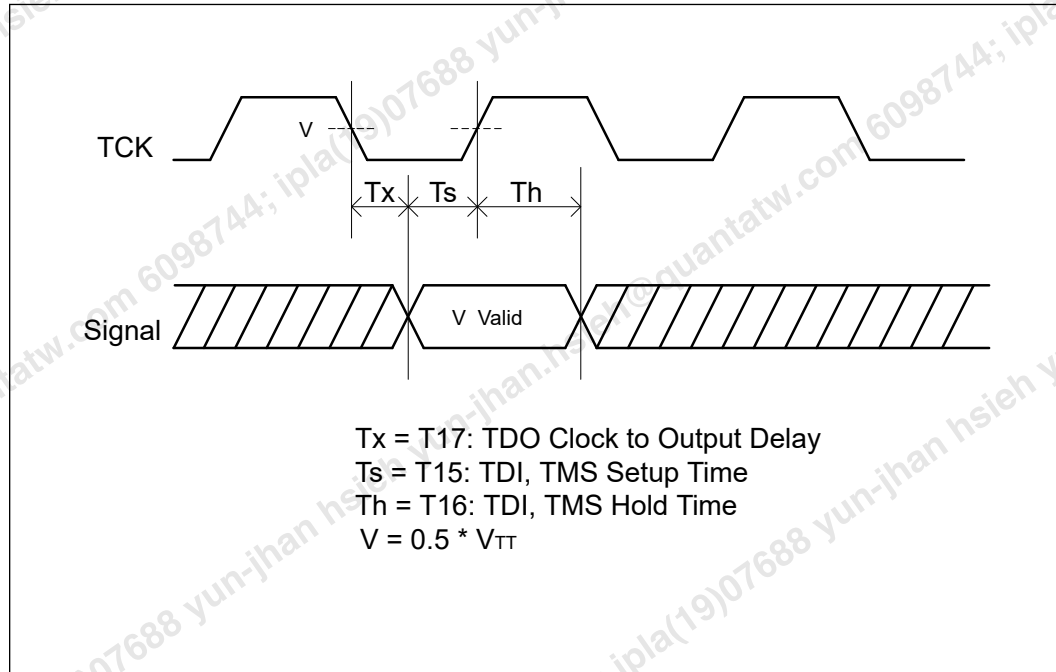
Figure 10. DDR Command / Control and Clock Timing Waveform**Figure 11. DDR Data Setup and Hold Timing Waveform**

Figure 12. TAP Valid Delay Timing Waveform



Refer to [Single Ended DC Spec](#) on page 98 for DC specifications.

Figure 13. Test Reset (PROC_JTAG_TRST#), Async Input, and FORCEPR# Output Timing Waveform

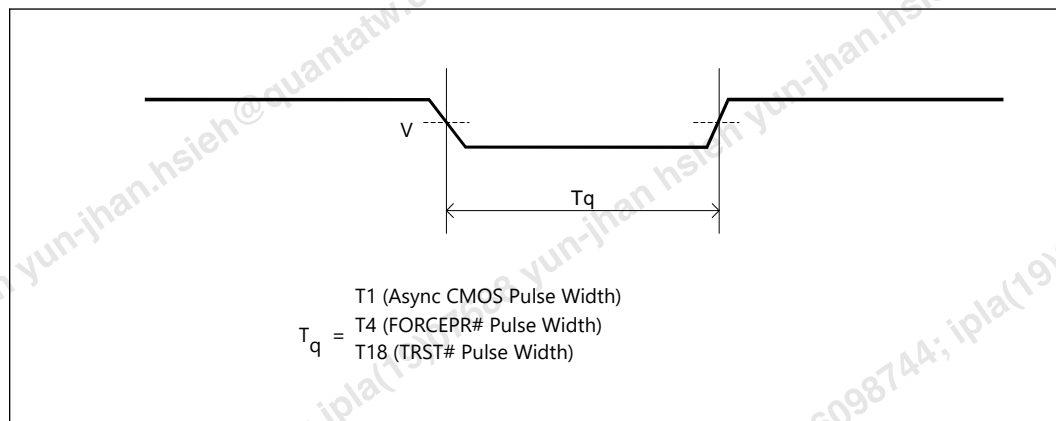
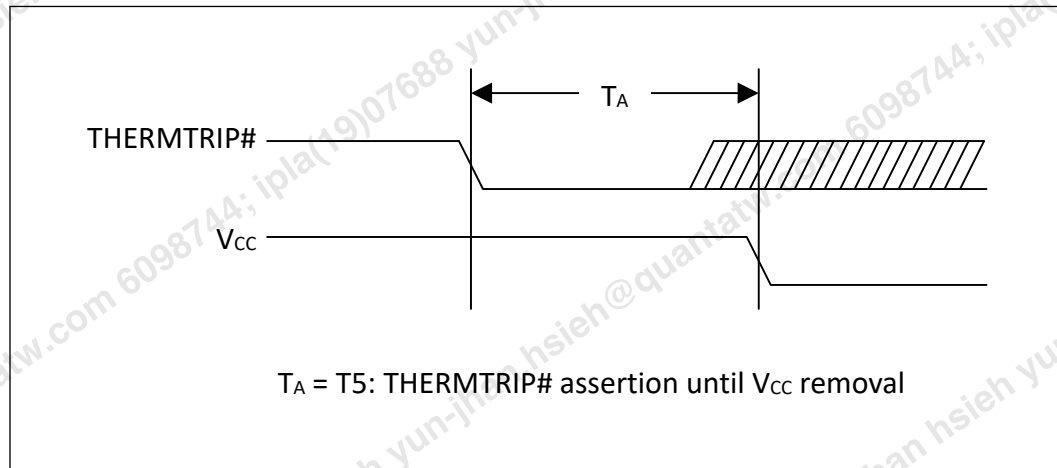


Figure 14. THERMTRIP# Power Down Sequence

13.6 Signal Quality

Data transfer requires the clean reception of data signals and cloak signals. Ringing below receiver thresholds, non-monotonic signal edges, and excessive voltage swings will adversely affect system timings. Ringback and signal non-monotonicity cannot be tolerated since these phenomena may inadvertently advance receiver state machines. Excessive signal swings (overshoot and undershoot) are detrimental to silicon gate oxide integrity, and can cause device failure if absolute voltage limits are exceeded. Overshoot and interference (ISI) effects.

For these reasons, it is crucial that the designer work towards a solution that provides acceptable signal quality across all systematic variations encountered in volume manufacturing.

This section documents signal quality metrics used to derive topology and routing guidelines through simulation. All specifications are specified at the processor die (pad measurements).

Specifications for signal quality are for measurements at the processor IA core only and are only observable through simulation. Therefore, proper simulation is the only way to verify proper timing and signal quality.

13.6.1 Input Reference Clock Signal Quality Specifications

Overshoot/Undershoot and Ringback specifications for BCLKP/BCLKN are in [Overshoot Undershoot Guidelines](#) on page 110. Overshoot/Undershoot and Ringback specifications for the System Memory Reference Clocks are specified by the DIMM.

13.6.2 System Memory Signal Quality Specifications

Signal Quality specifications for Differential signals are included as part of the DC specifications and AC specifications. Various scenarios have been simulated to generate a set of layout guidelines. These are available in the platform design guide.

13.7 Overshoot Undershoot Guidelines

Overshoot (or undershoot) is the absolute value of the maximum voltage above or below VSS. The overshoot/undershoot specifications limit transitions beyond the operation voltage due to the fast signal edge rates. The processor can be damaged by single and/or repeated overshoot or undershoot events on any input, output, or I/O buffer if the charge is large enough (that is, if the over/undershoot is great enough). Baseboard designs that meet signal integrity and timing requirements and that do not exceed the maximum overshoot or undershoot limits will ensure reliable I/O performance for the lifetime of the processor.

13.7.1 VCC Overshoot Specification

VCC specifications are included in [VCC_CORE DC Specifications](#) on page 88

13.7.2 Overshoot Undershoot Magnitude

Magnitude describes the maximum potential difference between a signal and its voltage reference level. For the processor, both are referenced to VSS. It is important to note that the overshoot and undershoot conditions are separate and their impact should be determined independently.

The pulse magnitude and duration should be used to determine if the overshoot/undershoot pulse is within specifications.

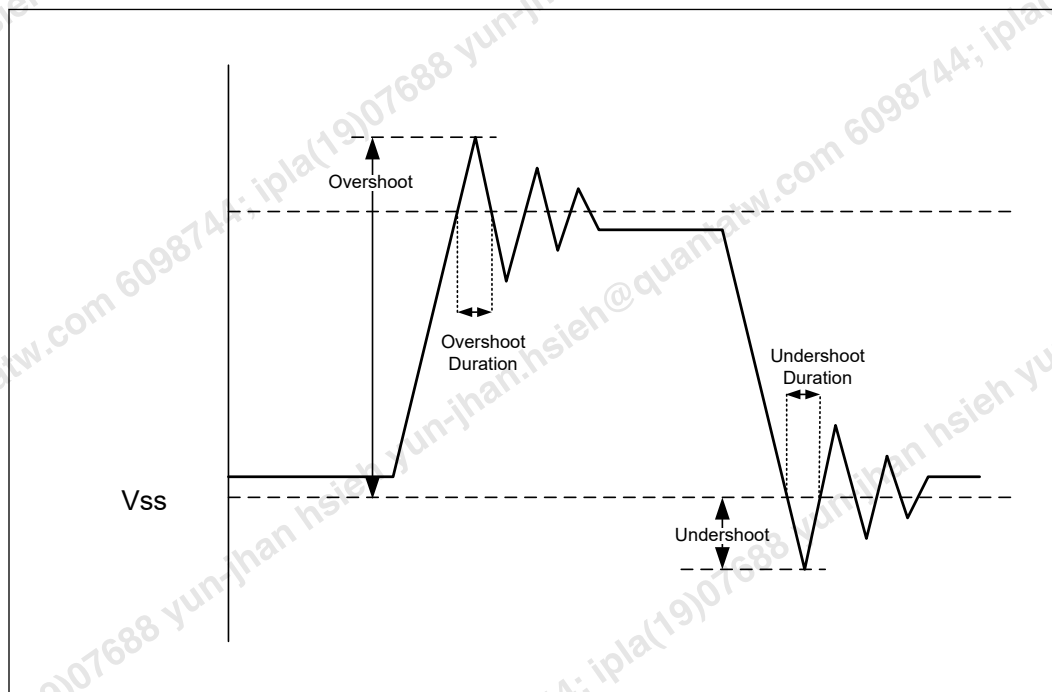
13.7.3 Overshoot Undershoot Pulse Duration

Pulse duration describes the total amount of time that an overshoot/undershoot event exceeds the overshoot/undershoot reference voltage. The total time could encompass several oscillations above the reference voltage. Multiple overshoot/undershoot pulses within a single overshoot/undershoot event may need to be measured to determine the total pulse duration.

Oscillations below the reference voltage cannot be subtracted from the total overshoot/undershoot pulse duration.

Table 45. Processor Overshoot / Undershoot Specifications

Signal Group	Maximum Overshoot [V]	Overshoot Duration [ns]	Minimum Undershoot [V]	Undershoot Duration [ns]
Operation Voltage 1.8 V	(1.9 + 0.5) V	1.7 ns	(0 - 0.5) V	1.7 ns
Operation Voltage 1.25 V	1.32+0.35	1.7 ns	(0 - 0.35) V	1.7 ns
<i>Notes:</i> 1. These specifications are measured at the processor pin. 2. These overshoot/undershoot values are absolute worst case, and measured under the assumption of full compliance with PDG spec. 3. Refer to the figure below for description of allowable Overshoot/Undershoot magnitude and duration.				

Figure 15. Maximum Acceptable Overshoot / Undershoot Waveform

13.8 Direct eSPI Timing Specifications

Table 46. Direct eSPI (Source Synchronous Mode) - 100MHz

Symbol	Parameter	Minimum	Maximum	Unit	Figure	Notes
t180	Serial Clock Frequency	n/a	100	MHz		
t183	Tco of eSPI I/O with respect to serial DIR_ESPI_CLK clock falling edge at the processor	-1.28	1.28	ns	Figure 16 on page 112 Figure 17 on page 113	Tco value is based on simulation with a test load of 2pF.
t184	Setup of eSPI I/O with respect to serial DIR_ESPI_R CLK clock falling edge at the processor	6.25	n/a	ns	Figure 16 on page 112	Measurement point for low time and high time is taken at 0.5(VCCSPI).
t185	Hold of eSPI I/O with respect to serial DIR_ESPI_R CLK clock	-0.8	n/a	ns	Figure 16 on page 112	Measurement point for low time and high time is taken at 0.5(VCCSPI).

continued...

Symbol	Parameter	Minimum	Maximum	Unit	Figure	Notes
	falling edge at the processor					
t186	Setup of eSPI CS# assertion with respect to serial clock rising edge at the host	1.5*CP	n/a	ns	Figure 16 on page 112 Figure 17 on page 113	CP means clock period
t187	Hold of eSPI CS# assertion with respect to serial clock falling edge at the host	1*CP	n/a	ns	Figure 16 on page 112 Figure 17 on page 113	CP means clock period
t188	eSPI CLK High time	4.75	5.25	ns	Figure 16 on page 112 Figure 17 on page 113	
t189	eSPI CLK Low time	4.75	5.25	ns	Figure 16 on page 112 Figure 17 on page 113	

Figure 16. Direct eSPI (Source Synchronous Mode) Timings

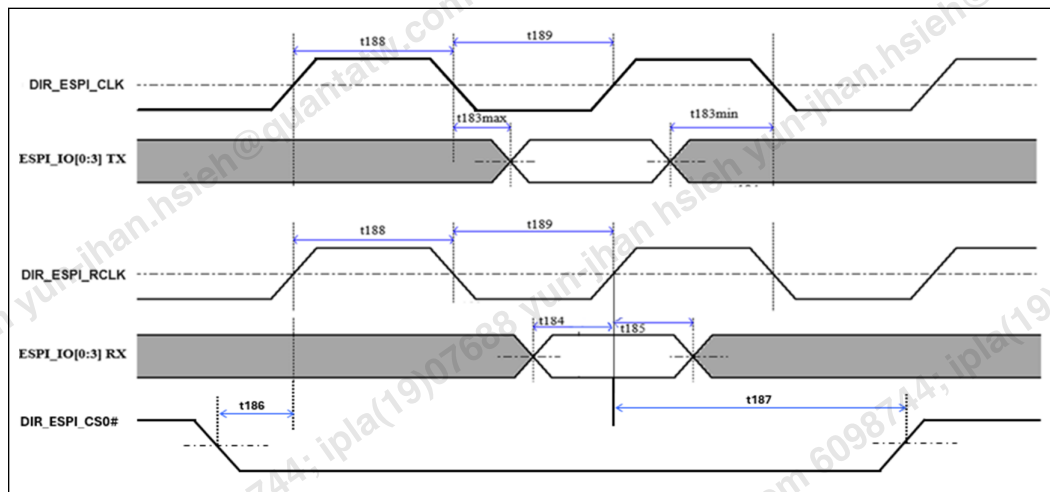
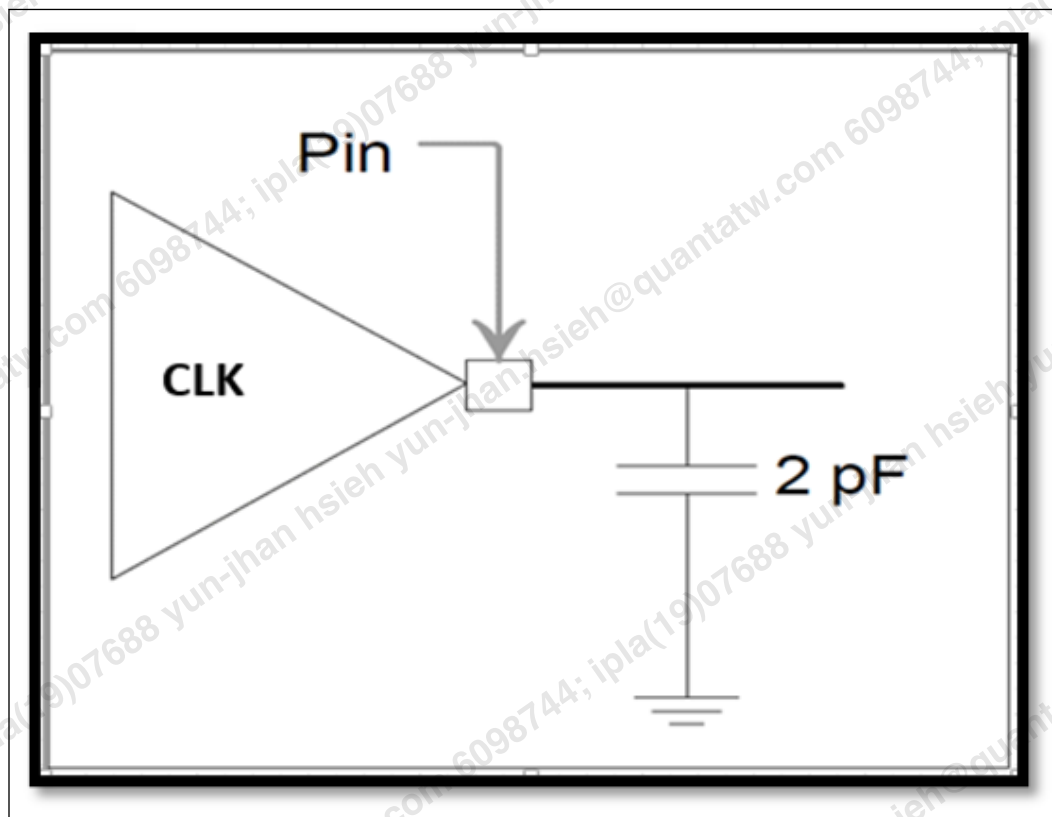


Figure 17. Test Load



14.0 Thermal Management

Table 47. Definitions/Acronyms

Acronyms	Description
Max Operating Temperature	This is the maximum operating temperature allowed as reported by temperature sensors. Instantaneous temperature may exceed this value for short durations. <i>Note:</i> Maximum observable temperature is configurable by system vendor and can be design specific.

14.1 Processor Thermal Management

The thermal solution provides both component-level and system-level thermal management. To allow optimal operation and long-term reliability of Intel processor-based systems, the system/processor thermal solution should be designed so that the processor:

- Remains below the maximum operating temperature specification at the maximum Processor Base Power.
- Conforms to system constraints, such as system acoustics, system skin-temperatures, and exhaust-temperature requirements.

CAUTION

Thermal specifications given in this chapter are on the component and package level and apply specifically to the processor. Operating the processor outside the specified limits may result in permanent damage to the processor and potentially other components in the system.

14.1.1 Thermal Considerations

The Processor Base Power is the assured sustained power that should be used for the design of the processor thermal solution, design to a higher thermal capability will get more Turbo residency. Processor Base Power is the time-averaged power dissipation that the processor is validated to not exceed during manufacturing while executing an Intel-specified high complexity workload at Base Frequency and at the maximum operating temperature as specified in the External Design Specification (EDS) for the SKU segment and configuration.

NOTE

The System on Chip processor integrates multiple compute cores and I/O on a single package. Platform support for specific usage experiences may require additional concurrency power to be considered when designing the power delivery and thermal sustained system capability

The processor integrates multiple processing IA cores, graphics cores and for some SKUs a chipset on a single package. This may result in power distribution differences across the package and should be considered when designing the thermal solution. Refer to the appropriate Platform Thermal Mechanical Design Guide for more details.

Intel® Turbo Boost Technology allows processor IA cores to run faster than the base frequency. It is invoked opportunistically and automatically as long as the processor is conforming to its temperature, power delivery, and current control limits. When Intel® Turbo Boost Technology is enabled:

- Applications are expected to run closer to Processor Base Power more often as the processor will attempt to maximize performance by taking advantage of estimated available energy budget in the processor package.
- The processor may exceed the Processor Base Power for short durations to utilize any available thermal capacitance within the thermal solution. The duration and time of such operation can be limited by platform runtime configurable registers within the processor. Refer to the appropriate processor Turbo Implementation Guide and processor family BIOS Specification for more details.
- Graphics peak frequency operation is based on the assumption of only one of the graphics domains (GT/GTx) being active. This definition is similar to the IA core Turbo concept, where peak turbo frequency can be achieved when only one IA core is active. Depending on the workload being applied and the distribution across the graphics domains the user may not observe peak graphics frequency for a given workload or benchmark.
- Thermal solutions and platform cooling that is designed to less than thermal design guidance may experience thermal and performance issues. For more details, refer to the appropriate processor turbo implementation guide and processor Platform Thermal Mechanical Design Guide.

NOTE

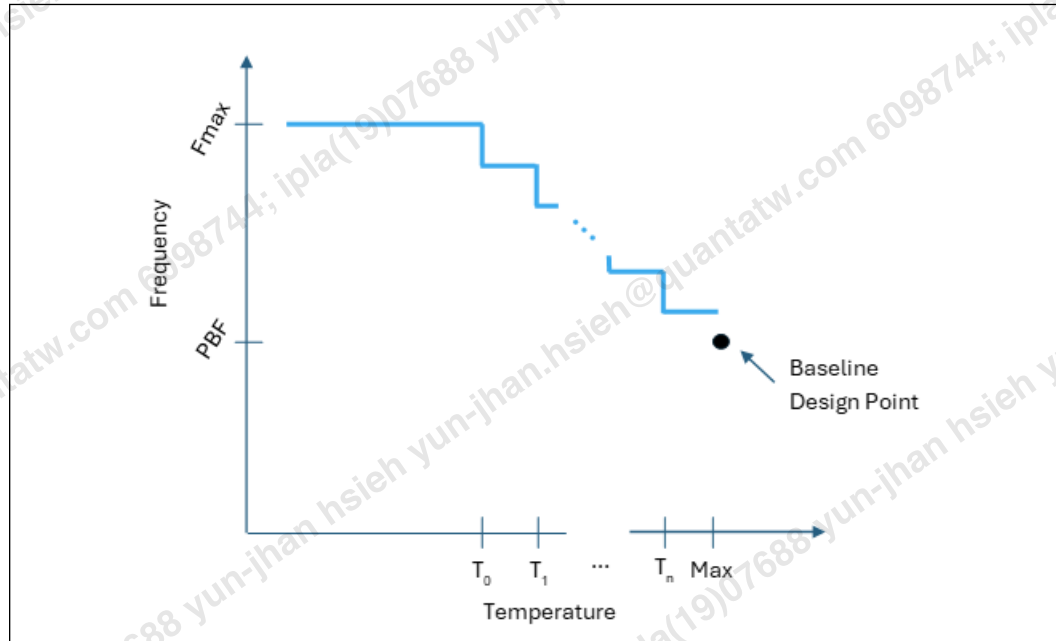
Intel® Turbo Boost Technology availability may vary between the different SKUs. (Refer to the appropriate processor Turbo Implementation Guide for more information.)

14.1.1.1 Processor Frequency vs. Temperature

Processor Power Management F/W manages processor frequencies to control fast temperature changes, caused by operating the processor at high frequencies.

This technology may proactively limit frequencies at temperatures below the maximum operating temperature while ensuring Processor Base Frequency (PBF) is achievable at the maximum operating temperature. Figure below describes this technology:

Figure 18. Processor Frequency vs. Temperature



Platform Thermal Mechanical Design Guide (TMDG) provides cooling guidance that ensures sustained operation at Processor Base Power (PBP) at the maximum operating temperature. Depending on environmental conditions and workload, this guidance allows the processor to operate at Turbo frequencies for a period of time before settling to the PBF. To increase maximum turbo frequency residency, customer is recommended to design the thermal solution to have a lower thermal resistance than the Baseline Design Point (see figure above). For more details, refer to Nova Lake Desktop Platform Thermal Mechanical Design Guide (#853008).

14.1.1.2 Package Power Control

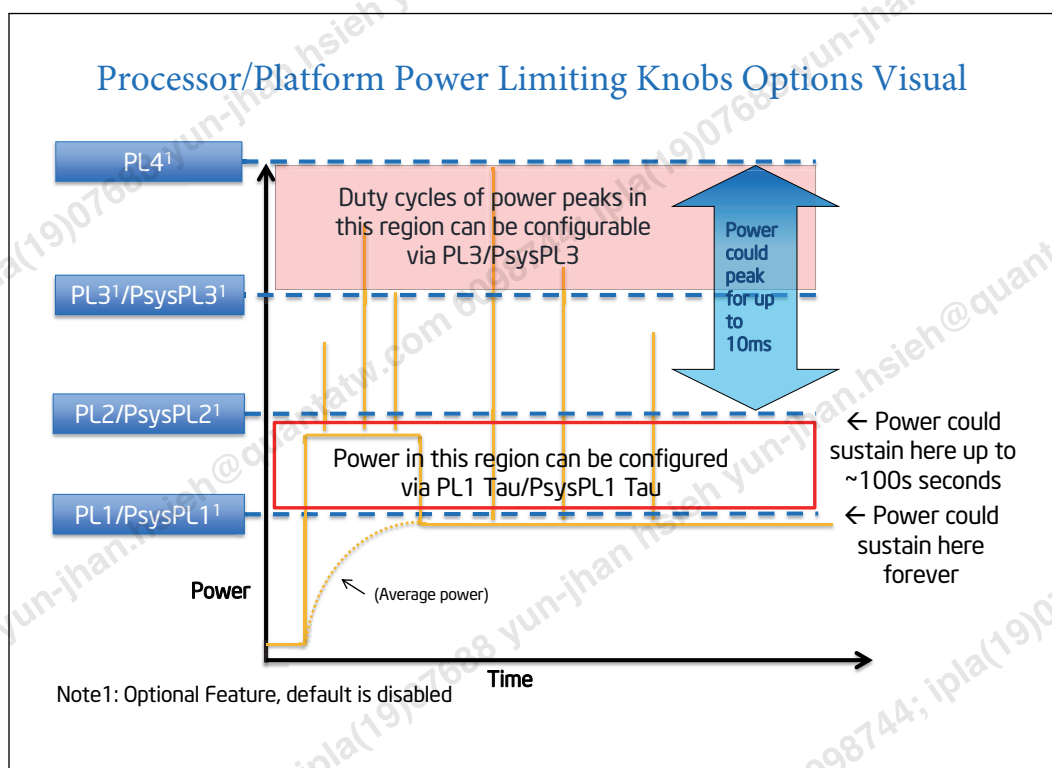
The package power control settings of PL1, PL2, PL3, PL4, and Tau allow the designer to configure Intel® Turbo Boost Technology to match the platform power delivery and package thermal solution limitations.

- **Power Limit 1 (PL1):** A threshold for average power that will not exceed - recommend to set to equal Processor Base Power power. PL1 should not be set higher than thermal solution cooling limits.
- **Power Limit 2 (PL2):** A threshold that if exceeded, the PL2 rapid power limiting algorithms will attempt to limit the spike above PL2.
- **Power Limit 3 (PL3):** A threshold that if exceeded, the PL3 rapid power limiting algorithms will attempt to limit the duty cycle of spikes above PL3 by reactively limiting frequency. This is an optional setting.
- **Power Limit 4 (PL4):** A limit that will not be exceeded, the PL4 power limiting algorithms will preemptively limit frequency to prevent spikes above PL4.
- **Turbo Time Parameter (Tau):** An averaging constant used for PL1 exponential weighted moving average (EWMA) power calculation.

NOTES

1. Implementation of Intel® Turbo Boost Technology 2.0 only requires configuring PL1, PL1, Tau and PL2.
2. The Turbo Implementation guide and BIOS Specification.
3. PL3 and PL4 are disabled by default.
4. NVL Platform Power Map Design Guide for default and extreme Power Limits (PL2, PL4) can be found in Nova Lake-S Platform Design Guide (#830661).
5. Intel® Dynamic Tuning Technology (DTT) is recommended for performance improvement in platforms. Dynamic Tuning is configured by system manufacturers dynamically optimizing the processor power based on the current platform thermal and power delivery conditions. Contact Intel Representatives for enabling details.

Figure 19. Package Power Control



14.1.1.3 Platform Power Control

The processor introduces Psys (Platform Power) to enhance processor power management. The Psys signal needs to be sourced from a compatible charger circuit and routed to the IMVP (voltage regulator). This signal will provide the total thermally relevant platform power consumption (processor and rest of platform) via SVID to the processor.

When the Psys signal is properly implemented, the system designer can utilize the package power control settings of PsysPL1/Tau, PsysPL2, and PsysPL3 for additional manageability to match the platform power delivery and platform thermal solution

limitations for Intel® Turbo Boost Technology. The operation of the PsysPL1/tau, PsysPL2 and PsysPL3 are analogous to the processor power limits described in [Package Power Control](#) on page 116.

- **Platform Power Limit 1 (PsysPL1):** A threshold for average platform power that will not be exceeded - recommend to set to equal platform thermal capability.
- **Platform Power Limit 2 (PsysPL2):** A threshold that if exceeded, the PsysPL2 rapid power limiting algorithms will attempt to limit the spikes above PsysPL2.
- **Platform Power Limit 3 (PsysPL3):** A threshold that if exceeded, the PsysPL3 rapid power limiting algorithms will attempt to limit the duty cycle of spikes above PsysPL3 by reactively limiting frequency.
- **PsysPL1 Tau:** An averaging constant used for PsysPL1 exponential weighted moving average (EWMA) power calculation.
- The Psys signal and associated power limits / Tau are optional for the system designer and disabled by default.
- The Psys data will not include power consumption for charging.
- Refer to the Turbo Implementation guide and BIOS Specification for additional details on use in your system.
- The Intel® Dynamic Tuning Technology (DTT/DPTF) is recommended for performance improvement in platforms. Dynamic Tuning is configured by system manufacturers dynamically optimizing the processor power based on the current platform thermal and power delivery conditions. Contact Intel Representatives for enabling details.

14.1.1.4 Turbo Time Parameter (Tau)

Turbo Time Parameter (Tau) is a mathematical parameter (units of seconds) that controls the Intel® Turbo Boost Technology algorithm. During a maximum power turbo event, the processor could sustain PL2 for a duration longer than the Turbo Time Parameter. If the power value and/or Turbo Time Parameter is changed during runtime, it may take some time based on the new Turbo Time Parameter level for the algorithm to settle at the new control limits. The time varies depending on the magnitude of the change, power limits and other factors. There is an individual Turbo Time Parameter associated with Package Power Control and Platform Power Control.

Refer to the appropriate processor Platform Thermal Mechanical Design Guide and processor Turbo Implementation Guide for more information.

14.1.2 Assured Power (cTDP)

Assured Power (cTDP) form a design option where the processor's behavior and package Processor Base Power are dynamically adjusted to a desired system performance and power envelope. Assured Power (cTDP) technologies offer opportunities to differentiate system design while running active workloads on select processor SKUs through scalability, configuration and adaptability. The scenarios or methods by which each technology is used are customizable but typically involve changes to PL1 and associated frequencies for the scenario with a resultant change in performance depending on system's usage. Either technology can be triggered by (but are not limited to) changes in OS power policies or hardware events such as docking a system, flipping a switch or pressing a button. Assured Power (cTDP) are designed to be configured dynamically and do not require an operating system reboot.

NOTES

- FORCEPR# events should be triggered after BIOS active. Triggering FORCEPR# after BIOS is active should be ensured as it is essential for system stability.
- Assured Power technologies are not battery life improvement technologies.

14.1.2.1 Assured Power (cTDP) Modes**NOTE**

Assured Power (cTDP) availability may vary between the different SKUs.

With cTDP, the processor is now capable of altering the maximum sustained power with an alternate processor IA core base frequency. Assured Power (cTDP) allows operation in situations where extra cooling is available or situations where a cooler and quieter mode of operation is desired. Refer to the appropriate processor family BIOS Specification for more enabling details.

cTDP consists of three modes as shown in the following table.

Table 48. Assured Power (cTDP)

Processor Power Characteristic	Description of Characteristic	Processor Design Considerations
Maximum Turbo Power	The maximum sustained (>1s) power dissipation of the processor as limited by current and/or temperature controls. Instantaneous power may exceed Maximum Turbo Power for short durations (<=10ms). Maximum Turbo Power is configurable by system vendor and can be system specific.	Intel performance advocacy for power delivery and transient thermal solution design (PL2)
Processor Base Power	The time-averaged power dissipation that the processor is validated to not exceed during manufacturing while executing an Intel-specified high complexity workload at Base Frequency and at the operating temperature as specified in the EDS for the SKU segment and configuration.	Intel reference performance advocacy for sustained thermal solution design (PL1)
Minimum Assured Power	Min Assured Power is a performance advocacy determined by running a complex scenario defined by Intel. Every product SKU stack has guidance on Min Assured Power for thermal chassis design. Represents Intel specified min PL1 that needs to be taken for thermal design to get the advocated performance experience. Min Assured Power is	Intel minimum performance advocacy for sustained thermal solution design (PL1)

continued...

Processor Power Characteristic	Description of Characteristic	Processor Design Considerations
	the performance vs power cross-over point across the product SKU stack.	
High Concurrency Power	High Concurrency Power is a functional characteristic determined by running a complex scenario defined by Intel. Every processor consumes a minimum power during a maximum connected case. This is a functional characteristic, not intended to indicate performance floor. Scenario takes into consideration IO ports, compute IPs concurrency (CPU, GPU, NPU) along with memory BW. Temperature assumption of spec limit. Manufacturing screening is done to exclude parts that don't meet the target power for the high concurrency scenario. The processor may not honor PL1 values set lower than high concurrency power during the high concurrency scenario.	Thermal solution capability required to support the high concurrency scenario as described

In each mode, the Intel® Turbo Boost Technology 2.0 power limits are reprogrammed along with a new OS controlled frequency range. The Intel® Dynamic Tuning driver assists in Processor Base Power operation by adjusting processor PL1 dynamically. The CTPD mode does not change the maximum per-processor IA core turbo frequency.

14.1.3 Thermal Management Features

Occasionally the processor may operate in conditions that are near to its maximum operating temperature. This can be due to internal overheating or overheating within the platform. In order to protect the processor and the platform from thermal failure, several thermal management features exist to reduce package power consumption and thereby temperature in order to remain within normal operating limits. Furthermore, the processor supports several methods to reduce memory power.

14.1.3.1 Adaptive Thermal Monitor

The purpose of the Adaptive Thermal Monitor is to reduce processor IA core power consumption and temperature until it operates below its maximum operating temperature. Processor IA core power reduction is achieved by:

- Adjusting the operating frequency (using the processor IA core ratio multiplier) and voltage.
- Modulating (starting and stopping) the internal processor IA core clocks (duty cycle).

The Adaptive Thermal Monitor can be activated when the package temperature, monitored by any Digital Thermal Sensor (DTS), meets its maximum operating temperature.

Reaching the maximum operating temperature activates the Thermal Control Circuit (TCC). When activated the TCC causes both the processor IA core and graphics core to reduce frequency and voltage adaptively. The Adaptive Thermal Monitor will remain

active as long as the package temperature remains at its specified limit. Therefore, the Adaptive Thermal Monitor will continue to reduce the package frequency and voltage until the TCC is de-activated.

Maximum Operating Temperature is factory calibrated and is not user configurable. The default value is software visible in the TEMPERATURE_TARGET (1A2h) MSR, bits [23:16].

The Adaptive Thermal Monitor does not require any additional hardware, software drivers, or interrupt handling routines. It is not intended as a mechanism to maintain processor thermal control to PL1 = Processor Base Power. The system design should provide a thermal solution that can maintain normal operation when PL1 = Processor Base Power within the intended usage range.

Adaptive Thermal Monitor protection is always enabled.

TCC Activation Offset

TCC Activation Offset can be set as an offset from maximum operating temperature to lower the onset of TCC and Adaptive Thermal Monitor. In addition, there is an optional time window (Tau) to manage processor performance at the TCC Activation offset value via an EWMA (Exponential Weighted Moving Average) of temperature. For more information on TCC Activation offset, refer to the appropriate processor family BIOS Specification and Turbo Implementation Guide.

TCC Activation Offset with Tau=0

An offset (degrees Celsius) can be written to the TEMPERATURE_TARGET (1A2h) MSR, bits [29:24], the offset value will be subtracted from the value found in bits [23:16]. When the time window (Tau) is set to zero, there will be no averaging, the offset, will be subtracted from the Maximum Operating Temperature value and used as a new maximum temperature set point for Adaptive Thermal Monitoring. This will have the same behavior as in prior products to have TCC activation and Adaptive Thermal Monitor to occur at this lower target silicon temperature.

If enabled, the offset should be set lower than any other passive protection such as ACPI _PSV trip points.

TCC Activation Offset with Tau

To manage the processor with the EWMA (Exponential Weighted Moving Average) of temperature, an offset (degrees Celsius) is written to the TEMPERATURE_TARGET (1A2h) MSR, bits [29:24], and the time window (Tau) is written to the TEMPERATURE_TARGET (1A2h) MSR [6:0]. The Offset value will be subtracted from the value found in bits [23:16] and be the temperature.

The processor will manage to this average temperature by adjusting the frequency of the various domains. The instantaneous Operating Temperature can briefly exceed the average temperature. The magnitude and duration of the overshoot is managed by the time window value (Tau).

This averaged temperature thermal management mechanism is in addition, and not instead of Maximum Operating Temperature thermal management. That is, whether the TCC activation offset is 0 or not, TCC Activation will occur at Maximum Operating Temperature.

Frequency / Voltage Control

Upon Adaptive Thermal Monitor activation, the processor attempts to dynamically reduce processor temperature by lowering the frequency and voltage operating point. The operating points are automatically calculated by the processor IA core itself and do not require the BIOS to program them as with previous generations of Intel processors. The processor IA core will scale the operating points such that:

- The voltage will be optimized according to the temperature, the processor IA core bus ratio and the number of processor IA cores in deep C-states.
- The processor IA core power and temperature are reduced while minimizing performance degradation.

Once the temperature has dropped below the trigger temperature, the operating frequency and voltage will transition back to the normal system operating point.

Once a target frequency/bus ratio is resolved, the processor IA core will transition to the new target automatically.

- On an upward operating point transition, the voltage transition precedes the frequency transition.
- On a downward transition, the frequency transition precedes the voltage transition.
- The processor continues to execute instructions. However, the processor will halt instruction execution for frequency transitions.

If a processor load-based Enhanced Intel SpeedStep® Technology/P-state transition (through MSR write) is initiated while the Adaptive Thermal Monitor is active, there are two possible outcomes:

- If the P-state target frequency is higher than the processor IA core optimized target frequency, the P-state transition will be deferred until the thermal event has been completed.
- If the P-state target frequency is lower than the processor IA core optimized target frequency, the processor will transition to the P-state operating point.

Clock Modulation

If the frequency/voltage changes are unable to end an Adaptive Thermal Monitor event, the Adaptive Thermal Monitor will utilize clock modulation. Clock modulation is done by alternately turning the clocks off and on at a duty cycle (ratio between clock “on” time and total time) specific to the processor. The duty cycle is factory configured to 25% on and 75% off and cannot be modified. The period of the duty cycle is configured to 32 microseconds when the Adaptive Thermal Monitor is active. Cycle times are independent of processor frequency. A small amount of hysteresis has been included to prevent excessive clock modulation when the processor temperature is near its maximum operating temperature. Once the temperature has dropped below the maximum operating temperature, and the hysteresis timer has expired, the Adaptive Thermal Monitor goes inactive and clock modulation ceases. Clock modulation is automatically engaged as part of the Adaptive Thermal Monitor activation when the frequency/voltage targets are at their minimum settings. Processor performance will be decreased when clock modulation is active. Snooping and interrupt processing are performed in the normal manner while the Adaptive Thermal Monitor is active.

Clock modulation will not be activated by the Package average temperature control mechanism.

Thermal Throttling

As the processor approaches Maximum Operating Temperature a throttling mechanisms will engage to protect the processor from over-heating and provide control thermal budgets.

Achieving this is done by reducing IA and other subsystem agent's voltages and frequencies in a gradual and coordinated manner that varies depending on the dynamics of the situation. IA frequencies and voltages will be directed down as low as LFM (Lowest Frequency Mode). Further restricts are possible via Thermal Throttling point (TT1) under conditions where thermal budget cannot be re-gained fast enough with voltages and frequencies reduction alone. TT1 keeps the same processor voltage and clock frequencies the same yet skips clock edges to produce effectively slower clocking rates. This will effectively result in observed frequencies below LFM on the Windows PERF monitor.

14.1.3.2 Digital Thermal Sensor

Each processor has multiple on-tile Digital Thermal Sensor (DTS) that detects the instantaneous temperature of processor IA, GT and other areas of interest.

Temperature values from the DTS can be retrieved through:

- A software interface using processor Model Specific Register (MSR).
- A processor hardware interface.

When the temperature is retrieved by the processor MSR, it is the instantaneous temperature of the given DTS. When the temperature is retrieved using PECI, it is the average of the highest DTS temperature in the package over a 256 ms time window. Intel recommends using the PECI reported temperature for platform thermal control that benefits from averaging, such as fan speed control. The average DTS temperature may not be a good indicator of package Adaptive Thermal Monitor activation or rapid increases in temperature that triggers the Out of Specification status bit within the PACKAGE_THERM_STATUS (1B1h) MSR and IA32_THERM_STATUS (19Ch) MSR.

Code execution is halted in C1 or deeper C-states. Package temperature can still be monitored through PECI in lower C-states.

Unlike traditional thermal devices, the DTS outputs a temperature relative to the maximum supported operating temperature of the processor, regardless of TCC activation offset. It is the responsibility of software to convert the relative temperature to an absolute temperature. The absolute reference temperature is readable in the TEMPERATURE_TARGET (1A2h) MSR. The temperature returned by the DTS is an implied negative integer indicating the relative offset from Maximum Operating Temperature. The DTS does not report temperatures greater than Maximum Operating Temperature. Refer to appropriate processor family BIOS Specification for specific register details. The DTS-relative temperature readout directly impacts the Adaptive Thermal Monitor trigger point. When a package DTS indicates that it has reached the TCC activation (a reading of 0h, except when the TCC activation offset is changed), the TCC will activate and indicate an Adaptive Thermal Monitor event. A TCC activation will lower both processor IA core and graphics core frequency, voltage, or both. Changes to the temperature can be detected using two programmable thresholds located in the processor thermal MSRs. These thresholds have the capability of generating interrupts using the processor IA core's local APIC. Refer to the *Intel 64 Architectures Software Developer's Manual* for specific register and programming details.

Digital Thermal Sensor Accuracy (T_{accuracy})

The error associated with DTS measurements will not exceed ± 5 °C within the entire operating range.

Fan Speed Control with Digital Thermal Sensor

Digital Thermal Sensor based fan speed control (T_{FAN}) is a recommended feature to achieve optimal thermal performance. At the T_{FAN} temperature, Intel recommends full cooling capability before the DTS reading reaches Maximum Operating Temperature.

14.1.3.3 FORCEPR# Signal

The FORCEPR# (Force Power Reduction) is an input signal to the CPU. It is used to reduce processor's electrical load following power and thermal events. FORCEPR# should not be used as an output indication.

The FORCEPR# signal can be configured to the following mode:

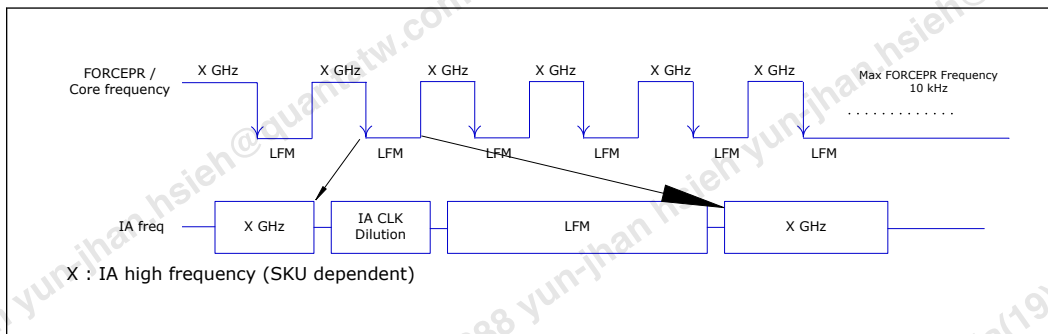
- **Input Only:** FORCEPR# is driven by an external force.

FORCEPR Input Only

Following FORCEPR assertion, Fast FORCEPR feature will instantly reduce frequency and CPU will limit the frequency to lowest frequency according to FORCEPR_RESPONSE configuration that will remain until FORCEPR de-assertion.

FORCEPR Demotion Algorithm will be activated in case of multiple FORCEPR events

Figure 20. FORCEPR Demotion Description



14.1.3.4 FORCEPR Demotion Algorithm

FORCEPR demotion algorithm is designed to improve system performance following multiple Platform FORCEPR consecutive assertions. During each FORCEPR assertion processor will eventually transition to the lowest P-State (P_n) supported by the processor IA cores and graphics cores (LFM). When detecting several FORCEPR consecutive assertions the processor will reduce the max frequency in order to reduce the FORCEPR assertions events. The processor will keep reducing the frequency until no consecutive assertions detected. The processor will raise the frequency if no consecutive FORCEPR assertion events will occur.

14.1.3.5 Voltage Regulator Protection using FORCEPR#

FORCEPR# may be used for thermal protection of voltage regulators (VR). System designers can create a circuit to monitor the VR temperature and assert FORCEPR# and, if enabled, activate the TCC when the temperature limit of the VR is reached. When FORCEPR# is configured as a bi-directional or input only signal, if the system assertion of FORCEPR# is recognized by the processor, results in power reduction. Power reduction down to LFM and duration of the platform FORCEPR# assertion, supported by the processor IA cores and graphics cores. Systems should still provide proper cooling for the VR and rely on bi-directional FORCEPR# only as a backup in case of system cooling failure. Overall, the system thermal design should allow the power delivery circuitry to operate within its temperature specification even while the processor is operating at its Adaptive Thermal Monitor protection is always enabled. Refer to Platform Design Guide and VR specifications, for example, IMVP9.3 Pulse Width Modulation VR Vendor Enabling Specification (#637348).

NOTE

During FORCEPR demotion, the core frequency may be reduced below LFM for several uSec.

14.1.3.6 Thermal Solution Design and FORCEPR# Behavior

With a properly designed and characterized thermal solution, it is anticipated that FORCEPR# will only be asserted for very short periods of time when running the most power intensive applications. The processor performance impact due to these brief periods of TCC activation is expected to be so minor that it would be immeasurable. However, an under-designed thermal solution that is not able to prevent excessive assertion of FORCEPR# in the anticipated ambient environment may:

- Cause a noticeable performance loss.
- Result in prolonged operation at or above the specified maximum operating temperature and affect the long-term reliability of the processor.
- May be incapable of cooling the processor even when the TCC is active continuously (in extreme situations).

14.1.3.7 Low-Power States and FORCEPR# Behavior

Depending on package power levels during package C-states, outbound FORCEPR# may de-assert while the processor is idle as power is removed from the signal. Upon wake up, if the processor is still hot, the FORCEPR# will re-assert, although typically package idle state residency should resolve any thermal issues. The PECI interface is fully operational during all C-states and it is expected that the platform continues to manage processor IA and E core and package thermals even during idle states by regularly polling for thermal data over PECI.

14.1.3.8 THERMTRIP# Signal

Regardless of enabling the automatic or on-demand modes, in the event of a catastrophic cooling failure, the package will automatically shut down when the silicon has reached an elevated temperature that risks physical damage to the product. At this point, the THERMTRIP# signal will go active.

14.1.3.9 Critical Temperature Detection

Critical Temperature detection is performed by monitoring the package temperature. This feature is intended for graceful shutdown before the THERMTRIP# is activated. However, the processor execution is not guaranteed between critical temperature and THERMTRIP#. If the Adaptive Thermal Monitor is triggered and the temperature remains high, a critical temperature status and sticky bit are latched in the PACKAGE_THERM_STATUS (1B1h) MSR and the condition also generates a thermal interrupt, if enabled. For more details on the interrupt mechanism, refer to *Intel® 64 Architectures Software Developer's Manual* or appropriate processor family BIOS Specification.

14.1.3.10 Software Controlled Clock Modulation (On-Demand Mode)

The processor provides an auxiliary mechanism that allows system software to force the processor to reduce its power consumption using clock modulation. This mechanism is referred to as "On-Demand" mode and is distinct from Adaptive Thermal Monitor and bi-directional FORCEPR#. The processor platforms should not rely on software usage of this mechanism to limit the processor temperature. On-Demand Mode can be accomplished using processor MSR or chipset I/O emulation. On-Demand Mode may be used in conjunction with the Adaptive Thermal Monitor. However, if the system software tries to enable On-Demand mode at the same time the TCC is engaged, the factory configured the duty cycle of the TCC will override the duty cycle selected by the On-Demand mode. If the I/O based and MSR-based On-Demand modes are in conflict, the duty cycle selected by the I/O emulation-based On-Demand mode will take precedence over the MSR-based On-Demand Mode.

14.1.4 Intel® Memory Thermal Management

DRAM Thermal Aggregation

P-Unit firmware is responsible for aggregating DRAM temperature sources into a per-DIMM reading as well as an aggregated virtual 'max' sensor reading. At reset, MRC communicates to the MC the valid channels and ranks as well as DRAM type. At that time, Punit firmware sets up a valid channel and rank mask that is then used in the thermal aggregation algorithm to produce a single maximum temperature.

DRAM Thermal Monitoring

- DRAM thermal sensing Periodic DDR thermal reads from DDR.
- DRAM thermal calculation Punit reads of DDR thermal information direct from the memory controller (MR4 or MPR) Punit estimation of a virtual maximum DRAM temperature based on per-rank readings. Application of thermal filter to the virtual maximum temperature.

DRAM Refresh Rate Control

The MRC will natively interface with MR4 or MPR readings to adjust DRAM refresh rate as needed to maintain data integrity. This capability is enabled by default and occurs automatically. Direct override of this capability is available for debug purposes, but this cannot be adjusted during runtime.

DRAM Bandwidth Throttling (Change to DDR Bandwidth Throttling)

Control for bandwidth throttling is available through the memory controller. Software may program a percentage bandwidth target at the current operating frequency and that used to throttle read and write commands based on the maximum memory MPR/MR4 reading.

14.2 Processor Base Power Thermal and Power Specifications

The following notes apply to [Table 50](#) on page 128.

Table 49. General Notes

Note	Definition
1	The Processor Base Power and Assured Power (cTDP) values are the average power dissipation in operating temperature condition limit, for the SKU Segment and Configuration, for which the processor is validated during manufacturing when executing an associated Intel-specified high-complexity workload at the processor IA core frequency corresponding to the configuration and SKU.
2	Processor Base Power workload may consist of a combination of processor IA core intensive and graphics core intensive applications.
3	Can be modified at runtime by MSR writes, with MMIO and with PECI commands.
4	'Turbo Time Parameter' is a mathematical parameter (units of seconds) that controls the processor turbo algorithm using a moving average of energy usage. Do not set the Turbo Time Parameter to a value less than 0.1 seconds. Refer to Platform Power Control on page 117 for further information.
5	The shown limit is a time averaged-power, based upon the Turbo Time Parameter. Absolute product power may exceed the set limits for short durations or under virus or uncharacterized workloads.
6	The Processor will be controlled to a specified power limit. If the power value and/or 'Turbo Time Parameter' is changed during runtime, it may take a short period of time (approximately 3 to 5 times the 'Turbo Time Parameter') for the algorithm to settle at the new control limits.
7	This is a hardware default setting and not a behavioral characteristic of the part.
8	For controllable turbo workloads, the PL2 limit may be exceeded for up to 10ms.
9	LPM power level is an opportunistic power and is not a guaranteed value as usages and implementations may vary.
10	Power limits may vary depending on if the product supports the Minimum Assured Power (cTDP Down) and/or Maximum Assured Power (cTDP Up) modes. Default power limits can be found in the PKG_PWR_SKU MSR (614h).
11	The processor tile do not reach maximum sustained power simultaneously since the sum of all active circuit's estimated power budget is controlled to be equal to or less than the specified PL1 limit. For additional information, refer to the appropriate Mobile TMDG for more information.
12	Minimum Assured Power(cTDP Down) is based on Xe equivalent graphics configuration. Minimum Assured Power(cTDP Down) does not decrease the number of active Processor Graphics EUs but relies on Power Budget Management (PL1) to achieve the specified power level.
13	May vary based on SKU.
14	- The formula of $PL2 = PL1 * 1.25$ is the hardware. - PL2- Processor opportunistic higher Average Power with limited duration controlled by Tau_PL1 setting, the larger the Tau, the longer the PL2 duration. - PL2 recommended value can be found in the PDG/Power Map.
continued...	

Note	Definition
15	Processor Base Power workload does not reflect various I/O connectivity cases such as Thunderbolt. Refer to the Platform Design Guide, Thermal Power Consideration section for adjustments to the Processor Base Power required to preserve base frequency associated to the sustained long-term thermal capability.
16	Hardware default of PL1 Tau=1s, By including the benefits available from power and thermal management features the recommended is to use PL1 Tau=28s.
17	PL1 Tau max recommendation value is the default value in the BIOS/BKC and this value is been tested.

Table 50. Processor Base Power Specifications (S Processor Line)

Segment and Package	Processor P/E cores, Graphics Configuration and Processor Base Power	Configuration			Processor P/ E core Frequency [GHz]	Thermal Design Power (Processor Base Power) [w]	General Notes	
S-Process or Line LGA	16P+32E Core 175W ¹	IA Core Frequency	Processor Base power	P-Core	TBD	175	1,9,10,11,12, 15	
				E-Core	TBD			
			Low Frequency Mode - LFM			TBD		N/A
		Graphics Core Frequency	Graphics Frequency			TBD		N/A
			Low Frequency Mode - LFM			TBD		
S-Process or Line LGA	8P+16E Core 125W ¹	IA Core Frequency	Processor Base power	P-Core	TBD	125	1,9,10,11,12, 15	
				E-Core	TBD			
			Low Frequency Mode - LFM			TBD		N/A
		Graphics Core Frequency	Graphics Frequency			TBD		N/A
			Low Frequency Mode - LFM			TBD		
S-Process or Line LGA	8P+16E Core 65W ¹	IA Core Frequency	Processor Base power	P-Core	TBD	65	1,9,10,11,12, 15	
				E-Core	TBD			
			Low Frequency Mode - LFM			TBD		N/A
		Graphics Core Frequency	Graphics Frequency			TBD		N/A
			Low Frequency Mode - LFM			TBD		
S-Process or Line LGA	4P+8E Core 65W ¹	IA Core Frequency	Processor Base power	P-Core	TBD	65	1,9,10,11,12, 15	
				E-Core	TBD			
			Low Frequency Mode - LFM			TBD		N/A
		Graphics Core Frequency	Graphics Frequency			TBD		N/A
			Low Frequency Mode - LFM			TBD		
S-Process or Line LGA	4P+8E Core 35W ¹	IA Core Frequency	Processor Base power	P-Core	TBD	35	1,9,10,11,12, 15	
				E-Core	TBD			
			Low Frequency Mode - LFM			TBD		N/A
		Graphics Core Frequency	Graphics Frequency			TBD		N/A
			Low Frequency Mode - LFM			TBD		

continued...

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Segment and Package	Processor P/E cores, Graphics Configuration and Processor Base Power	Configuration			Processor P/ E core Frequency [GHz]	Thermal Design Power (Processor Base Power) [w]	General Notes
S-Process or Line LGA	4P+0E Core 65W ¹	IA Core Frequency	Processor Base power	P-Core	TBD	65	1,9,10,11,12, 15
				E-Core	TBD		
		Low Frequency Mode - LFM			TBD	N/A	
		Graphics Core Frequency	Graphics Frequency		TBD	N/A	
			Low Frequency Mode - LFM		TBD		
S-Process or Line LGA	4P+0E Core 35W ¹	IA Core Frequency	Processor Base power	P-Core	TBD	35	1,9,10,11,12, 15
				E-Core	TBD		
		Low Frequency Mode - LFM			TBD	N/A	
		Graphics Core Frequency	Graphics Frequency		TBD	N/A	
			Low Frequency Mode - LFM		TBD		

Note: Refer to [General Notes](#)

14.3 Processor Line Thermal and Power Specifications

Table 51. Package Turbo Specifications (S Processor Lines)

No Specifications for Min/Max PL1/PL2 values, refer to the PDG/Power Map (Platform Design Guide) for PL2 recommendation.

Hardware default of PL1 Tau=1s, By including the benefits available from power and thermal management features the recommended is to use PL1 Tau based on table below.

PL2- Processor opportunistic higher Average Power – Reactive, Limited Duration controlled by Tau_PL1 setting.

PL1 Tau - PL1 average power is controlled via PID algorithm with this Tau, The larger the Tau, the longer the PL2 duration.

System cooling solution and designs found to not being able to support the Performance TauPL1, adjust the TauPL1 to cooling capability.

Segment and Package	Processor IA Cores, Graphics, Configuration and Processor Base Power	Parameter	Minimum	Tau MSR Max Value	Recommended Value	Units	General Notes
S-Processor Line LGA	16P+32E Core 175W ¹	Power Limit 1 Time (PL1 Tau)	0.1	448	56	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	N/A	125	W	
		Power Limit 2 (PL2)	N/A	N/A	Refer to NVL-S PDG (#830661) for PL2 recommendation.	W	

continued...

Segment and Package	Processor IA Cores, Graphics, Configuration and Processor Base Power	Parameter	Minimum	Tau MSR Max Value	Recommended Value	Units	General Notes
S-Processor Line LGA	8P+16E Core 125W ¹	Power Limit 1 Time (PL1 Tau)	0.1	448	56	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	N/A	125	W	
		Power Limit 2 (PL2)	N/A	N/A	Refer to NVL-S PDG (#830661) for PL2 recommendation.	W	
S-Processor Line LGA	8P+16E Core 65W ¹	Power Limit 1 Time (PL1 Tau)	0.1	448	28	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	N/A	65	W	
		Power Limit 2 (PL2)	N/A	N/A	Refer to NVL-S PDG (#830661) for PL2 recommendation.	W	
S-Processor Line LGA	4P+8E Core 65W ¹	Power Limit 1 Time (PL1 Tau)	0.1	448	28	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	N/A	65	W	
		Power Limit 2 (PL2)	N/A	N/A	Refer to NVL-S PDG (#830661) for PL2 recommendation.	W	
S-Processor Line LGA	4P+8E Core 35W ¹	Power Limit 1 Time (PL1 Tau)	0.1	448	28	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	N/A	35	W	
		Power Limit 2 (PL2)	N/A	N/A	Refer to NVL-S PDG (#830661) for PL2 recommendation.	W	
S-Processor Line LGA	4P+0E Core 65W ¹	Power Limit 1 Time (PL1 Tau)	0.1	448	28	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	N/A	65	W	
		Power Limit 2 (PL2)	N/A	N/A	Refer to NVL-S PDG (#830661) for PL2 recommendation.	W	
S-Processor Line LGA	4P+0E Core 35W ¹	Power Limit 1 Time (PL1 Tau)	0.1	448	28	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	N/A	35	W	
		Power Limit 2 (PL2)	N/A	N/A	Refer to NVL-S PDG (#830661) for PL2 recommendation.	W	
Note: Refer to General Notes							

Table 52. Operating Temperature Specifications (S Processor Line)

Segment	Package Turbo Parameter	Temperature Range		Processor Base Power Specification Temperature Range		Units	Notes
		Minimum	Maximum	Minimum	Maximum		
S-Processor Line LGA	Operating temperature	0	100	35	100	°C	1, 2
Notes: 1. The thermal solution needs to ensure that the processor temperature does not exceed the Processor Base Power Specification Temperature. 2. The processor operating temperature is monitored by Digital Temperature Sensors (DTS). For DTS accuracy, refer to Digital Thermal Sensor on page 123.							

Table 53. Low Power and Thermal Test Vehicle (TMTV) Specifications (S Processor Line LGA)

Processor IA Cores, Graphics Configuration and Processor Base Power	Maximum Power Package C8 (W) ^{1,4,5}	TMTV Processor Base Power (W) ^{6,7}	Min TCASE (°C)	Maximum TMTV TCASE (°C)
16P+32E Core 175W	N/A	175	0	TBD
8P+16E Core 125W	N/A	125	0	TBD
8P+16E Core 65W	N/A	65	0	TBD
4P+8E Core 65W	N/A	65	0	TBD
4P+8E Core 35W	N/A	35	0	TBD
4P+0E Core 65W	N/A	65	0	TBD
4P+0E Core 35W	N/A	35	0	TBD
Notes: 1. The package C-state power is the worst case power in the system configured as follows: a. DMI and PCIe links are at L1 2. Specification at DTS = 50 °C and minimum voltage loadline. 3. Specification at DTS = 35 °C and minimum voltage loadline. 4. These DTS values in Notes 2 - 3 are based on the TCC Activation MSR having a value of 100. 5. These values are specified at VCC_MAX and VNOM for all other voltage rails for all processor frequencies. Systems should be designed to ensure the processor is not to be subjected to any static VCC and ICC combination wherein VCCP exceeds VCCP_MAX at specified ICCP. Refer to the loadline specifications. 6. Processor Base Power should be used for processor thermal solution design targets. Processor Base Power is not the maximum power that the processor can dissipate. Processor Base Power is measured at DTS = -1. Processor Base Power is achieved with the Memory configured for DDR. 7. Platform Compatibility Guide (PCG) (previously known as FMB) provides a design target for meeting all planned processor frequency requirements. 8. Not 100% tested. Specified by design characterization.				

Table 54. TCONTROL Offset Configuration (S Processor Line LGA - Client)

Segment	TEMP_TARGET (TCONTROL) [°C]
16P+32E Core 175W	TBD
8P+16E Core 125W	TBD
8P+16E Core 65W	TBD
4P+8E Core 65W	TBD
4P+8E Core 35W	TBD
continued...	

Segment	TEMP_TARGET (TCONTROL) [°C]
4P+0E Core 65W	TBD
4P+0E Core 35W	TBD
Notes: 1. Digital Thermal Sensor (DTS) based fan speed control is recommended to achieve optimal thermal performance. 2. Intel recommends full cooling capability at approximately the DTS value of -1, to minimize TCC activation risk. 3. For example, if TCONTROL = 20 °C, Fan acceleration operation will start at 80 °C (100 °C - 20 °C).	

14.4 Error and Thermal Protection Signals

Table 55. Error and Thermal Protection Signals

Signal Name	Description	Dir.	Link Type
CATERR#	Catastrophic Error: This signal indicates that the system has experienced a catastrophic error and cannot continue to operate. The processor will set this signal for non-recoverable machine check errors or other unrecoverable internal errors. CATERR# is used for signaling the following types of errors: Legacy MCERRs, CATERR# is asserted for 16 BCLKs. Legacy IERRs, CATERR# remains asserted until warm or cold reset.	O	SE
FORCEPR#	Processor Hot: FORCEPR# goes active when the processor temperature monitoring sensor(s) detects that the processor has reached its maximum safe operating temperature. This indicates that the processor Thermal Control Circuit (TCC) has been activated, if enabled. This signal can also be driven to the processor to activate the TCC.	I	I:GTL
THERMTRIP#	Thermal Trip: The processor protects itself from catastrophic overheating by use of an internal thermal sensor. This sensor is set well above the normal operating temperature to ensure that there are no false trips. The processor will stop all executions when the operating temperature exceeds approximately 125 °C. This is signaled to the system by the THERMTRIP# pin. Refer to the appropriate platform design guide for termination requirements.	O	SE

14.5 Fan Speed Control Scheme with Digital Thermal Sensor (DTS)

With Digital Thermal Sensor (DTS) 1.1

To correctly use DTS 1.1, the designer must first select a worst case scenario T_{AMBIENT} , and ensure that the Fan Speed Control (FSC) can provide a Ψ_{CA} that is equivalent or greater than the Ψ_{CA} specification.

The DTS 1.1 implementation consists of two points:

- a Ψ_{CA} at T CONTROL
- a Ψ_{CA} at DTS = -1
- The Ψ_{CA} point at DTS = -1 defines the minimum Ψ_{CA} required at Processor Base Power considering the worst case system design T_{AMBIENT} design point:

$$\Psi_{\text{CA}} = (T_{\text{CASE-MAX}} - T_{\text{AMBIENT-TARGET}} - 1) / \text{Processor Base Power}$$

For example, for a 125 W Processor Base Power part, the T_{CASE} maximum is 62.0 °C and at a worst case design point of 40 °C local ambient this will result in:

$$\Psi_{CA} = (62.0 - 40 - 1) / 125 = 0.168 \text{ °C/W}$$

Similarly for a system with a design target of 45 °C ambient, the Ψ_{CA} at DTS = -1 needed will be 0.128 °C/W.

- The second point defines the thermal solution performance (Ψ_{CA}) at $T_{CONTROL}$. The following table lists the required Ψ_{CA} for the various Processor Base Power processors.

These two points define the operational limits for the processor for DTS 1.1 implementation. At $T_{CONTROL}$ the fan speed must be programmed such that the resulting Ψ_{CA} is better than or equivalent to the required Ψ_{CA} listed in the following table. Similarly, the fan speed should be set at DTS = -1 such that the thermal solution performance is better than or equivalent to the Ψ_{CA} requirements at $T_{AMBIENT-MAX}$.

The fan speed controller must linearly ramp the fan speed from processor DTS = $T_{CONTROL}$ to processor DTS = -1.

Figure 21. Digital Thermal Sensor (DTS) 1.1 Definition Points

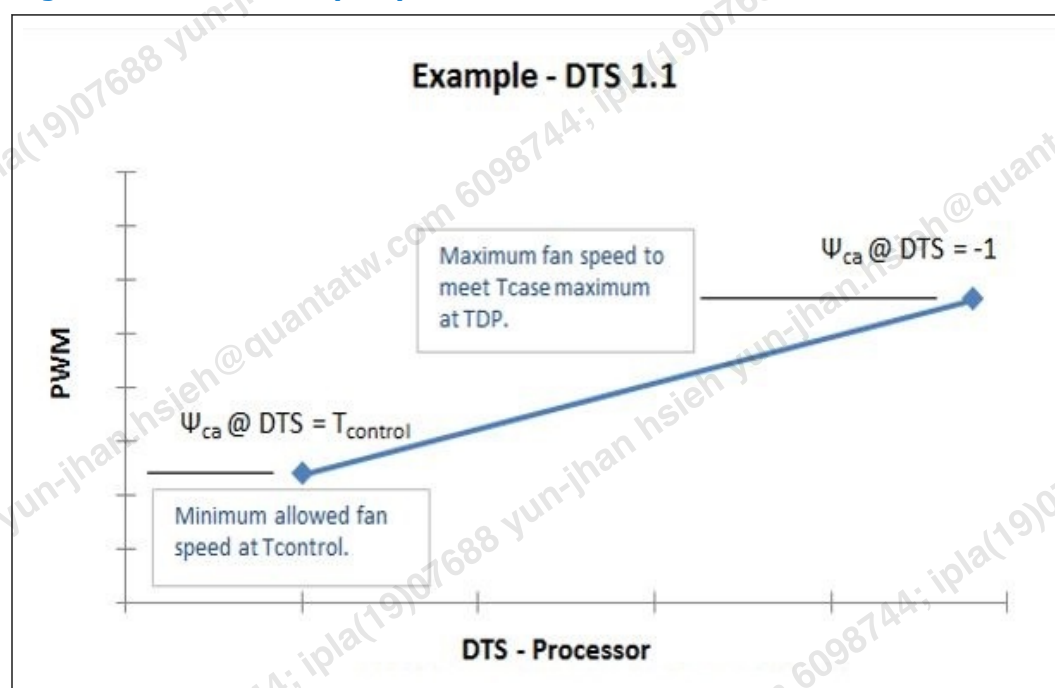


Table 56. Digital Thermal Sensor (DTS) 1.1 Thermal Solution Performance Above T_{CONTROL}

Die Core Configuration	Processor Base Power [W]	PL1[W]	Ψ_{CA} at DTS = $T_{CONTROL}^{1, 2}$ At System $T_{AMBIENT_MAX} = 30\text{ }^{\circ}\text{C}$	Ψ_{CA} at DTS = -1 At System $T_{AMBIEN} T_{_MAX} = 40\text{ }^{\circ}\text{C}$	Ψ_{CA} at DTS = -1 At System $T_{AMBIEN} T_{_MAX} = 45\text{ }^{\circ}\text{C}$	Ψ_{CA} at DTS = -1 At System $T_{AMBIEN} T_{_MAX} = 50\text{ }^{\circ}\text{C}$
16P+32E	175	175	TBD	TBD	TBD	TBD
		180	TBD	TBD	TBD	TBD
		185	TBD	TBD	TBD	TBD
		190	TBD	TBD	TBD	TBD
8P+16E	125	125	0.34	0.12	0.08	0.03
		130	0.34	0.12	0.08	0.03
		140	0.32	0.12	0.08	0.03
		150	0.34	0.12	0.08	0.03
8P+16E	65	65	0.70	0.53	0.45	0.37
		70	0.70	0.53	0.45	0.37
		80	0.70	0.53	0.45	0.37
		90	0.73	0.53	0.45	0.37
4P+8E	65	65	0.81	1.19	1.04	0.90
		70	0.84	1.19	1.04	0.90
		80	0.85	1.19	1.04	0.90
		90	0.84	1.19	1.04	0.90
4P+8E	35	35	TBD	TBD	TBD	TBD
		40	TBD	TBD	TBD	TBD
		50	TBD	TBD	TBD	TBD
		60	TBD	TBD	TBD	TBD
4P+0E	65	65	TBD	TBD	TBD	TBD
		70	TBD	TBD	TBD	TBD
		80	TBD	TBD	TBD	TBD
		90	TBD	TBD	TBD	TBD
4P+0E	35	35	TBD	TBD	TBD	TBD
		40	TBD	TBD	TBD	TBD
		50	TBD	TBD	TBD	TBD
		60	TBD	TBD	TBD	TBD

Notes: 1. Ψ_{CA} at “DTS = $T_{CONTROL}$ ” is applicable to systems that have an internal T_{RISE} (T_{ROOM} temperature to Processor cooling fan inlet) of less than 10 °C. In case the expected T_{RISE} is greater than 10 °C, a correction factor should be used as explained below. For each 1 °C T_{RISE} above 10 °C, the correction factor (CF) is defined as $CF = 1.7 / (\text{Processor Base Power})$.

2. The table data match for GT0.

With Digital Thermal Sensor (DTS) 2.0

To simplify processor thermal specification compliance, the processor calculates the DTS Thermal Profile from TCONTROL Offset, TCC Activation Temperature, Processor Base Power, and the Thermal Margin Slope provided in the following table.

NOTE

TCC Activation Offset is 0 for the processors.

Using the DTS Thermal Profile, the processor can calculate and report the Thermal Margin, where a value less than 0 indicates that the processor needs additional cooling, and a value greater than 0 indicates that the processor is sufficiently cooled.

Figure 22. Digital Thermal Sensor (DTS) 2.0 Definition Points

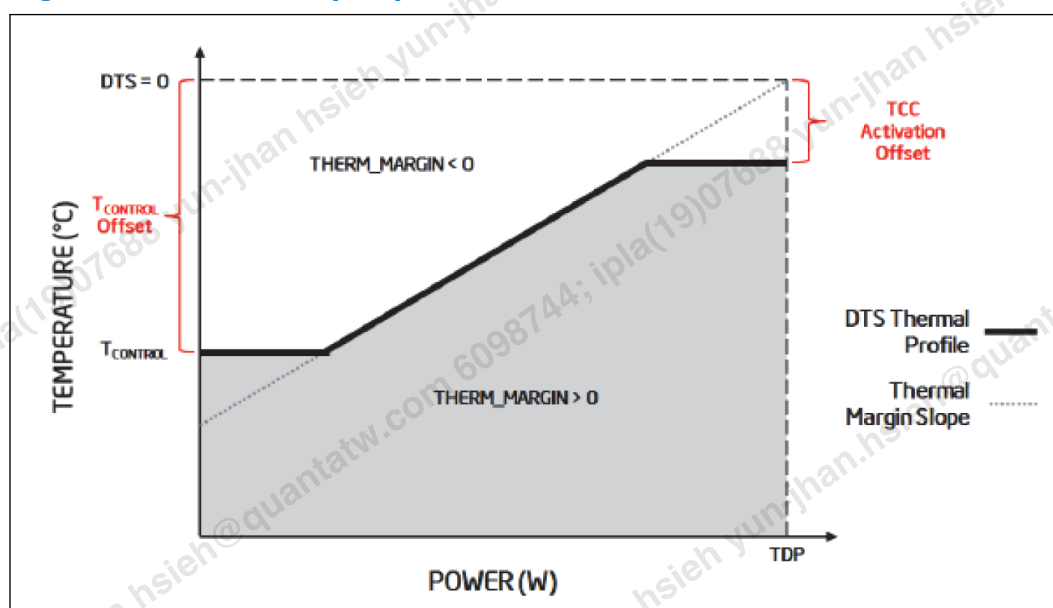


Table 57. Thermal Margin Slope

Die Core Configuration	Processor Base Power [W]	PL1 [W]	Temperature Control Offset	TCC Activation [°C]	Thermal Margin Slope [°C/W]
16P+32E	175	175	20	90	TBD
		180			TBD
		185			TBD
8P+16E	125	125	20	90	0.393
		130			0.383
		135			0.373
8P+16E	65	65	20	90	0.606
		70			0.566
		75			0.536

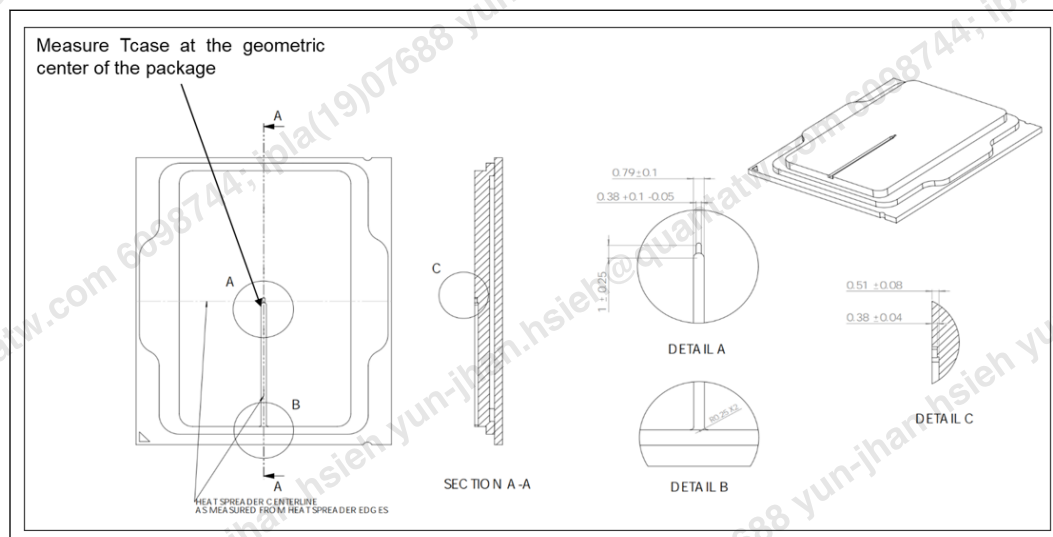
continued...

Die Core Configuration	Processor Base Power [W]	PL1 [W]	Temperature Control Offset	TCC Activation [°C]	Thermal Margin Slope [°C/W]
		80			0.506
4P+8E	65	65	20	90	TBD
		70			TBD
		75			TBD
		80			TBD
4P+8E	35	35	20	90	TBD
		40			TBD
		45			TBD
		50			TBD
4P+0E	65	65	20	90	TBD
		65			TBD
		75			TBD
		80			TBD
4P+0E	35	35	20	90	TBD
		40			TBD
		45			TBD
		50			TBD
Note: 1. The default BIOS settings for this SKU is 10C TCC offset.					

14.6 Thermal Metrology

The maximum Thermal Mechanical Test Vehicle (TMTV) case temperatures ($T_{\text{CASE-MAX}}$) can be derived from the data in the appropriate TMTV thermal profile earlier in this chapter. The TMTV T_{CASE} is measured at the geometric top center of the TMTV integrated heat spreader (IHS). Below figure illustrates the location where T_{CASE} temperature measurements should be made.

Figure 23. Thermal Mechanical Test Vehicle (TMTV) Case Temperature (T_{CASE}) Measurement Location



The following supplier can machine the groove and attach a thermocouple to the IHS. The following supplier is listed as a convenience to Intel's general customers and may be subject to change without notice.

THERM-X OF CALIFORNIA, 3200 Investment Blvd,

Hayward, Ca 94544. George Landis +1-510-441-7566 Ext. 368 george@therm-x.com.

The vendor part number is XTMS1565.

15.0 Thermal Sensor

The processor incorporates an on-tile Digital Thermal Sensor (DTS) for thermal management.

15.1 Modes of Operation

The DTS has two usages when enabled:

1. One use is to provide the temperature of the CPU in units of 1 °C. There is a 9 bit field for the temperature, with a theoretical range from -256 °C to +256 °C. Practically the operational range for TS would be between -40 °C and 110 °C.
2. The second use is to allow programmed trip points to cause alerts to SW or in the extreme case shutdown. Temperature may be provided without having any SW alerts set.

There are two thermal alert capabilities. One is for the catastrophic event (thermal runaway) which results in an immediate system power down (S5 state). The other alert provides an indication to the platform that a particular temperature has been caused. This second alert needs to be routed to SMI or SCI based on SW programming.

15.2 Temperature Trip Point

The internal thermal sensor reports three trip points: Cool, Hot, and Catastrophic trip points in the order of increasing temperature.

Crossing the cool trip point when going from higher to lower temperature may generate an interrupt. Crossing the hot trip point going from lower to higher temp may generate an interrupt. Each trip point has control register bits to select what type of interrupt is generated.

Crossing the cool trip point while going from low to higher temperature or crossing the hot trip point while going from high to lower temperature will not cause an interrupt.

When triggered, the catastrophic trip point will transition the system to S5 unconditionally.

15.3 Thermal Sensor Accuracy (Taccuracy)

The processor thermal sensor accuracy is:

- ± 5 °C over the temperature range from 50 °C to 110 °C.
- ± 7 °C over the temperature range from 30 °C to 50 °C.
- ± 10 °C over the temperature range from -10 °C to 30 °C.
- No accuracy is specified for temperature range beyond 110 °C or below -10 °C.

15.4 Thermal Reporting to EC

To support a platform EC that is managing the system thermals, the processor provides the ability for the EC to read the processor temperature over SMBus and/or over eSPI. If enabled, Power Management will drive the temperature directly to the SMBus and eSPI units. The EC will issue an SMBus read or eSPI OOB Channel request and receives a single byte of data, indicating a temperature between 0°C and 127°C, where 255 (0xFF) indicates that the sensor isn't enabled yet. The EC must be connected to either SMLink1 or eSPI for thermal reporting support.

15.5 Thermal Sensor Programming

Refer to the Platform BIOS specifications for recommendations and details.

16.0 Interrupt Timer Subsystem (ITSS)

Table 58. Acronyms

Acronym	Description
ITSS	Interrupt Timer Subsystem
HPET	High Precision Event Timer
8254 PIT	Legacy 8254 Programmable Interrupt Timer
INTR	Interrupt
NMI	Non-maskable Interrupt
INIT	Processor Initialization
SERR	System Error

Table 59. References

Specification	Document Number/Location
ACPI Specification	https://uefi.org/specifications

16.1 Feature Overview

ITSS supports following features:

- It houses the HPET, Legacy 8254 Timers, Legacy 8259 Interrupt Controllers, Legacy 8259 interrupt controller and APIC Interrupt Controllers.
- Fully synchronous-based design adopted for 8254 PIT.
- Functions as a simple Internal Host Space Error Collector and Reporting Block.
- 8254 PIT - Consists of 3 16-bit Timers capable of supporting up to 6 different modes.
- 8259 PIC - Consists of 1 Master and 1 Slave 8259 PIC, where Slave 8259 PIC is cascaded to Master 8259 PIC with support of up to 15 IRQs.
- APIC - Supports up to 120 IRQs.
- HPET - Contains 8 Timer Blocks and a single always running 64-bit counter. Each Timer is interrupt capable, with option to route to APIC or directly to hose using MSI. Improved resolution, reduced overhead in comparison to Legacy 8254, IOxAPIC & RTC Timers.

16.2 Functional Description

The ITSS (Interrupt Timer Sub System) have below sub blocks:

- **ITSS** : Consists of the HPET, 8254, 8259, Processor interfaces and APIC.

16.2.1 8254 Timers

There are three counters that have fixed uses. All registers and functions associated with these timers are in the Primary well. The 8254 unit is clocked by a 1.193 MHz periodic timer tick, which is functional only in S0 states. The 1.193 MHz periodic timer tick is generated off the XTAL clock.

Counter 0, System Timer

This counter functions as the system timer by controlling the state of IRQ0 and is typically programmed for Mode 3 operation. The counter produces a square wave with a period equal to the product of the counter period (838 ns) and the initial count value. The counter loads the initial count value 1 counter period after software writes the count value to the counter I/O address. The counter initially asserts IRQ0 and decrements the count value by two each counter period. The counter negates IRQ0 when the count value reaches 0. It then reloads the initial count value and again decrements the initial count value by two each counter period. The counter then asserts IRQ0 when the count value reaches 0, reloads the initial count value, and repeats the cycle, alternately asserting and negating IRQ0.

16.2.1.1 Timer Programming

The counter/timers are programmed in the following fashion:

1. Write a control word to select a counter.
2. Write an initial count for that counter.
3. Load the least and/or most significant bytes (as required by Control Word bits 5, 4) of the 16 bit counter.
4. Repeat with other counters.

Only two conventions need to be observed when programming the counters. First, for each counter, the control word must be written before the initial count is written. Second, the initial count must follow the count format specified in the control word (least significant Byte only, most significant Byte only, or least significant Byte, and then most significant Byte).

A new initial count may be written to a counter at any time without affecting the counter's programmed mode. Counting is affected as described in the mode definitions. The new count must follow the programmed count format.

If a counter is programmed to read/write 2-byte counts, the following precaution applies – a program must not transfer control between writing the first and second Byte to another routine, which also writes into that same counter. Otherwise, the counter will be loaded with an incorrect count.

The Control Word Register at port 43h controls the operation of counter. Several commands are available:

- **Control Word Command:** Specifies which counter to read or write, the operating mode, and the count format (binary or BCD).
- **Counter Latch Command:** Latches the current count so that it can be read by the system. The countdown process continues.
- **Read Back Command:** Reads the count value, programmed mode, the current state of the OUT pins, and the state of the Null Count Flag of the selected counter.

The table below lists the six operating modes for the interval counters:

Table 60. Counter Operating Modes

Mode	Function	Description
0	Out signal on end of count (=0)	Output is 0. When count goes to 0, output goes to 1 and stays at 1 until counter is reprogrammed.
1	Hardware retriggerable one-shot	Output is 0. When count goes to 0, output goes to 1 for one clock time.
2	Rate generator (divide by n counter)	Output is 1. Output goes to 0 for one clock time, then back to 1 and counter is reloaded.
3	Square wave output	Output is 1. Output goes to 0 when counter rolls over, and counter is reloaded. Output goes to 1 when counter rolls over, and counter is reloaded, and so on
4	Software triggered strobe	Output is 1. Output goes to 0 when count expires for one clock time.
5	Hardware triggered strobe	Output is 1. Output goes to 0 when count expires for one clock time.

16.2.1.2 Reading from Interval Timer

It is often desirable to read the value of a counter without disturbing the count in progress. There are three methods for reading the counters—a simple read operation, counter Latch command, and the Read-Back command. Each one is explained below:

With the simple read and counter latch command methods, the count must be read according to the programmed format; specifically, if the counter is programmed for 2-byte counts, 2-bytes must be read. The 2-bytes do not have to be read one right after the other. Read, write, or programming operations for other counters may be inserted between them.

Simple Read

The first method is to perform a simple read operation. The counter is selected through Port 40h (Counter 0).

NOTE

Performing a direct read from the counter does not return a determinate value, because the counting process is asynchronous to read operations.

Counter Latch Command

The Counter Latch command, written to Port 43h, latches the count of a specific counter at the time the command is received. This command is used to ensure that the count read from the counter is accurate, particularly when reading a 2-byte count. The count value is then read from each counter's Count register as was programmed by the Control register.

The count is held in the latch until it is read or the counter is reprogrammed. The count is then unlatched. This allows reading the contents of the counters on the fly without affecting counting in progress. Multiple Counter Latch Commands may be used to latch more than one counter. Counter Latch commands do not affect the programmed mode of the counter in any way.

If a Counter is latched and then, sometime later, latched again before the count is read, the second Counter Latch command is ignored. The count read is the count at the time the first Counter Latch command was issued.

Read Back Command

The Read Back command, written to Port 43h, latches the count value, programmed mode, and current states of the OUT pin and Null Count flag of the selected counter or counters. The value of the counter and its status may then be read by I/O access to the counter address.

The Read Back command may be used to latch multiple counter outputs at one time. This single command is functionally equivalent to several counter latch commands, one for each counter latched. Each counter's latched count is held until it is read or reprogrammed. Once read, a counter is unlatched. The other counters remain latched until they are read. If multiple count Read Back commands are issued to the same counter without reading the count, all but the first are ignored.

The Read Back command may additionally be used to latch status information of selected counters. The status of a counter is accessed by a read from that counter's I/O port address. If multiple counter status latch operations are performed without reading the status, all but the first are ignored.

Both the count and status of the selected counters may be latched simultaneously. This is functionally the same as issuing two consecutive, separate Read Back commands. If multiple count and/or status Read Back commands are issued to the same counters without any intervening reads, all but the first are ignored.

If both the count and status of a counter are latched, the first read operation from that counter returns the latched status, regardless of which was latched first. The next one or two reads, depending on whether the counter is programmed for one or two type counts, returns the latched count. Subsequent reads return unlatched count.

16.2.2 8259 Interrupt Controller

The interrupt controller is composed of two 8259 interrupt controllers. One functions as the 8259 Host, and the other serves as the 8259 Target.

Table 61. 8259 interrupt controllers Functions

8259 Input	Typical Interrupt Source	Connected Pin / Function
8259 Host		
0	Internal	Internal Timer / Counter 0 output or Multimedia Timer #0
1	Keyboard	IRQ1 via SERIRQ. Option for configurable sources including eSPI, GPIO, internal ACPI devices.
2	Internal	Target Controller INTR output
3	Serial Port A	IRQ3 from configurable sources including PIRQx, SERIRQ, eSPI, GPIO, internal ACPI devices.
4	Serial Port B	IRQ4 from configurable sources including PIRQx, SERIRQ, eSPI, GPIO, internal ACPI devices.
continued...		

8259 Input	Typical Interrupt Source	Connected Pin / Function
5	Parallel Port / Generic	IRQ5 from configurable sources including PIRQx, SERIRQ, eSPI, GPIO, internal ACPI devices.
6	Floppy Disk	IRQ6 from configurable sources including PIRQx, SERIRQ, eSPI, GPIO, internal ACPI devices.
7	Parallel Port / Generic	IRQ7 from configurable sources including PIRQx, SERIRQ, eSPI, GPIO, internal ACPI devices.
8259 Target		
0	Real Time Clock (ALL)	Inverted IRQ8# from internal RTC or Multimedia Timer #1
1	Generic	IRQ9 from configurable sources including PIRQx, SERIRQ, eSPI, GPIO, internal ACPI devices, SCI, TCO.
2	Generic	IRQ10 from configurable sources including PIRQx, SERIRQ, eSPI, GPIO, internal ACPI devices, SCI, TCO.
3	Generic	IRQ11 from configurable sources including PIRQx, SERIRQ, eSPI, GPIO, internal ACPI devices, SCI, TCO or HPET #2
4	PS/2 Mouse	IRQ12 from configurable sources including PIRQx, SERIRQ, eSPI, GPIO, internal ACPI devices, SCI, TCO or HPET #3.
5	Internal	IRQ13 from configurable sources including PIRQx, eSPI, GPIO, internal ACPI devices.
6	Internal	IRQ14 from configurable sources including PIRQx, SERIRQ, eSPI, GPIO, internal ACPI devices.
7	Internal	IRQ15 from configurable sources including PIRQx, SERIRQ, eSPI, GPIO, internal ACPI devices.

Interrupts can individually be programmed to be edge or level, except for IRQ0, IRQ1, IRQ2 and IRQ8 which are always default to edge.

Interrupt Handling

The controller interrupt sequence involves three bits, from the IRR, ISR, and IMR, for each interrupt level. These bits are used to determine the interrupt vector returned, and status of any other pending interrupts. The IRR and ISR registers are accessed on OCW3. The IMR register is accessed on OCW1.

Table 62. Interrupt Status Registers

Bit	Description
IRR	Interrupt Request Register. This bit is set on a low to high transition of the interrupt line in edge mode, and by an active high level in level mode. This bit is set whether or not the interrupt is masked. However, a masked interrupt will not generate INTR.
ISR	Interrupt Service Register. This bit is set, and the corresponding IRR bit cleared, when an interrupt acknowledge cycle is seen, and the vector returned is for that interrupt.
IMR	Interrupt Mask Register. Determines whether an interrupt is masked. Masked interrupts will not generate INTR.

Acknowledging Interrupts

The CPU issues an interrupt acknowledge cycle, which is translated into a special cycle. The interrupt controller converts this into two internal INTA# pulses for the 8259 cores. The first pulse freezes the interrupt state for priority resolution. On the second pulse, the 8259 host or 8259 target controller sends the interrupt vector to the processor.

Hardware/Software Interrupt Sequence

- An IRQ line is asserted, setting the corresponding IRR bit.
- If not masked, the PIC signals the CPU with an INTR.
- The CPU acknowledges with an interrupt acknowledge cycle, which is translated over DMI.
- The 8259 PIC receives this and generates two INTA# pulses for the master-slave pair.
- On the first INTA# pulse, the highest priority ISR bit is set and the IRR bit is cleared; the master PIC sends a slave ID code if needed.
- On the second INTA# pulse, the PIC returns the interrupt vector, or vector 7 if no request is present.
- The interrupt cycle completes. In AEOI mode, the ISR bit is cleared automatically; otherwise, it remains set until an EOI command is issued.

Initialization Command Words (ICW)

Each 8259 must be initialized with a four-byte sequence: ICW1, ICW2, ICW3, and ICW4. These commands are sent to fixed I/O addresses: 20h for the master controller and A0h for the slave controller.

ICW1- Writing to the master or slave controller base address with data bit 4 set to 1 is recognized as an ICW1 write, starting the initialization sequence. The interrupt controller then expects three more bytes at 21h (host) or A1h (target).

During initialization:

- IRQ inputs require a low-to-high transition to trigger interrupts.
- The Interrupt Mask Register is cleared.
- IRQ7 is set to priority 7.
- The slave mode address is set to 7.

- Special Mask Mode is cleared, and Status Read is set to IRR.

ICW2 - The second write, ICW2, sets bits [7:3] of the interrupt vector to be sent during an interrupt acknowledge. Each interrupt controller is assigned a different base value.

ICW3 - The third write, ICW3, serves different purposes for each controller:

- For the host controller, ICW3 specifies which IRQ line is used to connect the target; this is IRQ2 (so bit 2 is set to 1).
- For the target controller, ICW3 stores the target identification code. During an interrupt acknowledge, the target compares this code to the one broadcast by the host and, if it matches, sends the interrupt vector.

ICW4 - The final write, ICW4, must be programmed for both controllers. At minimum, bit 0 should be set to 1 to indicate operation in an Intel Architecture-based system.

Operation Command Words (OCW)

- OCW1: Masks or unmasks interrupt lines.
- OCW2: Manages interrupt priority rotation and controls the End of Interrupt (EOI) function.
- OCW3: Configures ISR/IRR reads, enables or disables SMM, and toggles polled interrupt mode.

Modes of Operation

- **Fully Nested Mode:** IRQs are prioritized from 0 (highest) to 7 (lowest). When an interrupt is acknowledged, the highest priority is serviced and its ISR bit is set. Lower or equal priority interrupts are blocked until the ISR bit is cleared by an EOI command or automatically in AEOI mode. Higher priority interrupts can still be serviced.
- **Special Fully Nested Mode:** Used in cascaded systems to maintain priority within each target. When a target interrupt is in service, higher priority interrupts from the same target can still be recognized. Software must check and clear the target's ISR after servicing.
- **Automatic Rotation Mode:** All devices have equal priority. After servicing, a device moves to the lowest priority. This ensures fair access among devices. Rotation is managed using OCW2 commands.
- **Specific Rotation Mode:** Software can set any IRQ as the lowest priority, making the next one the highest. This is done via OCW2. Priority changes are independent of EOI but can also be combined with EOI commands.
- **Poll Mode:** Allows multiple interrupts to share a single vector, reducing vector table usage. The CPU polls the PIC to determine which device needs service. INTR is not used; instead, software issues a Poll Command (P=1 in OCW3) and reads the highest priority pending interrupt.
- **Edge and Level Triggered Mode:** Determines how interrupts are detected:
 - Edge-triggered: Interrupts are recognized on a low-to-high transition.
 - Level-triggered: Interrupts are recognized when the line is held high. In PCH, edge/level is set per input using Edge/Level control Registers ELCR1/ELCR2. The IRQ must stay active until after the first INTA# pulse; otherwise, a default IRQ7 vector is returned.

End of Interrupt (EOI)

- **Normal EOI:** Software issues an EOI command before exiting the interrupt service routine to signal completion.
 - **Non-Specific EOI:** Clears the highest priority ISR bit set to 1.
 - **Specific EOI:** Clears a specific ISR bit, used when maintaining a fully nested structure. In Special Mask Mode, masked ISR bits are not cleared by Non-Specific EOI. EOI commands must be sent to both host and target controllers.
- **Automatic EOI (AEIOI):** The interrupt controller automatically clears the ISR bit at the end of the interrupt acknowledge cycle. This mode is only for the host controller and should not be used if nested interrupts are needed.

Masking Interrupts

- **Individual Masking:** Each interrupt can be masked individually using the IMR, programmed via OCW1. Each IMR bit masks a specific IRQ channel. Masking IRQ2 on the host also masks all target controller requests.
- **Special Mask Mode:** This mode allows selective enabling of interrupts during an interrupt service routine, even if lower priority interrupts would normally be blocked. It is enabled by setting OCW3.SSMM and OCW3.SMM, and cleared by resetting them. The Mask Register pattern determines which interrupts are enabled.

16.2.3 APIC Advanced Interrupt Controller

The APIC is accessed via an indirect addressing scheme. These registers are mapped into memory space. These are programmable through PCI Config IOAC register. Refer to the EDS Vol.2 for more details.

16.2.4 Processor Interface

The Virtual Wire (VLW) messages such as SMI#, NMI, INIT#, and INTR. or the external signal such as PLTRST# or THERMTRIP# are sent to the processor to indicate interrupt based on an event.

INIT# Functionality

The INIT# is asserted internally for 16 PCI clocks when triggered by specific events and is edge-triggered via VLW. When triggered, an INIT# assert message is sent.

For Secure Boot systems, INIT# must not be sent upstream to the Processor to avoid Processor-only resets, which could create security risks. CSE firmware is responsible for preventing INIT# from reaching the Processor.

For test/debug scenarios needing INIT# (e.g., to speed up test cycles), modified CSE firmware must be used to allow INIT# to reach the Processor.

NMI Functionality

NMIs can be triggered by several sources:

- SERR# activation from internal device errors.
- Do_SERR message from the CPU (via Legacy Host SMBus Controller)
- NMI_NOW bit assertion in the TCO1_CNT register (SMBus controller)

- GPIO configured as input and routed as NMI (enabled by GPI_NMI_EN)
- ME firmware setting ME_CSR[7] (ME_NMI_STS bit)

VLW Functionality

The VLW messages report the status of legacy sideband signals (INTR, SMI#, INIT#, NMI) to the Processor. These inbound messages are sent as Vendor Defined Messages (VDM) over the DMI link. The INIT# signal status can be selectively communicated via VLW, controlled by a BIOS register. When enabled, INIT# status is sent through VLW; when disabled, it is not. Legacy Processor signals (INTR, SMI#, INIT#, NMI) are only delivered via VLW VDM and not through the I/O (X) APIC controller. By default, INIT# VLW is enabled after reset, but releasing INIT# VLW messages to the host requires BIOS configuration, as mastering is disabled by default.

VLW Behavior

- VLW messages are sent upstream to the Processor Complex and can update the state of multiple signals in a single message.
- The VLW format supports both edge- and level-triggered signals:
 - **Level mode:** Uses two bits per signal—a Level bit (0 = inactive, 1 = active) and a Change bit (indicates which Level bits are valid in the message).
 - **Edge mode:** Uses a single bit to indicate if an active edge occurred (1 = active edge, 0 = no edge). The Change bit is reserved and not used.
- All signals use '1' for active and '0' for inactive. Active-low signals (like SMI and INIT) are inverted before being sent.
- All virtual signals start as inactive before the first VLW message.
- If a VLW message is triggered by a non-posted Processor cycle (e.g., synchronous SMI# from an IO write), the VLW must be delivered before the completion response to maintain correct ordering—unlike standard PCIe rules.
- Reserved bits in VLW messages must be set to 0 by the sender and ignored by the receiver.
- VLW messages can only be sent after the CPU_RESET_DONE_ACK message is sent on DMI; sending VLW before this is not allowed.
- The VLW logic is reset by PLTRST#, setting all virtual signals to inactive. No VLW message is sent for signals going inactive due to PLTRST#.
- The VLW logic is not reset by a virtual INIT# signal; the Processor must maintain VLW state across INIT#.
- For level-triggered signals, the state should not change until the corresponding VLW is delivered.

16.2.5 High Precision Event Timer (HPET)

This function provides a set of timers that can be used by the operating system. The timers are defined such that the operating system may assign specific timers to be used directly by specific applications. Each timer can be configured to cause a separate interrupt.

The processor provides eight timers. The timers are implemented as a single counter with a set of comparators. Each timer has its own comparator and value register. The counter increases monotonically. Each individual timer can generate an interrupt when the value in its value register matches the value in the main counter.

Timer 0 supports periodic interrupts.

The registers associated with these timers are mapped to a range in memory space (much like the I/O APIC). However, it is not implemented as a standard PCI function. The BIOS reports to the operating system the location of the register space using ACPI. The hardware can support an assignable decode space; however, BIOS sets this space prior to handing it over to the operating system. It is not expected that the operating system will move the location of these timers once it is set by BIOS.

Table 63. References

Specification	Location
IA-PC HPET (High Precision Event Timers) Specification, Revision 1.0a	https://www.intel.com/content/dam/www/public/us/en/documents/technical-specifications/software-developers-hpet-spec-1-0a.pdf

16.2.5.1 Timer Accuracy

The timers are accurate over any 1 ms period to within 0.05% of the time specified in the timer resolution fields.

Within any 100 us period, the timer reports a time that is up to two ticks too early or too late. Each tick is less than or equal to 100 ns; thus, this represents an error of less than 0.2%.

The timer is monotonic. It does not return the same value on two consecutive reads (unless the counter has rolled over and reached the same value).

The main counter uses the XTAL as its clock. The accuracy of the main counter is as accurate as the crystal that is used in the system. The XTAL clock frequency is determined by the pin strap that is sampled on RSMRST#.

16.2.5.2 Timer Off-load

The timer off-load feature allows the HPET timers to remain operational during very low power S0 operational modes when the XTAL clock is disabled. The clock source during this off-load is the Real Time Clock's 32.768 kHz clock. This clock is calibrated against the XTAL clock during boot time to an accuracy that ensures the error introduced by this off-load is less than 10 ppb (0.000001%).

When the XTAL clock is active, the 64 bit counter will increment by one each cycle of the XTAL clock when enabled. When the XTAL clock is disabled, the timer is maintained using the RTC clock. The long-term (> 1 ms) frequency drift allowed by the HPET specification is 500 ppm. The off-load mechanism ensures that it contributes < 1 ppm to this, which will allow this specification to be easily met given the clock crystal accuracies required for other reasons.

Timer off-load is prevented when there are HPET comparators active.

The HPET timer runs typically on the XTAL crystal clock and is off-loaded to the 32 kHz clock once the processor enters C10. This is the state where there are no C10 wake events pending and when the off-load calibrator is not running. HPET timer re-uses

this 28 bit calibration value calculated by PMC when counting on the 32 kHz clock. During C10 entry, PMC sends an indication to HPET to off-load and keeps the indication active as long as the processor is in C10 on the 32 kHz clock. The HPET counter will be off-loaded to the 32 kHz clock domain to allow the XTAL clock to shut down when it has no active comparators.

Theory of Operation

The Off-loadable Timer Block consists of a 64 bit fast clock counter and an 82 bit slow clock counter. During fast clock mode the counter increments by one on every rising edge of the fast clock. During slow clock mode, the 82 bit slow clock counter will increment by the value provided by the Off-load Calibrator.

The Off-loadable Timer will accept an input to tell it when to switch to the slow RTC clock mode and provide an indication of when it is using the slow clock mode. The switch will only take place on the slow clock rising edge, so for the 32 kHz RTC clock the maximum delay is around 30 us to switch to or from slow clock mode. Both of these flags will be in the fast clock domain.

When transitioning from fast clock to slow clock, the fast clock value will be loaded into the upper 64 bits of the 82 bit counter, with the 18 LSBs set to zero. The actual transition though happens in two stages to avoid metastability. There is a fast clock sampling of the slow clock through a double flop synchronizer. Following a request to transition to the slow clock, the edge of the slow clock is detected and this causes the fast clock value to park. At this point the fast clock can be gated. On the next rising edge of the slow clock, the parked fast clock value (in the upper 64 bits of an 82 bit value) is added to the value from the Off-load Calibrator. On subsequent edges while in slow clock mode the slow clock counter increments its count by the value from the Off-load Calibrator.

When transitioning from slow clock to fast clock, the fast clock waits until it samples a rising edge of the slow clock through its synchronizer and then loads the upper 64 bits of the slow clock value as the fast count value. It then de-asserts the indication that slow clock mode is active. The 32 kHz clock counter no longer counts. The 64 bit MSB will be over-written when the 32 kHz counter is reloaded once conditions are met to enable the 32 kHz HPET counter but the 18 bit LSB is retained and it is not cleared out during the next reload cycle to avoid losing the fractional part of the counter.

After initiating a transition from fast clock to slow clock and parking the fast counter value, the fast counter no longer tracks. This means if a transition back to fast clock is requested before the entry into off-load slow clock mode completes, the Off-loadable Timer must wait until the next slow clock edge to restart. This case effectively performs the fast clock to slow clock and back to fast clock on the same slow clock edge.

16.2.5.3 Periodic Versus Non-Periodic Modes

Non-Periodic Mode

This mode can be thought of as creating a one-shot timer.

When a timer is set up for non-periodic mode, it will generate an interrupt when the value in the main counter matches the value in the timer's comparator register. Another interrupt will be generated when the main counter matches the value in the timer's comparator register after a wrap around.

During run-time, the value in the timer's comparator value register will not be changed by the hardware. Software can of course change the value.

The Timer 0 Comparator Value register cannot be programmed reliably by a single 64 bit write in a 32 bit environment except if only the periodic rate is being changed during run-time. If the actual Timer 0 Comparator Value needs to be reinitialized, then the following software solution will always work regardless of the environment:

- Set `TIMER0_VAL_SET_CNF` bit
- Set the lower 32 bits of the Timer0 Comparator Value register
- Set `TIMER0_VAL_SET_CNF` bit
- Set the upper 32 bits of the Timer0 Comparator Value register

Timer 0 is configurable to 32 (default) or 64 bit mode, whereas Timers 1:7 only support 32 bit mode.

WARNING

Software must be careful when programming the comparator registers. If the value written to the register is not sufficiently far in the future, then the counter may pass the value before it reaches the register and the interrupt will be missed. The BIOS should pass a data structure to the operating system to indicate that the operating system should not attempt to program the periodic timer to a rate faster than 5 us.

All of the timers support non-periodic mode.

Refer to *IA-PC HPET Specification* for more details of this mode.

Periodic Mode

When a timer is set up for periodic mode, the software writes a value in the timer's comparator value register. When the main counter value matches the value in the timer's comparator value register, an interrupt can be generated. The hardware will then automatically increase the value in the comparator value register by the last value written to that register.

To make the periodic mode work properly, the main counter is typically written with a value of 0 so that the first interrupt occurs at the right point for the comparator. If the main counter is not set to 0, interrupts may not occur as expected.

During run-time, the value in the timer's comparator value register can be read by software to find out when the next periodic interrupt will be generated (not the rate at which it generates interrupts). Software is expected to remember the last value written to the comparator's value register (the rate at which interrupts are generated).

If software wants to change the periodic rate, it should write a new value to the comparator value register. At the point when the timer's comparator indicates a match, this new value will be added to derive the next matching point.

If the software resets the main counter, the value in the comparator's value register needs to reset as well. This can be done by setting the `TIMERn_VAL_SET_CNF` bit. Again, to avoid race conditions, this should be done with the main counter halted. The following usage model is expected:

1. Software clears the `ENABLE_CNF` bit to prevent any interrupts.
2. Software Clears the main counter by writing a value of 00h to it.

3. Software sets the TIMER0_VAL_SET_CNF bit.
4. Software writes the new value in the TIMER0_COMPARATOR_VAL register.

Software sets the ENABLE_CNF bit to enable interrupts.

NOTE

As the timer period approaches zero, the interrupts associated with the periodic timer may not get completely serviced before the next timer match occurs. Interrupts may get lost and/or system performance may be degraded in this case.

Each timer is NOT required to support the periodic mode of operation. A capabilities bit indicates if the particular timer supports periodic mode. The reason for this is that supporting the periodic mode adds a significant amount of gates.

Only timer 0 will support the periodic mode. This saves a substantial number of gates.

16.2.5.4 Enabling the Timers

The BIOS or operating system PnP code should route the interrupts. This includes the Legacy Rout bit, Interrupt Rout bit (for each timer), and interrupt type (to select the edge or level type for each timer).

The Device Driver code should do the following for an available timer:

1. Set the Overall Enable bit (Offset 10h, bit 0).
2. Set the timer type field (selects one-shot or periodic).
3. Set the interrupt enable.
4. Set the comparator value.

16.2.5.5 Interrupt Levels

Interrupts directed to the internal 8259s are active high. Refer to the **Advanced Programmable Interrupt Controller (APIC) (D31:F0)** for information regarding the polarity programming of the I/O APIC for detecting internal interrupts.

If the interrupts are mapped to the 8259 or I/O APIC and set for level-triggered mode, they can be shared with legacy interrupts. They may be shared although it is unlikely for the operating system to attempt to do this.

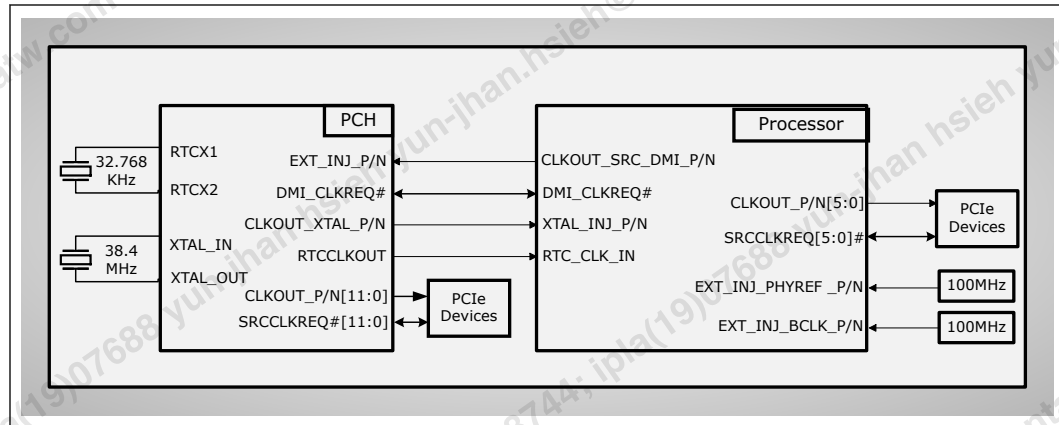
If more than one timer is configured to share the same IRQ (using the TIMERn_INT_ROUT_CNF fields), then the software must configure the timers to level-triggered mode. Edge-triggered interrupts cannot be shared.

For handling interrupts and issues related to 64 bit timers with 32 bit processors, refer to IA-PC HPET Specification.

17.0 System Clocks

17.1 Integrated Clock Controller (ICC)

Figure 24. ICC Diagram



17.2 Signal Description

Table 64. Signal Description

Signal Name	Type	SSC	Description
CLKOUT_P[5:0] CLKOUT_N[5:0]	O	Yes	PCIe Clock Outputs: Serial Reference 100 MHz PCIe specification compliant differential output clocks to PCIe devices. CLKOUT_P/N[5:0]= Can be used for PCIe 1.0, 2.0, 3.0, 4.0, and 5.0 support. Any un-used Clock differential pair output not being routed on a platform should be configured as "Disabled."
SRCCLKREQ[5:0]#	IO D		PCIe Clock Request: Serial Reference Clock request signals for PCIe 100 MHz differential clocks. The SRCCLKREQ[5:0]# signals can be configured to map to any of the PCIe Root Ports while using any of the CLKOUT_P/N[5:0] differential pairs.
CLKOUT_SRC_DMI_P CLKOUT_SRC_DMI_N	O	Yes	DMI Clock Output: Serial Reference 100 MHz differential clock from Processor to PCH for DMI interfacing.
DMI_CLKREQ#	IO D		DMI Clock Request : Serial Reference Clock request signal for DMI interfacing.
RTC_CLK_IN	I		32.768 Khz RTC input from PCH.
XTAL_INJ_P XTAL_INJ_N	I		38.4 MHz crystal inputs from PCH.
EXT_INJ_BCLK_P	I		External 100 MHz Differential bus clock inputs to the processor.

continued...



Signal Name	Type	SSC	Description
EXT_INJ_BCLK_N			
EXT_INJ_PHYREF_P EXT_INJ_PHYREF_N	I		External 100 MHz Differential bus clock input for PCIe and DMI Over Clocking.
RTCX1 RTCX2	I		32.768 KHz RTC Clock

18.0 Memory

18.1 System Memory Interface

18.1.1 Processor SKU Support Matrix

Table 65. DDR Support Matrix Table

Technology	DDR5	
Processor	S	S
Configuration	1SPC ¹	2SPC
Maximum Frequency [MT/s]	S-Processor: CUDIMM/CSODIMM 8000 ⁵ UDIMM/SODIMM 5600	S-Processor : CUDIMM 1DIMM 1R 6400 / 2R 5600 ⁷ CUDIMM 2DIMMs 1R 5600 / 2R 5200 ⁸ UDIMM 1DIMM 1R/2R 5600 ⁷ UDIMM 2DIMM 1R/2R 5600 ⁸
VDDQ [V] ⁴	5	5
Maximum RPC ²	2	4
Die Density [Gb]	16, 24, 32	16, 24, 32
Ballmap Mode	IL	IL

Notes: 1. 1SPC refers to one DIMM Slot per x64 Channel.
 2. RPC = Rank Per Channel
 3. VDD2 is Processor and DRAM voltage, and VDDQ is DRAM voltage.
 4. DIMM PMIC input voltage is 5V, DRAM input voltage is 1.1V.
 5. POR top speed refers to Processor top SKU. Other SKUs may use lower memory speed, refer to ark.intel.com for top memory speed.
 6. DDR5 ECC is supported only when all memory populated in system supports ECC
 7. Far memory slot to be populated, in case single DIMM is placed on a 2 slots Channel.
 8. Maximum 2DPC frequency supported when same DIMM part number populated within 2 channels. Frequency is not guaranteed when mix DIMM's populated.
 9. DDR5 CUDIMM/COSDIMM requires to define a main clock out of 2 clock deferential pairs for each 32-bit channel.
 10. Refer to the Nova Lake DDR5 CUDIMM/COSDIMM CLK Setting Technical Advisory (#TBD) that provides guidance how to set the new BIOS policies for DDR5 CUDIMM/COSDIMM CLK connectivity that may be different than RVP design.

Table 66. DDR Technology Support Matrix

Technology	Form Factor	Ball Count	Processor
DDR5	SoDIMM/CSODIMM	262	S
DDR5	UDIMM/CUDIMM	288	S

18.1.2 Supported Memory Modules and Devices

Table 67. Supported DDR5 Non-ECC SoDIMM/CSoDIMM Module Configurations

Raw Card Version	DIMM Capacity [GB]	DRAM Device Technology [Gb]	DRAM Organization	# of DRAM Devices	# of Ranks	# of Row/Col Address Bits	# of Banks Inside DRAM	Page Size [K]
A , F	16	16	2048M x 8	8	1	17/10	16	8
C , H	8	16	1024M x 16	4	1	17/10	8	8
B , G	32	16	2048M x 8	16	2	17/10	16	8
A , F	24	24	3072M x 8	8	1	17/10	32	8
C , H	12	24	1536M x 16	4	1	17/10	16	8
B , G	48	24	3072M x 8	16	2	17/10	32	8
A , F	32	32	4096M x 8	8	1	17/10	32	8
C , H	16	32	2048M x 16	4	1	17/10	16	8
B , G	64	32	4096M x 8	16	2	17/10	32	8

Table 68. Supported DDR5 ECC SoDIMM/CSoDIMM Module Configurations

Raw Card Version	DIMM Capacity [GB]	DRAM Device Technology [Gb]	DRAM Organization	# of DRAM Devices	# of Ranks	# of Row/Col Address Bits	# of Banks Inside DRAM	Page Size [K]
D , J	16	16	2048M x 8	10	1	17/10	16	8
E , K	32	16	2048M x 8	20	2	17/10	16	8
D , J	24	24	3072M x 8	10	1	17/10	32	8
E , K	48	24	3072M x 8	20	2	17/10	32	8
D , J	32	32	4096M x 8	10	1	17/10	32	8
E , K	64	32	4096M x 8	20	2	17/10	32	8

Table 69. Supported DDR5 Non-ECC UDIMM/CUDIMM Module Configurations

Raw Card Version	DIMM Capacity [GB]	DRAM Device Technology [Gb]	DRAM Organization	# of DRAM Devices	# of Ranks	# of Row/Col Address Bits	# of Banks Inside DRAM	Page Size [K]
A , F	16	16	2048M x 8	8	1	17/10	16	8
C , H	8	16	1024M x 16	4	1	17/10	8	8
B , G	32	16	2048M x 8	16	2	17/10	16	8
A , F	24	24	3072M x 8	8	1	17/10	32	8
C , H	12	24	1536M x 16	4	1	17/10	16	8
B , G	48	24	3072M x 8	16	2	17/10	32	8
A , F	32	32	4096M x 8	8	1	17/10	32	8
C , H	16	32	2048M x 16	4	1	17/10	16	8
B , G	64	32	4096M x 8	16	2	17/10	32	8

Table 70. Supported DDR5 ECC UDIMM/CUDIMM Module Configurations

Raw Card Version	DIMM Capacity [GB]	DRAM Device Technology [Gb]	DRAM Organization	# of DRAM Devices	# of Ranks	# of Row/Col Address Bits	# of Banks Inside DRAM	Page Size [K]
D, J	16	16	2048M x 8	10	1	17/10	16	8
E, K	32	16	2048M x 8	20	2	17/10	16	8
D, J	24	24	3072M x 8	10	1	17/10	32	8
E, K	48	24	3072M x 8	20	2	17/10	32	8
D, J	32	32	4096M x 8	10	1	17/10	32	8
E, K	64	32	4096M x 8	20	2	17/10	32	8

18.1.3 System Memory Timing Support

The IMC supports the following DDR Speed Bin, CAS Write Latency (CWL), and command signal mode timings on the main memory interface:

- tCL = CAS Latency
- tRCD = Activate Command to READ or WRITE Command delay
- tRP = PRECHARGE Command Period
- tRPb = per-bank PRECHARGE time
- tRPab = all-bank PRECHARGE time
- CWL = CAS Write Latency
- Command Signal modes:
 - 2N indicates a new DDR5 command may be issued every 2 clocks
 - 1N indicates a new DDR5 command may be issued every clock

Table 71. DDR5 System Memory Timing Support

DRAM Device	Transfer Rate (MT/s)	tCL (tCK)	tRCD (ns)	tRP (ns)	CWL (tCK)	CMD Mode
DDR5	4800	40	16.00	16.00	38	2N
DDR5	5600	46	16.00	16.00	44	2N
DDR5	6000	48	16.00	16.00	46	2N
DDR5	6400	52	16.00	16.00	50	2N
DDR5	7200		16.00	16.00		2N
DDR5	8000		16.00	16.00		2N

18.1.3.1 SAGV Points

SAGV (System Agent Geysserville) is a way by which the processor can dynamically scale the work point (V/F), by applying DVFS (Dynamic Voltage Frequency Scaling) based on memory bandwidth utilization and/or the latency requirement of the various workloads for better energy efficiency at System-Agent. Pcode heuristics are in charge of providing request for Qclock work points by periodically evaluating the utilization of the memory and IA stalls.

Table 72. SA Speed Enhanced Speed Steps (SA-GV) and Gear Mode Frequencies

Processor	Technology	Rank Config	DDR Maximum Rate [MT/s]	SAGV-LowBW	SAGV-MedBW	SAGV-HighBW	SAGV- High Performance
S	DDR5	TBD	8000	TBD	TBD	TBD	TBD
	DDR5	TBD	5600	TBD	TBD	TBD	TBD
Notes: 1. Nova Lake Processors supports dynamic gearing technology where the Memory Controller can run at 1:2 (Gear-2 mode) or 1:4 (Gear-4 mode) ratio of DRAM speed. The gear ratio is the ratio of DRAM speed to Memory Controller Clock . MC Channel Width equal to DDR Channel width multiply by Gear Ratio. 2. SA-GV modes: a. LowBW- Low frequency point, Minimum Power point. Characterized by low power, low BW, high latency. The system will stay at this point during low to moderate BW consumption. b. MedBW - Tuned for balance between power & performance. c. HighBW - Characterized by high power, low latency, moderate BW also used as RFI mitigation point. d. MaxBW/Lowest latency Lowest Latency point, peak BW and highest power.							

DDR Frequency Shifting

DDR interfaces emit electromagnetic radiation which can couple to the antennas of various radios that are integrated in the system, and cause radio frequency interference (RFI). The DDR Radio Frequency Interference Mitigation (DDR RFIM) feature is primarily aimed at resolving narrowband RFI from DDR5 technologies for the Wi-Fi* high and ultra-high bands (~5-7 GHz) . By changing the DDR data rate, the harmonics of the clock can be shifted out of a radio band of interest, thus mitigating RFI to that radio. This feature is working with SAGV on, the 3rd SAGV point is used as RFI mitigation point

18.1.4 Memory Controller

The integrated memory controller is responsible for transferring data between the processor and the DRAM as well as the DRAM maintenance. There are two instances of MC, one per memory slice. Each controller is capable of supporting up to two channels of DDR5.

The two controllers are independent and have no means of communicating with each other, they need to be configured separately.

In a symmetric memory population, each controller provides access to half of the total physical memory address space.

18.1.5 System Memory Controller Organization Mode

The IMC supports two memory organization modes, single-channel and dual-channel. Depending upon how the DDR Schema and DIMM Modules are populated in each memory channel, a number of different configurations can exist.

Single-Channel Mode

In this mode, all memory accesses are directed to a single Memory Controller. Single-Channel mode is used when either the MC0 or MC1 are populated in any order, but not both.

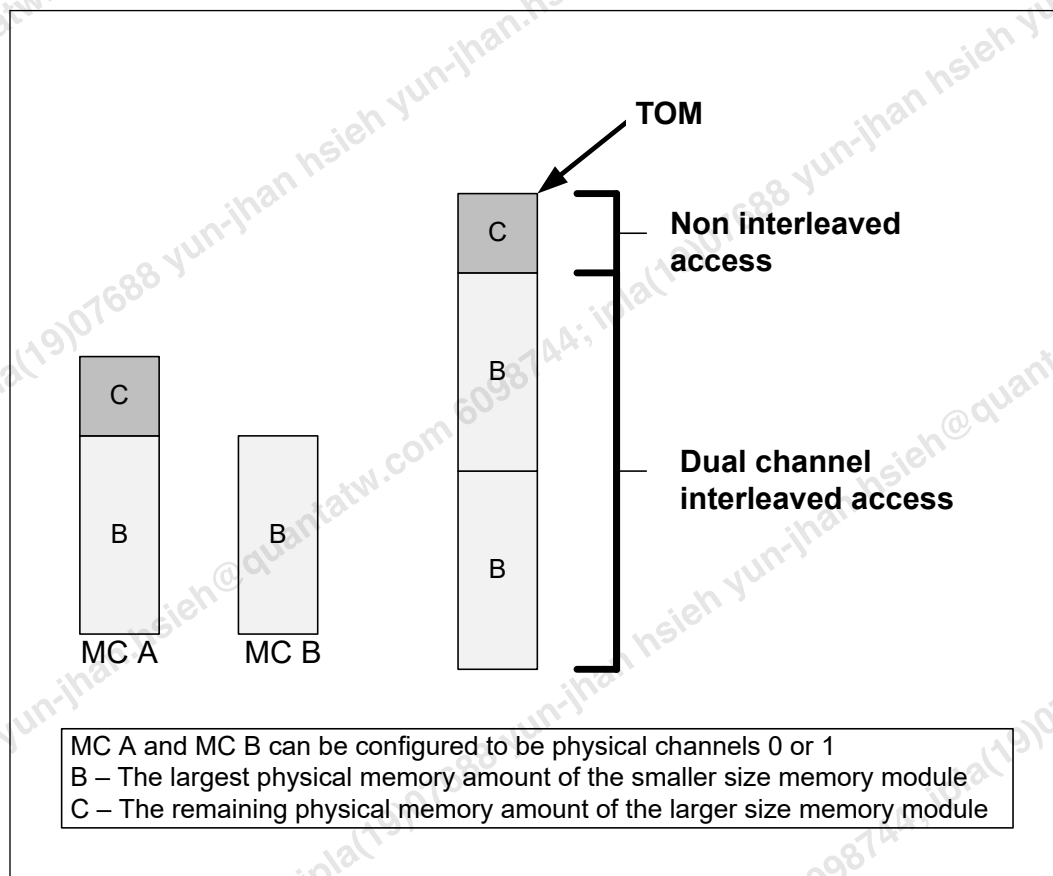
Dual-Channel Mode – Intel® Flex Memory Technology Mode

The IMC supports Intel Flex Memory Technology Mode. Memory is divided into a symmetric and asymmetric zone. The symmetric zone starts at the lowest address in each MC and is contiguous until the asymmetric zone begins or until the top address of the channel with the smaller capacity is reached. In this mode, the system runs with one zone of dual-channel mode and one zone of single-channel mode, simultaneously, across the whole memory array.

NOTE

MC A and MC B can be mapped for physical MC0 and MC1 respectively or vice versa; however, Channel A size should be greater or equal to Channel B size.

Figure 25. Intel® DDR5 Flex Memory Technology Operations



Dual-Channel Symmetric Mode (Interleaved Mode)

Dual-Channel Symmetric mode, also known as interleaved mode, provides maximum performance on real world applications. Addresses are ping-ponged between the channels. If there are two requests, and the second request is to an address on the opposite channel from the first, that request can be sent before data from the first request has returned. If two consecutive cache lines are requested, both may be retrieved simultaneously. Use Dual-Channel Symmetric mode when both MC0 and MC1 are populated in any order, with the total amount of memory in each channel being the same.

When both MCs are populated with the same memory capacity and the boundary between the dual channel zone and the single channel zone is the top of memory, IMC operates completely in Dual-Channel Symmetric mode.

NOTE

- The DDR5 DRAM device technology and width may vary from one channel to another.
-

18.1.6 System Memory Frequency

In all modes, the frequency of system memory is the lowest frequency and highest latency of all memory modules placed in the system, as determined through the SPD registers on the memory modules. The system Memory Controller supports a two DIMM connectors per channel. If DIMMs with different latency are populated across the MCs, the BIOS will use the slower of the two latencies for both MCs. For Dual-Channel modes, both MCs should have a DIMM connector populated. For Single-Channel mode, only a single MC is populated.

18.1.7 Technology Enhancements of Intel Fast Memory Access

The following sections describe the Just-in-Time Scheduling, Command Overlap, and Out-of-Order Scheduling Intel® FMA technology enhancements.

Just-in-Time Command Scheduling

The memory controller has an advanced command scheduler where all pending requests are examined simultaneously to determine the most efficient request to be issued next. The most efficient request is picked from all pending requests and issued to system memory Just-in-Time to make optimal use of Command Overlapping. Thus, instead of having all memory access requests go individually through an arbitration mechanism forcing requests to be executed one at a time, they can be started without interfering with the current request allowing for concurrent issuing of requests. This allows for optimized bandwidth and reduced latency while maintaining appropriate command spacing to meet system memory protocol.

Command Overlap

Command Overlap allows the insertion of the DRAM commands between the Activate, Pre-charge, and Read/Write commands normally used, as long as the inserted commands do not affect the currently executing command. Multiple commands can be issued in an overlapping manner, increasing the efficiency of system memory protocol.

Out-of-Order Scheduling

While leveraging the Just-in-Time Scheduling and Command Overlap enhancements, the IMC continuously monitors pending requests to system memory for the best use of bandwidth and reduction of latency. If there are multiple requests to the same open page, these requests would be launched in a back to back manner to make optimum use of the open memory page. This ability to reorder requests on the fly allows the IMC to further reduce latency and increase bandwidth efficiency.

18.1.8 Data Scrambling

The system memory controller incorporates a Data Scrambling feature to minimize the impact of excessive di/dt on the platform system memory VRs due to successive 1s and 0s on the data bus. Past experience has demonstrated that traffic on the data bus is not random and can have energy concentrated at specific spectral harmonics creating high di/dt which is generally limited by data patterns that excite resonance between the package inductance and on die capacitances. As a result, the system memory controller uses a data scrambling feature to create pseudo-random patterns on the system memory data bus to reduce the impact of any excessive di/dt.

18.1.9 ECC H-Matrix Syndrome Codes

Syndrome Value	Flipped Bit	Syndrome Value	Flipped Bit	Syndrome Value	Flipped Bit	Syndrome Value	Flipped Bit
0				No Error			
1	64	37	26	81	2	146	53
2	65	38	46	82	18	148	4
4	66	41	61	84	34	152	20
7	60	42	9	88	50	161	49
8	67	44	16	97	21	162	1
11	36	47	23	98	38	164	17
13	27	49	63	100	54	168	33
14	3	50	47	104	5	176	44
16	68	52	14	112	52	193	8
19	55	56	30	128	71	194	24
21	10	64	70	131	22	196	40
22	29	67	6	133	58	200	56
25	45	69	42	134	13	208	19
26	57	70	62	137	28	224	11
28	0	73	12	138	41	241	7
31	15	74	25	140	48	242	31
32	69	76	32	143	43	244	59
35	39	79	51	145	37	248	35
Notes: 1. All other syndrome values indicate unrecoverable error (more than one error). 2. This table is relevant only for S-Processor ECC supported SKUs.							

18.1.10 Data Swapping

By default, the processor supports on-board data swapping in two manners (for all segments and DRAM technologies):

- Bit swapping is allowed within each Byte for all DDR technologies.
- DDR5 x32 sub-channels can be swizzle within their x64 MC.
- DDR5: Byte swapping is allowed within each x32 Channel.

- ECC bits swap is allowed within ECC byte/nibble: DDR5 ECC [4][3..0].

NOTE

All DRAM devices sharing ZQ resistor must be connected to the same MC channel.

18.1.11 DDR IO Interleaving

The processor supports I/O interleaving, which has the ability to swap DDR bytes for routing considerations. BIOS configures the I/O interleaving mode before DDR.

18.1.12 DRAM Clock Generation

Each support rank has a differential clock pair for DDR5.

18.1.13 DRAM Reference Voltage Generation

Read Vref is generated by the memory controller in all technologies. Write Vref is generated by the DRAM in all technologies. In all cases, it has small step sizes and is trained by MRC.

18.1.14 Data Swizzling

All Processor Series have no die-to-package DDR swizzling.

18.1.15 Post Package Repair

PPR is supported according to JEDEC Spec.

BIOS can identify a single Row failure per Bank in DRAM and perform Post Package Repair (PPR) to exchange failing Row with spare Row.

PPR can be supported only with DRAM that supports PPR according to Jedec spec.

18.2 Integrated Memory Controller Power Management

The main memory is power managed during normal operation and in low-power ACPI C-states.

18.2.1 Disabling Unused System Memory Outputs

Any system memory (SM) interface signal that goes to a memory in which it is not connected to any actual memory devices (such as SODIMM connector is unpopulated, or is single-sided) is tri-stated. The benefits of disabling unused SM signals are:

- Reduced power consumption.
- Reduced possible overshoot/undershoot signal quality issues seen by the processor I/O buffer receivers caused by reflections from potentially unterminated transmission lines.

When a given rank is not populated, the corresponding control signals (CLK_P/CLK_N/CS) are not driven.

At reset, all ranks should be assumed to be populated, until it can be proven that they are not populated. This is due to the fact that when CS is tri-stated with a DRAMs present, the DRAMs are not ensured to maintain data integrity. CS tri-state should be enabled by BIOS where appropriate, since at reset all ranks should be assumed to be populated.

18.2.2 DRAM Power Management and Initialization

The processor implements extensive support for power management on the memory interface.

The DRAM Powerdown is one of the power-saving means. When DRAM is in Powerdown state, the internal DDR clock is disabled and the DDR power is reduced. The power-saving differs according to the selected mode and the DDR type used. For more information, refer to the IDD table in the DDR specification.

The processor supports three different types of power-down modes in package C0 state. The different power-down modes can be enabled through configuring PM PDWN config register.

The different power-down modes supported are:

- **No power-down**
- **Pre-charged Power-down (PPD):** This mode is entered if all banks in DDR are pre-charged when entering Powerdown state. Power-saving in this mode is intermediate. Power consumption is defined by IDD2P. Exiting this mode is defined by tXP. In this mode when waking-up, all page-buffers are empty.

The Powerdown state is determined per rank, whenever it is inactive. Each rank has an idle counter. The idle-counter starts counting as soon as the rank has no accesses, and if it expires, the rank may enter power-down while no new transactions to the rank arrive to queues. It is important to understand that since the power-down decision is per rank, the IMC can find many opportunities to power down ranks, even while running memory intensive applications; the savings are significant (may be few Watts, according to DDR specification). This is significant when each channel is populated with more ranks.

Selection of power modes should be according to power-performance or a thermal trade-off of a given system:

- When trying to achieve maximum performance and power or thermal consideration is not an issue: use no power-down
- In a system which tries to minimize power-consumption, try using the deepest power-down mode possible
- In high-performance systems with dense packaging (that is, tricky thermal design) the power-down mode should be considered in order to reduce the heating and avoid DDR throttling caused by the heating.

The idle timer expiration count defines the # of DCLKs that a rank is idle that causes entry to the selected power mode. As this timer is set to a shorter time the IMC will have more opportunities to put the DDR in power-down. There is no BIOS hook to set this register. Customers choosing to change the value of this register can do it by changing it in the BIOS. For experiments, this register can be modified in real time if BIOS does not lock the IMC registers.

18.2.2.1 Conditional Self-Refresh

During S0 idle state, system memory may be conditionally placed into self-refresh state when the processor is in package C3 or deeper power state. Refer to [Intel® Rapid Memory Power Management \(Intel® RMPM\)](#) on page 70 for more details on conditional self-refresh with Intel® HD Graphics enabled.

When entering the S3 – Suspend-to-RAM (STR) state or S0 conditional self-refresh, the processor IA core flushes pending cycles and then enters SDRAM ranks that are not used by the processor graphics into self-refresh. The CKE signals remain LOW so the SDRAM devices perform self-refresh.

The target behavior is to enter self-refresh for package C3 or deeper power states as long as there are no memory requests to service.

18.2.2.2 Dynamic Power-Down

Dynamic power-down of memory is employed during normal operation. Based on idle conditions, a given memory rank may be powered down. The IMC implements aggressive CKE control to dynamically put the DRAM devices in a power-down state.

The processor IA core controller can be configured to put the devices in active power down or pre-charge power-down. Pre-charge power-down provides greater power savings but has a bigger performance impact, since all pages will first be closed before putting the devices in power-down mode.

If dynamic power-down is enabled, all ranks are powered up before doing a refresh cycle and all ranks are powered down at the end of the refresh.

18.2.2.3 DRAM IO Power Management

Unused signals should be disabled to save power and reduce electromagnetic interference. Clocks and CS signals are controlled per DIMM rank and will be powered down for unused ranks.

The I/O buffer for an unused signal should be tri-stated (output driver disabled), the input receiver (differential sense-amp) should be disabled. The input path should be gated to prevent spurious results due to noise on the unused signals (typically handled automatically when input receiver is disabled).

18.2.3 DDR Electrical Power Gating

The DDR I/O of the processor supports Electrical Power Gating (DDR-EPG) while the processor is at C3 or deeper power state.

In C3 or deeper power state, the processor internally gates VDDQ and VDD2 for the majority of the logic to reduce idle power while keeping all critical DDR pins such as CKE in the appropriate state.

In C8 or deeper power state, the processor internally gates VCCSA for all non-critical state to reduce idle power.

In S3 or C-state transitions, the DDR does not go through training mode and will restore the previous training information.

18.2.4 Power Training

BIOS MRC performing Power Training steps to reduce DDR I/O power while keeping reasonable operational margins still guaranteeing platform operation. The algorithms attempt to weaken ODT, driver strength and the related buffers parameters both on the MC and the DRAM side and find the best possible trade-off between the total I/O power and the operating margins using advanced mathematical models.

18.3 Signal Description

Table 73. DDR5 Memory Interface

Signal Name	Description	Dir.	Buffer Type	Link Type
DDR0_DQ[3:0][7:0] DDR0_DQ[4][3:0] DDR1_DQ[3:0][7:0] DDR1_DQ[4][3:0] DDR2_DQ[3:0][7:0] DDR2_DQ[4][3:0] DDR3_DQ[3:0][7:0] DDR3_DQ[4][3:0]	Data Buses: Data signals interface to the SDRAM data buses. Example: DDR0_DQ[2][5] refers to DDR channel 0, Byte 2, Bit 5.	I/O	DDR5	SE
DDR0_DQS_P[4:0] DDR0_DQS_N[4:0] DDR1_DQS_P[4:0] DDR1_DQS_N[4:0] DDR2_DQS_P[4:0] DDR2_DQS_N[4:0] DDR3_DQS_P[4:0] DDR3_DQS_N[4:0]	Data Strobes: Differential data strobe pairs. The data is captured at the crossing point of DQS during reading and write transactions. Example: DDR0_DQSP0 refers to DQSP of DDR channel 0, Byte 0.	I/O	DDR5	Diff
DDR0_CLK[3:0]_P DDR0_CLK[3:0]_N DDR1_CLK[3:0]_P DDR1_CLK[3:0]_N DDR2_CLK[3:0]_P DDR2_CLK[3:0]_N DDR3_CLK[3:0]_P DDR3_CLK[3:0]_N	SDRAM Differential Clock: Differential clocks signal pairs, pair per rank. The crossing of the positive edge and the negative edge of their complement are used to sample the command and control signals on the SDRAM.	O	DDR5	Diff
DDR0_CS[3:0] DDR1_CS[3:0] DDR2_CS[3:0] DDR3_CS[3:0]	Chip Select: (1 per rank). These signals are used to select particular SDRAM components during the active state. There is one Chip Select for each SDRAM rank. The Chip select signal is Active Low.	O	DDR5	SE
DDR0_CA[12:0] DDR1_CA[12:0] DDR2_CA[12:0] DDR3_CA[12:0]	Command Address: These signals are used to provide the multiplexed command and address to the SDRAM.	O	DDR5	SE
DRAM_RESET#	Memory Reset: Refer to the appropriate platform design guide for implementation details.	O	CMOS	SE
DDR0_ALERT DDR1_ALERT	Alert: This signal is used at command training only. It is getting the Command and Address Parity error flag during training. CRC feature is not supported.	I	DDR5	SE

19.0 USB-C Sub System

USB Type-C* is a cable and connector specification defined by USB-IF.

The USB Type-C sub-system supports USB 3.2, USB4v2, DPoC (DisplayPort over Type-C) protocols. The USB Type-C sub-system can also be configured as native HDMI 2.1/DP2.1 interfaces, for more information refer to [Display](#) chapter.

Thunderbolt™ 5 is a USB Type-C solution brand which requires the following elements:

- USB 2.0, USB 3.2 (2x10 Gb/s), USB 3.2/DP implemented at the connector.
- In addition, it requires USB4v2 implemented up to 80 Gbps, including Thunderbolt 3 compatibility as defined by USB4v2/USB-PD specs and 15 W of bus power.
- Thunderbolt™ 5 solutions use (and prioritize) the USB4v2 PD entry mode (while still supporting Thunderbolt™ 3 alt mode).
- Refer to the Thunderbolt™ 5 Brand Requirement Details (#764566) and other related documentation for details.

This product is compliant with TBT5 brand requirements.

Mainstream SKU supports only single TBT4 port.

19.1 USB-C Sub-System General Capabilities

- xHCI (USB 3.2 host controller) implemented in the processor.
- Intel® AMT/vPro over Thunderbolt™ docking.
- Support RTD3 on TCP port when link is idle.
- Wake capable on each host port from S0i3, Sx.

Table 74. USB Type-C* Port Configuration

Port	Supported Protocols	S PCH SKUs
TCP 0	USB4v2 , DisplayPort , USB 3.2 , HDMI	Q970, Z990,W980, B960
TCP 1		Q970, Z990,W980

Notes:

1. TCP1 is available as fixed HDMI/DP only with PCH SKUs B960
2. Supported on Type-C or Native connector (Fixed DP up to UHBR20 link rate)
3. Supported only on Native connector.
4. USB 3.2 supported link rates:
 - a. USB 3.2 Gen 1x1 (5 Gbps)
 - b. USB 3.2 Gen 2x1 (10 Gbps)
 - c. USB 3.2 Gen 2x2 (20 Gbps)
5. USB4v2 operating link rates (including both rounded and non-rounded modes for Thunderbolt™ 4 compatibility):
 - a. USB4 Gen 3x2 (40 Gbps)
 - b. USB4 Gen 4x2 (80 Gbps)
 - c. B960 support TBT4 only
 - d. 10.3125 Gbps, 20.625 Gbps per lane - Compatible to Thunderbolt™ 3 non-rounded modes.
6. USB 2.0 interface supported over Type-C connector.
7. The two ports share USB4v2 router which supports up to two display interfaces
8. Display interface can be connected directly to a DP/HDMI/Type-C port or through USB4v2 router on a Type-C connector.
9. If two ports are configured, one as USB4v2 and the other as DP/HDMI fixed connection, each port will support single display interface when monitors are attached.

Table 75. USB Type-C* Lanes Configuration

Lane1	Lane2	Comments
USB4v2 / TBT5	USB4v2 / TBT5	Both lanes operate at same speed, one of (40 Gbps/20.6 Gbps/10.3 Gbps/20 Gbps/10 Gbps)
USB4v2 / TBT5	No connect	20.6 Gbps/10.3 Gbps/20Gbps/10 Gbps
No connect	USB4v2 / TBT5	
USB 3.2	USB 3.2	Multi-Lane USB 3.2 (Host Only), 2x10Gbps = 20Gbps
USB 3.2	No connect	Any combination of: USB 3.2 Gen 1x1 (5Gbps) USB 3.2 Gen 2x1 (10Gbps)
No connect	USB 3.2	
USB 3.2	DPx2	Any of HBR3/HBR2/HBR1/HRBR for DP1.4a, DP2.1 (2x10/20 Gbps) , and USB 3.2 (10 Gbps)
DPx2	USB 3.2	
DPx4	Both lanes at same DP rate - no support for 2x DPx2 USB Type-C connector	Any of HBR3/HBR2/HBR1/HRBR for DP1.4a, DP2.1 (4x10/20 Gbps)

19.2 USB4v2 Router

USB4v2 architecture is a transformational high-speed, dual protocol I/O, and it provides flexibility and simplicity by encapsulating both data (PCIe* & USB 3.2) and video (DisplayPort*) on a single cable connection that can daisy-chain up to five devices. USB4v2/Thunderbolt™ controllers act as a point of entry or a point of exit in the USB4v2 domain. The USB4v2 domain is built as a daisy chain of USB4v2/

Thunderbolt™ enabled products for the encapsulated protocols - PCIe, USB 3.2 and DisplayPort. These protocols are encapsulated into the USB4v2 fabric and can be tunneled across the domain.

USB4v2 controllers can be implemented in various systems such as PCs, laptops and tablets, or devices such as storage, docks, displays, home entertainment, cameras, computer peripherals, high end video editing systems, and any other PCIe based device that can be used to extend system capabilities outside of the system's box.

The integrated connection maximum data rate is 40 Gbps per lane but supports also 20.625 Gbps, 20.0 Gbps, 10.3125 Gbps, and 10.0 Gbps and is compatible with older Thunderbolt™ device speeds.

19.2.1 USB 4 Host Router Implementation Capabilities

The integrated USB Type-C sub-system implements the following interfaces via USB4v2:

- Up to two DisplayPort* sink interfaces each one capable of:
 - DisplayPort 2.1 specification for tunneling
 - 1.62 Gbps or 2.7 Gbps or 5.4 Gbps or 8.1 Gbps or UHBR10 or UHBR20 link rates
 - x1, x2 or x4 lane operation
 - Support for DSC compression
- Up to two PCI Express* Root Port interfaces each one capable of:
 - PCI Express* 4.0 x4 compliant @ 16.0 GT/s
- Up to two xHCI Port interfaces each one capable of:
 - USB 3.2 Gen 2x1 (10 Gbps)
 - USB 3.2 Gen 2x2 (20 Gbps)
- USB4v2 Host Interface:
 - PCI Express* 4.0 x4 compliant endpoint
 - Supports simultaneous transmit and receive on 12 paths
 - Raw mode and frame mode operation configurable on a per-path basis
 - MSI and MSI-X support
 - Interrupt moderation support
- USB4v2 Time Management Unit (TMU):
- Up to two Interfaces to USB Type-C* connectors, each one supports:
 - USB4v2 PD entry mode, as well as TBT 3 compatibility mode, each supporting:
 - 20 paths per port
 - Each port support 40/20.625/20.0 Gbps or 10.3125/10.0 Gbps link rates per lane.
 - 16 counters per port

19.3 USB-C Sub-system xHCI Controllers

The processor supports xHCI controllers. The native USB 3.2 path proceeds from the memory directly to PHY.

19.3.1 USB 3 Controllers

19.3.1.1 Extensible Host Controller Interface

Extensible Host Controller Interface (xHCI) is an interface specification that defines Host Controller for a universal Serial Bus (USB 3.2), which is capable of interfacing with USB 1.x, 2.0, and 3.x compatible devices.

In case that a device (example, USB 3.2 Flash Drive) was connected to the computer, the computer will work as Host and the xHCI will be activated inside the processor.

The xHCI controller support link rate of up to USB 3.2 Gen 2x2 (20G).

19.4 USB-C Sub-System Display Interface

Refer to [Display](#) on page 184.

19.5 USB Type-C Signals

Signal Name	Description	Dir.	Link Type	Availability
TCP[1:0]_TX[1:0]_P TCP[1:0]_TX[1:0]_N	TX Data Lane.	O	Diff	S-Series Processor
TCP[1:0]_TXRX[1:0]_P TCP[1:0]_TXRX[1:0]_N	RX Data Lane, also serves as the secondary TX data lane.	I/O	Diff	S-Series Processor
TCP[1:0]_AUX_P TCP[1:0]_AUX_N	Common Lane AUX-PAD.	I/O	Diff	S-Series Processor

19.6 LSx

LSx interface supports two ports. Each port of the LSx controller has two bi-directional signals configured either as Tx (Output) or Rx (Input). Operating voltage of the LSx interface is 1.8 V. LSx controller is responsible for link initialization/management of HSIO in the Thunderbolt subsystem.

19.6.1 LSx_Signals

Signal Name	Type	Description
GPP_SB13/DDP0_CTRLCLK/ TBT_LSx0_RXD	I	LSx 0 Receive Data
GPP_SB12/DDP0_CTRLCLK/ TBT_LSx0_TXD	O	LSx 0 Transmit Data
GPP_SB16/DDP1_CTRLCLK/ TBT_LSx1_RXD	I	LSx 1 Receive Data
GPP_SB15/DDP1_CTRLCLK/ TBT_LSx1_TXD	O	LSx 1 Transmit Data

19.6.2 Integrated Pull-Ups and Pull-Downs

None.

19.6.3 IO Signal Planes and States

Signal Name	Power Plane	During Reset ¹	Immediately after Reset ¹	S3/S4/S5
TBT_LSx[0:1]_RXD	Primary	Undriven	Undriven	Undriven
TBT_LSx[0:1]_TXD	Primary	Undriven	Undriven	Undriven

Note: 1. Reset reference for primary well pins is RSMRST#.

19.7 AUS BIAS/Orientation/Isolation Control

In USB/DP retimer-less configuration, the AUX BIAS/Orientation/Isolation Control is supported by 3rd party PD Controllers. The platform relies on the SBU XBAR mux capability inside the PD controllers. The PD controllers are responsible for switching the mux based on orientation.

20.0 Intel® Volume Management Device (Intel® VMD) Technology

Objective

Standard Operating Systems generally recognize individual PCIe Devices and load individual drivers. This is undesirable in some cases such as, for example, when there are several PCIe-based hard-drives connected to a platform where the user wishes to configure them as part of a RAID array. The Operating System current treats individual hard-drives as separate volumes and not part of a single volume.

In other words, the Operating System requires multiple PCIe devices to have multiple driver instances, making volume management across multiple host bus adapters (HBAs) and driver instances difficult.

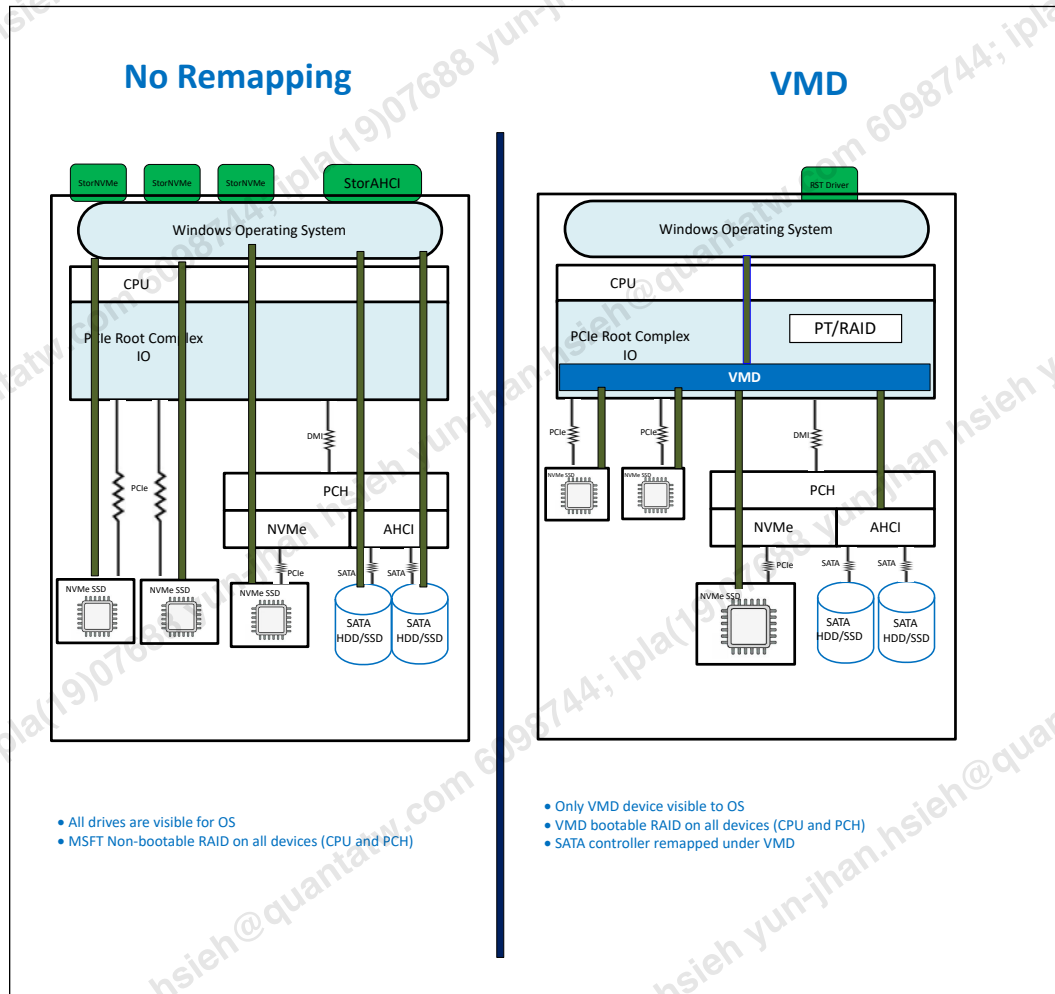
Intel® Volume Management Device (Intel® VMD) technology provides a means to provide volume management across separate PCI Express HBAs and SSDs without requiring operating system support or communication between drivers. For example, the OS will see a single RAID volume instead of multiple storage volumes, when Volume Management Device is used.

Technology Description

Intel® Volume Management Device technology does this by obscuring each storage controller from the OS, while allowing a single driver to be loaded that would control each storage controller.

Intel® Volume Management technology requires support in BIOS and driver, memory and configuration space management.

Figure 26. Technology Description



A Volume Management Device (VMD) exposes a single device to the operating system, which will load a single storage driver. The VMD resides in the processor's PCIe root complex and it appears to the OS as a root bus integrated endpoint. In the processor, the VMD is in a central location to manipulate access to storage devices which may be attached directly to the processor or indirectly. Instead of allowing individual storage devices to be detected by the OS and therefore causing the OS to load a separate driver instance for each, VMD provides configuration settings to allow specific devices and root ports on the root bus to be invisible to the OS.

Access to these hidden target devices is provided by the VMD to the single, unified driver.

Features Supported

Supports MMIO mapped Configuration Space (CFGBAR):

- Supports MMIO Low
- Supports MMIO High
- Supports Register Lock or Restricted Access

- Supports Device Assign
- Function Assign
- MSI Remapping Disable

21.0 PCI Express

Table 76. Acronym

Acronyms	Description
PCIe*	PCI Express* (Peripheral Component Interconnect Express*)

Table 77. Reference Table

Specification	Location
PCI Express® Base Specification Revision 5.0, Version 1.0 (May 28, 2019)	https://pcisig.com/
PCI Express® M.2 Specification Revision 5.1 (May 20, 2024)	https://pcisig.com/
PCI Express® Card Electromechanical Specification, Revision 5.1, Version 1.0 (August 17, 2023)	https://pcisig.com/

21.1 Functional Description

Table 78. Features Supported

PCIe Controller Feature	PCIe Controllers	
	A	B
L1 Sub-States (L1.0, L1.1, L1.2)	Yes	Yes
L0s Link State (RX/TX)	Yes	Yes
S4/S5 Sleep States (Sx)	Yes	Yes
S3 Sleep States (Sx)	No	No
Common Clock Mode	Yes	Yes
Separate Reference Clock with Independent SSC (SRIS)	No	No
Separate Reference Clock with No SSC (SRNS)	No	No
Precision Time Management (PTM)	Yes	Yes
Advanced Error Reporting (AER)	Yes	Yes
Lane Reversal	Yes	Yes
Latency Tolerance Reporting (LTR)	Yes	Yes
LTR Programmable by OS	No	No
PCIe TX Half Swing	No	No
PCIe TX Full Swing	Yes	Yes
Run Time D3 (RTD3)	Yes	Yes
RTD3 through PFET_EN to remove power from PCIe Device	Yes	Yes
Access Control Services (ACS)	Yes	Yes
continued...		

PCIe Controller Feature	PCIe Controllers	
	A	B
Alternative Routing-ID Interpretation (ARI)	Yes	Yes
Port 80h I/O Cycle Decode Forwarding	No	No
Lane Polarity Inversion	Yes	Yes
PCIe Controller Root Port Hot-Plug Connector Hot-Plug via CLKREQ#	Yes	Yes
Downstream Port Containment (DPC)	Yes	Yes
Enhanced Downstream Port Containment (eDPC)	Yes	Yes
Virtual Channel (VC)	VC0/1	VC0/1
NVMe Cycle Router	No	No
Volume Management Device (Intel® VMD)	Yes	Yes
RAID[0] and RAID[1] Mode Support ^{1,2,3}	Yes	Yes
RAID[5] Mode Support ^{1,2,3}	No	No
RAID[10] Mode Support ^{1,2,3}	No	No
PCIe Controller (PC) Root Port (RP) Peer-2-Peer (P2P) Mem Write Transactions	RPs within PCA or within PCB = Yes RPs between PCA and PCB = Yes	
PCIe Controller (PC) Root Port (RP) Peer-2-Peer (P2P) Mem Read Transactions	No	
PCIe Controller (PC) Root Port (RP) Peer-2-Peer (P2P) MCTP VDM Transactions	RPs within PCA or within PCB = Yes RPs between PCA and PCB = Yes	
Flattening Portal Bridge (FPB)	No	No
PCIe Root Port Initiated Dynamic Width Change	No	No
PCIe Root Port Initiated Dynamic Speed Change	Yes	Yes
End Point Device Initiated Dynamic Width Change	Yes	Yes
End Point Device Initiated Dynamic Speed Change	Yes	Yes
Max Payload Size (MPS)	512B	512B

NOTES

- 1. No restrictions on PCIe Controller. PCIe RAID is expected to work across all Root Ports within a PCIe Controller and between Root Ports from different PCIe Controllers (Processor and PCH).
- 2. No RAID support between PCIe and SATA storage devices.
- 3. The Intel® Rapid Storage Technology (RST) does not restrict RAID modes so if any unsupported RAID mode is enabled it is up to the motherboard designer to validate any non-supported RAID mode.

21.2 PCIe Signals

Signal Name	Type	Description
PCIE_A[7:0]_TX_N PCIE_A[7:0]_TX_P	O	PCI Express* Controller A Differential Transmit Pairs These are the PCI Express* based outbound high-speed differential signals.
PCIE_A[7:0]_RX_N PCIE_A[7:0]_RX_P	I	PCI Express* Controller A Differential Receive Pairs These are the PCI Express* based inbound high-speed differential signals.
PCIE_B[15:0]_TX_N PCIE_B[15:0]_TX_P	O	PCI Express* Controller B Differential Transmit Pairs These are the PCI Express* based outbound high-speed differential signals.
PCIE_B[15:0]_RX_N PCIE_B[15:0]_RX_P	I	PCI Express* Controller B Differential Receive Pairs These are the PCI Express* based inbound high-speed differential signals.
GPP_SD17/PCIE_LINK_DOWN/ SRCCLKREQ3#	O	PCI Express* Link Down Debug Signal PCIe link failure debug signal. PCIe Root Port(s) will assert this signal when a link down event occurs and is detected. For example when a link fails to train during an L1 sub-state exit event.

21.3 IO Signal Planes and States

Table 79. Power Plane and States for PCI Express* Signals

Signal Name	Type	Power Plane	During Reset ²	Immediately After Reset ²	S4/S5	Deep Sx
PCIE_A[7:0]_TX_N PCIE_A[7:0]_TX_P	O	Primary	Internal Pull-down	Internal Pull-down	Internal Pull-down	OFF
PCIE_A[7:0]_RX_N PCIE_A[7:0]_RX_P	I	Primary	Internal Pull-down	Internal Pull-down	Internal Pull-down	OFF
PCIE_B[15:0]_TX_N PCIE_B[15:0]_TX_P	O	Primary	Internal Pull-down	Internal Pull-down	Internal Pull-down	OFF
PCIE_B[15:0]_RX_N PCIE_B[15:0]_RX_P	I	Primary	Internal Pull-down	Internal Pull-down	Internal Pull-down	OFF

Notes: 1. PCIE_RX_N/PCIE_RX_P pins transition from un-driven to Internal Pull-down during Reset.
2. Reset reference for primary well pins is RSMRST#.

21.4 PCI Express Root Port Support Feature Details

Table 80. PCI Express* Root Port Feature Details

PCIe Data Rate	Encoding	Transfer Rate (MT/s)	Theoretical Max Bandwidth (GB/s)				
			x1	x2	x4	x8	x16
1.0	8b/10b	2500	0.25	0.50	1.00	2.00	4.00
2.0	8b/10b	5000	0.50	1.00	2.00	4.00	8.00
3.0	128b/130b	8000	1.00	2.00	3.94	7.88	15.75

continued...

PCIe Data Rate	Encoding	Transfer Rate (MT/s)	Theoretical Max Bandwidth (GB/s)				
			x1	x2	x4	x8	x16
4.0	128b/130b	16000	1.97	3.94	7.88	15.75	31.51
5.0	128b/130b	32000	3.94	7.88	15.75	31.51	63.02

Note: 1. Theoretical Maximum Bandwidth (GB/s) = ((Transfer Rate * Encoding * # PCIe Lanes) / 8) / 1000

- Gen5 with 16 PCIe Lanes Example = ((32000 * 128/130 * 16) / 8) / 1000 = 63.02 GB/s

Figure 27. Supported PCI Express* Link Configurations

NVL-S																									
Flex I/O Lane	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	
PCIe Controllers	A								B																
PCIe Max Rate	5.0								5.0																
x8 Effective Throughput ³	1								1								1								
Bi-Furcation Straps ^{4,5}	Soft Straps ⁴								GPP_SA[16:14] Pin Straps + Soft Straps ⁵																
Lane Reversal Type	Per Root Port								Per Root Port																
Root Port #	1				2				3					4				5					6		
PCIe (TX/RX) Lanes	A0	A1	A2	A3	A4	A5	A6	A7	B0	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12	B13	B14	B15	
PCIe Lane Count	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	
PCIe Configurations (Bi-Furcation) ²	1px8/1px8(LR)								1px16/1px16(LR)																
	1px4/1px4(LR)				1px4/1px4(LR)				1px8/1px8(LR)								1px8/1px8(LR)								
									1px8/1px8(LR)								1px4/1px4(LR)				1px4/1px4(LR)				
									1px4/1px4(LR)				1px4/1px4(LR)				1px4/1px4(LR)				1px4/1px4(LR)				
Logical Link Lanes	0/7	1/6	2/5	3/4	4/3	5/2	6/1	7/0	0/15	1/14	2/13	3/12	4/11	5/10	6/9	7/8	8/7	9/6	10/5	11/4	12/3	13/2	14/1	15/0	
	0/3	1/2	2/1	3/0	0/3	1/2	2/1	3/0	0/7	1/6	2/5	3/4	4/3	5/2	6/1	7/0	0/7	1/6	2/5	3/4	4/3	5/2	6/1	7/0	
									0/7	1/6	2/5	3/4	4/3	5/2	6/1	7/0	0/3	1/2	2/1	3/0	0/3	1/2	2/1	3/0	
									0/3	1/2	2/1	3/0	0/3	1/2	2/1	3/0	0/3	1/2	2/1	3/0	0/3	1/2	2/1	3/0	
Assigned Root Ports	RP1								RP2								RP3/RP6								
	RP1				RP2				RP3								RP5								
									RP3				RP5/RP6				RP6/RP5								
									RP3/RP4				RP4/RP3				RP5/RP6				RP6/RP5				
Bus - Dev - Func (BDF) Assignments ¹	RP	Bus	Device	Function	RP	Bus	Device	Function	RP	Bus	Device	Function	RP	Bus	Device	Function	RP	Bus	Device	Function	RP	Bus	Device	Function	
	1	0h	6h	0h	3	0h	6h	2h	5	0h	6h	2h	5	0h	6h	2h	5	0h	6h	2h	5	0h	6h	4h	
	2	0h	6h	1h	4	0h	6h	3h	6	0h	6h	3h	6	0h	6h	3h	6	0h	6h	3h	6	0h	6h	5h	

NOTES

1. Device (BDF) groupings have multiple functions, the lowest active Root Port within the Device (BDF) grouping will always be assigned Function 0 while any remaining active Root Port within the Device (BDF) grouping will be assigned their mapped Function # as shown.
2. Reduced Root Port width configurations, within Bi-Furcation configurations, are supported (example: x2 PCIe End Point Device populated in a PCIe Controller set as 1px4/1px16 will result in a 1px2 PCIe Root Port configuration or x1 PCIe End Point Device populated in a PCIe Controller set as 1px4/1px16 will result in a 1px1 PCIe Root Port configuration).
3. x8 Effective Throughput = "1" signifies the highest ranking. The effective throughput variations identified in the PCD-S PCIe Controllers are qualitatively based on hardware architecture, without any power management constraints. PCD-S PCIe Controllers with a lower "x8 Effective Throughput" value may offer a slight advantage over other PCD-S PCIe Controllers within the processor. The PCD-S PCIe Controller B root port, configured as 1px16 and operating at PCIe 5.0 rate, may achieve the highest Effective Throughput compared to all other PCD-S PCIe Controller root port configurations. The PCD-S PCIe Controllers will provide superior effective throughput in comparison to the PCH PCIe Controllers, attributed to the Direct Media Interface (DMI) facilitating the connection between the processor and the downstream PCH PCIe Controllers.
4. The PCIe Controller A Bi-Furcation Configuration Mode is set through Soft Straps.
5. The PCIe Bi-Furcation configuration mode for PCIe Controller B is defined by Pin Strap and Soft Strap settings. These Pin Strap settings utilize the GPP_SA[16:14] signal pins, which motherboard designs may implement through board switches, jumpers, or pull-up and pull-down resistors. Pin Straps and Soft Straps must correspond with the motherboard's implementation accurately. Understanding the impact of configuring these straps between the GPP_SA[16:14] Pin Straps and Soft Straps is vital; any incorrect or mismatched settings may impede a Root Port and link from achieving the desired data width or result in an endpoint device not being detected.
 - If the GPP_SA[16:15] Pin Straps and Soft Straps are not aligned, the PCIe Configuration prioritizes the settings with the lowest data width, applying them to determine the Bi-Furcation for the associated PCIe Controller Root Port.
 - If the GPP_SA[16:15] Pin Straps align with the Soft Straps, the corresponding data width value is uniformly prioritized, thereby setting the Bi-Furcation for the associated PCIe Controller Root Port.
 - The GPP_SA[14] Pin Strap serves as the Override Setting and, when enabled (1b), it grants the Pin Strap GPP_SA[16:15] Configuration settings precedence over the Soft Strap Configuration settings, with the exception of Lane Reversal. Lane Reversal is exclusively configured via the Lane Reversal Soft Strap settings.
6. The PCIe* Link Configuration support will vary depending on the SKU. Refer to the SKU details covered in the [Introduction](#) on page 15.
7. LR = Lane Reversal
8. $\text{PCIe Configuration (\#p) \times (\#)} = (\text{Number of PCIe Root Ports}) \times (\text{Number of Data Lane Pairs per PCIe Root Port})$
9. RP# refers to a specific PCI Express* Root Port #; for example RP3 = PCI Express* Root Port 3

10. A PCIe* Lane is composed of a single pair of Transmit (TX) and Receive (RX) differential pairs. A connection between two PCIe* devices is known as a PCIe* Link, and is built up from a collection of one or more PCIe* Lanes which make up the width of the link (such as bundling 2 PCIe* Lanes together would make a x2 PCIe* Link). A PCIe* Link is addressed by the lowest number PCIe* Lane it connects to and is known as the PCIe* Root Port (such as a x2 PCIe* Link connected to PCIe* Lanes 3 and 4 would be called x2 PCIe* Root Port 3).
 11. The PCIe* Lanes can be configured independently from one another but the max number of configured Root Ports (Devices) must not be exceeded
 12. Unidentified lanes within a PCIe* Link Configuration are disabled but their physical lanes are used for the identified Root Port
-

22.0 Graphics

22.1 Processor Graphics

The processor graphics is based on Xe3 graphics core architecture that enables substantial gains in performance and lower-power consumption over prior generations. Xe3 architecture supports up to 2 Xe3-core depending on the processor SKU.

The processor graphics architecture delivers high dynamic range of scaling to address segments spanning low power to high power, increased performance per watt, support for next generation of APIs. Xe3 scalable architecture is partitioned by usage domains along Render/Geometry, Media, and Display. The architecture also delivers very low-power video playback and next generation analytics and filters for imaging related applications. The new Graphics Architecture includes 3D compute elements, Multi-format HW assisted decode/encode pipeline, and Mid-Level Cache (MLC) for superior high definition playback, video quality, and improved 3D performance and media.

22.1.1 Media Support (Intel® QuickSync and Clear Video Technology HD)

The Media Engine is Xe3P which implements multiple media video codecs in hardware as well as a rich set of image processing algorithms.

22.1.1.1 Hardware Accelerated Video Decode

Xe3P implements a high-performance and low-power HW acceleration for video decoding operations for multiple video codecs.

The HW decode is exposed by the graphics driver using the following APIs:

- Direct3D11 Video API
- Direct3D12 Video API
- Intel Media SDK
- MFT (Media Foundation Transform) filters¹
- Intel VA API ²
- Intel one VPL

NOTES

1. Only for JPEG Decoder
 2. Only for Linux*
-

Xe3P supports full HW accelerated video decoding for MPEG2/AVC/HEVC/VP9/JPEG/AV1.

Table 81. Hardware Accelerated Video Decoding

Codec	Profile	Level	Maximum Resolution
MPEG2	Main	Main - 15Mbps High - 40Mbps	FHD
AVC/H264	4:2:0 8b/10b 4:2:2 10bit	L5.2	4K @ 430 4K @ 330
JPEG/MJPEG	Baseline	Unified level	16K x16K
HEVC/H265	Main12 420, 422, 444 - 8b/10b/12b SCC 420, 444 - 8b/10b	L6.1	8K @ 60
VP9	0 (420 8b) 1 (444 8b) 2 (420 10b/12b) 3 (444 10b/12b)	Unified level	8K @ 60 16K x 4K
AV1	Main10 (420, 422, 444 8b/10b)	L6.1	8K @ 60 16K x 16K (still picture)
VVC/H266	Main10 (420 8b/10b)	L6.1	8K @ 60 16K x 16K (Still Picture)
JPEGXS	High444.12 422, 444 - 10b/12b YCbCr HighBayer - 12b/14b/16b	Unified level	422 8K@60 444/Bayer 8K@30 16K x 16K

Expected Performance: More than 16 simultaneous decode streams @ 1080p.

NOTE

Actual performance depends on the processor SKU, content bit rate, and memory frequency. Hardware decode for H264 SVC is not supported.

22.1.1.2 Hardware Accelerated Video Encode

Xe3P implements a low-power low-latency fixed function encoder which supports AVC, HEVC, VP9, JPEG, and AV1.

The HW encode is exposed by the graphics driver using the following APIs:

- Direct3D12 Video API
- Intel® one VPL
- MFT (Media Foundation Transform) filters [Only for AVC/HEVC/JPEG/AV1 Encoder]

Xe3P supports full HW accelerated video encoding for AVC/HEVC/VP9/JPEG/AV1/ BT2020.

Table 82. Hardware Accelerated Video Encode

Codec	Profile	Level	Maximum Resolution
AVC/H264	High Main Constrained Baseline	L5.2	4K@60
JPEG			16Kx16K
<i>continued...</i>			

Codec	Profile	Level	Maximum Resolution
HEVC/H265	Main10 422 - 8b/10b	L5.2	4K@60
	Main Main10 420, 444 - 8b/10b SCC 420 444 - 8b/10b	L6.1	4320p(8K) @60 16Kx12K
AV1	Main10 (420, 422, 444 8b/10b)	L6.1	8K@60

NOTE

Hardware encode for H264 SVC is not supported.

22.1.1.3 Hardware Accelerated Video Processing

There is hardware support for image processing functions such as De-interlacing, Film cadence detection, detail enhancement, gamut compression, Adaptive contrast enhancement, skin tone enhancement, total color control, De-noise, SFC (Scalar and Format Conversion), memory compression, 16 bpc support for de-noise/de-mosaic, HDR10 and Dolby Vision Tone Mapping HW acceleration.

Xe3P newly adds support for Tetrahedral 3DLUT and FP16 input/output.

The HW video processing is exposed by the graphics driver using the following APIs:

- Direct3D* 11 Video API
- Intel® One VPL
- Intel® Graphics Control Library
- Intel VA API

NOTE

Not all features are supported by all the above APIs. Refer to the relevant documentation for more details.

22.1.1.4 Hardware Accelerated Transcoding

Transcoding is a combination of decode, video processing (optional) and encode. Using the above hardware capabilities can accomplish a high-performance transcode pipeline. There is not a dedicated API for transcoding.

The processor graphics supports the following transcoding features:

- High performance high quality flexible encoder for video editing, video archiving.
- Low-power low latency encoder for video conferencing, wireless display, and game streaming.
- Low power Scaler and Format Converter.

22.1.2 Graphics Core Cache

The Xe3 Graphics Core architecture has a hierarchy of caches which contains first, second and third level caches.

First and Second Level Cache

The first and second level cache is a lower-level Instruction and the Data caches. They are implemented close to the Xe3/3D compute elements, decode/encode and media pipelines. These cache units are not shared between the different units.

Third Level Cache

Third level cache is a central memory cache that is implemented as higher cache hierarchy. The device cache is a multi-way set-associative that allow memory pages to be cached either coherently or non-coherently with respect to an external memory system.

22.2 Platform Graphics Hardware Feature

22.2.1 Hybrid Graphics

Microsoft* Windows* 11 operating system enables the Windows*11 Hybrid graphics framework wherein the GPUs and their drivers can be simultaneously utilized to provide users with the benefits of both performance capability of discrete GPU (dGPU) and low-power display capability of the processor GPU (iGPU). For instance, when there is a high-end 3D gaming workload in progress, the dGPU will process and render the game frames using its graphics performance, while iGPU continues to perform the display operations by compositing the frames rendered by dGPU. We recommend that OEMS should seek further guidance from Microsoft* to confirm that the design fits all the latest criteria defined by Microsoft* to support HG.

Microsoft* Hybrid Graphics definition includes the following:

1. The system contains a single integrated GPU and a single discrete GPU.
2. It is a design assumption that the discrete GPU has a significantly higher performance than the integrated GPU.
3. Both GPUs shall be physically enclosed as part of the system.
 - a. Microsoft* Hybrid DOES NOT support hot-plugging of GPUs
 - b. OEMS should seek further guidance from Microsoft* before designing systems with the concept of hot-plugging
4. Starting with Windows*11 (WDDM 2.0), a previous restriction that the discrete GPU is a render-only device, with no displays connected to it, has been removed. A render-only configuration with NO outputs is still allowed, just NOT required.

23.0 Display

This chapter provides information on the following topics:

- Display Technologies Support
- Display Configuration
- Display Features

23.1 Display Technologies Support

Technology	Standard
eDP* 1.5	VESA* Embedded DisplayPort* Standard 1.5
DisplayPort* 2.1	VESA* DisplayPort* Standard 2.1
HDMI* 2.1	High-Definition Multimedia Interface Specification Version 2.1

Table 83. Display Ports Availability and Link Rate

Port	Supported Technologies
DDI 3	eDP up to HBR3, DP up to UHBR20, HDMI 12 Gbps
DDI 2	DP up to UHBR20, HDMI 12 Gbps
TCP 1	DP up to UHBR20, HDMI 12 Gbps
TCP 0	DP up to UHBR20, HDMI 12 Gbps
Note: 1. DDI3 is internally muxed to DDIA. 2. DDI2 may also be denoted in other documentation as DDI4.	

23.2 Display Interfaces Signals

This section provides information on the following topics:

- Digital Display Interface (DDI) Signals
- Digital Display Audio Signals
- Digital Display Interface TCP Signals

23.2.1 Digital Display Interface DDI Signals

Table 84. Digital Display Interface DDI Signals

Signal Name	Type	Description
GPP_SD09/ VDDEN ¹ (Option 1)	0	Digital Display Interface A (DDI3): eDP Panel power control enable signal.
GPP_SA21/ VDDEN ¹ (Option 2)		
GPP_SD10/ BKLTEN ¹ (Option 1)	0	Digital Display Interface A (DDI3): eDP Panel back-light control enable signal.
continued...		

Signal Name	Type	Description
GPP_SA22/ BKLTEN ¹ (Option 2)	O	Digital Display Interface A (DDI3): eDP Panel back-light control Pulse Wide Modulation (PWM) signal.
GPP_SD11/ BKLTCTL ¹ (Option 1)		
GPP_SA23/RSVD/ BKLTCTL ¹ (Option 2)		
DDI2_TX_P[3:0] DDI2_TX_N[3:0] DDI3_TX_P[3:0] DDI3_TX_N[3:0]	O	Digital Display Interface x (DDIx): Digital Display Interface main link transmitter lanes.
DDI2_AUX_P DDI2_AUX_N DDI3_AUX_P DDI3_AUX_N	I/O	Digital Display Interface x (DDIx): DisplayPort Auxiliary: Half-duplex, bidirectional channel consist of one differential pair.
GPP_SB04/ DDP2_CTRLCLK GPP_SB02/ DDP3_CTRLCLK GPP_SB05/ DDP2_CTRLDATA GPP_SB03/ DDP3_CTRLDATA	I/O	Digital Display Interface x (DDIx): HDMI Graphics Management Bus (GMBUS).
GPP_SD14/ DDSP_HPD2# /DISP_MISC2 ¹ (Option 1) GPP_SD13/ DDSP_HPD3# /DISP_MISC3 ¹ (Option 1) or GPP_SA20/ DDSP_HPD2# /DISP_MISC2 ¹ (Option 2) GPP_SA16/ DDSP_HPD3# /DISP_MISC3 ¹ (Option 2)	I	Digital Display Interface x (DDIx): Hot Plug Detect (HPD).
GPP_SD15/DDSP_HPD0#/ DISP_MISC0 GPP_SD16/DDSP_HPD1#/ DISP_MISC1 GPP_SD14/DDSP_HPD2#/ DISP_MISC2 GPP_SD13/DDSP_HPD3#/ DISP_MISC3	O	DDI Misc signals.
Notes: 1. These signals are alternate signals with identical functionality, either Option 1 or Option 2 should be used. 2. Auxiliary Channel (AUX CH) is a half-duplex bidirectional channel used for link management and device control. AUX CH is an AC coupled differential signal. 3. GMBUS follows I2C Protocol.		

23.2.2 Digital Display Interface TCP Signals

Table 85. Digital Display Interface TCP Signals

Signal Name	Type	Description
TCP0_TXRX[1:0]_P TCP0_TXRX[1:0]_N TCP0_TX[1:0]_P TCP0_TX[1:0]_N	O	Digital Display Interface 0 (TCP0): Digital Display Interface main link transmitter lanes.
TCP0_AUX_P TCP0_AUX_N	I/O	Digital Display Interface 0 (TCP0): DisplayPort Auxiliary: Half-duplex, bidirectional channel consist of one differential pair.
GPP_SB13/ DDP0_CTRLDATA / TBT_LSX0_RXD GPP_SB12/ DDP0_CTRLCLK / TBT_LSX0_TXD	I/O	Digital Display Interface 0 (TCP0): HDMI Graphics Management Bus (GMBUS).
GPP_SD15/ DDSP_HPD0# / DISP_MISC0 ¹ (Option 1) Or	I	Digital Display Interface 0 (TCP0): Hot Plug Detect (HPD).
continued...		

Signal Name	Type	Description
GPP_SA14/ DDSP_HPD0# / DISP_MISC0/TIME_SYNC0 ¹ (Option 2)		
TCP1_TXRX[1:0]_P TCP1_TXRX[1:0]_N TCP1_TX[1:0]_P TCP1_TX[1:0]_N	O	Digital Display Interface 1 (TCP1): Digital Display Interface main link transmitter lanes.
TCP1_AUX_P TCP1_AUX_N	I/O	Digital Display Interface 1 (TCP1): DisplayPort Auxiliary: Half-duplex, bidirectional channel consist of one differential pair.
GPP_SB16/ DDP1_CTRLDATA / TBT_LSX1_RXD GPP_SB15/ DDP1_CTRLCLK / TBT_LSX1_TXD	I/O	Digital Display Interface 1 (TCP1): HDMI Graphics Management Bus (GMBUS).
GPP_SD16/ DDSP_HPD1# / DISP_MISC1 ¹ (Option 1) Or GPP_SA15/ DDSP_HPD1# / DISP_MISC1/TIME_SYNC1 ¹ (Option 2)	I	Digital Display Interface 1 (TCP1): Hot Plug Detect (HPD).
Notes: 1. These signals are alternate signals with identical functionality, either Option 1 or Option 2 should be used. - Auxiliary Channel (AUX CH) is a half-duplex bidirectional channel used for link management and device control. AUX CH is an AC coupled differential signal. - GMBUS follows I2C Protocol.		

23.2.3 Digital Display Audio Signals

Table 86. Digital Display Audio Signals

Signal Name	Dir.	Description	Link Type
GPP_SA26/ SNDW_DATA0_RX / AUDOUT	O	Serial Data Output for display audio interface.	SE
GPP_SA25/ SNDW_DATA0_TX / AUDIN	I	Serial Data Input for display audio interface.	SE
GPP_SA24/ SNDW_CLK / AUDCLK	I	Serial Data Clock	SE
GPP_SA24/ SNDW_CLK / AUDCLK	O	SoundWire Clock: Serial bit clock used to control the timing of a transfer.	
GPP_SA25/ SNDW_DATA0_TX / AUDIN	O	SoundWire Data 0 TX: Serialized data line containing framing and data being transmitted for Display audio.	
GPP_SA26/ SNDW_DATA0_RX / AUDOUT	I	SoundWire Data 0 RX: Serialized data line containing framing and data being received for Display audio.	
GPP_SA27/ SNDW_DATA1_TX	O	SoundWire Data 1 TX: Serialized data line containing framing and data being transmitted for Display audio.	
GPP_SA28/ SNDW_DATA1_RX	I	SoundWire Data 1 RX: Serialized data line containing framing and data being received for Display audio.	

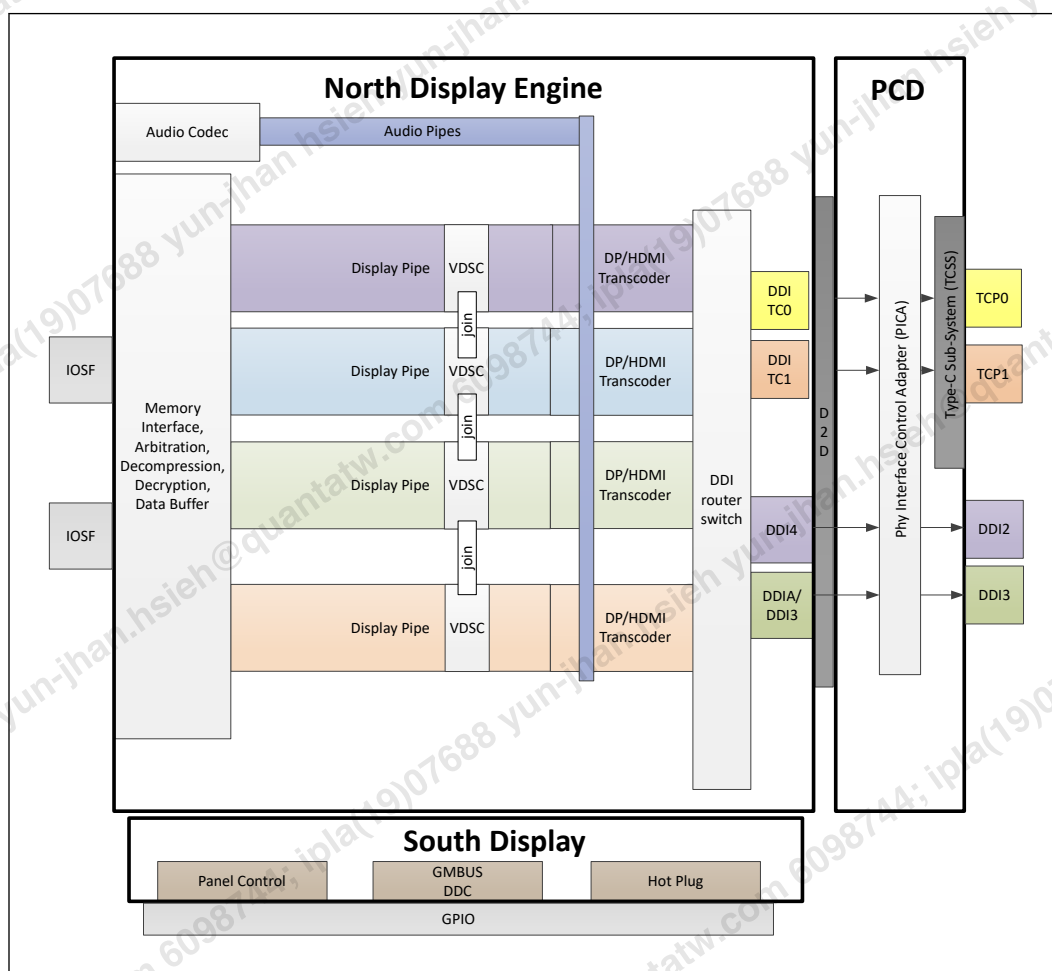
23.3 Display Features

This section provides information on the following topics:

- General Capabilities
- Multiple Display Configurations
- High-bandwidth Digital Content Protection (HDCP)
- DisplayPort*
- High-Definition Multimedia Interface (HDMI*)
- embedded DisplayPort* (eDP*)
- Integrated Audio

23.3.1 General Capabilities

Figure 28. Processor Display Architecture



NOTE

For port availability in each of the processor series, refer to [Table 83](#) on page 184.

Max Display Concurrency:

Up to 4 simultaneous displays, combinations as follows:

Table 87. Display Combinations

	Internal	External
Option 1	3840x2400p120 HDR	3 x 4K144 HDR (3840x2160)
Option 2	3840x2400p120 HDR	3 x 6K60 HDR (6144x3456)
Option 3	3840x2400p120 HDR	1 x 8K60 HDR (7680x4320) + 1 x 4K144 HDR (3840x2160)
Option 4	-	4 x 4K144 HDR (3840x2160)
Option 5	-	4 x 6K60 HDR (6144x3456)
Option 6	-	2 x 8K60 HDR (7680x4320)
Notes: - Standard shipping displays timings (Hight/Vert/Refresh-rate), other timings may have more possible combinations/options - Single external panel up to 8K60Hz, supported by joining two pipes over single port		

- Display interfaces supported:
 - DDI interfaces supports DP*, HDMI*, eDP*
 - TCP interfaces supports DP*, HDMI*, Display Alt Mode over Type-C and DP* tunneled.
- Audio stream support on external ports.
- HDR (High Dynamic Range) support.
- Four Display Pipes - Supporting blending, color adjustments, scaling and dithering.
- Transcoder - Containing the Timing generators supporting eDP*, DP*, HDMI* interfaces.
- One Low Power optimized pipes supporting Embedded DisplayPort*
 - LACE (Localized Adaptive Contrast Enhancement)
 - 3D LUT - power efficient pixel modification function for color processing.

23.3.2 Multiple Display Configurations

The following multiple display configuration modes are supported (with appropriate driver software):

- Single Display is a mode with one display port activated to display the output to one display device.
- Display Clone is a mode with up to four display ports activated to drive the display content of same color depth setting but potentially different refresh rate and resolution settings to all the active display devices connected.
- Extended Desktop is a mode with up to four display ports activated to drive the content with potentially different color depth, refresh rate, and resolution settings on each of the active display devices connected.

23.3.3 High-bandwidth Digital Content Protection

HDCP is the technology for protecting high-definition content against unauthorized copy or unrecap between a source (computer, digital set top boxes, and so on) and the sink (panels, monitor, and TVs). The processor supports HDCP 2.3 content protection over wired displays (HDMI* and DisplayPort*).

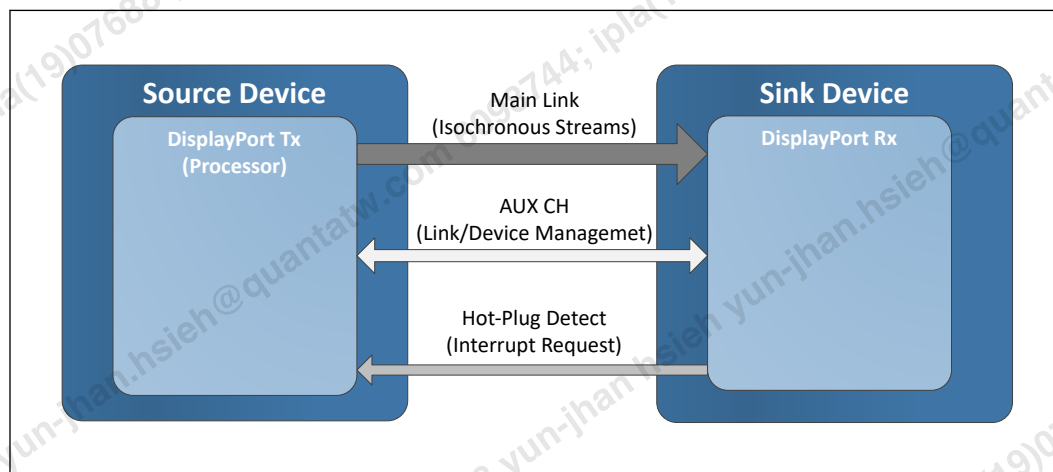
23.3.4 DisplayPort*

The DisplayPort* is a digital communication interface that uses differential signaling to achieve a high-bandwidth bus interface designed to support connections between PCs and monitors, projectors, and TV displays.

A DisplayPort* consists of a Main Link (four lanes), Auxiliary channel, and a Hot-Plug Detect signal. The Main Link is a unidirectional, high-bandwidth, and low-latency channel used for transport of isochronous data streams such as uncompressed video and audio. The Auxiliary Channel (AUX CH) is a half-duplex bi-directional channel used for link management and device control. The Hot-Plug Detect (HPD) signal serves as an interrupt request from the sink device to the source device.

The processor is designed in accordance with VESA* DisplayPort* specification.

Figure 29. DisplayPort* Overview



- Supports main link of 1, 2, or 4 data lanes.
- Link rate supports up to UHBR20 (UHBR13.5 is not supported)
- Aux channel for Link/Device management
- Hot Plug Detect
- Supports up to 36 bpp (Bit Per Pixel)
- Supports SSC
- Supports YCbCr 4:4:4, YCbCr 4:2:0, YCbCr 4:2:2, and RGB color format
- Supports MST (Multi-Stream Transport)
- Supports VESA DSC 1.2b
- Supports panel replay
- Adaptive Sync

23.3.4.1 Multi-Stream Transport

- The processor supports Multi-Stream Transport (MST), enabling multiple monitors to be used via a single DisplayPort connector.

Table 88. DisplayPort Maximum Resolution

Standard	S-Series Processor
DP*	8K60Hz (2 pipes), 6K120Hz (2 pipes); 4K144Hz (Single pipe)
Notes: 1. bpp - bit per pixel. 2. Resolution support is subject to memory BW availability. 3. High resolutions will consume two display pipes.	

23.3.5 High-Definition Multimedia Interface (HDMI*)

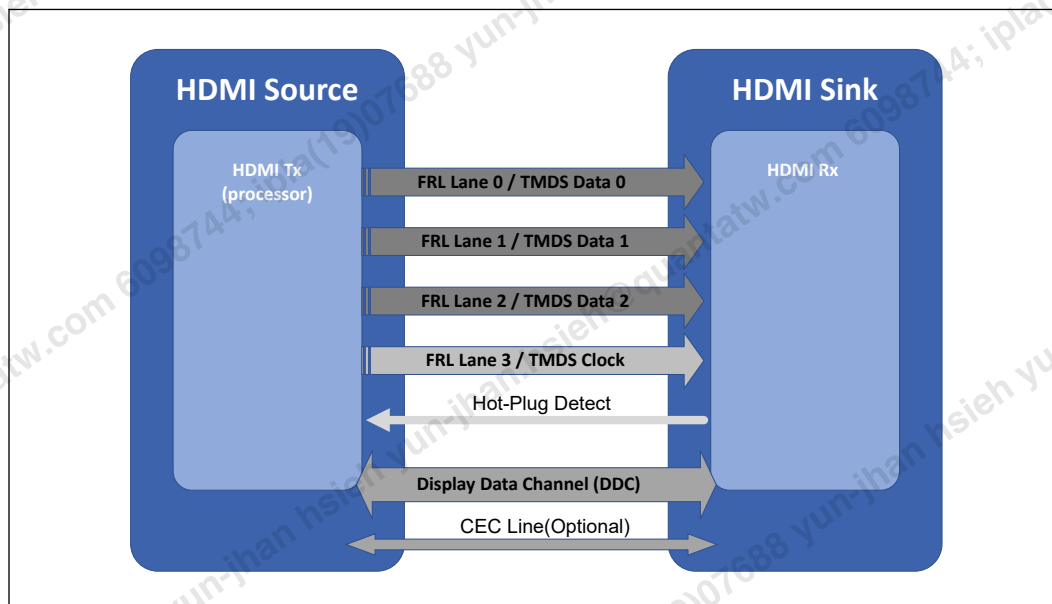
The High-Definition Multimedia Interface (HDMI*) is provided for transmitting digital audio and video signals from DVD players, set-top boxes, and other audio-visual sources to television sets, projectors, and other video displays. It can carry high-quality multi-channel audio data and all standard and high-definition consumer electronics video formats. The HDMI display interface connecting the processor and display devices uses transition minimized differential signaling (TMDS) or Fixed Rate Link (FRL) to carry audiovisual information through the same HDMI cable.

HDMI* includes three separate communications channels: TMDS or FRL, DDC/GMBUS, and the optional CEC (consumer electronics control). CEC is not supported on the processor. As shown in the following figure, the HDMI* cable carries four differential pairs that make up the TMDS data and clock channels or FRL lanes. These channels are used to carry video, audio, and auxiliary data. In addition, HDMI carries a VESA DDC. The DDC/GMBUS is used by an HDMI* Source to determine the capabilities and characteristics of the Sink.

Audio, video, and auxiliary (control/status) data is transmitted across the three TMDS data channels.

TMDS architecture has 3 Data lane and 1 Clock lane

FRL architecture has 4 Data lane, and no clock lane

Figure 30. HDMI* Overview

- Supports up to 6Gbps TMDS link rates
- Supports up to 12Gbps FRL link rates
- Support YCbCR 4:4:4, YCbCR 4:2:0, YCbCR 4:2:2, and RGB color format
- Supports up to 36 bpp (Bit Per Pixel)
- Supports VESA DSC 1.2a in FRL mode
- Hot Plug Detect
- Adaptive Sync supported in FRL mode

Table 89. HDMI Maximum Resolution

Standard	S-Series Processor
HDMI 2.1 (Up to 6Gbps)	4K60 Hz 24 bpp
HDMI 2.1 (Up to 12 Gbps)	8K60Hz Compressed, 6K120Hz compressed 4K144Hz Compressed
Notes: 1. bpp - bit per pixel. 2. Resolution support is subject to memory BW availability. 3. Compressed mean DSC only	

23.3.6 embedded DisplayPort

The embedded DisplayPort* (eDP*) is an embedded version of the DisplayPort standard oriented towards applications such as notebook and All-In-One PCs. Like DisplayPort, embedded DisplayPort* also consists of the Main Link, Auxiliary channel, and an optional Hot-Plug Detect signal.

- Supports Low power optimized pipes
- Supports up to HBR3 link rate
- Supports Backlight PWM control and enable signals, and power enable

- Supports VESA DSC 1.2a
- Supports SSC
- Panel Self Refresh 1
- Panel Self Refresh 2
- Panel Replay
- MSO 2x2, 4x1(Multi Segment Operation)
- Dedicated Aux channel
- Adaptive Sync

Table 90. Embedded DisplayPort Maximum Resolution

Standard	S-Series Processor ¹ eDP1.4a	S-Series Processor ¹ eDP1.5
eDP*	8k 60Hz compressed 4k 120Hz compressed	6k 60Hz compressed 4k 144Hz compressed
Notes: 1. Maximum resolution is based on the implementation of 4 lanes at HBR3 link data rate. 2. 5k120Hz cannot work with PSR*. 3. Resolution support is subject to memory BW availability.		

23.3.7 Integrated Audio

- HDMI* and DisplayPort interfaces can carry audio along with video.
- The processor supports up to four High Definition Audio streams on four digital ports simultaneously.

Table 91. Processor Supported Audio Formats over HDMI* and DisplayPort*

Audio Formats	HDMI*	DisplayPort*
AC-3 Dolby* Digital	Yes	Yes
Dolby* Digital Plus	Yes	Yes
DTS-HD*	Yes	Yes
LPCM, 32 KHz, 44.1 KHz, 48 KHz, 88.2 KHz, 96 KHz, 176.4 KHz, and 192 KHz, 16/24 bit, 2/4/6/8 channels	Yes	Yes
Dolby* TrueHD, DTS-HD Master Audio* (Lossless Blu-Ray Disc* Audio Format)	Yes	Yes

The processor will continue to support Silent stream. A Silent stream is an integrated audio feature that enables short audio streams, such as system events to be heard over the HDMI* and DisplayPort* monitors. The processor supports silent streams over the HDMI and DisplayPort interfaces at 32 KHz, 44.1 KHz, 48 KHz, 88.2 KHz, 96 KHz, 176.4 KHz, and 192 KHz sampling rates and silent multi-stream support.

24.0 General Purpose Input and Output

The General Purpose Input/Output (GPIO) signals are grouped into multiple groups (such as GPP_SA, GPP_SB, and so on). All GPIO groups are powered by the Primary well.

The high level features of GPIO:

- 1.8 V operation (including the muxed functions on the pin).
- Integrated pull-up / pull-down.
- Configurable as GPIO input, GPIO output, or native function signal.
- Configurable GPIO pad ownership by host, CSME, or ISH.
- SCI (GPE) and IOAPIC interrupt capable on all GPIOs
- NMI and SMI capability capable (on selected GPIOs).

Table 92. Acronyms

Acronyms	Description
GPI	General Purpose Input
GPO	General Purpose Output
GPP	General Purpose I/O in Primary Well

24.1 Functional Description

24.1.1 Interrupt IRQ via GPIO Requirement

A GPIO, as an input, can be used to generate an interrupt / IRQ to the processor. In this case, it is required that the pulse width on the GPIO must be at least 100 us for the processor to recognize the interrupt.

24.1.2 Integrated Pull-ups and Pull-downs

All GPIOs have programmable internal pull-up/pull-down resistors. Most of the GPIOs have the internal pull-up/pull-down being off by default. The internal pull-up/pull-down for each GPIO can be enabled by BIOS programming the corresponding PAD_CFG_DW1 register, TERM bit. Refer to Volume 2 (Register Information) and Nova Lake S GPIO Implementation Summary (#853123) for more details.

24.1.3 SCI SMI and NMI

SCI capability is available on all GPIOs, while SMI and NMI capability is available on only select GPIOs.

Below are the GPIOs that can be routed to generate SMI# or NMI:

- GPP_SB16, GPP_SD[04:00]

24.1.4 Timed GPIO

The processor supports two Timed GPIOs as native function (TIME_SYNC) that is multiplexed on GPIO pins. The intent usage of the Timed GPIO function is for time synchronization purpose.

Timed GPIO can be an input or an output:

- As an input, a GPIO input event triggers the HW to capture the processor Always Running Timer (ART) time in the Time Capture register. The GPIO input event must be asserted for at least two crystal oscillator clocks period in order for the event to be recognized.
- As an output, a match between the ART time and the software programmed time value triggers the HW to generate a GPIO output event and capture the ART time in the Time Capture register. If periodic mode is enabled, HW generates the periodic GPIO events based on the programmed interval. The GPIO output event is asserted by HW for at least two crystal oscillator clock periods.

NOTE

TIME_SYNC can be set as input when both Direction (DIR) bit and Enable (EN) bit in Timed GPIO Control Register are set to 1 (refer to EDS Vol2 for the register info). When EN bit is set to 0, TIME_SYNC will default to output low regardless of DIR bit setting.

Timed GPIO supports event counter. When Timed GPIO is configured as input, event counter increments by one for every input event triggered. When Timed GPIO is configured as output, event counter increments by one for every output event generated. The event counter provides the correlation to associate the Timed GPIO event (the nth event) with the captured ART time. The event counter value is captured when a read to the Time Capture Value register occurs.

NOTE

When Timed GPIO is enabled, the crystal oscillator will not be shut down as crystal clock is needed for the Timed GPIO operation. As a result, SLP_S0# will not be asserted. This has implication to platform power (such as IDLE or S0ix power). Software should only enable Timed GPIO when needed and disable it when Timed GPIO functionality is not required.

24.1.5 GPIO Blink and Serial Blink

Certain GPIOs are capable of supporting blink (BK, or also referred to as PWM), and serial blink (SBK, or also referred to as Serial POST Codes). The BK and SBK are implemented as native functions muxed on selected GPIOs. To enable BK or SBK on a GPIO having the capability, BIOS needs to select the BK or SBK native function on the GPIO.

BK provides a programmable PWM wave. The Blink/PWM frequency and duty cycle is programmable through the PWM Control register. Refer to the EDS Volume 2 for more info.

NOTE

While 5 pins are Blink/PWM capable, only a single PWM module is implemented in the GPIO Controller, hence the same Blink/PWM output is made available on the 5 pins, if Blink/PWM is selected.

SBK allows system software to serialize POST or other messages on GPIO to a serial monitor. The Serial Blink messages is programmed through the Serial Blink Command/Status and Serial Blink Data registers. Refer to the EDS Volume 2 for more info.

NOTE

While 5 pins are Serial Blink capable, only a single serializer module is implemented in the GPIO Controller, hence the same Serial Blink output is made available on the 5 pins, if Serial Blink is selected.

24.1.6 GPIO Ownership

A GPIO can be owned by the host, the Intel® CSME, depending on how the pin ownership being programmed. The programmed agent will then own the pin exclusively. For example, when a GPIO pad ownership is programmed to Intel® CSME, the host software no longer has access to the pin programming.

24.1.7 Native Function and TERM Bit Setting

Certain native function signals that are muxed onto GPIO pins support dynamic termination override, which allows the native controller to dynamically control the integrated pull-up / pull-down resistors on the signals. For those native function signals, when used, software must program the TERM bit field in the corresponding GPIO's Pad Configuration DW1 to 1111b. Refer to Volume 2 for information on the PAD configuration DW1 register and the TERM bit field.

24.2 Signal Description

For GPIO pin implementation including multiplexed native functions, default values, signal states, and other characteristics, refer to the Nova Lake S GPIO Implementation Summary (#853123) and Nova Lake S PCH GPIO Implementation Summary (#852643).

NOTE

The new Power Management Substates feature (S*ix.y) enhances power savings and requires configuring the I/O Standby State (IOSSTATE) to "IO standby masked" mode for all GPIO pins, including ESPI-related pins. This configuration is mandatory. For detailed register configuration instructions, refer to EDS Vol 2.

25.0 Direct Media Interface

Direct Media Interface (DMI) connects the processor and the PCH.

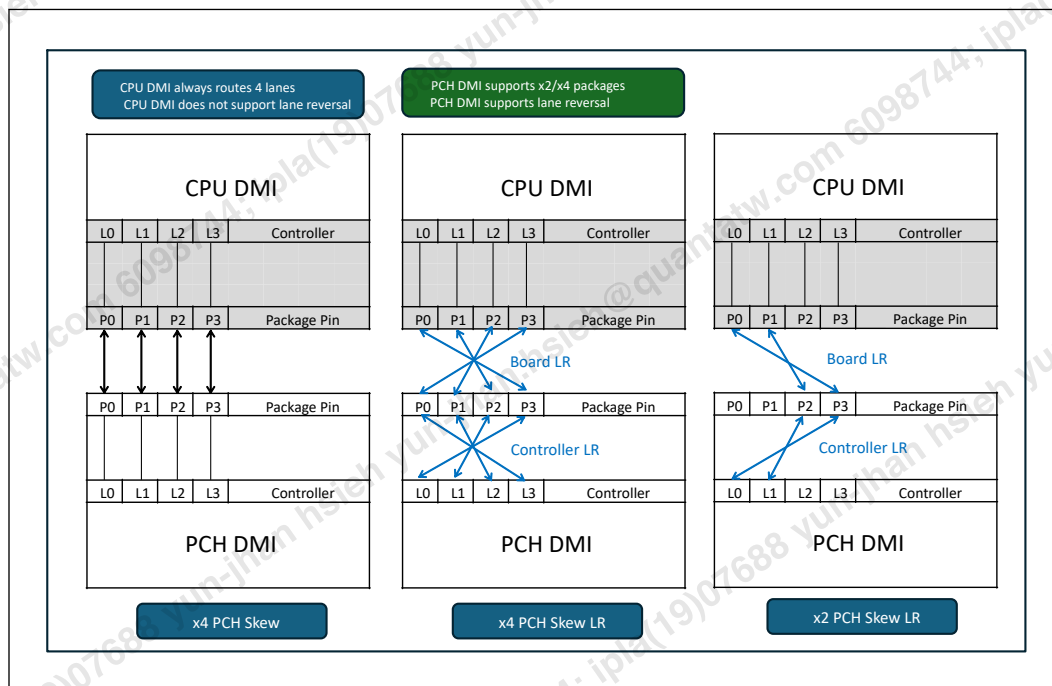
The main characteristics are as follows:

- 4 lanes Gen 5 DMI support
- 2 lanes Gen 5 Reduced DMI support
- 32 GT/s point-to-point DMI interface to PCH
- DC coupling - no capacitors between the processor and the PCH
- PCH end-to-end lane reversal across the link
- L0 (Active) and L1 (Low power) states support
- Half-Swing support (low-power/low-voltage)

25.1 DMI Lane Reversal and Polarity Inversion

NOTE

Polarity Inversion and Lane Reversal on DMI Link are not allowed in S-Processor segment. Lane reversal can only be allowed on the PCH side.

Figure 31. Example for DMI Lane Reversal Connection

1. DMI Lane Reversal is supported only on PCH-S and not on the Processor.
2. L[3:0] - Processor and PCH DMI Controller Logical Lane Numbers.
3. P[3:0] - Processor and PCH DMI Package Pin Lane Numbers.

Table 93. Processor DMI Link Mapping

Negotiated DMI Link Width	Processor DMI Physical Lanes			
	0	1	2	3
	Processor DMI Logical Lanes			
x2	0	1		
x4	0	1		

25.2 DMI Error Flow

DMI can only generate SERR in response to errors; never SCI, SMI, MSI, PCI INT, or GPE. Any DMI related SERR activity is associated with Device 0.

25.3 DMI Link Down

The DMI link going down is a fatal, unrecoverable error. If the DMI data link goes to data link down, after the link was up, then the DMI link hangs the system by not allowing the link to retrain to prevent data corruption. This link behavior is controlled by the PCH.

Downstream transactions that had been successfully transmitted across the link prior to the link going down may be processed as normal. No completions from downstream, non-posted transactions are returned upstream over the DMI link after a link down event.

25.4 DMI Signals

Signal Name	Type	Description
DMI_TX_P[0] DMI_TX_N[0]	O	DMI transmit lane 0
DMI_RX_P[0] DMI_RX_N[0]	I	DMI receive lane 0
DMI_TX_P[1] DMI_TX_N[1]	O	DMI transmit lane 1
DMI_RX_P[1] DMI_RX_N[1]	I	DMI receive lane 1
DMI_TX_P[2] DMI_TX_N[2]	O	DMI transmit lane 2
DMI_RX_P[2] DMI_RX_N[2]	I	DMI receive lane 2
DMI_TX_P[3] DMI_TX_N[3]	O	DMI transmit lane 3
DMI_RX_P[3] DMI_RX_N[3]	I	DMI receive lane 3
DMI_PERST#	O	DMI PERST Strobe

26.0 Direct eSPI

26.1 Direct eSPI Functional Description

The Direct eSPI interface is used in order to support the secure eSPI link between the processor and the PCH.

26.1.1 Processor-PCH eSPI Return Clock Support

In order to improve the eSPI bandwidth between the processor and the PCH, an eSPI return clock is added to increase the eSPI clock frequency up to 100 MHz. The processor Direct eSPI controller samples the return data using the return clock. The Direct eSPI interface supports 100 MHz.

26.1.2 PCH Multiple External eSPI Device Support

For the Direct eSPI interface, there is only one eSPI chip select between the processor and the PCH. The processor Direct eSPI controller initializes the PCH Direct eSPI target and the four channels during the initialization flow. However, the processor Direct eSPI controller supports multiple external eSPI devices connected to the PCH eSPI controller including the peripheral channel, OOB channel, and VW channel. The Flash channel is only supported on CS0 on the PCH eSPI controller.

26.1.3 Processor Direct eSPI Channel Support

Peripheral Channel

PCH eSPI Attached External EC/Device IO/Memory Cycles

The processor routes the eSPI peripheral channel fixed IO, generic IO/memory cycles to the processor Direct eSPI controller through the peripheral channel to the PCH Direct eSPI target. Then the BIOS sends these IO/memory cycles to different external eSPI devices attached to the PCH. The PCH eSPI switch will route them to different eSPI devices based on the IO Enable/IO Decode/LPC Generic Memory Range register setting on the PCH eSPI controller.

The PCH eSPI switch forwards the EC bus mastering memory cycles and LTR messages between the processor Direct eSPI controller and an EC attached to the PCH.

PCH RTC Port

The processor Direct eSPI controller access and sends the IO cycle to the RTC IO port through the PCH on the peripheral channel and the completion is converted back to the processor.

PCH IP Register Access

The PCH IP register space for SPI/eSPI, GPIO, RTC, and IOxAPIC can be accessed by the processor BIOS using the reserved window of the processor Direct eSPI controller peripheral channel device 0 Generic memory range(LGMR). The BIOS access can be locked using a lock bit in the PCH eSPI private register.

Virtual Wire Channel

The processor PMC, ITSS, and GPIO transmit/receive virtual wires to/from the PCH attached EC. The VWs are forwarded to/from the EC by the PCH eSPI switch.

The GPIO VWs are also supported between the processor GPIO controller and the PCH attached EC in both directions.

A new upstream VW is added to support PCH NMI reporting. The processor Direct eSPI controller forwards this VW to the processor ITSS.

The processor Direct eSPI controller does not broadcast the power management VWs (e.g. PLTSRT/SLP_Sx etc) using the PCH Direct eSPI target. The PCH takes responsibility to broadcast the power management VWs if there are multiple eSPI devices attached to the PCH.

OOB Channel

The processor PMC, Intel® CSME, Intel® Silicon Security Engine, TCSS, DfX tracing, and the RTC uses the Direct eSPI to communicate to the PCH via the OOB channel. The processor PMC also uses the legacy OOB interface with PCH attached EC for other OOB services including PECI over eSPI.

Flash Channel

The processor BIOS, CSME, and ESE use the processor SPI controller for flash access in SAF mode. The BIOS also uses the processor SPI controller for RPMC access in SAF mode. The PCH CSME uses PCH SPI controller for RPMC access.

To improve the flash access/boot time in the desktop platform, the processor BIOS initializes the DMA. During the DMA flow, it sends multiple read requests for the processor and the PCH to the flash device to optimize the performance.

The processor descriptor contains the region definitions and descriptor based permission control. The processor SPI controller accesses its descriptor on PCH Flash or EC via the SAF channel between the processor and the PCH.

The PCH CSME share region 2 and the processor ESE and the PCH ESE share region 10. The other region assignment is listed in the table below.

Table 94. Region Entries in the Descriptor

Register Name	Region Name	Flash Master	Special Notes
FLREG0	Descriptor	Processor/PCH descriptor	Located at the first 12KB of the flash.
FLREG1	BIOS	Processor BIOS region	
FLREG2	CSME	PCH CSME region	
FLREG3	GbE	PCH Gbe region	
continued...			

Register Name	Region Name	Flash Master	Special Notes
FLREG4	Platform Data		
FLREG5	Device Expansion #1	Reserved	
FLREG6	Secondary BIOS	Processor 2nd BIOS	
FLREG7	Reserved	Reserved	
FLREG8	Embedded Controller	PCH EC	
FLREG9	2nd Descriptor Region	Processor/PCH SPI controller use this region for the 2nd descriptor region.	For the usages of the 2nd descriptor. Always located at flash physical address 12KB-24KB.
FLREG10	Processor/PCH ESE	Processor/PCH ESE shared region	
FLREG11	Reserved		
FLREG12	Reserved		
FLREG13	Reserved		
FLREG14	Reserved		
FLREG15	Reserved		

Processor BIOS FRACC (Flash Region Access Permission) Control

The BIOS FRACC and SFRACC (Secondary FRACC) grant permission to ESE (in Processor and the PCH) to read/write access to the BIOS/secondary BIOS region in the flash. To provide BIOS to access the PCH SPI controller's control registers FRACC and SFRACC, the PCH SPI controller's BIOS PCI configuration and MMIO registers are mapped to the processor's Direct eSPI peripheral channel generic memory range.

The processor and the PCH expose the PCH SPI controller to BIOS through DMI so that BIOS can directly program the BIOS FRACC from the PCH SPI controller to grant access to the BIOS in the PCH flash.

Processor BIOS Software Sequencing

The processor only supports SAF mode on the Direct eSPI. The BIOS software sequencing is not supported in SAF mode. To use this feature on the platform, the processor and the PCH exposes the PCH SPI controller to BIOS through DMI so that BIOS can directly program the software sequencing from PCH SPI controller to send software sequencing cycles.

26.1.4 PECCI over eSPI

The Embedded Controller (EC), can access Processor's Platform Environmental Control Interface (PECCI) and supported commands through the eSPI interface, eliminating the need for the physical PECCI pin.

PECCI over eSPI is not supported in Sx states. During these states, the EC or Baseboard Management Controller (BMC) is prohibited from sending PECCI commands via eSPI.

EC can only send PECCI requests after the VW PLT_RST# signal is de-asserted.

At S0ix state, upon receiving a PECI command, the Platform Power Management Controller (PMC) will wake the processor from its Cx state and respond with the necessary data.

26.2 Direct eSPI Signal

Signal Name	Type	Description
DIR_ESPI_IO0	I/O	Direct eSPI Signal 0 (Processor): Direct eSPI bi-directional command or data between Processor and PCH.
DIR_ESPI_IO1	I/O	Direct eSPI Signal 1 (Processor): Direct eSPI bi-directional command or data between Processor and PCH.
DIR_ESPI_IO2	I/O	Direct eSPI Signal 2 (Processor): Direct eSPI bi-directional command or data between Processor and PCH.
DIR_ESPI_IO3	I/O	Direct eSPI Signal 3 (Processor): Direct eSPI bi-directional command or data between Processor and PCH.
DIR_ESPI_CS0#	O	Direct eSPI Chip Select (Processor): Driving CS# signal low to enable eSPI PCH for the transaction.
DIR_ESPI_CLK	O	Direct eSPI Clock (Processor): eSPI clock output from the Processor to PCH.
DIR_ESPI_RESET#	O	Direct eSPI Reset (Processor): Reset signal from the Processor to PCH.
DIR_ESPI_RCLK	I	Direct eSPI Return Clock (Processor): Direct eSPI Clock from PCH to Processor.

27.0 Testability and Monitoring

This section contains information regarding the testability signals that provides access to JTAG, run control, system control, and Platform Monitoring Technology.

NOTE

JTAG Debug DBX/XDP will not function on ES1/ES2 samples without implementing onboard terminations workaround. Refer to Nova Lake JTAG On Die Termination Workaround Technical Advisory (#862716) for details and most recent update.

27.1 Signal Description

Table 95. Testability Signals

Signal Name	Type	Description
PRDY#	IOD	Probe Mode Ready: PRDY# is a processor output used by debug tools to determine processor debug readiness
PREQ#	IOD	Probe Mode Request: PREQ# is used by debug tools to request debug operation of the processor.
SOC JTAG Signals		
SOC_JTAG_TCK	I/O	Test Clock Input (TCK): The test clock input provides the clock for the JTAG test logic.
SOC_JTAG_TMS	IOD	Test Mode Select (TMS): The signal is decoded by the Test Access Port (TAP) controller to control test operations.
SOC_JTAG_TDI	IOD	Test Data Input (TDI): Serial test instructions and data are received by the test logic at TDI.
SOC_JTAG_TDO	IOD	Test Data Output (TDO): TDO is the serial output for test instructions and data from the test logic defined in this standard.
SOC_JTAG_TRST#	I/O	Test Reset (TRST): Resets the Test Access Port (TAP) logic. This signal should be driven low during power on Reset.
Breakpoint and Performance Monitor Signals		
BPM[0]	I/O	Breakpoint and Performance Monitor Signals (BPM): Outputs from the processor that indicate the status of breakpoints and programmable counters used for monitoring processor performance.
BPM[1]	I/O	Breakpoint and Performance Monitor Signals (BPM): Outputs from the processor that indicate the status of breakpoints and programmable counters used for monitoring processor performance.
Note: Refer to Depopulation Guidelines for Debug Port section in Nova Lake-S Platform Design Guide (#830661) for more information.		
Boot Halt Signal		
BOOTHALT#	IOD	Boot Halt : This signal is used for platform boot halt.

27.2 IO Signal Planes and States

Table 96. Power Planes and States for Testability Signals

Signal Name	Power Plane ²	Resistors ^{2 3}	During Reset ¹	Immediately after Reset ¹	S4/S5
SOC_JTAG_TCK	Primary	Internal Pull-down	Driven Low	Driven Low	Driven Low
SOC_JTAG_TMS	Primary	Internal Pull-Up	Driven High	Driven High	Driven High
SOC_JTAG_TDI	Primary	Internal Pull-Up	Driven High	Driven High	Driven High
SOC_JTAG_TDO	Primary	External Pull-Up	Undriven	Undriven	Undriven
SOC_JTAG_TRST#	Primary	Internal Pull-down	Driven Low	Driven Low	Driven Low
PRDY#	Primary	External Pull-Up	Driven High	Driven High	Undriven
PREQ#	Primary	External Pull-Up	Driven High	Driven High	Undriven
BPM[1:0]	Primary	External Pull-Up	Undriven	Undriven	Undriven

Notes: 1. Reset reference for primary well pins is RSMRST#.
2. For more information, refer to Open and Closed Chassis Debug (#730814) for more information related to debug technologies.
3. It is strongly recommended to reserve pads for PU\PD resistor in parallel to the internal resistor.

27.3 Debug Technologies

Refer to the Open and Closed Chassis Debug User Guide (#730814) for more information related to debug technologies.

27.3.1 Platform CrashLog

- The CrashLog feature is intended for use by system builders (OEMs) as a means to triage and perform first level debug of failures.

For more information related to CrashLog technology, refer to Intel® Client Crash Log Technology User Guide #791950.

27.3.2 Platform Monitoring Technology for Production Systems

The Platform Monitoring Technology (PMT) serves as an architectural and discoverable interface to hardware telemetry:

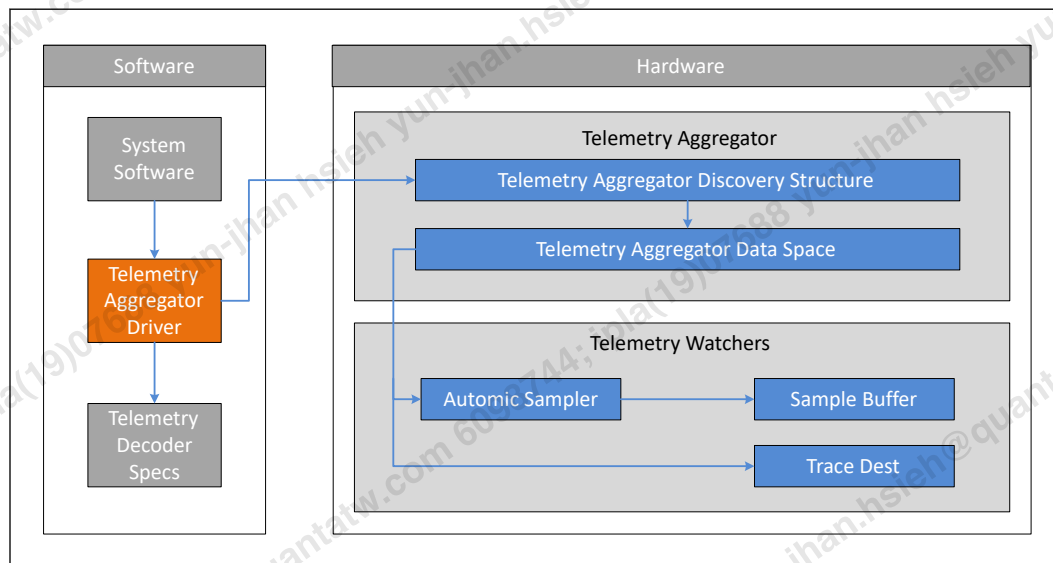
- Standardized PCIe discovery solution that enables software to discover and manage telemetry across products
- Standardized definitions for telemetry decode, including data type definitions
- Exposure of commonly used telemetry for power and performance debug including:
 - P-State status, residency and counters
 - C-State status, residency and counters
 - Energy monitoring

- Device state monitoring (for example, PCIe L1)
- Interconnect/bus bandwidth counters
- Thermal monitoring

Exposure of processor state snapshot for atomic monitoring of package power states, uninterrupted by software that reads.

The PMT is a stand alone device and feature to enumerate and access telemetry data from multiple IPs in the CPU. For more information about PMT, refer to Intel® Platform Monitoring Technology (Intel® PMT) - External Specification ([#710389](#))

Figure 32. Platform Monitoring Technology



For more information about PMT and CrashLog, refer to Intel® Platform Monitoring Technology (Intel® PMT) - External Specification ([#710389](#)).

28.0 Miscellaneous Signals

28.1 Signal Description

Table 97. Signal Descriptions

Signal Name	Type	Description
GPP_SD00/ TIME_SYNC0 or GPP_SA14/DDSP_HPDP0#/ DISP_MISC0/ TIME_SYNC0	I	Time Synchronization GPIO 0: Timed GPIO event for time synchronization for interfaces that do not support time synchronization natively.
GPP_SD01/ TIME_SYNC1 or GPP_SA15/DDSP_HPDP1#/ DISP_MISC1/ TIME_SYNC1	I	Time Synchronization GPIO 1: Timed GPIO event for time synchronization for interfaces that do not support time synchronization natively.
SKTOCC#	O	Socket Occupied: Pulled down directly in the processor package to the ground. System board designers may use this signal to determine if the processor is present for safety purposes, it helps to avoid accidentally applying power to the socket while nothing is installed into the socket. If the customers do not want to use or do not need to use the pin(PKG without socket), they can leave it floating.
EKEY	N/A	Socket Electronic Key: Used to distinguish between packages with different pin assignments. Connect this pin to the Enable signal of the first VR in the power sequence, or as appropriate, to shut down complete power to the processor/platform when an incorrect package is installed. This signal must be integrated into the platform power and sequencing control circuitry to ensure the platform does not attempt to exit the S5 power state when an incompatible package is installed in the LGA1700 socket.

28.2 Integrated Pull-Ups and Pull-Downs

None

28.3 Ground and Reserved Signals

The following are the general types of reserved (RSVD) signals and connection guidelines:

- RSVD – These signals should not be connected
- RSVD_TP – These signals should be routed to a test point
- RSVD_PULLDOWN - These signals should be pull down with resistor, refer to Nova Lake-S Platform Design Guide (#830661) for the resistors value.

Arbitrary connection of these signals to VCC, VDD2, VSS, or to any other signal (including each other) may result in component malfunction or incompatibility with future processors. Refer to the table below.

For reliable operation, always connect unused inputs or bi-directional signals to an appropriate signal level. Unused active high inputs should be connected through a resistor to ground (VSS). Unused outputs may be left unconnected however, this may interfere with some Test Access Port (TAP) functions, complicate debug probing and prevent boundary scan testing. A resistor should be used when tying bi-directional signals to power or ground. When tying any signal to power or ground the resistor can also be used for system testability. Resistor values should be within $\pm 20\%$ of the impedance of the baseboard trace, unless otherwise noted in the appropriate platform design guidelines.

Table 98. GND, RSVD, and NCTF Signals

Signal Name	Description
VSS	Ground: Processor ground node
VSS_NCTF	Non-Critical To Function: These signals are for package mechanical reliability and should be connected to VSS on the board.
RSVD	Reserved: All signals that are RSVD should not be connected on the board.
RSVD_TP	Test Point: Intel recommends to route each RSVD_TP to an accessible test point. Intel may require these test points for platform specific debug. Leaving these test points inaccessible could delay debug by Intel.
RSVD_PULLDOWN	All signals that are RSVD_PULLDOWN should be connected on the board to the ground with a resistor